NID DAT 2027 (B.Des)
Complete Preparation Package

Research findings · Theory notes · Drawing & creative preparation
1,100-question bank · 5 model papers with solutions · Current affairs ·
Revision book · Study plans

Admission cycle: NID Admissions 2027-28 · DAT Prelims: Sunday, 20 December 2026

Prepared: 7 October 2026

Practice material and predictions are not official NID content. Verify dates and rules at admissions.nid.edu.

RESEARCH FINDINGS & EXAM BLUEPRINT — NID DAT (B.Des) 2027

1. Correct admission cycle
2. What you are applying to (2027-28)
3. Eligibility & age (2027-28)
4. Exam stages & official pattern
5. Previous-paper trends (summary; full analysis in Volume 1 §6 and Volume 5)
6. Most important topics (ranked)
7. Most likely question categories [PREDICTION]
8. Major changes from older NID papers
9. Recommended preparation priorities (for ~10 weeks to 20 Dec 2026)
10. Sources consulted (full log in `99-SOURCES-AND-SELF-AUDIT.md`)

VOLUME 1 — NID DAT EXAM GUIDE (B.Des / Integrated Pathway, Cycle 2027-28)

1. The exam in one page
2. Stage-by-stage walkthrough
3. Eligibility, reservation and documents (2027-28)
4. Marking and evaluation: what examiners reward
5. The complete NID DAT syllabus [INFERRED: NID publishes none]
6. What previous NID papers teach us
7. Topic frequency & priority table
8. Exam-day strategy & time management
9. How to use this package

VOLUME 2A — THEORY NOTES: DESIGN APTITUDE, DESIGN FUNDAMENTALS & VISUAL REASONING

PART 1: DESIGN FUNDAMENTALS

1.1 Elements of Design
1.2 Principles of Design
1.3 Colour Theory
1.4 Typography
1.5 Composition
1.6 Perspective
1.7 Gestalt Principles of Visual Perception
1.8 Visual Communication: Pictograms, Signage, Posters, Logos, Packaging
1.9 Human-Centred Design (HCD) & Design Thinking
1.10 Creative Thinking Techniques
1.11 Ergonomics, Usability & Sustainability (quick reference)

PART 2: VISUAL REASONING

2.1 Mirror Images
2.2 Rotation
2.3 Paper Folding & Cutting
2.4 Cubes, Dice & Nets
2.5 Counting Figures (triangles, squares)
2.6 Pattern & Series Completion
2.7 Embedded / Hidden Figures, Silhouettes, Shadows

PART 3: DESIGN APTITUDE — OBSERVATION & SENSITIVITY

3.1 Observation
3.2 Visual Sensitivity / Aesthetic Judgement
Quick self-test for Volume 2A (answers below)

VOLUME 2B — THEORY NOTES: LOGICAL REASONING, QUANTITATIVE APTITUDE, VERBAL ABILITY

PART 1: LOGICAL REASONING

1.1 Series (number and letter)
1.2 Analogies & Classification (Odd One Out)
1.3 Coding–Decoding
1.4 Syllogisms (Deduction)
1.5 Statements, Assumptions, Inferences, Cause–Effect

1.6 Arrangements & Puzzles
1.7 Directions

PART 2: QUANTITATIVE APTITUDE (NID-relevant only)
2.1 Percentages
2.2 Ratio & Proportion; Scale
2.3 Averages
2.4 Fractions & Unit Conversion
2.5 Time, Speed, Distance
2.6 Geometry & Mensuration
2.7 Logical-Mathematical Puzzles

PART 3: VERBAL ABILITY
3.1 Vocabulary for design (learn these; they also help in writing)
3.2 Synonyms & Antonyms
3.3 Idioms (common in older GAT sections)
3.4 Reading Comprehension: method
3.5 Descriptive & Persuasive Writing (HIGH priority: used in the subjective part)

VOLUME 2C — THEORY NOTES: DESIGN-FOCUSED GENERAL AWARENESS, INDIAN ART & CULTURE, ARCHITECTURE
1. About NID (know your institute; interview favourite)
2. Indian Folk & Tribal Painting Traditions
3. Indian Textiles (high value for NID; Textile Design is one of the three main discipline groups)
4. Indian Handicrafts
5. Indian Architecture
6. World Architecture & Architects
7. Designers: Master Tables
8. Art Movements (world)
9. Famous Logos & Brands (frequently tested in MCQ years)
10. Iconic Products (often asked: "who designed / which brand")
11. Museums, Institutions & Awards
12. Sustainability, Technology & AI in Design (concepts)

VOLUME 3 — DRAWING & CREATIVE PREPARATION
1. What NID examiners are likely looking for [INFERRED]
2. Tools & materials (exam kit)
3. Core drawing skills
4. Perspective scenes (VERY HIGH priority)
5. Viewpoints & imagination
6. Human figures, faces, hands, expressions, gestures
7. Light, shadow, texture & material representation
8. Product sketching & redesign (VERY HIGH priority)
9. Visual communication
10. Sequential drawing & visual storytelling (VERY HIGH priority)
11. Memory drawing
12. Mains preparation
13. THE 110-PROMPT SUBJECTIVE BANK

VOLUME 4A — QUESTION BANK: DESIGN APTITUDE (150 Questions)
A. Elements & Principles of Design (1–25)
B. Colour (26–45)
C. Typography & Visual Communication (46–65)
D. Perception & Composition (66–80)
E. Human-Centred Design, Usability & Ergonomics (81–110)
F. Materials, Sustainability & Products (111–130)
G. Scenario-Based Design Judgement (131–150)

VOLUME 4B — QUESTION BANK: VISUAL REASONING (150 Questions)
A. Mirror & Water Images (1–20)

B. Rotation & Symmetry (21–40)
C. Paper Folding & Cutting (41–60)
D. Cubes, Dice & Nets (61–85)
E. Counting Figures (86–100)
F. Series, Matrices & Visual Analogies (101–120)
G. Orthographic Views, Shadows & Perspective (121–135)
H. Spatial Relations, Gears, Maps & Mixed (136–150)

VOLUME 4C — QUESTION BANK: LOGICAL REASONING (100 Questions)
A. Number & Letter Series (1–15)
B. Analogies (16–30)
C. Classification / Odd One Out (31–42)
D. Coding–Decoding (43–52)
E. Syllogisms & Deduction (53–65)
F. Statements, Assumptions, Inferences, Cause–Effect (66–78)
G. Arrangements, Ranking, Directions & Puzzles (79–100)

VOLUME 4D — QUESTION BANK: QUANTITATIVE / MATHEMATICAL APTITUDE (100 Questions)
A. Percentages, Ratio, Proportion (1–20)
B. Averages, Statistics & Data Interpretation (21–32)
C. Units, Paper & Digital Measures (33–45)
D. Time, Speed, Distance, Work (46–58)
E. Geometry & Mensuration (59–85)
F. Design-Context Word Problems & Puzzles (86–100)

VOLUME 4E — QUESTION BANK: VERBAL ABILITY (100 Questions)
A. Synonyms (1–15)
B. Antonyms (16–30)
C. Words in Context & One-Word Substitution (31–45)
D. Idioms & Phrases (46–55)
E. Grammar & Sentence Correction (56–70)
F. Fill in the Blanks (71–80)
G. Reading Comprehension (81–100)

VOLUME 4F — QUESTION BANK: GENERAL AWARENESS, DESIGN-FOCUSED (200 Questions)
A. National Symbols & Landmarks (1–20)
B. GI Tags & Regional Products (21–45)
C. UNESCO, Currency Motifs & Performing Arts (46–70)
D. Institutions, Museums, Awards (71–95)
E. Brands, Logos & Products (96–125)
F. Science, Technology & Digital Design (126–150)
G. Environment & Sustainability (151–175)
H. Indian Culture, Festivals & Media (176–200)

VOLUME 4G — QUESTION BANK: ART, DESIGN, ARCHITECTURE & CULTURE (200 Questions)
A. Indian Painting Traditions & Artists (1–40)
B. Indian Textiles & Crafts (41–70)
C. Indian Architecture (71–100)
D. World Architecture & Architects (101–130)
E. Art Movements & World Artists (131–160)
F. Designers: Graphic, Typography, Industrial, Fashion (161–200)

VOLUME 5 — MOST EXPECTED QUESTION TYPES FOR THE UPCOMING NID DAT (20 Dec 2026)
1. Two scenarios for the paper
2. Tier A — VERY HIGH probability patterns
3. Tier B — HIGH probability patterns
4. Tier C — MODERATE probability patterns (mainly if the objective Part I returns)
5. Expected "twists" to be ready for
6. A high-yield 20-question drill set (do one a day for 20 days)
7. Pre-solved model approaches for 5 Tier-A questions

VOLUME 6 — MODEL PAPER 1 (Difficulty: Easy → Moderate)

INSTRUCTIONS
PART I — OBJECTIVE (30 marks)
PART II — SUBJECTIVE (70 marks)
ANSWER KEY — PART I
DETAILED SOLUTIONS — PART I
DETAILED SOLUTIONS — PART II (model approaches + marking rubrics)
SCORING INTERPRETATION — MODEL PAPER 1

VOLUME 7 — MODEL PAPER 2 (Difficulty: Moderate)

INSTRUCTIONS
PART I — OBJECTIVE (30 marks)
PART II — SUBJECTIVE (70 marks)
ANSWER KEY — PART I
DETAILED SOLUTIONS — PART I
DETAILED SOLUTIONS — PART II
SCORING INTERPRETATION — MODEL PAPER 2

VOLUME 8 — MODEL PAPER 3 (Difficulty: Moderate → Difficult)

INSTRUCTIONS
PART I — OBJECTIVE (30 marks)
PART II — SUBJECTIVE (70 marks)
ANSWER KEY — PART I
DETAILED SOLUTIONS — PART I
DETAILED SOLUTIONS — PART II
SCORING INTERPRETATION — MODEL PAPER 3

VOLUME 9 — MODEL PAPER 4 (Difficulty: Difficult) · 2026-STYLE ALL-SUBJECTIVE FORMAT

INSTRUCTIONS
QUESTIONS
ANSWER KEY (Compact: what must appear for full credit)
DETAILED SOLUTIONS & MODEL APPROACHES
SCORING INTERPRETATION — MODEL PAPER 4

VOLUME 10 — MODEL PAPER 5 · FULL MOCK EXAM SIMULATION (Toughest Realistic Paper)

MOCK-EXAM PROTOCOL (follow it exactly)
INSTRUCTIONS (as printed on the paper)
SUGGESTED SECTION TIMING
ANSWER SHEET — PART I (copy onto paper or print)
ANSWER SHEET — PART II (one A4 per question)
PART I — OBJECTIVE (30 marks)
PART II — SUBJECTIVE (70 marks)
ANSWER KEY — PART I
DETAILED SOLUTIONS — PART I
DETAILED SOLUTIONS — PART II (model approaches + rubrics)
SELF-EVALUATION SYSTEM
SCORING INTERPRETATION — FULL MOCK (Model Paper 5)

VOLUME 11 — DESIGN-ORIENTED CURRENT AFFAIRS (≈ mid-2025 to Oct 2026) + 100 MCQs

PART A — EVENT FILE (each entry gives: what happened, when, who, why it matters, design relevance, and a possible question)
PART B — HOW TO USE CURRENT AFFAIRS IN YOUR ANSWERS
PART C — CONFIRMED BACKGROUND FACTS OFTEN LINKED TO NEWS (older, stable)
PART D — 100 CURRENT-AFFAIRS MCQs
PART E — YOUR UPDATE LOG (fill in weekly until the exam)

VOLUME 12 — NID DAT LAST-MINUTE REVISION BOOK

1. ONE-PAGE EXAM STRATEGY (read this the night before)
2. VISUAL & DESIGN PRINCIPLES (core list)
3. COLOUR (essentials)

4. TYPOGRAPHY & VISUAL COMMUNICATION
5. PERSPECTIVE & DRAWING RULES
6. REASONING FORMULAS & RULES
7. INDIAN ART, CRAFTS & TEXTILES (rapid list)
8. ARCHITECTURE (rapid list)
9. DESIGNERS & MOVEMENTS (rapid list)
10. LOGOS & PRODUCTS (rapid list)
11. KEY TERMS (one line each)
12. FLASHCARDS (160)
13. TOP 100 NID DAT MISTAKES STUDENTS MAKE (and the fix for each)

VOLUME 13 — STUDY PLANS, DAILY ROUTINES, REVISION SYSTEM & ERROR LOG

1. THE DATED 10-WEEK CALENDAR (7 Oct → 20 Dec 2026)
2. 30-DAY PLAN (late-start or compressed version) · 5–6 hours/day
3. 60-DAY PLAN (recommended core) · 4–5 hours/day
4. 90-DAY PLAN (for the Mains phase or the next cycle) · 3–4 hours/day
5. DAILY ROUTINES
6. ERROR LOG SYSTEM
7. 15-DAY REVISION PLAN (use 5–20 Dec, or any 15-day window)
8. 7-DAY REVISION PLAN (if you only have one week)
9. FINAL 7-DAY STRATEGY (13–19 Dec) & FINAL 24-HOUR STRATEGY (19–20 Dec)

SOURCES USED, RESEARCH LOG & QUALITY SELF-AUDIT

A. Research method & limitations (read this)
B. SOURCES
C. QUALITY-CONTROL SELF-AUDIT

RESEARCH FINDINGS & EXAM BLUEPRINT — NID DAT (B.Des) 2027

Research date: 7 October 2026
Prepared for: a Class 12 student taking the NID B.Des Design Aptitude Test (DAT) independently

> **How to read the labels used throughout this package**
>
> - **[OFFICIAL]**: stated in an NID document (Admissions Handbook, sample paper, notification). Where I could not open the NID PDF myself, I say what I relied on.
> - **[REPORTED]**: reported by several education news sites or coaching analyses after the exam, usually from candidates' memory. Probably right, but not confirmed by NID.
> - **[TREND-BASED / INFERRED]**: my own analysis of patterns across years.
> - **[PREDICTION]**: an informed guess. It is never a guarantee.
>
> **Access note (please read):** my research environment could not open `admissions.nid.edu` directly; the network proxy blocked it. Every "official" fact below comes from search-engine extracts of NID's own PDFs (the 2026-27 B.Des Handbook and the 2027-28 B.Des / Integrated Pathway Handbook), cross-checked against at least two independent education portals. **Before you rely on any date, fee or rule, check it once on https://admissions.nid.edu. NID highlights all changes there.**

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1. Correct admission cycle

Item	Finding	Status
Today's date	7 Oct 2026	—
Cycle your brother is applying for	**NID Admissions 2027-28**, called "NID DAT 2027", for students joining in July/Aug 2027	[OFFICIAL] Handbook title "Admissions 2027-28 Handbook"
Why not "NID 2026"?	DAT 2026 (the 2026-27 cycle) is **finished**. Prelims were on 21 Dec 2025, Studio Test in April 2026 and In-Person Test in May 2026.	[REPORTED]
Application opened	10 Sept 2026	[OFFICIAL]
Last date to apply (online only)	**30 Nov 2026, 11:59 pm**	[OFFICIAL]
Correction window	1–3 Dec 2026	[REPORTED]
Prelims admit card	~10 Dec 2026	[REPORTED]
DAT Prelims	**Sunday, 20 December 2026**, pen-and-paper, 3 hours (last cycle ran 10:00 am–1:00 pm)	[OFFICIAL] date; timing [REPORTED]
Prelims result (B.Des)	~16 Mar 2027 (tentative)	[REPORTED]
DAT Mains	Dates to be announced after Prelims; last cycle had the Studio Test in April and the In-Person Test in May	[REPORTED]

2. What you are applying to (2027-28)

Two undergraduate routes are open through the same DAT:

1. **4-year B.Des**: at NID Ahmedabad and at the NIDs in Andhra Pradesh (Amaravati), Assam (Jorhat), Haryana (Kurukshetra) and Madhya Pradesh (Bhopal). [REPORTED, consistent with handbook extracts]
2. **NEW: 5.5-year "Professional Education Master of Design (Integrated Pathway)"**: 11 semesters, 15 sector-specific disciplines grouped under 5 design streams, at the Ahmedabad, Gandhinagar and Bengaluru campuses. Reports cite about 135 seats (15 disciplines × 9). [OFFICIAL: programme named in the 2027-28 Handbook title; seat figure REPORTED]

Seat matrix in the last cycle (2026-27), for scale: **428 B.Des seats**. Ahmedabad had 128, and AP, Assam, Haryana and MP had 75 each (25 each in Communication, Industrial, and Textile & Apparel Design), plus up to 15% supernumerary overseas seats. [OFFICIAL 2026-27; the 2027-28 matrix may differ, so check the new Handbook]

Context: the Union Budget 2026-27 proposed a **new NID in eastern India**, to be set up through a "challenge route". [REPORTED, govt budget coverage]

3. Eligibility & age (2027-28)

Category	Born on or after	Status
General / Gen-EWS (Indian)	**1 July 2006**	[REPORTED from Handbook extracts]
OBC-NCL, SC, ST	1 July 2003 (3-year relaxation)	[REPORTED]
PwD	1 July 2001 (5-year relaxation)	[REPORTED]
Overseas / foreign nationals	1 July 2006	[REPORTED]

- **Qualification:** passed, or appearing in 2026-27 for, Class 12 (10+2) in **any stream**: Science, Commerce, Arts or Vocational. No minimum percentage has been reported for B.Des. [REPORTED]
- **Application fee (B.Des):** ₹3,000 for General/EWS/OBC-NCL and ₹1,500 for SC/ST. One portal reports a lower fee for female candidates; verify this. [REPORTED: confirm on portal]

4. Exam stages & official pattern

Stage 1: DAT Prelims (shortlisting only)

What the official Handbook text says [OFFICIAL, per handbook extracts for 2026-27 and quoted again for 2027-28]:

Part	Questions	Marks
Part I: objective	Q1–14 at 1 mark each (14) + Q15–22 at 2 marks each (16)	**30**
Part II: subjective	Q23 = 15, Q24 = 20, Q25 = 15, Q26 = 20	**70**
Total	26 questions, **3 hours**	**100**, **no negative marking**

- **The Prelims score is used only to shortlist for Mains. It does NOT carry weightage into the final merit list.** [OFFICIAL: "Prelims score will not be carried forward as any weightage to the DAT Mains stage"]
- Last cycle shortlisted roughly **1.5 candidates per seat** for B.Des Mains. [REPORTED]

What actually happened in Dec 2025 (DAT 2026) [REPORTED by Careers360, Shiksha, GetMyUni and others]:

- **There was no objective / General Aptitude section.** The paper was fully subjective.
- It had **6 main questions with 2–3 sub-parts each, about 14 drawing/writing tasks** in 3 hours. Students found it lengthy rather than conceptually hard ("moderate to difficult").
- Memory-based questions include:
  - a sequential drawing of helping an elderly person who falls during a morning walk
  - sketching the items to carry when travelling with your grandmother
  - drawing a product you love and writing about it in 200 words
  - illustrating shikanji / thandai (an Indian drink)
  - designing 5 signboards for narrow streets
  - also reported: a three-storey scene, completing an everyday object, redesigning a product, a perspective drawing, a process drawing

Bottom line for 2027 [INFERRED]: the Handbook's printed scheme (22 objective + 4 subjective) and the real 2026 paper (all subjective) do **not** match. NID has the right to change the paper format. **Prepare for both.** The safe strategy is 75–80% of prep time on drawing, ideation and design thinking, with the objective skills (visual reasoning, design GK) kept warm. Those skills also feed the subjective answers: you cannot draw a good "Warli-inspired signboard" without knowing Warli.

Stage 2: DAT Mains (decides final merit)

Component	What it involves	Weight (last cycle)
Studio Sensitivity Test	Hands-on: sketching, doodling, material handling, 3D model-making with given materials (clay, paper, wire, found objects), sometimes audio-visual prompts. 2026: ~2.5 hours (9:30–12:00).	**60%** [REPORTED]
In-Person Sensitivity Test	Face-to-face interaction / interview-style assessment of awareness, communication, motivation and portfolio-type thinking	**40%** [REPORTED]

- For 2027-28, one portal says the "exact split [is] to be announced". Treat 60:40 as the most recent known split, not a confirmed one. [REPORTED]
- **Final merit = 100% Mains.** [OFFICIAL in principle, from the "Prelims not carried forward" rule]

Official syllabus

NID does not publish a topic-wise syllabus. [REPORTED by several portals; consistent with the Handbooks] The Handbook describes the test only in broad terms: design aptitude, creativity, visualisation, observation and problem-solving. Every "syllabus" in this package is therefore **[INFERRED]** from papers.

Cutoffs

NID does not publish official subject-wise cutoffs in a form I could verify. Coaching estimates for 2026 (General ~50–55, OBC ~45, SC/ST ~40 out of 100) are **[UNVERIFIED]** and are not used as targets anywhere in this package.

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5. Previous-paper trends (summary; full analysis in Volume 1 §6 and Volume 5)

Year (exam held)	Structure	Character	Source status
DAT 2019–2022	MCQ-heavy Part A (GK, visual reasoning, design awareness, some maths) + drawing Part B	GAT and CAT roughly balanced	[REPORTED]
DAT 2024 (Dec 2023)	31 Qs: Section 1 = Q1–27 MCQ (40 marks); Section 2 = Q28–31 subjective (60 marks)	easy–moderate; visual reasoning and design awareness	[REPORTED]
DAT 2025 (Dec 2024)	22 objective + 4 subjective = 100	CAT gained weight; logo identification; a poster for an orange-juice corner from a **crow's point of view**; a **neem branch overhanging two neighbours** problem; a train-related problem question	[REPORTED]
DAT 2026 (Dec 2025)	**6 subjective Qs (~14 tasks); no MCQs**	lengthy; empathy, everyday Indian life, sequence, signage, product love, food illustration	[REPORTED]
DAT 2027 (20 Dec 2026)	**Unknown.** Handbook still prints the 22+4 scheme.	Prepare for a subjective-heavy paper	[PREDICTION]

Clear long-run trend [TREND-BASED]: the paper moves away from recalling facts and towards **observation of everyday Indian life + empathy + problem-solving, communicated through quick, clear drawings and short writing.**

6. Most important topics (ranked)

1. **Scene drawing with people in perspective** (markets, streets, homes, stations, multi-storey views)
2. **Sequential / storyboard drawing** (3–6 frames showing an event or process, often with empathy: elderly people, children, accidents, help)
3. **Product redesign / problem-solving** (identify user problems, then sketch the solution with labels)
4. **Visual communication** (posters, signage, pictograms, symbols, logos, packaging) with colour and typography

5. **Memory and observation drawing** (objects, food, vehicles, an Indian context)
6. **Creative / lateral thinking** (unusual points of view such as a crow's eye or an ant's eye; "what if"; combining objects)
7. **Short descriptive or persuasive writing** (100–200 words about a product or experience)
8. **Visual reasoning** (folding, cubes, mirror images, patterns, counting): needed if Part I returns
9. **Design & cultural awareness** (Indian crafts, textiles, art forms, architecture, logos, designers)
10. **Current design affairs**: the low-probability MCQ topic, but useful for the interview

7. Most likely question categories [PREDICTION]

- **Tier A (very high):** a scene with people; a sequential story with an empathy angle; redesign or solve an everyday problem; a poster or signage set; drawing from memory plus a description
- **Tier B (high):** pictograms or symbols; packaging; unusual viewpoints; object completion or transformation; colour-scheme application; writing 150–200 words
- **Tier C (moderate):** an MCQ block (visual reasoning, logos, crafts, designers); maths-lite puzzles; current affairs

8. Major changes from older NID papers

1. **Objective GK/aptitude shrank, then vanished (in 2026).**
2. **Prelims lost its weightage** in the final merit; it is now only a gate.
3. **Questions became longer and multi-part.** Time management now decides results.
4. **Indian everyday context dominates:** grandparents, street food, narrow lanes, neighbours, trains.
5. **Writing appears alongside drawing** (justify, describe, label).
6. **A new 5.5-year Integrated Pathway** joins B.Des in 2027-28, using the same Prelims.

9. Recommended preparation priorities (for ~10 weeks to 20 Dec 2026)

Priority	Share of time	What
1	40%	Daily drawing: people, perspective scenes, objects from memory
2	20%	Ideation and problem-solving drills (timed), redesigns, posters, signage
3	15%	Full timed papers (Vol 6–10) plus self-evaluation
4	10%	Visual reasoning MCQs (Vol 4B)
5	10%	Design / culture GK (Vol 2C, 4F–G, 12)
6	5%	Current affairs (Vol 11)

10. Sources consulted (full log in `99-SOURCES-AND-SELF-AUDIT.md`)

- **Official:** NID Admissions portal https://admissions.nid.edu ; *B.Des Admissions Handbook 2026-27* (`admissions.nid.edu/NIDA2026/download/BDes_AdmissionsHandbook2026-27.pdf`) ; *Admissions 2027-28 Handbook: B.Des & 5.5-yr M.Des Integrated Pathway* (`admissions.nid.edu/NIDA2027/download/BDES_IntegratedPathway_AdmissionsHandbook2027_28.pdf`) ; *B.Des DAT Prelims Sample Test Paper* (`admissions.nid.edu/NIDA2026/download/SamplePaper_DATPrelims_BDes.pdf`)
- **Previous-paper analyses:** Careers360 (NID DAT 2026 Prelims Question Paper with Solution, Parts A/B); Shiksha (NID DAT 2026 B.Des Exam Analysis); GetMyUni (2026 analysis); Careers360 and PW (2025 analysis); Careers360 and Shiksha (2024)
- **Dates & eligibility:** PW Live (NID DAT Notification 2027), Shiksha, Toprankers, Careers360, CareerIndia

- **Current affairs:** La Biennale di Venezia, Dezeen, Domus, ArchDaily, UNESCO-based coverage, PIB / NeGD (AI Impact Summit), Serpentine Galleries

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NEXT VOLUME: Volume 1 — NID DAT Exam Guide (`01-VOL1-EXAM-GUIDE.md`)

VOLUME 1 — NID DAT EXAM GUIDE (B.Des / Integrated Pathway, Cycle 2027-28)

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1. The exam in one page

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APPLY (10 Sep – 30 Nov 2026, online)
        │
        ▼
DAT PRELIMS  ── Sun 20 Dec 2026 · 3 hrs · 100 marks · pen & paper · no negative marking
        │        (Handbook: 22 objective + 4 subjective; 2026 actual: 6 subjective Qs)
        │        → SHORTLISTING ONLY, score not carried forward
        ▼
DAT MAINS (Apr–May 2027, expected)
   ├── Studio Sensitivity Test  (hands-on: sketch, model, material)   ~60%
   └── In-Person Sensitivity Test (interaction / interview)            ~40%
        │
        ▼
FINAL MERIT (Mains only) → seat allotment by campus / discipline preference
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Labels: dates and the "not carried forward" rule are [OFFICIAL]; the 60/40 split and the 2026 paper shape are [REPORTED].

2. Stage-by-stage walkthrough

2.1 Application

- Apply online at admissions.nid.edu. You need a photo, a signature, the Class 10 certificate (for date of birth), a category certificate if applicable, and a PwD certificate if applicable.
- Choose your programme(s) and campus preferences carefully. **Read the 2027-28 Handbook's rules on whether you can apply to both the B.Des and the Integrated Pathway, and how preferences are filled.** [Verify on portal]
- Choose a test city you can reach easily. Prelims is offline.

2.2 DAT Prelims (20 Dec 2026)

Materials usually allowed [REPORTED from past instructions; verify on admit card]: pencils (HB, 2B, 4B, 6B), eraser, sharpener, scale, colour pencils, crayons or oil pastels, sketch pens, and a black pen. Water colours or poster colours have been permitted in some years. **Always follow the admit card.** Blank papers are generally not allowed; you work in the answer booklet.

Two formats to be ready for:

Format A: Handbook scheme	Format B: 2026 actual
Part I: 22 objective (30 marks)	No objective part
Part II: 4 subjective (70 marks)	6 subjective questions, ~14 sub-tasks (100 marks)
Time split: ~25 min Part I, ~150 min Part II	~12 min per sub-task, ~10 min buffer

2.3 DAT Mains: Studio Sensitivity Test

What candidates have reported across years [REPORTED]:

- Make a 3D model from clay, paper, card, thread, wire, foam or found objects on a theme (e.g. "movement", "a creature from your imagination", "a toy for a blind child").
- Sketch an object placed in front of you; observe and draw it from different views.
- Respond to an audio clip or video by drawing or writing.
- Material exploration: "Make as many textures as you can with this paper."
- Group or individual tasks that show the process as well as the result.

What it tests: hand skill, material sensitivity, 3D thinking, originality, and the ability to explain your idea.

2.4 DAT Mains: In-Person Sensitivity Test

- An interaction with a faculty panel. Common themes [REPORTED]:
  - why design
  - observations from your hometown
  - a product you admire and why
  - crafts of your state
  - current design news
  - your hobbies
  - your Studio Test model
- Preparation is covered in Volume 3 §12 and Volume 11.

3. Eligibility, reservation and documents (2027-28)

Requirement	Detail	Status
Education	Class 12, any stream, passed or appearing in 2027	[REPORTED]
Age, General/EWS	born on or after 1 Jul 2006	[REPORTED]
Age, OBC-NCL/SC/ST	on or after 1 Jul 2003	[REPORTED]
Age, PwD	on or after 1 Jul 2001	[REPORTED]
Reservation	As per Govt of India norms (SC 15%, ST 7.5%, OBC-NCL 27%, EWS 10%, PwD 5% horizontal). Some NIDs have state-domicile seats for their home state; check the Handbook for each campus.	[Govt norms; campus domicile REPORTED]
Overseas seats	supernumerary, up to 15%	[OFFICIAL 2026-27]
Fee	₹3,000 (Gen/EWS/OBC) · ₹1,500 (SC/ST)	[REPORTED]

4. Marking and evaluation: what examiners reward

NID has not published a rubric. The criteria below are **[INFERRED]** from NID's stated aims (design aptitude, creativity, visualisation, observation, problem-solving) and from patterns across years.

Criterion	What it looks like on paper	Common way to lose marks
Understanding the question	You answer exactly what is asked: the right number of frames, signboards, the viewpoint	Drawing 3 signboards when 5 are asked; ignoring "write 200 words"
Idea quality / originality	A fresh but sensible interpretation	The first cliché idea (e.g. "save trees" with a crying tree)
Observation	Believable details from real life: the bucket under the tap, the scooter's mirror, slippers outside the door	Generic, empty scenes
Communication	A viewer understands the drawing in about 5 seconds; labels and arrows where they help	Cluttered or ambiguous drawings
Composition	Clear focal point, use of the whole space, varied scale	A tiny drawing in the centre of a huge box
Proportion & perspective	People sized correctly against doors and vehicles; a consistent horizon	Floating people, tilted floors
Empathy / user focus	The solution fits the user (elderly people, children, vendors)	"Cool" ideas that ignore the user
Craft (secondary)	Clean lines, purposeful shading and colour	Over-rendering one corner and leaving the rest unfinished
Completion	Every task attempted	Leaving the last question blank

Photorealism is NOT required. A clear, confident, well-observed sketch with a good idea scores higher than a beautiful drawing of a weak idea. [INFERRED from consistent NID messaging and the nature of the questions]

5. The complete NID DAT syllabus [INFERRED: NID publishes none]

Priority key: **VH** = very high, **H** = high, **M** = medium, **L** = low (for the Dec 2026 Prelims).

A. Design Aptitude (VH)

- **Observation**: noticing details, proportion and context. VH
- **Visual perception**: figure–ground, gestalt (proximity, similarity, closure, continuity), illusions. H
- **Visual reasoning**: reading meaning from images. H
- **Design thinking**: empathise → define → ideate → prototype → test. VH
- **Problem solving**: finding the real problem, constraints, trade-offs. VH
- **Creative thinking**: fluency, flexibility, originality, elaboration. VH
- **Lateral thinking**: unusual viewpoints, reversal, random stimulus. VH
- **Divergent vs convergent thinking**: many ideas, then select one. H
- **Functional thinking**: form follows function; affordances. H
- **User-centred thinking**: personas, accessibility, inclusivity. VH
- **Innovation**: incremental vs radical; frugal design (jugaad, used critically). M
- **Aesthetics**: harmony, proportion, colour, style. H
- **Usability**: learnability, efficiency, error prevention, feedback. H
- **Ergonomics**: anthropometry, reach, posture, grip. M
- **Sustainability**: reduce, reuse, recycle, repair; life-cycle; materials. H

B. Visual Reasoning (H if Part I appears; M otherwise)

- **Patterns, series, figure completion, odd one out, visual analogy**: H
- **Mirror and water images; rotation; reflection**: H
- **Paper folding, cutting and unfolding**: H
- **2D↔3D: cubes, nets, dice, isometric↔orthographic views**: H
- **Spatial relations; hidden and embedded figures; counting figures**: M
- **Perspective, shadows, silhouettes**: H (these also feed drawing)

C. Quantitative Aptitude (L–M)

Basic arithmetic, percentages, ratio, averages, fractions, units, time and distance, area, perimeter, volume, number series, logical puzzles.
How much maths does NID need? Very little. Class 8–10 level is enough. In MCQ years there were usually only 1–4 maths-flavoured items, mostly geometry, area or proportion embedded in a visual context (e.g. "how many tiles cover this floor?"). **Do not** prepare CAT/JEE-level maths. Spend at most 5% of your time here.

D. Logical Reasoning (M)

Sequences, arrangements, analogies, classification, coding-decoding, syllogisms, statement–assumption, cause–effect, visual logic, puzzles.

E. Verbal Ability (L–M)

Vocabulary, comprehension, idioms, interpreting statements, and **descriptive writing** (H for the subjective part).

F. General Awareness, design-focused (M)

Indian crafts and textiles, folk and tribal art, architecture, designers, artists, logos, iconic products, design movements, museums, institutions, awards, UNESCO / GI, sustainability, AI and design.

G. Current Affairs (L for Prelims, M for the interview)

The last 12–18 months of design, architecture, art, tech, AI, environment and culture.

H. Drawing & Visualisation (VH): the core of the paper

Line, shape, form, proportion, perspective (1-, 2- and 3-point), human figures and gestures, memory drawing, scenes, storyboards, product sketching, light and shadow, texture, materials, colour application, lettering.

6. What previous NID papers teach us

6.1 Pattern evolution: older → recent

```
2016–2019  ──►  MCQ-heavy GAT (GK, maths, logic, English, design awareness)
                + 3–5 drawing questions (memory drawing, poster, symbol)
2020–2022  ──►  Fewer pure-GK MCQs; more visual reasoning & design awareness;
                drawing still ~60–70 marks; "what if" + empathy prompts rise
2023–2025  ──►  22–27 objective items (30–40 marks), heavily visual (logos,
                spatial, observation); 4 big subjective tasks (60–70 marks)
                Ex (2025): poster from a crow's viewpoint; neem-branch neighbour dispute
2026       ──►  NO objective part. 6 long subjective questions (~14 tasks):
                empathy sequences (elderly fall), travel kit for grandmother,
                "product you love" + 200 words, Indian drink illustration,
                5 signboards for narrow streets, scene/perspective, redesign
2027       ──►  [PREDICTION] Subjective-dominant. Objective part may or may
                not return. Everyday India + empathy + communication.
```

All year-specific details are [REPORTED] (memory-based). The arrows describe a [TREND-BASED] reading.

6.2 Recurring question types, with an analysis of each

1. Scene drawing with people (markets, stations, homes, festivals)

- *Frequency:* almost every year. [TREND-BASED]
- *How it changed:* "draw a railway platform" has become "draw a three-storey scene", "a narrow street with signboards" or "a view from a specific viewpoint".
- *Skill tested:* perspective, proportion, observation, composition, storytelling.
- *Common misunderstanding:* students think it is about drawing skill alone. It is about **believable life**: activity, interaction, details.
- *How to prepare:* 1 scene a day from observation or memory; learn 1- and 2-point perspective; draw people in 30-second gestures.

2. Sequential drawing / storyboard

- *Frequency:* very frequent, and growing.
- *How it changed:* from neutral processes ("making tea") to empathy and emotion ("helping an elderly person who falls").
- *Skill:* narrative, continuity (same characters and setting across frames), camera angles, emotion.
- *Misunderstanding:* drawing unrelated frames, or changing the character's look between frames.
- *Prepare:* 3/4/6-frame stories twice a week. Vary the shots (wide, medium, close-up).

3. Product redesign / problem-solving

- *Frequency:* very frequent. Examples: the neem branch dispute (2025), a train-related problem (2025), redesigning a product (2026).
- *Skill:* problem identification, user empathy, feasible ideas, explaining with labelled drawings.
- *Misunderstanding:* jumping to a gadget. Examiners want **problem → insight → solution**.
- *Prepare:* the "Problem–User–Constraint–Idea–Sketch" drill (Vol 3 §8).

4. Visual communication: posters, signage, symbols, logos, packaging

- *Frequency:* very frequent. Examples: an orange-juice poster from a crow's viewpoint (2025); 5 signboards for narrow streets (2026).
- *Skill:* hierarchy, colour, typography, simplification, clarity at a distance.
- *Misunderstanding:* adding too much text or detail. Signage must work in 2 seconds.
- *Prepare:* learn pictogram rules (Vol 2A §6) and design 3 posters or symbols a week.

5. Memory drawing + writing

- *Frequency:* frequent. Example: "a product you love + 200 words" (2026).
- *Skill:* recall accuracy, descriptive language, reasoning about design qualities.
- *Prepare:* draw 5 everyday objects from memory each week, then check them against the real thing.

6. Unusual viewpoints / imagination ("what if", "if you were an ant")

- *Frequency:* frequent.
- *Skill:* lateral thinking, scale manipulation, worm's-eye and bird's-eye perspective.
- *Prepare:* the viewpoint drills in Vol 3 §5.

7. Visual reasoning MCQs: folding, cubes, mirror images, counting, patterns

- *Frequency:* every year with Part I (to 2025); absent in 2026.
- *Skill:* spatial visualisation.
- *Prepare:* Vol 4B. Practise until accuracy is above 85%.

8. Logos / design GK MCQs

- *Frequency:* frequent in Part I years (logo identification in 2025).
- *Prepare:* the logo table in Vol 2C and the flashcards in Vol 12.

9. Colour application

- *Frequency:* moderate; often embedded ("use only warm colours", "monochrome").
- *Prepare:* the colour theory section of Vol 2A, plus colouring practice with limited palettes.

10. Creative writing / captions / slogans

- *Frequency:* moderate and rising.
- *Prepare:* timed 150–200-word descriptions; taglines of 3–6 words.

7. Topic frequency & priority table

Frequency is a [TREND-BASED] estimate across roughly 2016–2026 B.Des Prelims from reported memory-based papers. "Expected" is a [PREDICTION] for 20 Dec 2026.

Topic	Frequency	Recent trend	Difficulty	Importance	Expected for DAT 2027
Scene / environment drawing with people	Very high	Rising (multi-level, specific viewpoints)	M–H	Core	**VERY HIGH PRIORITY**
Sequential / storyboard	High	Rising, with an empathy angle	M	Core	**VERY HIGH PRIORITY**
Product redesign / problem-solving	Very high	Rising	M–H	Core	**VERY HIGH PRIORITY**
Poster / signage / pictogram	High	Stable–rising	M	High	**VERY HIGH PRIORITY**
Memory / object drawing	High	Stable	E–M	High	**HIGH PRIORITY**
Descriptive writing (100–200 words)	Medium	Rising	E–M	Medium	**HIGH PRIORITY**
Unusual viewpoint / imagination	Medium–high	Stable	M	High	**HIGH PRIORITY**
Logo / symbol design	Medium	Stable	M	Medium	**HIGH PRIORITY**

Topic	Frequency	Recent trend	Difficulty	Importance	Expected for DAT 2027
Packaging design	Medium	Stable	M	Medium	**MEDIUM PRIORITY**
Colour theory (applied)	Medium	Embedded in drawing	E	Medium	**MEDIUM PRIORITY**
Visual reasoning MCQ (folding, cubes, mirror)	High until 2025	Absent in 2026	M	Conditional	**MEDIUM PRIORITY**
Observation / hidden figures / counting	Medium	Absent in 2026	M	Conditional	**MEDIUM PRIORITY**
Logo identification / brand GK	Medium	Absent in 2026	E	Conditional	**MEDIUM PRIORITY**
Indian crafts / textiles / art GK	Medium	Feeds subjective work	E–M	Medium	**MEDIUM PRIORITY**
Architecture / designers GK	Low–medium	Declining in Prelims	E–M	Interview	**LOW PRIORITY** (Prelims)
Logical reasoning (verbal)	Low–medium	Declining	E–M	Low	**LOW PRIORITY**
Quantitative aptitude	Low	Declining	E	Low	**LOW PRIORITY**
English grammar / vocabulary	Low	Declining	E	Low	**LOW PRIORITY**
Current affairs (design)	Low	Interview-relevant	E	Low (Prelims)	**LOW PRIORITY** (Prelims) / MEDIUM (Mains)
Studio: 3D model / material	Mains every year	Stable	M–H	Mains core	**VERY HIGH PRIORITY** (after Prelims)

8. Exam-day strategy & time management

8.1 Before the exam

- Sleep 7–8 hours. Pack the admit card, photo ID, and a pencil pouch with spares: 2 HB, 2B, 4B, 6B, eraser, kneaded eraser, sharpener, 30 cm scale, 24-shade colour pencils, oil pastels, sketch pens and 2 black fine-liners. Bring a watch if allowed.
- Reach the centre 45 minutes early.

8.2 The first 10 minutes (both formats)

1. Read the **whole paper** once. Circle the counts ("5 signboards", "4 frames", "200 words") and the constraints ("only 3 colours", "bird's-eye view").
2. Write a time budget next to each question (marks ÷ 100 × 165 minutes).
3. Note 2–3 quick idea keywords beside each subjective question. Your brain keeps working on them in the background.

8.3 Time budgets

Format A (22 objective + 4 subjective, 180 min):

Block	Minutes
Read the paper	5
Part I, Q1–22 (30 marks)	25 (skip any item over 90 sec, return later)
Q23 (15)	28
Q24 (20)	38
Q25 (15)	28
Q26 (20)	38
Buffer: finishing, labels, unanswered MCQs	18

Format B (6 questions, ~14 sub-tasks, 100 marks):

- Minutes per sub-task ≈ 165 ÷ number of sub-tasks (≈ 12 min each for 14).
- Use the **70% rule**: once a drawing is 70% done, move on. Come back to polish only if time remains.
- **Never leave a task blank.** A 3-minute rough sketch with labels earns marks; a blank earns zero.

8.4 Guessing policy

No negative marking (officially), so **answer every MCQ**. Eliminate options first, then guess.

8.5 The five golden rules for subjective answers

1. **Answer the question asked**: count, viewpoint, medium, user.
2. **Idea first, rendering second.** Spend 2–3 minutes thinking before drawing.
3. **Fill the box with purpose**: a clear focal point and a sensible scale.
4. **Label smartly**: short labels and arrows for mechanisms and materials. Write a 1–2 line note explaining the idea if allowed.
5. **Finish everything**, then add colour or shading to your strongest piece.

9. How to use this package

Volume	File	Use it when
0	`00-RESEARCH-FINDINGS-AND-EXAM-BLUEPRINT.md`	First, for orientation
1	this file	To understand the exam
2	`02A`, `02B`, `02C` theory notes	Weeks 1–6, daily topic study
3	`03-VOL3-DRAWING-CREATIVE.md`	Every day: drawing drills + 100+ prompts
4	`04A`–`04G` question bank	Daily practice by topic
5	`05-VOL5-EXPECTED-QUESTIONS.md`	Weeks 4+
6–10	model papers 1–5	One per week from week 4 (timed)
11	current affairs	Weekly, plus before the interview
12	revision book + flashcards + top-100 mistakes	Daily 15 min; final week
13	study plans + error log	Day 1, to pick your plan
99	sources + self-audit	Verification

NEXT VOLUME: Volume 2A — Complete Theory Notes: Design Aptitude, Design Fundamentals & Visual Reasoning (`02A-VOL2-THEORY-DESIGN-AND-VISUAL.md`)

VOLUME 2A — THEORY NOTES: DESIGN APTITUDE, DESIGN FUNDAMENTALS & VISUAL REASONING

Each topic follows the same 12-point format:
Concept · Simple explanation · Key facts · Rules/principles · Common traps · How NID tests it · Example · Practice Qs · Answers · Explanations · Difficulty · Priority

Practice-question answers are given right below each set, so you can self-check immediately.

PART 1: DESIGN FUNDAMENTALS

1.1 Elements of Design

Concept: the elements are the basic "ingredients" every visual is built from. There are eight: **point, line, shape, form, colour, texture, space, value**.

Simple explanation & key facts

Element	What it is	What it does	Design example
Point	The smallest mark; a position	Attracts the eye; creates focus; many points make a pattern or tone (stippling)	A bindi on a blank wall; a map pin
Line	A point in motion; has direction	Guides the eye; divides; outlines; expresses mood. Horizontal = calm; vertical = strength/height; diagonal = action; curved = softness; zig-zag = tension	Road markings; the Nike swoosh
Shape	An enclosed 2D area	Geometric shapes (circle, square, triangle) feel man-made and stable; organic shapes feel natural and friendly	Traffic signs: triangle = warning, circle = mandatory/prohibitory, rectangle = information (Indian road-sign convention)
Form	3D shape with volume (sphere, cube, cone, cylinder)	Shown on paper through shading and perspective	A pressure cooker as cylinder + dome
Colour	Light reflected; hue, saturation, value	Mood, meaning, hierarchy, identification	Red for stop or danger
Texture	Surface quality, actual (felt) or visual (implied)	Realism, material identity, interest	Jute bag vs silk saree
Space	The area around and between objects. Positive space = the object; negative space = the background	Breathing room; figure–ground; depth	The FedEx arrow hidden in negative space
Value	Lightness or darkness	Creates form, depth and contrast	A pencil shading scale from 1 to 9

Rules / principles

- A line's *weight* communicates. Thick lines come forward or carry importance; thin lines recede or carry detail.

- Shapes carry meaning. Circles suggest unity or continuity; triangles suggest direction or danger; squares suggest stability or trust.
- Negative space is not empty; it is a design tool.

Common traps

- Confusing **shape (2D)** with **form (3D)**.
- Thinking texture must be real. **Visual texture** (drawn wood grain) counts.
- Thinking value only applies to black and white. Every colour has a value.

How NID tests it: MCQs ("Which element dominates this image?"); drawing tasks that use only one element ("Create a composition using only lines to express chaos"); negative-space logos.

Example: "Express *anger* using only lines." A good answer uses jagged zig-zags, sharp angles, heavy weight, dense clustering and diagonal direction.

Practice

1. Which element creates a sense of depth through gradation from light to dark? (a) Line (b) Value (c) Shape (d) Point
2. A logo where a hidden arrow sits between two letters relies mainly on: (a) texture (b) negative space (c) colour (d) form
3. Horizontal lines generally convey: (a) excitement (b) calm/rest (c) danger (d) growth
4. Drawing wood grain on a flat paper box is an example of: (a) actual texture (b) visual texture (c) form (d) value

Answers: 1-b, 2-b, 3-b, 4-b.
Explanations: (1) gradation of value models form and depth. (2) The FedEx arrow is formed by the space between E and x. (3) A horizontal line echoes the horizon and a body lying down: rest. (4) It looks textured but is smooth to touch, so it is visual texture.
Difficulty: Easy · **Priority:** HIGH

1.2 Principles of Design

Concept: principles are the "recipes": the ways elements are arranged to communicate.

Principle	Meaning	How to create it	Indian / everyday example
Balance	Visual weight evenly distributed. *Symmetrical* (formal), *asymmetrical* (informal but balanced), *radial* (from a centre)	Large light object vs small dark object; position from the centre	Taj Mahal (symmetrical); rangoli (radial); a modern poster (asymmetrical)
Contrast	Difference that creates interest and readability	Light/dark, big/small, warm/cool, thick/thin, smooth/rough	White text on a dark-green highway sign
Emphasis	One focal point dominates	Contrast, isolation, size, colour pop, converging lines	A red "SALE" on a grey poster
Hierarchy	Order of importance; what the viewer reads 1st, 2nd, 3rd	Size, weight, colour, position, spacing	A newspaper: headline → subheading → body
Proportion	Size relationship of parts to the whole	Golden ratio (~1:1.618), human proportions (adult ≈ 7.5–8 heads tall)	Correctly sized door vs person
Scale	Size relative to a reference or to the human body	Exaggerate for drama; keep true scale for realism	The Statue of Unity looks huge next to people

Principle	Meaning	How to create it	Indian / everyday example
Rhythm	Organised repetition that creates visual tempo: regular, alternating, progressive, flowing	Repeat elements with intervals	Pillars of a temple corridor; saree borders
Repetition	Using the same element again	Consistency, pattern, branding	Block-print motifs
Movement	The path the eye takes	Lines, diagonals, direction of gaze, gradients	A comic panel order; a figure's pointing hand
Unity	All parts feel like one whole	Common colour palette, style, alignment, proximity	A brand identity system
Variety	Differences that add interest	Change size, colour or shape within a theme	Different fonts used thoughtfully
Harmony	Parts are compatible and pleasing together	Related colours, similar shapes	Analogous earth tones in a Warli wall

Rules

- **Unity without variety is boring; variety without unity is chaos.**
- There should be **one** dominant focal point per composition. Two equal focal points compete.
- **Hierarchy** = emphasis applied to several levels.

Common traps: confusing **proportion** (part-to-part, within the object) with **scale** (relative to another object or the human); confusing **rhythm** (repetition *with tempo*) with plain repetition.

How NID tests it: "Identify the principle in the image" MCQs; "redesign this poster to improve hierarchy"; compositions with constraints ("show rhythm using only circles").

Practice

1. A small red dot placed off-centre on a large white page balancing a large grey square on the other side shows: (a) symmetrical balance (b) asymmetrical balance (c) radial balance (d) no balance
2. Size → weight → colour used to order headline, sub-head and body is: (a) unity (b) rhythm (c) hierarchy (d) harmony
3. A rangoli built around a central point shows: (a) radial balance (b) asymmetry (c) proportion (d) texture
4. A poster with seven fonts, ten colours and no alignment lacks: (a) variety (b) unity (c) contrast (d) movement
5. Gradually increasing circle sizes across a page creates: (a) regular rhythm (b) progressive rhythm (c) radial balance (d) unity

Answers: 1-b, 2-c, 3-a, 4-b, 5-b.
Explanations: (1) Different objects are balanced by visual weight: a saturated small dot vs a large neutral mass. (2) Ordering by importance is hierarchy. (3) Elements radiate from a centre. (4) Too much variety destroys unity. (5) Progressive rhythm changes in steps.
Difficulty: Easy–Medium · **Priority:** HIGH

1.3 Colour Theory

Concept: a systematic way to choose and combine colours.

Key facts

Colour wheel (RYB, the painter's/art-school wheel used in NID-style questions)

```
                YELLOW (P)
        Y-Orange(T)       Y-Green(T)
   ORANGE(S)                   GREEN(S)
        R-Orange(T)       B-Green(T)
                RED(P)  ...  BLUE(P)
        R-Violet(T)       B-Violet(T)
                VIOLET(S)
P = primary   S = secondary   T = tertiary
```

- **Primary (RYB):** red, yellow, blue. **Secondary:** orange (R+Y), green (Y+B), violet (B+R). **Tertiary:** primary + adjacent secondary, e.g. red-orange, yellow-green, blue-violet.
- **Light (additive, screens): RGB.** R+G = yellow; G+B = cyan; R+B = magenta; R+G+B = white.
- **Print (subtractive): CMYK.** Cyan, magenta, yellow, key (black). C+M+Y in theory gives black (in practice a muddy brown, hence the K).
- **Hue** = the colour family name. **Saturation / chroma** = intensity or purity. **Value** = lightness or darkness.
- **Tint** = hue + white (pastel). **Shade** = hue + black. **Tone** = hue + grey.

Colour schemes

Scheme	Definition	Effect	Example
Monochromatic	One hue in different tints, tones and shades	Calm, unified	All blues
Analogous	2–5 neighbours on the wheel	Harmonious, natural	Yellow, yellow-orange, orange
Complementary	Opposite on the wheel: R–G, B–O, Y–V	Maximum contrast, vibrant	Indian cricket kit: blue + orange accents
Split-complementary	One hue + the two neighbours of its complement	Contrast with less tension	Blue + red-orange + yellow-orange
Triadic	3 hues equally spaced (120°)	Lively, balanced	R, Y, B or O, G, V
Tetradic / square	4 hues (2 complementary pairs)	Rich; hard to balance	R, G, O, B
Warm	Reds, oranges, yellows	Energy, appetite, advance visually	Food brands
Cool	Blues, greens, violets	Calm, recede visually	Hospitals, banks

Colour psychology (culture-dependent; Indian context noted)

- **Red:** energy, danger, love. In India it is auspicious, bridal and sindoor.
- **Saffron / orange:** spirituality, courage; it appears on the national flag.
- **Yellow:** happiness, caution, haldi, spring (Basant).
- **Green:** nature, safety, go, Islam, prosperity.
- **Blue:** trust, calm, technology; associated with Krishna.
- **White:** purity, peace. In India it is also the colour of mourning.
- **Black:** power, elegance, mourning (West), protection from the evil eye (*nazar* in India).
- **Purple:** luxury, royalty, creativity.

Rules

- **60-30-10 rule:** 60% dominant colour, 30% secondary, 10% accent.
- Warm colours advance and cool colours recede. Use this for depth in landscapes (**atmospheric perspective**: distant hills look bluer and paler).
- For readability, **value contrast** matters more than hue contrast. Red text on green has the same value and fails, especially for colour-blind viewers.

- About 8% of men have red–green colour blindness. Never use colour as the only signal; add a shape or label.

Common traps

- Mixing up the RYB wheel and the RGB light model. Red + green paint makes brown; red + green light makes yellow.
- "Tone" in everyday speech means many things, but in colour theory it means **hue + grey**.
- Complementary pairs depend on the wheel; NID-style questions assume RYB (red–green, blue–orange, yellow–violet).

How NID tests it: MCQs on complements and schemes; "colour this scene using only analogous colours"; "show the same street in a warm palette (morning festival) and a cool palette (rainy night)".

Example: "Design a poster for a hospital blood-donation drive." Use red as the 10% accent (the drop), lots of white for cleanliness (60%), and calm blue-grey for text (30%).

Practice

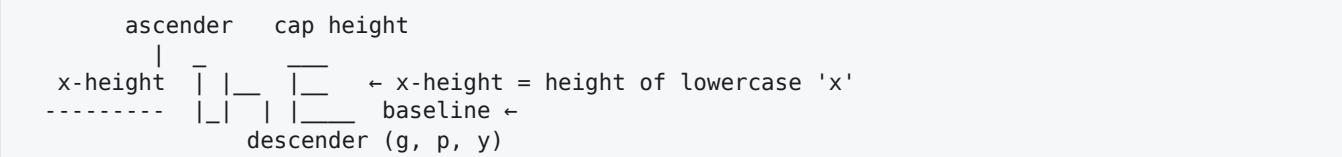
1. The complement of blue on the RYB wheel is: (a) yellow (b) orange (c) green (d) violet
2. Red + white produces: (a) shade (b) tint (c) tone (d) hue
3. Mixing red, green and blue light in equal intensity produces: (a) black (b) brown (c) white (d) grey
4. Yellow-green is: (a) primary (b) secondary (c) tertiary (d) complementary
5. A palette of light blue, mid blue and navy is: (a) analogous (b) monochromatic (c) triadic (d) complementary
6. In a landscape, distant mountains are painted bluish and paler because of: (a) linear perspective (b) atmospheric perspective (c) foreshortening (d) chiaroscuro

Answers: 1-b, 2-b, 3-c, 4-c, 5-b, 6-b.
Explanations: (1) Blue is opposite orange. (2) Adding white gives a tint (pink). (3) Additive mixing gives white. (4) A primary plus an adjacent secondary is tertiary. (5) One hue with value changes is monochromatic. (6) Air scatters light, so far-off objects lose contrast and turn bluish.
Difficulty: Easy · **Priority:** HIGH

1.4 Typography

Concept: the art of arranging type to make language readable, legible and expressive.

Key facts & terminology



Term	Meaning
Serif	Small strokes at the ends of letters (Times New Roman, Garamond, Georgia). Traditional, bookish; good for long print text.
Sans serif	No serifs (Helvetica, Arial, Futura, Gill Sans). Modern, clean; good on screens and signage.
Slab serif	Thick block serifs (Rockwell, Courier). Sturdy; typewriter feel.
Script	Handwriting-like (Brush Script). Elegant or personal.
Display / decorative	For headlines only; low legibility at small sizes.

Term	Meaning
Monospace	Every character the same width (Courier). Used for code.
Typeface vs font	A typeface is the design family (Helvetica). A font is one specific weight/style/size (Helvetica Bold 12 pt).
Kerning	Space between a *specific pair* of letters (AV, To)
Tracking	Uniform spacing across a *whole word or line*
Leading (say "ledding")	Vertical space between lines (baseline to baseline); from strips of lead in metal type
Alignment	Left (ragged right), right, centred, justified
Legibility	How easily individual *characters* can be told apart (a property of the typeface design)
Readability	How easily *text blocks* can be read (a property of the layout: size, leading, line length, contrast)
Hierarchy	Using size, weight and colour to order the information
Widow / orphan	A lonely word or line at the end or start of a paragraph or column
Point size	1 pt ≈ 1/72 inch

Rules

- Ideal body-text line length is about **45–75 characters**.
- Use a maximum of **2 typefaces** per layout (one display + one text).
- Leading of about **120–145%** of the type size suits body text.
- Use all caps for short labels only; long all-caps text is hard to read.
- Signage: sans serif, high contrast, large x-height. Highway signs use mixed case because word shapes help recognition at speed.

Common traps: kerning vs tracking; legibility vs readability; "font" vs "typeface".

How NID tests it: "Which is more legible?" MCQs; "write the word FRAGILE so it looks fragile" (expressive typography); signage design; lettering in posters.

Example: expressive typography. **MELT**: letters drip downward. **HEAVY**: thick, bottom-weighted, sinking. **ECHO**: repeated, fading letters.

Practice

1. Adjusting the space between "A" and "V" only is: (a) tracking (b) kerning (c) leading (d) alignment
2. Which typeface classification is generally preferred for wayfinding signage? (a) Script (b) Decorative (c) Sans serif (d) Blackletter
3. Leading refers to: (a) letter width (b) space between lines (c) space between words (d) stroke thickness
4. Helvetica Bold Italic 14 pt is best described as a: (a) typeface (b) font (c) type family (d) glyph set

Answers: 1-b, 2-c, 3-b, 4-b.
Explanations: (1) Kerning adjusts one pair. (2) Sans serif stays clear at a distance and at speed. (3) Leading is the line spacing. (4) A specific style and size is a font.
Difficulty: Easy–Medium · **Priority:** MEDIUM (HIGH for poster and signage tasks)

1.5 Composition

Concept: the arrangement of elements inside a frame.

Key techniques

- **Rule of thirds:** divide the frame 3×3 and place subjects on the lines or intersections ("power points"). This feels more dynamic than dead centre.

```
+-------+-------+-------+
|       |       |       |
+-------X-------X-------+   X = power points
|       |       |       |
+-------X-------X-------+
|       |       |       |
+-------+-------+-------+
```

- **Symmetry:** formal, stable, ceremonial (temple fronts, logos).
- **Asymmetry:** dynamic and modern; balance it with visual weight.
- **Focal point:** where the eye lands first.
- **Negative space:** empty space around the subject that gives it importance.
- **Grid:** an invisible structure for alignment (columns, modules).
- **Leading lines:** roads, rails or rivers that pull the eye to the subject.
- **Framing:** doors, windows or arches that frame the subject.
- **Foreground–middle ground–background:** layering creates depth.
- **Odd numbers:** groups of 3 or 5 feel more natural than 2 or 4.
- **Golden ratio / spiral:** 1:1.618.
- **Visual weight:** large, dark, saturated, textured, isolated or high-placed elements feel heavier.

Rules for NID drawing boxes

1. Decide the viewpoint and horizon *before* drawing.
2. Put the main action on a thirds intersection.
3. Use overlapping to show depth.
4. Crop boldly. Letting objects run off the edge makes the scene feel larger.
5. Leave some breathing space; don't fill every millimetre with equal detail.

Common traps: centring everything; tangents (edges that just touch, creating confusion); a horizon line that cuts the drawing exactly in half; all objects the same size.

How NID tests it: every scene and poster is judged on composition; MCQs ask "which layout follows the rule of thirds?"

Practice

1. Placing the horizon on the lower third line emphasises: (a) the ground (b) the sky (c) the people (d) nothing
2. A railway track converging toward a distant station is an example of: (a) framing (b) leading lines (c) radial balance (d) texture
3. Why are tangents avoided? (a) They waste ink (b) They create visual confusion about depth (c) They are hard to draw (d) They add too much colour

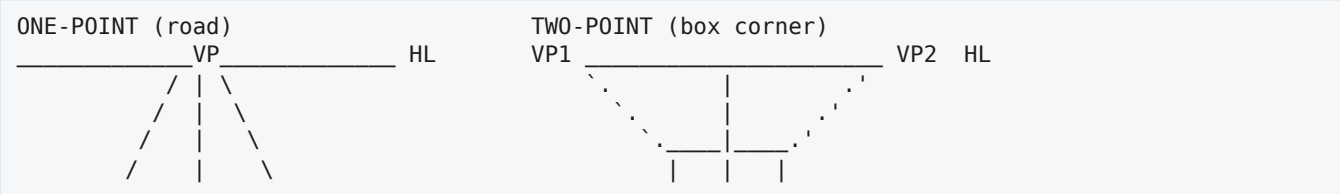
Answers: 1-b, 2-b, 3-b.
Explanations: (1) Two-thirds of the frame becomes sky. (2) The converging rails guide the eye. (3) When edges touch exactly, it is unclear which object is in front.
Difficulty: Easy · **Priority:** VERY HIGH (applied in every drawing)

1.6 Perspective

Concept: a way to draw 3D space on a 2D surface so that it looks real.

Key facts

- **Horizon line (HL)** = the viewer's eye level. Everything above the HL is seen from below; everything below it is seen from above.
- **Vanishing point (VP)** = the point on the HL where parallel lines appear to meet.
- **One-point perspective:** one VP. Used for frontal views of roads, corridors, railway tracks and rooms. Lines are horizontal, vertical, or converge to the VP.
- **Two-point perspective:** two VPs on the HL. Used for a building corner or a box seen edge-on. Verticals stay vertical.
- **Three-point perspective:** two VPs on the HL plus a third above (worm's-eye) or below (bird's-eye). Used for tall buildings or dramatic views.
- **Foreshortening:** objects pointing toward the viewer look compressed (an outstretched arm).
- **Atmospheric perspective:** distance reduces contrast and detail and shifts colour toward blue.
- **Size diminution:** identical objects get smaller with distance. **Overlap:** nearer objects hide farther ones.



Rules for people in perspective (very important for NID scenes)

- If the viewer is standing on flat ground, **the eye level (HL) passes through the heads of all standing adults of similar height, at any distance.** Draw farther people smaller, but keep their heads on the HL.
- If the HL is at waist height (viewer sitting), the HL passes through the waists of standing adults.
- Doors are about 2.1 m tall; an adult is about 1.6–1.75 m. Use doors and vehicles as scale references.

Common traps: several horizon lines in one scene; verticals that tilt in 1- or 2-point perspective; VPs placed too close together (heavy distortion); people floating above the ground plane.

How NID tests it: "Draw a street, platform or market from a given viewpoint"; "draw the view from a balcony or the top of a Ferris wheel" (bird's-eye); "the view of an ant" (worm's-eye); MCQs ("where is the VP?").

Practice

1. In a standing-eye-level street scene, where do the heads of distant standing adults fall? (a) Above the HL (b) On the HL (c) Below the HL (d) At the VP only
2. A tall skyscraper drawn looking straight up needs: (a) 1-point (b) 2-point (c) 3-point (d) isometric
3. Which perspective keeps all vertical lines vertical and uses two VPs? (a) one-point (b) two-point (c) three-point (d) aerial
4. Isometric drawing differs from perspective because: (a) parallel lines stay parallel (b) it uses three VPs (c) objects shrink with distance (d) it has no 3D effect

Answers: 1-b, 2-c, 3-b, 4-a.
Explanations: (1) The eye-height rule. (2) A third VP is needed for the vertical convergence. (3) The two-point definition. (4) Isometric is a parallel projection; nothing converges.
Difficulty: Medium · **Priority:** VERY HIGH

1.7 Gestalt Principles of Visual Perception

Concept: the mind organises parts into wholes ("the whole is other than the sum of its parts"). These principles come from early-20th-century German psychologists (Wertheimer, Köhler, Koffka).

Principle	Meaning	Example
Proximity	Things near each other are seen as a group	Grouped buttons on a remote
Similarity	Things that look alike (colour, shape) form a group	A team in the same jersey colour
Closure	The mind completes incomplete shapes	The WWF panda, IBM stripes
Continuity	The eye follows smooth paths	A curving road
Figure–ground	We separate an object from its background	Rubin's vase (faces or vase)
Common fate	Elements moving together are grouped	A flock of birds
Symmetry	Symmetrical elements are seen as one unit	Brackets [ ]
Prägnanz (simplicity)	We read the simplest interpretation	The Olympic rings read as 5 circles, not complex shapes

Trap: closure vs continuity. Closure fills **gaps in a shape**; continuity follows **lines or paths**.
How NID tests it: identify the principle used in a logo; design a logo using closure.

Practice: 1. The WWF panda logo uses: (a) proximity (b) closure (c) common fate (d) continuity. 2. Rubin's vase demonstrates: (a) similarity (b) figure–ground (c) symmetry (d) rhythm.
Answers: 1-b (the outline is incomplete and the mind fills it in); 2-b (it switches between two readings).
Difficulty: Easy · **Priority:** MEDIUM

1.8 Visual Communication: Pictograms, Signage, Posters, Logos, Packaging

Pictogram rules

1. Use a simple silhouette, ideally recognisable in under 1 second.
2. Keep one idea per sign.
3. Use consistent line weight, corner radius and stance across a set (think Otl Aicher's 1972 Munich Olympic pictograms).
4. Use a standard grid (e.g. a square with consistent margins).
5. Don't rely on text. A pictogram should work for non-readers and in many languages.
6. Use colour conventions: red = prohibition/fire, yellow = warning, green = safety/exit, blue = mandatory/information.
7. Prohibition = a red circle with a diagonal slash from top-left to bottom-right, over a black symbol.

Signage for narrow Indian streets (directly relevant: DAT 2026 asked for 5 signboards)

- **Placement:** projecting (blade) signs perpendicular to the wall, so they can be seen down a narrow lane; at a height clear of heads and vehicles (about 2.4 m+).
- **Visibility:** high contrast, big icons, short words; bilingual (local language + English) or icon-led; reflective or lit at night.
- **Content:**
  - one-way lane
  - two-wheelers only / no four-wheelers
  - dead end ("no through road")

    - pedestrian priority
    - shop direction
    - public toilet / water tap
    - temple / masjid ahead, "silence zone"
    - no parking
    - honking prohibited
    - low overhead wires
    - emergency exit route
    - house-number directory
- **Materials:** weather-resistant (powder-coated metal, enamel); compact.

Poster anatomy: headline (big, 3–6 words) → visual (one strong image) → supporting text → call to action → logo / date / venue. Use the Z-pattern or F-pattern of eye movement.

Logo types: wordmark (Google, Coca-Cola), lettermark (IBM, HP, LIC), pictorial (Apple, Twitter bird), abstract (Nike swoosh, Pepsi globe), mascot (Amul girl, Air India Maharaja), emblem (Harley-Davidson, Starbucks), combination mark (Burger King).
Good logo qualities: simple, memorable, timeless, versatile (works in black and white, small sizes, embroidered, on a phone icon), appropriate.

Packaging functions: protect, preserve, contain, communicate, convenience (open, pour, reseal, carry), sustainability (mono-material, compostable, reusable).

How NID tests it: signage sets, posters (2025: a crow's-viewpoint juice poster), symbols, logos, packaging redesign.
Difficulty: Medium · **Priority:** VERY HIGH

1.9 Human-Centred Design (HCD) & Design Thinking

Concept: solving problems by first understanding the people affected.

The 5-stage design thinking process (Stanford d.school model)

Stage	What you do	NID-paper equivalent	Example: grandmother's travel kit (DAT 2026 theme)
1. Empathise	Observe, interview, feel the user's situation	Imagine the user's day; list pains	Grandma: weak eyesight, knee pain, medicines at fixed times, needs a shawl, a fear of losing tickets
2. Define	Write a clear problem statement: "[User] needs [need] because [insight]"	Underline the real problem	"Grandma needs to find medicines and essentials quickly without bending or searching, because she tires easily and has poor eyesight."
3. Ideate	Generate many ideas without judging; then select	Quick thumbnails of 3–5 ideas	A colour-coded pouch organiser; a pill box with alarm; a trolley bag with a seat; a neck pouch for tickets
4. Prototype	Make it tangible: sketch, mock-up	The final labelled drawing	Drawing of the organiser with labels: "large-print tags", "magnetic flap", "reflective strip"
5. Test	Get feedback; iterate	Add a "how it helps" note or a scenario frame	Frame: grandma opening the pouch on the train easily

Other models: the Double Diamond (UK Design Council, 2005): **Discover → Define → Develop → Deliver**. It alternates divergent and convergent thinking. IDEO's HCD: Inspiration → Ideation → Implementation.

Key concepts

- **Persona:** a fictional but realistic user profile.
- **User journey map:** the steps the user goes through, with emotions at each step.
- **Pain points:** moments of frustration.
- **Affordance (Don Norman):** a property that shows how to use an object. A handle affords pulling; a flat plate affords pushing.
- **Signifier:** a perceivable cue (a "PUSH" label, an arrow).
- **Feedback:** the system's response (a click sound, a light).
- **Mapping:** the relationship between controls and effects (stove knobs laid out like the burners).
- **Constraints:** limits that prevent errors (a USB-C plug fits either way; an asymmetric plug fits one way only).
- **Poka-yoke (mistake-proofing):** from Japanese manufacturing.
- **Universal / inclusive design:** usable by the widest range of people (ramps, OXO Good Grips peelers, kerb cuts).
- **Accessibility:** design for disabilities (tactile paving: lines = direction, dots = warning/stop).

Don Norman's book: *The Design of Everyday Things* (originally *The Psychology of Everyday Things*, 1988). It popularised "Norman doors", doors whose design confuses people about whether to push or pull.

Dieter Rams' 10 principles of good design: good design

1. is innovative
2. makes a product useful
3. is aesthetic
4. makes a product understandable
5. is unobtrusive
6. is honest
7. is long-lasting
8. is thorough down to the last detail
9. is environmentally friendly
10. is as little design as possible ("Weniger, aber besser" = "less, but better")

Common traps

- Jumping to the solution without a problem statement.
- Designing for yourself instead of the user.
- Over-tech solutions (an app for everything) where a simple physical solution works better.
- Ignoring cost and context in India: electricity cuts, water scarcity, crowding, multiple languages, literacy levels.

How NID tests it: "Identify problems faced by X and propose a solution"; "redesign Y for elderly users"; the neem-branch neighbour problem (2025). Examiners look for a visible **problem → insight → solution** chain.

Practice

1. A flat metal plate on a door signals "push". This is an example of: (a) feedback (b) affordance (c) mapping (d) constraint
2. The Double Diamond's four stages are: (a) Plan, Do, Check, Act (b) Discover, Define, Develop, Deliver (c) Empathise, Define, Ideate, Test (d) Research, Sketch, Build, Sell
3. Stove knobs arranged in the same layout as the burners show good: (a) feedback (b) mapping (c) affordance (d) branding
4. Tactile paving with raised dots usually means: (a) go straight (b) warning / stop (c) turn left (d) a shop entrance

Answers: 1-b, 2-b, 3-b, 4-b.
Explanations: (1) The plate's form suggests the action, so it affords pushing (a signifier in Norman's later terminology; NID-level questions accept "affordance"). (2) The Design Council model. (3) The spatial correspondence is mapping. (4) Dots mark hazards or crossings; bars give direction.
Difficulty: Easy–Medium · **Priority:** VERY HIGH

1.10 Creative Thinking Techniques

Technique	How it works	Quick NID example
Brainstorming (Alex Osborn)	Quantity over quality; no judgement	20 uses of a brick in 3 minutes
SCAMPER (Bob Eberle)	**S**ubstitute, **C**ombine, **A**dapt, **M**odify/Magnify, **P**ut to other use, **E**liminate, **R**everse/Rearrange	Umbrella: *Combine* with a torch; *Reverse* it so it folds inside-out to keep water in
Lateral thinking (Edward de Bono)	Escape obvious patterns; provocation ("Po"), random entry	"What if trains had no seats?" leads to standing supports with lean pads
Six Thinking Hats (de Bono)	White = facts, Red = feelings, Black = caution, Yellow = benefits, Green = creativity, Blue = process	Evaluate a solution from 6 angles
Mind mapping (Tony Buzan)	Radial association	Theme "monsoon" → umbrellas, potholes, chai, frogs, floods, greenery
Forced connection	Combine two unrelated things	Kettle + bird = a whistling kettle shaped like a bird
Changing scale / viewpoint	See the problem as an ant, a giant, a crow	The 2025 crow's-viewpoint poster
Biomimicry	Learn from nature	Velcro (burrs); the Shinkansen nose (kingfisher beak); termite-mound cooling (Eastgate Centre, Harare, by Mick Pearce)
Reverse brainstorming	"How could we make this worse?" then flip the answers	"How to make a bus stop terrible?" → no shade, no info, so add both

Torrance's creativity dimensions: fluency (number of ideas), flexibility (variety of categories), originality (rarity), elaboration (detail). NID's "multiple uses" and "make as many as you can" tasks test fluency and flexibility.

Trap: listing 10 ideas from one category (e.g. ten kinds of container) shows fluency but low flexibility. Spread across categories: tool, toy, art, shelter, music, game.

Practice: List the SCAMPER letter for each. (i) Using a bicycle wheel as a pottery wheel. (ii) A pen with a built-in stylus. (iii) Removing the cord from headphones.
Answers: (i) P (put to other use); (ii) C (combine); (iii) E (eliminate).
Difficulty: Easy · **Priority:** HIGH

1.11 Ergonomics, Usability & Sustainability (quick reference)

Ergonomics

- Anthropometry is the measurement of human bodies. Design usually covers the 5th–95th percentile of users.
- Seat height ≈ knee height (about 40–45 cm for adults). Table ≈ 72–75 cm. Kitchen counter ≈ 85–90 cm (Indian standards often ~80–85 cm). Door ≈ 2.1 m high × 0.9 m wide. Stair riser ≈ 15–18 cm, tread ≈ 25–30 cm.

- Grip: a handle diameter of 30–40 mm suits a power grip.
- Wheelchair ramp slope: about 1:12 is the widely used maximum.

Usability (Jakob Nielsen's heuristics, simplified)

- visibility of system status
- match with the real world
- user control
- consistency
- error prevention
- recognition over recall
- flexibility
- minimalist design
- error recovery
- help

Sustainability

- The R's: Refuse, Reduce, Reuse, Repair, Refurbish, Recycle, Rot (compost).
- Life-cycle thinking: raw material → manufacture → transport → use → end of life.
- **Cradle to Cradle** (McDonough & Braungart, 2002): products as nutrients, with no waste.
- **Circular economy** vs linear "take-make-dispose".
- **Design for disassembly**: screws, not glue; mono-materials.
- **Indian traditions:**
  - kulhads (clay cups)
  - leaf plates (pattal, sal or areca leaves)
  - steel tiffins
  - jute and cotton bags
  - stepwells and jaali for passive cooling
  - kantha quilting from old saris (upcycling)
- **Greenwashing:** a false or exaggerated environmental claim.

Practice

1. An adult chair seat height is closest to: (a) 25 cm (b) 43 cm (c) 65 cm (d) 80 cm
2. Kantha embroidery traditionally reuses: (a) plastic (b) old saris and cloth (c) leaves (d) metal
3. Designing a product with screws instead of glue supports: (a) obsolescence (b) design for disassembly (c) greenwashing (d) ergonomics only

Answers: 1-b, 2-b, 3-b.
Explanations: (1) Seat height roughly matches the lower-leg length. (2) Kantha layers old cloth with running stitch. (3) Screws allow repair and material separation.
Difficulty: Easy · **Priority:** MEDIUM–HIGH

PART 2: VISUAL REASONING

Each type below has: rules, a worked example, traps and practice. All practice items are text/ASCII-solvable. More practice is in Vol 4B.

2.1 Mirror Images

Rule: a mirror reverses the direction **perpendicular to the mirror**.

- **Vertical mirror** (on the right or left side of the object): left ↔ right swap; top and bottom stay the same.
- **Horizontal mirror** (below or above, as in a "water image"): top ↔ bottom swap; left and right stay the same.

Letters symmetrical about a vertical axis (unchanged in a vertical mirror): **A, H, I, M, O, T, U, V, W, X, Y**.
Letters symmetrical about a horizontal axis (unchanged in a water image): **B, C, D, E, H, I, K, O, X**.
Letters unchanged in both: H, I, O, X.

Worked example: the vertical-mirror image of **"NID"**. Reverse the order to D-I-N, then flip each letter left–right: D becomes a backward D, I stays I, N becomes a backward N. Result: **ᗡIИ** (written right-to-left). Students often forget to flip the individual letters.

Clock mirror trick: mirror time = **11:60 − actual time** (for a vertical mirror). 3:15 → 11:60 − 3:15 = **8:45**. If the actual time is between 11:00 and 12:59, use 23:60 − time instead. Water-image (horizontal mirror) time has no single-subtraction rule that works for all times, so sketch the clock and flip it top-to-bottom. Only the vertical-mirror formula is worth memorising.

Traps: forgetting to reverse the letter order; mixing up which axis flips; numbers 2, 3, 4, 5, 6, 7, 9 are asymmetric. 0, 1 (simple stroke) and 8 are symmetric about a vertical axis.

Practice

1. Which word looks the same in a vertical mirror? (a) MOM (b) DAD (c) NUN (d) BOB
2. A clock shows 2:20. Its mirror image shows: (a) 9:40 (b) 10:40 (c) 9:20 (d) 10:20
3. Which letter looks the same in a water image? (a) A (b) M (c) E (d) U

Answers: 1-a, 2-a, 3-c.
Explanations: (1) M and O are vertically symmetric, and the palindrome order keeps MOM unchanged; D and B are not vertically symmetric; N is not. (2) 11:60 − 2:20 = 9:40. (3) E is symmetric about a horizontal axis.
Difficulty: Easy–Medium · **Priority:** MEDIUM

2.2 Rotation

Rule: rotation turns a figure without flipping it. 90° clockwise: top → right → bottom → left → top. A rotated figure can **never** become its mirror image. "L" rotated is always some L-shape and can never become a "⅃".

Worked example: the letter "F" rotated 180° looks like an upside-down, backward F, but it is still not a mirror F. To test an option, mentally rotate it back.

Trap: choosing a mirror image when the question asks for a rotation.
Practice: An arrow points North. It is rotated 135° clockwise, then 45° anticlockwise. Where does it point?
Answer: net 90° clockwise → **East**.
Difficulty: Easy · **Priority:** MEDIUM

2.3 Paper Folding & Cutting

Rule: each fold doubles the layers. A hole punched through *n* layers gives *n* holes when unfolded, placed symmetrically about each fold line. **Unfold in reverse order.**

Worked example: a square sheet is folded in half (left over right), then folded in half again (top over bottom). One hole is punched near the folded corner. Unfolded, there are **4 holes**, arranged symmetrically about both the vertical and horizontal centre lines.

```
Unfolded result:
 +--------+--------+
 |        |        |
 |      o | o      |
 +--------+--------+
 |      o | o      |
 |        |        |
 +--------+--------+
```

(If the hole is near the corner where both folds meet, the four holes cluster near the centre, as shown.)

Traps: a hole punched *on* a fold line creates fewer separate holes (it becomes one elongated or merged hole); diagonal folds reflect across the diagonal; forgetting the reverse order.

Practice

1. A paper is folded in half 3 times. One hole is punched through all layers. How many holes appear when unfolded?
2. A square is folded once along its diagonal and a semicircle is cut on the folded edge (the cut sits on the fold). Unfolded, what is the cut shape?

Answers: 1 → **8** (2^3 layers). 2 → a **full circle** centred on the diagonal (the semicircle mirrors across the fold).
Difficulty: Medium · **Priority:** HIGH (if objective items return)

2.4 Cubes, Dice & Nets

Key facts

- A cube has 6 faces, 12 edges and 8 vertices.
- On a standard die, opposite faces sum to 7 (1–6, 2–5, 3–4).
- **Net rule:** in a straight row of faces in a net, faces **one apart** (with one face between them) are **opposite** in the folded cube. There are **11 distinct nets** of a cube.

```
Net example (cross):
      [1]
  [2] [3] [4] [5]
      [6]
Opposites: 3–5 (one apart in the row), 2–4 (one apart), 1–6 (above/below the same square 3)
```

Painted-cube formulas (a big cube of edge n cut into n^3 small cubes, all outer faces painted):

- 3 faces painted = **8** (the corners)
- 2 faces painted = **$12(n-2)$** (edges)
- 1 face painted = **$6(n-2)^2$** (face centres)
- 0 faces painted = **$(n-2)^3$** (inside)

Check for n = 3: 8 + 12 + 6 + 1 = 27 ✓.

Two-position dice rule: if two views of the same die share **one** common visible face and that face is in the same position, the other faces in the same positions are opposite. Otherwise, use rotation reasoning.

Practice

1. A 4×4×4 painted cube is cut into 64 small cubes. How many have exactly 2 faces painted?
2. In the cross net above, which face is opposite 2?
3. How many small cubes have no paint in a 5×5×5 painted cube?

Answers: 1 → 12(4 − 2) = **24**. 2 → **4**. 3 → $(5 - 2)^3$ = **27**.
Difficulty: Medium · **Priority:** HIGH (if objective items return)

2.5 Counting Figures (triangles, squares)

Formulas

- Triangles from a single apex with *n* lines fanning down to a base: **n(n − 1)/2** (where n counts all lines from the apex, including the two outer sides).
- Squares in an n×n grid: $\mathbf{1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6}$. For a 4×4 grid: 30.
- Rectangles in an m×n grid: **C(m+1, 2) × C(n+1, 2)**. For a 3×3 grid: 6 × 6 = 36 (this includes squares).
- A square with both diagonals contains **8 triangles** (4 small + 4 made of two halves).

Practice

1. How many squares are in a 3×3 grid?
2. A triangle has 4 lines from its top vertex to the base (including the two sides). How many triangles?
3. How many rectangles (including squares) in a 2×3 grid?

Answers: 1 → 1 + 4 + 9 = **14**. 2 → 4×3/2 = **6**. 3 → C(3,2) × C(4,2) = 3 × 6 = **18**.
Difficulty: Medium · **Priority:** MEDIUM

2.6 Pattern & Series Completion

Common transformations:

- rotation by a fixed angle each step (45°, 90°)
- an element added or removed each step
- shading moving clockwise
- size increasing
- alternating pairs
- the number of sides increasing (triangle → square → pentagon)
- positions cycling through corners

Method: track **one** element at a time across the series. Write down its rule. Repeat for each element.

Practice: series of polygons: triangle (3), square (4), pentagon (5), hexagon (6), ? → **heptagon (7 sides)**.
Dot inside moving: top-left, top-right, bottom-right, bottom-left, ? → **top-left** (it cycles clockwise).
Difficulty: Easy–Medium · **Priority:** MEDIUM

2.7 Embedded / Hidden Figures, Silhouettes, Shadows

- **Embedded figures:** find a simple shape hidden within a complex one. Look for its distinctive angles first.
- **Silhouettes:** the outline shape alone must identify the object. Good silhouette = a characteristic profile (a teapot's spout and handle, side-on).
- **Shadows:** a shadow falls *away* from the light. Its length depends on the light's height (low sun = long shadow at morning or evening; noon = short). All shadows in one scene point consistently from one light source. The cast shadow touches the object where the object touches the ground.
- **Orthographic views:** front, top and side (plan, elevation). From an isometric object, picture what each view shows. Hidden edges are drawn as dashed lines in technical drawing.

Practice

1. In the morning, the sun is in the East. Which way does your shadow fall? → **West**.
2. A cylinder lying on its side: its top view is a **rectangle**; its end view is a **circle**.
3. A cone standing on its base: the top view is a **circle with a centre point**; the front view is a **triangle**.
   Difficulty: Easy–Medium · **Priority:** MEDIUM–HIGH (also needed for drawing)

PART 3: DESIGN APTITUDE — OBSERVATION & SENSITIVITY

3.1 Observation

Concept: noticing details others miss, such as wear patterns, colours, proportions and how people actually use things.

NID-style observation facts worth knowing (common MCQ / drawing checks)

- An Indian ₹ symbol has **two horizontal lines** at the top (designed by D. Udaya Kumar, adopted in 2010).
- The national flag ratio is **2:3**. The Ashoka Chakra has **24 spokes** and is navy blue.
- A standard 4-legged chair: front legs are often splayed; most plastic chairs (monobloc) are stackable.
- Traffic light order, top to bottom: **red, amber, green**.
- A ceiling fan (Indian) usually has **3 blades**. Table fans have a guard.
- On a QWERTY keyboard the top-left letter row starts **Q W E R T Y**. The F and J keys have bumps for touch-typing.
- Analogue clocks in adverts often show **10:10** (it frames the logo and looks like a smile).
- Pencil hardness: H = hard and light; B = black and soft. HB is the middle.
- Shoelaces: the eyelets are usually in two columns with pairs at matching heights.
- A bicycle's chain is on the **right** side (drive side).
- Indian vehicles drive on the **left**, so the steering wheel is on the right.

Drill: every day, choose one object. Draw it from memory, then observe the real object and correct the drawing in red. Track your errors.

Priority: HIGH

3.2 Visual Sensitivity / Aesthetic Judgement

- Recognising good vs poor proportion, alignment and spacing; consistency of style; appropriateness of colour.
- **NID-style task:** "Which of these 4 layouts is best for a book cover and why?" Judge it by hierarchy, contrast, negative space and appropriateness.

Priority: MEDIUM

Quick self-test for Volume 2A (answers below)

1. Name the three primaries of light.
2. What is the difference between proportion and scale?
3. Which design thinking stage writes the problem statement?
4. In two-point perspective, how many vanishing points lie on the horizon line?

5. What does "affordance" mean?
6. How many distinct nets does a cube have?
7. How many faces of a 3×3×3 painted cube's centre small cube are painted?
8. Name the four stages of the Double Diamond.
9. Which letters look the same in both mirror and water images?
10. What is the 60-30-10 rule?

Answers

1. Red, Green, Blue.
2. Proportion compares parts within an object; scale compares the object's size to another reference (usually the human body).
3. Define.
4. Two.
5. A property of an object that suggests how it is used.
6. 11.
7. Zero.
8. Discover, Define, Develop, Deliver.
9. H, I, O, X.
10. 60% dominant, 30% secondary, 10% accent colour.

NEXT VOLUME: Volume 2B — Logical Reasoning, Quantitative Aptitude & Verbal Ability Notes (`02B-VOL2-THEORY-REASONING-QUANT-VERBAL.md`)

VOLUME 2B — THEORY NOTES: LOGICAL REASONING, QUANTITATIVE APTITUDE, VERBAL ABILITY

> **Weight check:** these three areas are **LOW–MEDIUM priority** for DAT 2027. They appeared as a handful of MCQs in older papers and were absent in the reported 2026 paper. Keep this to about 10–15% of study time. They still help indirectly: logical structure helps problem-solving answers, and good writing helps the 200-word tasks.

PART 1: LOGICAL REASONING

1.1 Series (number and letter)

Concept: find the rule that generates each term.
Common rules:

- constant difference (+3)
- increasing difference (+1, +2, +3...)
- multiplication (×2)
- squares or cubes (1, 4, 9, 16; 1, 8, 27)
- primes (2, 3, 5, 7, 11)
- alternating two series
- Fibonacci (each term = sum of the previous two)

Letter positions: A1 B2 C3 D4 E5 F6 G7 H8 I9 J10 K11 L12 M13 N14 O15 P16 Q17 R18 S19 T20 U21 V22 W23 X24 Y25 Z26.
Memory aid **"EJOTY"** = 5, 10, 15, 20, 25. Opposite letters sum to 27 (A↔Z, B↔Y, M↔N).

Trap: stopping at the first rule that fits only 2–3 terms. Check that it holds for all terms.
How NID tests it: occasionally, often disguised visually (dots increasing in a series of figures).

Practice

1. 2, 6, 12, 20, 30, ? (a) 40 (b) 42 (c) 44 (d) 36
2. 1, 1, 2, 3, 5, 8, ? (a) 11 (b) 12 (c) 13 (d) 15
3. B, E, H, K, ? (a) M (b) N (c) O (d) L
4. 3, 9, 27, 81, ? (a) 162 (b) 243 (c) 324 (d) 108

Answers: 1-b, 2-c, 3-b, 4-b.
Explanations: (1) The differences 4, 6, 8, 10 give 12 next, so 30 + 12 = 42 (equivalently n(n + 1)). (2) Fibonacci: 5 + 8 = 13. (3) +3 each time: K(11) + 3 = N(14). (4) ×3: 81 × 3 = 243.
Difficulty: Easy · **Priority:** LOW

1.2 Analogies & Classification (Odd One Out)

Analogy: A : B :: C : ?. Find the relationship between A and B (part–whole, tool–user, product–material, cause–effect, worker–tool, degree), then apply it.

Design-flavoured analogies are typical for NID:

- Potter : Wheel :: Weaver : **Loom**
- Pashmina : Kashmir :: Muga : **Assam**
- Bauhaus : Gropius :: De Stijl : **Mondrian / van Doesburg**
- Pixel : Image :: Brick : **Wall** (unit : whole)

Classification: find the item that breaks the shared property. Check several properties: category, material, region, era, function.

Trap: finding a surface similarity and missing the deeper one, e.g. treating the order of terms as irrelevant (A:B relationship reversed).

Practice

1. Kalamkari : Andhra Pradesh :: Phulkari : ? (a) Gujarat (b) Punjab (c) Bihar (d) Odisha
2. Odd one out: Madhubani, Warli, Gond, Kanjeevaram (a) Madhubani (b) Warli (c) Gond (d) Kanjeevaram
3. Chisel : Sculptor :: Scalpel : ? (a) Butcher (b) Surgeon (c) Carpenter (d) Barber
4. Odd one out: Helvetica, Futura, Garamond, Arial

Answers: 1-b, 2-d, 3-b, 4 → Garamond.
Explanations: (1) Phulkari is a Punjabi embroidery. (2) Kanjeevaram is a silk sari (textile); the others are painting traditions. (3) The tool–user relation. (4) Garamond is a serif; the others are sans serif.
Difficulty: Easy–Medium · **Priority:** MEDIUM (design analogies)

1.3 Coding–Decoding

Types: letter shift (+1, −2), reverse alphabet (A↔Z), number substitution, symbol substitution.
Method: compare one word with its code letter by letter and write the shift.

Practice: If DESIGN is coded as EFTJHO, how is CRAFT coded?
Answer: each letter +1, so C→D, R→S, A→B, F→G, T→U = **DSBGU**.
Difficulty: Easy · **Priority:** LOW

1.4 Syllogisms (Deduction)

Rules:

- "All A are B" + "All B are C" → "All A are C" ✓.
- "Some A are B" does **not** give "Some B are not A".
- From two "some" statements, **no** definite conclusion follows.
- From "All A are B", "Some B are A" follows ✓.

Method: draw Venn diagrams. A conclusion must be true in **every** possible diagram.

Practice: Statements: All designers are observers. Some observers are painters. Conclusions: I. Some designers are painters. II. Some observers are designers.
(a) Only I (b) Only II (c) Both (d) Neither

Answer: (b). II follows: all designers are observers, so some observers are designers. I is only possible, not certain.
Difficulty: Medium · **Priority:** LOW

1.5 Statements, Assumptions, Inferences, Cause–Effect

- **Assumption:** an unstated idea the statement takes for granted.
- **Inference:** a conclusion that probably follows from the facts.
- **Cause–effect:** decide which event led to the other. A shared third cause is possible.

Practice: Statement: "Use biodegradable cups at the fest to reduce waste." Assumption: I. Biodegradable cups produce less lasting waste. II. All students will refuse plastic cups.
Answer: Only I is assumed. II is extreme and not needed.
Difficulty: Easy–Medium · **Priority:** LOW

1.6 Arrangements & Puzzles

Method:

1. Draw slots.
2. Place the fixed clues first.
3. Use "not" clues to eliminate.
4. Try cases for the remaining clues.

Practice: Five artists A, B, C, D, E sit in a row facing north. C is in the middle. A is at the left end. B is to the right of C, but not at the right end. E is not next to A. Who is at the right end?
Solution: positions 1–5. A = 1, C = 3. B is right of C and not at 5, so B = 4. The remaining spots are 2 and 5 for D and E. E is not next to A, so E ≠ 2. So E = 5 and D = 2.
Answer: E.
Difficulty: Medium · **Priority:** LOW

1.7 Directions

North is up. Turning right from North = East. Use Pythagoras for the shortest distance.

Practice: A student walks 3 km North, then 4 km East. How far is the student from the start?
Answer: √(9 + 16) = **5 km** (North-East).
Difficulty: Easy · **Priority:** LOW

PART 2: QUANTITATIVE APTITUDE (NID-relevant only)

How much maths does NID really need? Class 8–10 arithmetic and mensuration, used in **design contexts**: scaling a drawing, tiles on a floor, fabric needed, ratios on a map, percentages in a survey, packaging volume. A strong student needs about 6–8 hours of maths revision in total. Stop there.

2.1 Percentages

- x% of y = xy/100. Percentage change = (change ÷ original) × 100.
- Successive changes a% and b%: net = a + b + ab/100. For +20% then −20%: net = −4%.

Practice: A ₹800 lamp is discounted 15%. Find the price. → 800 × 0.85 = **₹680**.

2.2 Ratio & Proportion; Scale

- If a : b = 2 : 3 and the total is 50, the parts are 20 and 30.
- **Map/drawing scale:** 1 : 100 means 1 cm on paper = 100 cm (1 m) in reality.

Practice: On a 1 : 50 drawing, a table is 3 cm long. Find the real length. → 3 × 50 = 150 cm = **1.5 m**.

2.3 Averages

Average = sum ÷ count. If one value changes by d, the average changes by d/n.

Practice: The average of 5 marks is 60. One mark of 40 is replaced by 90. Find the new average. → the change is +50 over 5 = +10, so **70**.

2.4 Fractions & Unit Conversion

- 1 inch = 2.54 cm; 1 foot = 12 in = 30.48 cm; 1 m = 100 cm; 1 km = 1000 m; 1 L = 1000 cm^3; 1 m^2 = 10,000 cm^2; 1 hectare = 10,000 m^2.
- 1 kg = 1000 g. Paper: A4 = 210 × 297 mm. A3 = 297 × 420 mm (twice A4). The aspect ratio of A-series paper is 1 : √2 ≈ 1 : 1.414.

Practice: How many A4 sheets make one A1? → A1 = 2 A2 = 4 A3 = **8 A4**.

2.5 Time, Speed, Distance

- Distance = speed × time. To convert km/h to m/s, multiply by 5/18.
- Clock angle: θ = $|30H - 5.5M|$ degrees.

Practice

1. A train at 72 km/h. Find its speed in m/s. → 72 × 5/18 = **20 m/s**.
2. The angle between the hands at 3:30? → $|90 - 165|$ = **75°**.

2.6 Geometry & Mensuration

Shape	Perimeter / surface	Area / volume
Square (side a)	4a	a^2
Rectangle	2(l + b)	lb
Triangle	sum of sides	½ × base × height
Circle (radius r)	2πr	πr^2
Cube (a)	surface $6a^2$	volume a^3
Cuboid	surface 2(lb + bh + hl)	volume lbh
Cylinder	curved surface 2πrh; total 2πr(r + h)	volume $\pi r^2 h$
Cone	slant l = $\sqrt{(r^2 + h^2)}$; curved surface πrl	volume $\frac{1}{3}\pi r^2 h$
Sphere	surface $4\pi r^2$	volume $\frac{4}{3}\pi r^3$

- Angles in a triangle sum to 180°; in a quadrilateral to 360°. The sum of interior angles of an n-gon is (n − 2) × 180°. Each interior angle of a regular hexagon = 120°.

- **Tessellation:** among regular polygons, only equilateral triangles, squares and hexagons tile a plane alone (because 60°, 90° and 120° divide 360° exactly).
- Scaling: if lengths are scaled by k, area scales by k^2 and volume by k^3.

Practice

1. A 4 m × 3 m floor is tiled with 50 cm × 50 cm tiles. How many tiles? → (400 × 300) ÷ (50 × 50) = 120000 ÷ 2500 = **48**.
2. A model is built at half scale. How does its volume compare with the original? → $(½)^3$ = **⅛**.
3. Why do bee hives use hexagons? → Hexagons tile with no gaps and enclose the most area for a given perimeter among shapes that tile, so they save wax.

2.7 Logical-Mathematical Puzzles

- Handshakes among n people: n(n − 1)/2.
- Number of diagonals of an n-gon: n(n − 3)/2.
- Pigeonhole principle: with n boxes and n + 1 items, at least one box holds 2.

Practice

1. 10 people each shake hands with every other person once. How many handshakes? → 10 × 9 / 2 = **45**.
2. How many diagonals does a hexagon have? → 6 × 3 / 2 = **9**.

Difficulty (Part 2 overall): Easy · **Priority:** LOW

PART 3: VERBAL ABILITY

3.1 Vocabulary for design (learn these; they also help in writing)

Word	Meaning
Aesthetic	concerned with beauty or appearance
Ergonomic	designed for efficiency and comfort in use
Utilitarian	designed to be useful rather than attractive
Ornate	elaborately decorated
Minimalist	using few elements; simple
Juxtapose	place side by side for contrast
Motif	a repeated decorative design or idea
Vernacular	local and traditional (vernacular architecture = local materials and building methods)
Iconography	visual images and symbols used in a work and their meaning
Palette	a range of colours used
Tactile	relating to touch
Ephemeral	lasting a very short time
Ubiquitous	found everywhere
Frugal	economical; simple and low-cost
Intuitive	easy to understand without explanation
Prototype	an early model for testing

Word	Meaning
Iterate	repeat a process to improve
Monochrome	in one colour
Symmetry	identical on both sides of an axis
Kitsch	art or objects considered tasteless due to garish excess, though sometimes appreciated ironically
Avant-garde	new and experimental
Pastiche	an imitation of the style of another work or period
Ambient	relating to the surrounding environment
Patina	a surface sheen caused by age (e.g. green on bronze)
Filigree	delicate metal ornamental work

3.2 Synonyms & Antonyms

Practice

1. Synonym of *ubiquitous*: (a) rare (b) omnipresent (c) unique (d) hidden
2. Antonym of *ornate*: (a) decorative (b) plain (c) elaborate (d) intricate
3. Synonym of *ephemeral*: (a) permanent (b) transient (c) heavy (d) ancient

Answers: 1-b, 2-b, 3-b.

3.3 Idioms (common in older GAT sections)

Idiom	Meaning
Think outside the box	think creatively
Back to the drawing board	start again after a failure
A picture is worth a thousand words	visuals communicate efficiently
Form follows function	design should arise from purpose (Louis Sullivan, 1896)
Less is more	simplicity is powerful (associated with Mies van der Rohe)
Less is a bore	Robert Venturi's postmodern reply
Cut corners	do something poorly to save time or money
The devil is in the details	small things matter
Reinvent the wheel	waste effort recreating something that exists

3.4 Reading Comprehension: method

1. Read the questions first (30 sec).
2. Read the passage for its main idea and tone.
3. Answer from the text, not from your opinion.
4. Eliminate extreme options ("always", "never").

Mini passage: "Traditional Indian stepwells were not merely water sources; they were social spaces where women gathered, travellers rested and festivals were held. Their descending architecture kept the water cool and the air several degrees lower than the surface. Many fell into disuse when piped water arrived, and some became dumping grounds."

Questions & answers

1. The main idea is: (a) stepwells were only for water (b) stepwells served social and climatic functions and declined with piped water (c) piped water is harmful (d) stepwells are dumping grounds. **Ans: (b)**. It covers all three parts of the passage.
2. "Descending architecture" helped mainly by: **keeping the water and air cool**. This is stated directly in the passage.
3. Tone: **informative, mildly regretful**. "Fell into disuse" and "dumping grounds" signal loss.

3.5 Descriptive & Persuasive Writing (HIGH priority: used in the subjective part)

Template for "Draw a product you love and write about it in 200 words" (asked in DAT 2026):

1. **Hook (1–2 lines):** what it is and the emotional connection.
2. **Form & material (2–3 lines):** shape, size, colour, texture, material.
3. **Function & use (2–3 lines):** how it works; the user experience.
4. **Design qualities (2–3 lines):** what is good: ergonomics, durability, simplicity, cultural value.
5. **One improvement (1–2 lines):** this shows critical thinking.
6. **Closing (1 line):** why it matters.

Model answer (about 190 words): "My steel tiffin box"

> My three-tier stainless-steel tiffin has travelled with me to school for nine years, and it still looks almost new. Its form is simple: three stacked cylindrical containers held together by a single clip-on frame with a handle at the top. The brushed steel surface hides scratches, and the rounded edges make it easy to clean.
> What I love most is how honestly it works. Each tier keeps food separate, so dal never leaks into the rotis. The single clamp locks all three tiers at once, and the handle lets me carry the whole lunch with one hand on a crowded bus. Nothing is decorative; every part has a purpose.
> It is also sustainable. It needs no plastic wrap, it never breaks, and it can be repaired or recycled. It echoes the dabbawalas of Mumbai, who move thousands of similar tiffins every day with remarkable accuracy.
> If I could improve one thing, I would add a silicone gasket to each lid so that gravies do not spill when the bus brakes suddenly.
> For me, this tiffin shows that good design can be humble, durable and deeply connected to everyday Indian life.

Writing rules: use concrete nouns, sensory adjectives and active verbs; stay within ±10% of the word limit; write a short heading; keep handwriting legible.

Difficulty: Easy–Medium · **Priority:** HIGH (writing) · LOW (grammar MCQs)

NEXT VOLUME: Volume 2C — General Awareness, Design Awareness, Indian Art & Culture, Architecture (`02C-VOL2-THEORY-GK-ART-ARCHITECTURE.md`)

VOLUME 2C — THEORY NOTES: DESIGN-FOCUSED GENERAL AWARENESS, INDIAN ART & CULTURE, ARCHITECTURE

> This is **not** a UPSC GK book. Everything here was chosen because it (a) appeared in NID-style MCQs (logos, crafts, designers, architecture), (b) helps you draw authentic Indian context in subjective answers, or (c) is common material for the In-Person Sensitivity Test.
> These are **confirmed historical facts** unless marked otherwise. Recent events are in Volume 11.

1. About NID (know your institute; interview favourite)

Fact	Detail
Founded	**1961**, Ahmedabad
Origin	**"The India Report" (1958)** by **Charles and Ray Eames**, commissioned by the Government of India with Ford Foundation support. It recommended a design institute to address the needs of a modernising India.
Key founders	**Gira Sarabhai** and **Gautam Sarabhai** (of the Sarabhai family, Ahmedabad)
Status	Declared an **Institution of National Importance** by the **NID Act, 2014**
Campuses	Ahmedabad (main); Gandhinagar (PG); Bengaluru (R&D / PG)
Other NIDs (separate institutes)	NID Andhra Pradesh (Amaravati), NID Assam (Jorhat), NID Haryana (Kurukshetra), NID Madhya Pradesh (Bhopal). All were set up in the late 2010s, mentored by NID Ahmedabad.
Recent	Union Budget 2026-27 proposed a new NID in eastern India [REPORTED, 2026]
Other design schools in India	IDC School of Design (IIT Bombay, 1969), NIFT (1986), Srishti Manipal (Bengaluru), CEPT (Ahmedabad, architecture/planning), Sir J.J. School of Art (Mumbai, 1857), IIT Guwahati / IIT Hyderabad design departments, IIITDM Jabalpur, UCEED (the exam for IIT B.Des)

World design schools:

- **Bauhaus** (Germany, 1919–1933: Weimar → Dessau → Berlin)
- **Ulm School of Design / HfG Ulm** (1953–1968, Germany)
- **Royal College of Art** (London)
- **Rhode Island School of Design (RISD)** (USA)
- **Parsons School of Design** (New York)
- **Central Saint Martins** (London)
- **Design Academy Eindhoven** (Netherlands)
- **Pratt Institute** (New York)
- **Politecnico di Milano** (Italy)
- **ArtCenter College of Design** (Pasadena)
- **Stanford d.school** (design thinking)

2. Indian Folk & Tribal Painting Traditions

Art form	State / region	Key features	Notable names / facts
Madhubani / Mithila	Bihar (Mithila region)	Bold outlines, no empty space, double lines, fish, peacocks, gods; natural colours; styles include Bharni, Kachni, Tantrik, Godna, Kohbar	Sita Devi, Ganga Devi, Mahasundari Devi, Baua Devi (Padma Shri). GI tag.
Warli	Maharashtra (Thane, Palghar)	White rice-paste on a mud or red-ochre wall; **circle, triangle, square** (sun/moon, mountains, sacred enclosure); stick figures dancing in a spiral (Tarpa dance)	Jivya Soma Mashe (Padma Shri). Painted by the Warli tribe.
Gond	Madhya Pradesh	Dots, dashes and lines fill the forms; animals, trees, myths; vivid colours	**Jangarh Singh Shyam** (pioneer), **Bhajju Shyam** (*The London Jungle Book*), Durga Bai
Pithora	Gujarat / MP (Rathwa, Bhil tribes)	Ritual wall paintings, horses, painted for thanksgiving	—
Bhil	MP / Rajasthan / Gujarat	Dot patterns (like grains of maize)	Bhuri Bai (Padma Shri)
Saura	Odisha	Ritual wall paintings (*ittals*), similar to Warli but with more detailed figures	—
Sohrai & Khovar	Jharkhand (Hazaribagh)	Mural art for harvest (Sohrai) and marriage (Khovar), made with earth colours	GI tag
Pattachitra	Odisha (Raghurajpur) and West Bengal	Cloth-based scroll paintings; Jagannath themes; fine lines; natural colours; ornate borders	Raghurajpur heritage crafts village
Kalighat	Kolkata, West Bengal	19th-century bazaar paintings; bold sweeping brush lines; satire of the babu culture	Painted near the Kalighat Temple
Kalamkari	Andhra Pradesh	"Kalam" (pen) + "kari" (work). **Srikalahasti style** (freehand pen, mythological) and **Machilipatnam style** (block-printed, Persian motifs)	GI tags for both
Phad	Rajasthan (Bhilwara)	Long horizontal scrolls narrating the epics of folk deities **Pabuji** and **Devnarayan**; performed by Bhopa priests	Joshi family of painters
Tanjore / Thanjavur	Tamil Nadu	Gold foil, gems, relief work on wooden boards; Hindu deities; Maratha-period patronage	GI tag
Mysore painting	Karnataka	Like Tanjore but subtler, with less relief	—
Cheriyal scrolls	Telangana	Narrative scrolls in a comic-strip style; red backgrounds	GI tag
Thangka	Sikkim, Ladakh, Himachal	Buddhist painting on cotton or silk; mandalas	—
Manjusha	Bihar (Bhagalpur)	Snake motifs; Bihula–Bishahari legend; border-based	GI tag
Kalamezhuthu	Kerala	Ritual floor drawings made with coloured powders	—
Rangoli / Kolam / Alpana / Mandana	All India (Kolam = TN; Alpana = Bengal; Mandana = Rajasthan)	Floor patterns, often radial, for welcome and festivals	Kolam uses rice flour, a white line-and-dot grid

Indian Miniature Painting Schools

School	Period / region	Features
Pala	Bengal/Bihar, 8th–12th c.	Buddhist palm-leaf manuscripts; the earliest Indian miniatures
Mughal	16th–19th c.	Realism, court scenes, portraits, nature studies. Akbar (*Hamzanama*), Jahangir (Mansur's birds and animals), Shah Jahan. Persian influence.
Rajput / Rajasthani	16th–19th c.	Mewar, Bundi, Kota, Kishangarh, Marwar, Jaipur. Bright colours; Krishna and Radha. **Kishangarh: "Bani Thani" by Nihal Chand**, called the "Indian Mona Lisa".
Pahari	Himachal / Jammu, 17th–19th c.	Basohli (bold), Guler, **Kangra** (lyrical, delicate, Krishna themes)
Deccan	Bijapur, Golconda, Ahmednagar	Persian and Turkish influence; rich colours

Modern & Contemporary Indian Artists

Artist	Known for
Raja Ravi Varma (1848–1906)	Oil paintings of Hindu gods in a European academic style; oleographic printing press; *Shakuntala*, *Hamsa Damayanti*
Abanindranath Tagore	Bengal School founder; *Bharat Mata* (1905)
Rabindranath Tagore	Started painting at 60+; founded Santiniketan / Visva-Bharati (Santiniketan became a UNESCO site in 2023)
Nandalal Bose	Bengal School; illustrated the original **Constitution of India** manuscript; Haripura posters (1938)
Amrita Sher-Gil (1913–1941)	Hungarian-Indian modernist; *Three Girls*, *Bride's Toilet*
Jamini Roy	Drew on Kalighat and folk traditions; flat colours; bold outlines
M.F. Husain	Progressive Artists' Group; horses; *Mother Teresa* series
S.H. Raza	The **Bindu** (dot) series; geometric abstraction
F.N. Souza	Founder of the Progressive Artists' Group (1947, Bombay)
Tyeb Mehta	*Kali*, *Mahishasura*; diagonal splits
V.S. Gaitonde	Abstract, meditative works; record auction prices
Ram Kinkar Baij	Sculptor; *Santhal Family* (Santiniketan)
Satish Gujral	Painter, sculptor and architect; Belgian Embassy, Delhi
K.G. Subramanyan	Artist and educator (Baroda)
Bhupen Khakhar	Pop and narrative art (Baroda)
Anish Kapoor	British-Indian sculptor; *Cloud Gate* ("The Bean", Chicago)
Subodh Gupta	Steel utensil sculptures (*Very Hungry God*)
Bharti Kher	Bindi artworks
Atul Dodiya, **Jitish Kallat**, **Nalini Malani**	Contemporary installation and painting
R.K. Laxman	Cartoonist; *The Common Man* (Times of India); created the Asian Paints mascot **Gattu**
Mario Miranda	Goan cartoonist; crowded, detailed street scenes
Ram V. Sutar	Sculptor; **Statue of Unity** (182 m, Gujarat, 2018, the world's tallest statue)

3. Indian Textiles (high value for NID; Textile Design is one of the three main discipline groups)

3.1 Woven Saris & Fabrics

Textile	State	Identifying features
Banarasi	Uttar Pradesh (Varanasi)	Silk brocade with gold/silver zari; Mughal floral motifs (kalga, bel); GI
Kanjeevaram / Kanchipuram	Tamil Nadu	Heavy mulberry silk; contrasting borders woven separately and interlocked (**korvai** technique); temple motifs
Paithani	Maharashtra (Paithan, Yeola)	Peacock (muniya) and parrot motifs; oblique-square design; zari
Patola	Gujarat (Patan)	**Double ikat** silk: both warp and weft are tie-dyed before weaving. Extremely laborious; the Salvi family.
Pochampally Ikat	Telangana	Geometric ikat; GI
Sambalpuri / Bomkai	Odisha	Ikat (bandha) with tribal motifs
Chanderi	Madhya Pradesh	Sheer, light cotton-silk with buttis
Maheshwari	Madhya Pradesh	Cotton-silk; reversible border; Ahilyabai Holkar's patronage
Baluchari	West Bengal (Bishnupur)	Mythological and narrative scenes woven on the pallu
Jamdani	West Bengal / Bangladesh	Fine muslin with supplementary-weft motifs; inscribed on UNESCO's ICH list for Bangladesh (2013)
Muga silk	Assam	Naturally **golden** silk; GI; the mekhela chador
Eri silk	Assam / NE	"Ahimsa / peace silk" (the moth can emerge before the fibre is spun)
Pashmina	Kashmir / Ladakh	Fine wool from the Changthangi goat; Kani shawls (tapestry twill)
Kota Doria	Rajasthan	Small square checks (khat) in a light weave
Ilkal	Karnataka	Cotton with silk pallu; **tope teni** joining technique
Kasavu	Kerala	Off-white cotton with gold border; worn at Onam
Mangalagiri	Andhra Pradesh	Cotton with Nizam border
Bhagalpuri / Tussar	Bihar / Jharkhand	Wild (tasar) silk
Naga shawls	Nagaland	Tribe-specific colours and stripes

3.2 Embroidery

Embroidery	State	Features
Chikankari	Uttar Pradesh (Lucknow)	White-on-white shadow work on fine cotton; Nur Jahan legend; GI
Phulkari	Punjab / Haryana	"Flower work"; darn stitch with silk floss; Bagh = fully covered
Kantha	West Bengal / Odisha	Running stitch on layered old saris; upcycling
Kasuti	Karnataka	Geometric, counted-thread embroidery
Zardozi	UP, Delhi, Hyderabad	Metallic gold/silver thread; Persian origin
Kashida / Sozni	Kashmir	Fine needlework on shawls
Mirror work (Abhla bharat)	Gujarat (Kutch), Rajasthan	Small mirrors stitched in
Toda embroidery (Pukhoor)	Tamil Nadu (Nilgiris)	Red and black on white

Embroidery	State	Features
Chamba Rumal	Himachal Pradesh	Pahari-style embroidered handkerchiefs
Pipli appliqué	Odisha	Bright cut-cloth appliqué; Jagannath canopies
Lambani	Karnataka / Telangana	Mirrors, coins, bright colours (Banjara community)

3.3 Dyeing & Printing

Technique	Region	Features
Bandhani / Bandhej	Gujarat, Rajasthan	Tie-dye with tiny dots
Leheriya	Rajasthan	Diagonal wave-like stripes (*lehar* = wave)
Ajrakh	Gujarat (Kutch, Ajrakhpur) / Sindh	Resist block print; indigo + madder red; geometric
Sanganeri	Rajasthan	Fine floral block print on a white background
Bagru	Rajasthan	Dabu (mud-resist) printing; earthy colours
Dabu	Rajasthan	Mud-resist
Kalamkari (block)	AP (Machilipatnam)	See above
Bagh print	Madhya Pradesh	Red and black block print
Ikat	Odisha, Telangana, Gujarat	Yarn is resist-dyed *before* weaving (single or double)
Batik	(Indonesia origin), MP, WB	Wax-resist dyeing; crackle effect

Classic trap: ikat dyes the **yarn before weaving**; block printing and batik work on **finished cloth**.

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4. Indian Handicrafts

Craft	State	What it is
Blue Pottery	Rajasthan (Jaipur)	Quartz-based (not clay), cobalt-blue glaze; Turko-Persian origin
Bidriware	Karnataka (Bidar)	Blackened zinc-copper alloy inlaid with silver; GI
Dhokra / Dokra	Chhattisgarh, WB, Odisha, Jharkhand	Non-ferrous metal casting by the **lost-wax** method; tribal figurines
Channapatna toys	Karnataka	Lacquered wooden toys (originally ivory-wood); GI
Kondapalli toys	Andhra Pradesh	Light softwood (tella poniki) toys
Etikoppaka toys	Andhra Pradesh	Lacquer-turned toys with natural dyes
Nirmal toys / paintings	Telangana	—
Aranmula Kannadi	Kerala	A **metal-alloy mirror** (not glass); GI
Pietra dura / Parchinkari	UP (Agra)	Semi-precious stones inlaid in marble (as in the Taj Mahal)
Bankura terracotta horse	West Bengal	An iconic stylised horse
Longpi pottery	Manipur	Black pottery made **without a wheel**, from serpentine rock and clay
Thewa	Rajasthan (Pratapgarh)	Gold sheet fused onto glass
Meenakari	Rajasthan (Jaipur)	Enamelling on metal
Kashmiri papier-mâché	J&K	Painted paper pulp objects
Rogan art	Gujarat (Kutch, Nirona village)	Castor-oil-based paint drawn with a stylus
Sankheda furniture	Gujarat	Lacquered teak with painted designs

Craft	State	What it is
Kinnal craft	Karnataka	Wooden toys and figures
Bastar iron craft	Chhattisgarh	Wrought iron figures
Cane & bamboo	Assam, Tripura, Manipur	Furniture, baskets; the Assamese **jaapi** (hat)
Sikki grass craft	Bihar	Golden-grass baskets and figures
Tarkashi	Rajasthan	Brass wire inlay in wood
Kathputli	Rajasthan	String puppets
Leather puppets (Tholu Bommalata)	Andhra Pradesh	Shadow puppets
Bell metal (Kansa)	Odisha, Assam	Utensils
Khurja pottery	UP	Glazed ceramics
Shola pith	West Bengal	Decorations from shola plant pith (for Durga Puja)
Mata ni Pachedi	Gujarat	Temple cloth honouring the Mother Goddess
Coir craft	Kerala	Coconut fibre products

GI (Geographical Indication): a tag certifying that a product's quality or reputation comes from its place of origin. India's GI law is the *Geographical Indications of Goods (Registration and Protection) Act, 1999*, in force since 2003. The **first Indian GI was Darjeeling Tea (2004)**. The GI Registry is in **Chennai**.

UNESCO Intangible Cultural Heritage from India (selection):

- Kutiyattam
- Vedic chanting
- Ramlila
- Ramman
- Mudiyettu
- Kalbelia
- Chhau
- Buddhist chanting of Ladakh
- Sankirtana (Manipur)
- traditional brass and copper craft of the **Thatheras of Jandiala Guru** (Punjab)
- Yoga
- Nowruz
- Kumbh Mela
- **Durga Puja in Kolkata (2021)**
- **Garba of Gujarat (2023)**

India also had further nominations in the 2024–25 cycles; see Vol 11 for recent items.

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5. Indian Architecture

5.1 Timeline of Indian architecture (with icons)

Period	Key works	Features
Indus Valley (c. 2600–1900 BCE)	Mohenjo-daro (**Great Bath**), Harappa, **Dholavira** (UNESCO 2021), Lothal (dockyard)	Grid town planning, burnt bricks, covered drains

Period	Key works	Features
Mauryan (3rd c. BCE)	**Ashokan pillars**: the Lion Capital of **Sarnath**, India's national emblem; **Sanchi Stupa** (core by Ashoka)	Polished sandstone (the "Mauryan polish")
Buddhist rock-cut	**Ajanta** (murals, 2nd c. BCE–6th c. CE), **Karla chaitya**, **Ellora** (Buddhist, Hindu and Jain caves)	Chaitya (prayer hall) and vihara (monastery)
Gupta (4th–6th c.)	Dashavatara Temple (Deogarh), Udayagiri caves	The beginnings of the Hindu temple form
Early medieval temples	**Kailasa Temple, Ellora (Cave 16)**: carved top-down from a single rock under the Rashtrakuta king **Krishna I** (8th c.). **Pallava Mahabalipuram** (Shore Temple, Rathas, Arjuna's Penance).	Monolithic carving
Nagara style (North)	**Khajuraho** (Chandela), **Konark Sun Temple** (Odisha, 13th c., King Narasimhadeva I of the Eastern Ganga dynasty; designed as a chariot with 24 wheels and 7 horses), **Lingaraja** (Bhubaneswar), **Modhera Sun Temple** (Solanki)	Curvilinear **shikhara**; no boundary walls; the amalaka on top
Dravida style (South)	**Brihadeeswara Temple, Thanjavur** (Rajaraja Chola I, completed c. 1010 CE); Meenakshi Temple (Madurai); Hampi (Vijayanagara)	Pyramidal **vimana**; huge **gopurams** (gateways); enclosed courtyards
Vesara (Deccan hybrid)	**Hoysala temples**: Belur, Halebidu, Somanathapura (the **Sacred Ensembles of the Hoysalas**, UNESCO 2023); star-shaped plans; soapstone	Mixes Nagara and Dravida
Kakatiya	**Ramappa (Rudreswara) Temple**, Telangana (UNESCO 2021); floating bricks	—
Stepwells	**Rani ki Vav**, Patan (11th c., commissioned by Queen Udayamati; UNESCO 2014; shown on the ₹100 note); Adalaj (Gujarat); **Chand Baori** (Abhaneri, Rajasthan)	Water storage and a cooler microclimate
Indo-Islamic (Delhi Sultanate)	**Qutub Minar** (begun by Qutb-ud-din Aibak, completed by Iltutmish; about 73 m); **Alai Darwaza** (true arch, dome)	True arches, domes, calligraphy, geometric ornament
Provincial Islamic	**Gol Gumbaz** (Bijapur, Adil Shahi; a huge dome with a whispering gallery); **Charminar** (Hyderabad, 1591, Muhammad Quli Qutb Shah)	—
Mughal	**Humayun's Tomb** (Delhi, 1560s–70s, commissioned by Bega Begum, architect Mirak Mirza Ghiyas; the **first garden tomb** on the Indian subcontinent; charbagh). **Fatehpur Sikri** (Akbar; Buland Darwaza). **Taj Mahal** (Shah Jahan for Mumtaz Mahal, c. 1632–1653; chief architect usually credited as **Ustad Ahmad Lahauri**; white Makrana marble; pietra dura). **Red Fort** (Delhi). **Jama Masjid** (Delhi).	Charbagh (four-part garden), double domes, symmetry, red sandstone and white marble
Rajput	**Amber Fort**; **Hawa Mahal** (Jaipur, 1799, Lal Chand Ustad; 953 jharokhas); **Jantar Mantar** (Sawai Jai Singh II; UNESCO); Chittorgarh; Mehrangarh	Jharokha, chhatri, jaali
Colonial / Indo-Saracenic	**Chhatrapati Shivaji Maharaj Terminus** (Mumbai, F.W. Stevens, 1888, Victorian Gothic + Indian; UNESCO); **Gateway of India** (George Wittet, 1924); **Victoria Memorial** (Kolkata, William Emerson); **Rashtrapati Bhavan** (Edwin Lutyens); **North & South Blocks** (Herbert Baker); old **Parliament House** (Lutyens & Baker, circular)	Domes, arches, chhatris + European plans
Modern	**Chandigarh** (Le Corbusier with Pierre Jeanneret; Capitol Complex part of a UNESCO transnational serial site, 2016); **IIM Ahmedabad** (Louis Kahn; brick arches); **Mill Owners' Association building** and **Sanskar Kendra**, Ahmedabad (Le Corbusier)	Béton brut (raw concrete), brutalism

Period	Key works	Features
Indian modernists	**B.V. Doshi**: Sangath (his studio), IIM Bangalore, **Aranya low-cost housing** (Indore), Amdavad ni Gufa (with M.F. Husain); **Pritzker 2018** (first Indian). **Charles Correa**: Gandhi Smarak Sangrahalaya (Sabarmati Ashram), Jawahar Kala Kendra (Jaipur; navagraha plan), Kanchanjunga Apartments (Mumbai), Navi Mumbai planning. **Laurie Baker**: low-cost brick architecture in Kerala, rat-trap bond, jaali walls. **Raj Rewal**: Hall of Nations (Pragati Maidan, demolished 2017), Asian Games Village. **Achyut Kanvinde**. **Anant Raje**. **Joseph Allen Stein**: India Habitat Centre. **Hafeez Contractor**: high-rises.	Climate-responsive modernism
Contemporary	**Lotus Temple** (Delhi, 1986, Fariborz Sahba; Bahá'í House of Worship); **Akshardham** (Delhi, 2005); **Statue of Unity** (2018); **New Parliament Building** (2023, **Bimal Patel / HCP Design**; triangular plan); **Navi Mumbai International Airport** (Zaha Hadid Architects; lotus-inspired; inaugurated 8 Oct 2025) [REPORTED, Vol 11]	—

5.2 UNESCO World Heritage Sites in India: key facts

- India's count **reached 44 with the Maratha Military Landscapes (2025)**: 12 forts, 11 in Maharashtra and **Gingee in Tamil Nadu**. **Sarnath** was reported as the **45th**, inscribed at the 48th session in Busan, South Korea, in July 2026. [REPORTED, verify]
- Recent inscriptions to remember:
  - **Dholavira** and **Ramappa** (2021)
  - **Santiniketan** and **Hoysala temples** (2023)
  - **Moidams of Charaideo, Assam** (2024: Ahom royal mound burials)
  - **Maratha Military Landscapes** (2025)
  - **Sarnath** (2026, reported)
- First Indian **World Heritage City**: **Ahmedabad (2017)**. **Jaipur** followed (2019).
- **Khangchendzonga National Park** (Sikkim, 2016) is India's only **mixed** (natural + cultural) site.

5.3 Architectural terms to know

Term	Meaning
Shikhara	Curved tower over the sanctum (North Indian)
Vimana	Pyramidal tower over the sanctum (South Indian)
Gopuram	Monumental gateway tower (South Indian)
Garbhagriha	Sanctum sanctorum ("womb chamber")
Mandapa	Pillared hall
Jaali	Perforated stone or wood screen (light, air, privacy)
Jharokha	Overhanging enclosed balcony
Chhatri	Elevated dome-shaped pavilion
Charbagh	Persian four-quadrant garden
Pishtaq	Formal gateway / arched portal of a Mughal building
Stupa	Hemispherical Buddhist reliquary mound (anda, harmika, chhatra, torana gateways)
Torana	Ornamental gateway (Sanchi)
Baoli / Vav	Stepwell
Haveli	Traditional courtyard mansion (Shekhawati frescoes)
Courtyard (aangan)	Climate and social core of Indian homes
Brutalism	Raw concrete, massive forms
Cantilever	A beam supported at one end only (Fallingwater)

Term	Meaning
Flying buttress	External support of Gothic cathedrals
Geodesic dome	Buckminster Fuller's triangulated dome

6. World Architecture & Architects

Architect	Country	Iconic works	Notes
Antoni Gaudí	Spain	Sagrada Família (unfinished), Casa Batlló, Park Güell	Catalan Modernisme
Frank Lloyd Wright	USA	**Fallingwater** (1935–39), **Guggenheim Museum NY** (spiral, 1959)	Organic architecture; Prairie style
Le Corbusier	Switzerland/France	**Villa Savoye**, Unité d'Habitation, Notre-Dame-du-Haut (Ronchamp), **Chandigarh**	**5 points of architecture**: pilotis, free plan, free façade, horizontal windows, roof garden; the **Modulor** proportion system; "a house is a machine for living in"
Walter Gropius	Germany	**Bauhaus** founder (1919); Bauhaus Dessau building	—
Ludwig Mies van der Rohe	Germany/USA	**Barcelona Pavilion** (1929), Seagram Building, Farnsworth House	"**Less is more**"; last Bauhaus director
Louis Kahn	USA	**IIM Ahmedabad**, National Assembly (Dhaka), Salk Institute	Monumental brick and concrete
Jørn Utzon	Denmark	**Sydney Opera House** (opened 1973)	Pritzker 2003
I.M. Pei	China/USA	**Louvre Pyramid** (1989), Bank of China Tower (HK)	—
Frank Gehry	Canada/USA	**Guggenheim Bilbao** (1997, titanium), Walt Disney Concert Hall, Dancing House (Prague, with Vlado Milunić)	Deconstructivism; the "Bilbao effect"
Zaha Hadid	Iraq/UK	Heydar Aliyev Center (Baku), MAXXI (Rome), London Aquatics Centre, Guangzhou Opera House	**First woman to win the Pritzker (2004)**; "Queen of the Curve"; died 2016. Her firm (ZHA) designed Navi Mumbai Airport.
Norman Foster	UK	**30 St Mary Axe ("The Gherkin")**, Reichstag dome, HSBC HK, Apple Park (with Apple)	High-tech
Renzo Piano	Italy	**Centre Pompidou** (with Richard Rogers, 1977; services exposed outside), **The Shard** (London)	—
Richard Rogers	UK	Lloyd's Building, Millennium Dome	High-tech
Rem Koolhaas / OMA	Netherlands	CCTV HQ (Beijing), Seattle Central Library	—
Tadao Ando	Japan	**Church of the Light**, Row House in Sumiyoshi	Concrete, light, minimalism; a former boxer, self-taught
Kengo Kuma	Japan	**Japan National Stadium** (Tokyo 2020 Olympics), V&A Dundee	Timber, "anti-object"

Architect	Country	Iconic works	Notes
Shigeru Ban	Japan	Paper-tube emergency shelters, Cardboard Cathedral (Christchurch)	Pritzker 2014; humanitarian architecture
SANAA (Sejima & Nishizawa)	Japan	21st Century Museum (Kanazawa), Louvre-Lens	Pritzker 2010
Oscar Niemeyer	Brazil	**Brasília** (Cathedral, National Congress)	Curves in concrete
Buckminster Fuller	USA	**Geodesic dome** (Montreal Biosphère, Expo 67)	"Spaceship Earth"
Eero Saarinen	Finland/USA	**Gateway Arch** (St. Louis), TWA Flight Center	Also designed the Tulip chair
Alvar Aalto	Finland	Paimio Sanatorium; Villa Mairea	Humane modernism; bent plywood
Bjarke Ingels (BIG)	Denmark	8 House, VIA 57 West, CopenHill (a waste-to-energy plant with a ski slope)	"Hedonistic sustainability"
Alejandro Aravena	Chile	**Quinta Monroy "half-houses"** (incremental housing)	Pritzker 2016
Diébédo Francis Kéré	Burkina Faso/Germany	Gando Primary School	**Pritzker 2022, first African laureate**
Gustave Eiffel	France	Eiffel Tower (1889 World's Fair); internal structure of the Statue of Liberty	Engineer
SOM (Adrian Smith)	USA	**Burj Khalifa** (828 m, 2010, the world's tallest)	Its Y-shaped plan was inspired by the *Hymenocallis* desert flower
Moshe Safdie	Israel/Canada	Habitat 67, Marina Bay Sands, Jewel Changi Airport	—
Herzog & de Meuron	Switzerland	**Beijing National Stadium ("Bird's Nest")**, Tate Modern conversion	—

Pritzker Prize (since **1979**; first laureate Philip Johnson). Recent winners:

Year	Laureate
2018	B.V. Doshi (India)
2019	Arata Isozaki (Japan)
2020	Yvonne Farrell & Shelley McNamara (Ireland)
2021	Anne Lacaton & Jean-Philippe Vassal (France)
2022	Diébédo Francis Kéré (Burkina Faso)
2023	David Chipperfield (UK)
2024	Riken Yamamoto (Japan)
2025	**Liu Jiakun** (China)
2026	**Smiljan Radić Clarke** (Chile) [REPORTED 2026; designed the Serpentine Pavilion 2014 and the Biobío Regional Theatre]

Architecture movements

Movement	Era	Characteristics	Examples
Gothic	12th–16th c.	Pointed arches, ribbed vaults, flying buttresses, stained glass	Notre-Dame (Paris)

Movement	Era	Characteristics	Examples
Renaissance	15th–17th c.	Classical symmetry, domes	Florence Cathedral dome (Brunelleschi), St Peter's
Baroque	17th c.	Drama, curves, ornament	Versailles
Art Nouveau	1890–1910	Organic, flowing plant lines	Gaudí, Victor Horta, Paris Métro entrances (Hector Guimard)
Art Deco	1920s–30s	Geometric, streamlined, luxurious	Chrysler Building; **Mumbai's Marine Drive** (Victorian Gothic and Art Deco Ensembles of Mumbai, UNESCO 2018)
Modernism / International Style	1920s–60s	Function, no ornament, steel, glass, concrete	Villa Savoye, Seagram Building
Brutalism	1950s–70s	Raw concrete, massive forms	Barbican; parts of Chandigarh
Postmodernism	1970s–90s	Playful, historic quotes, colour	Robert Venturi; Michael Graves
Deconstructivism	1980s+	Fragmented, non-rectilinear	Gehry, Hadid, Libeskind
High-tech	1970s+	Exposed structure and services	Centre Pompidou, Lloyd's
Parametric	2000s+	Computational curves	Zaha Hadid Architects
Sustainable / green	Now	Passive design, green roofs, net-zero	Bosco Verticale (Milan, Stefano Boeri)

7. Designers: Master Tables

7.1 Indian designers & design figures

Name	Field	Known for
D. Udaya Kumar	Graphic	Designed the **₹ rupee symbol** (adopted 2010)
Gira & Gautam Sarabhai	Institution-builders	Founded NID; Calico Museum of Textiles
M.P. Ranjan	Design educator (NID)	*Handmade in India*; bamboo and cane design research
Kirti Trivedi	Graphic / typography	IDC educator; Indian visual traditions
R.K. Joshi	Typography / calligraphy	Devanagari type design pioneer
Mukund Gokhale	Type designer	Devanagari typefaces
Sudarshan Dheer	Graphic	Pioneer of Indian corporate identities
Rajeev Sethi	Scenographer / craft revivalist	Festival of India exhibitions; Asian Heritage Foundation
Satyendra Pakhalé	Industrial	Amsterdam-based Indian product designer
Sandeep Sangaru	Product / furniture	Bamboo furniture
Ayush Kasliwal	Product / craft	Contemporary craft-based products
Ritu Kumar	Fashion	Revived Indian textiles; block prints
Sabyasachi Mukherjee	Fashion	Bridal couture; heritage textiles
Manish Malhotra	Fashion / costume	Bollywood costume design
Rohit Bal	Fashion	Lotus and peacock motifs (died 2024)
Rahul Mishra	Fashion	Hand embroidery; Paris Haute Couture Week
Anita Dongre	Fashion	Sustainable craft fashion (Grassroot)

Name	Field	Known for
Gaurav Gupta	Fashion	Sculptural couture
Masaba Gupta	Fashion	Quirky Indian prints
Bhanu Athaiya	Costume	**First Indian Oscar winner** (*Gandhi*, 1983, Best Costume Design)
Ashok Chatterjee	Former NID Director	Craft advocacy
Hafeez Contractor, **B.V. Doshi**, **Charles Correa**, **Laurie Baker**	Architecture	See §5

7.2 International graphic designers & typographers

Name	Country	Known for
Paul Rand	USA	Logos for **IBM**, ABC, UPS (1961), Westinghouse
Milton Glaser	USA	**I ♥ NY** (1977); the Bob Dylan poster
Saul Bass	USA	Film title sequences (*Vertigo*, *Psycho*); AT&T logo
Massimo Vignelli	Italy/USA	**NYC Subway map (1972)**; American Airlines identity
Harry Beck	UK	**London Underground diagram (1931/33)**, the topological map
Edward Johnston	UK	**Johnston typeface** (London Underground, 1916)
Otl Aicher	Germany	**Munich 1972 Olympic pictograms**; Lufthansa identity
Josef Müller-Brockmann	Switzerland	Grid systems; Swiss / International Typographic Style
Max Miedinger (with Eduard Hoffmann)	Switzerland	**Helvetica (1957)**, originally "Neue Haas Grotesk"
Adrian Frutiger	Switzerland	**Univers**, **Frutiger** (Charles de Gaulle Airport signage)
Paul Renner	Germany	**Futura (1927)**
Eric Gill	UK	**Gill Sans**, Perpetua
Stanley Morison	UK	**Times New Roman (1932)** for *The Times*
Matthew Carter	UK/USA	**Verdana, Georgia** (screen fonts for Microsoft)
Herb Lubalin	USA	ITC Avant Garde
Jan Tschichold	Germany	*Die neue Typographie* (1928); Penguin book redesign
Claude Garamond / **John Baskerville** / **Giambattista Bodoni**	France / UK / Italy	Classic serif typefaces
Johannes Gutenberg	Germany	Movable-type printing press (c. 1440)
Carolyn Davidson	USA	**Nike Swoosh (1971)**, famously for $35
Rob Janoff	USA	**Apple rainbow logo (1977)**
Lindon Leader	USA	**FedEx logo** (1994), with the hidden arrow
Susan Kare	USA	Original **Macintosh icons** and fonts
Paula Scher	USA	Citibank, The Public Theater identities
Chermayeff & Geismar	USA	Chase, NBC peacock, Mobil, PBS
Alan Fletcher	UK	V&A logo; co-founded Pentagram
Shepard Fairey	USA	Obama *"Hope"* poster (2008)

7.3 Industrial / product & furniture designers

Name	Country	Known for
Dieter Rams	Germany	Braun products (SK 4 record player, T3 radio); **10 principles**; Vitsœ 606 shelving
Jony Ive	UK	Apple: iMac G3 (1998), iPod, iPhone, MacBook
Raymond Loewy	France/USA	"Father of industrial design": Lucky Strike pack, Shell logo, Greyhound Scenicruiser, Studebaker; **MAYA principle** ("Most Advanced Yet Acceptable")
Charles & Ray Eames	USA	**Eames Lounge Chair (1956)**, moulded plywood chairs; *Powers of Ten* film; **The India Report (1958)**
Marcel Breuer	Hungary/Germany	**Wassily Chair** (1925, tubular steel, inspired by bicycle handlebars)
Mies van der Rohe	Germany	**Barcelona Chair** (1929)
Gerrit Rietveld	Netherlands	**Red and Blue Chair** (c. 1918); Rietveld Schröder House (De Stijl)
Michael Thonet	Germany/Austria	**No. 14 bistro chair (1859)**, steam-bent beechwood; flat-packed
Arne Jacobsen	Denmark	Egg Chair, Swan Chair, Series 7 chair
Hans Wegner	Denmark	Wishbone Chair
Alvar Aalto	Finland	Paimio Chair, Stool 60, **Savoy vase**
Eero Saarinen	Finland/USA	**Tulip Chair** (pedestal base)
Verner Panton	Denmark	**Panton Chair** (the first single-piece moulded plastic cantilever chair)
Philippe Starck	France	**Juicy Salif** citrus squeezer (Alessi, 1990), Louis Ghost chair
Ettore Sottsass	Italy	**Olivetti Valentine typewriter** (1969); founder of the **Memphis Group** (1981)
Marc Newson	Australia	Lockheed Lounge
Naoto Fukasawa	Japan	MUJI wall-mounted CD player; "without thought" design
Kenji Ekuan	Japan	**Kikkoman soy sauce dispenser (1961)**; Shinkansen designs
Sori Yanagi	Japan	**Butterfly Stool** (1954)
Isamu Noguchi	USA/Japan	**Akari** paper lamps; Noguchi coffee table
Ferdinand Porsche	Austria/Germany	**Volkswagen Beetle**
Dante Giacosa	Italy	**Fiat 500** (1957)
Alec Issigonis	UK	**Mini** (1959)
Corradino D'Ascanio	Italy	**Vespa** scooter (1946)
Karim Rashid	Egypt/Canada	Garbo trash can; "blobject" style
Konstantin Grcic	Germany	Chair_One
Yves Béhar	Switzerland/USA	Jawbone; One Laptop per Child XO

7.4 Fashion designers

Name	Country	Known for
Coco Chanel	France	The little black dress; Chanel No. 5; tweed suit; liberated women's fashion
Christian Dior	France	**"New Look" (1947)**
Yves Saint Laurent	France	*Le Smoking* tuxedo for women (1966); Mondrian dress (1965)
Cristóbal Balenciaga	Spain	Sculptural couture

Name	Country	Known for
Giorgio Armani	Italy	Unstructured suits (died 2025)
Issey Miyake	Japan	*Pleats Please*; A-POC (died 2022)
Rei Kawakubo	Japan	Comme des Garçons; deconstruction
Yohji Yamamoto	Japan	Black, oversized avant-garde
Vivienne Westwood	UK	Punk fashion (died 2022)
Alexander McQueen	UK	Theatrical shows
Karl Lagerfeld	Germany	Chanel creative director (1983–2019)
Ralph Lauren	USA	Polo; American preppy

8. Art Movements (world)

Movement	Period	Key artists	Features
Renaissance	14th–17th c.	Leonardo (*Mona Lisa*, *The Last Supper*), Michelangelo (Sistine Chapel, *David*), Raphael	Linear perspective (Brunelleschi), humanism, realism
Baroque	17th c.	Caravaggio, Rembrandt, Rubens	Drama; **chiaroscuro** (strong light and dark)
Romanticism	late 18th–19th c.	Delacroix, Turner, Goya	Emotion, nature's power
Realism	mid-19th c.	Courbet, Millet	Everyday life
Impressionism	1870s	**Monet** (*Impression, Sunrise* gave the movement its name), Renoir, Degas	Light, outdoor painting (en plein air), visible brushstrokes
Post-Impressionism	1880s–1900s	**Van Gogh** (*The Starry Night*), Cézanne, Gauguin, **Seurat** (pointillism: *A Sunday Afternoon on the Island of La Grande Jatte*)	Expressive colour, structure
Art Nouveau	1890–1910	Alphonse Mucha, Gustav Klimt (*The Kiss*)	Organic lines, decorative
Expressionism	1905–1920s	**Edvard Munch** (*The Scream*), Kirchner	Emotional distortion
Fauvism	1905	**Henri Matisse**	Wild, non-natural colour
Cubism	1907–1920s	**Picasso** (*Les Demoiselles d'Avignon*, *Guernica* 1937), **Georges Braque**	Multiple viewpoints, geometric fragmentation
Futurism	1909 (Italy)	Boccioni, Balla	Speed, machines, motion
Dada	1916 (Zurich)	**Marcel Duchamp** (*Fountain*, 1917, a urinal "readymade")	Anti-art, absurdity
Surrealism	1924+	**Salvador Dalí** (*The Persistence of Memory*, melting clocks), René Magritte (*The Treachery of Images*: "Ceci n'est pas une pipe"), Frida Kahlo (associated)	Dreams, the unconscious
De Stijl	1917 (Netherlands)	**Piet Mondrian**, Theo van Doesburg, Rietveld	Primary colours + black/white, horizontal and vertical grids
Bauhaus	1919–1933	Gropius, Kandinsky, Klee, Moholy-Nagy, Josef Albers, Marianne Brandt, Anni Albers	Unity of art, craft and technology; "form follows function" applied
Constructivism	1915+ (Russia)	Tatlin, Rodchenko, El Lissitzky	Art for social purpose; graphic posters

Movement	Period	Key artists	Features
Art Deco	1920s–30s	Cassandre (posters), Erté	Geometry, luxury, streamlining
Abstract Expressionism	1940s–50s (NY)	**Jackson Pollock** (drip painting), Mark Rothko (colour fields), de Kooning	Gesture, emotion, large scale
Pop Art	1950s–60s	**Andy Warhol** (*Campbell's Soup Cans*, Marilyn), Roy Lichtenstein (comic dots), Richard Hamilton	Consumer culture, mass media
Op Art	1960s	**Bridget Riley**, **Victor Vasarely**	Optical illusions
Minimalism	1960s	Donald Judd, Dan Flavin, Agnes Martin	Simple geometric forms
Memphis Design	1981–88	Ettore Sottsass	Bold colours, kitsch geometry, anti-modernist
Street art	1980s+	Banksy, Keith Haring, Jean-Michel Basquiat	Public, political

Design movements (summary):

- **Arts and Crafts** (William Morris, 1860s–1910, UK): a reaction against industrial mass production; handcraft.
- Art Nouveau.
- Deutscher Werkbund (1907).
- Bauhaus.
- Streamlining (1930s USA, Loewy).
- **Mid-century modern** (1945–1970: Eames, Saarinen).
- **Scandinavian design** (functional, natural materials).
- Swiss / International Typographic Style.
- Memphis.
- Postmodern.
- **Flat design** (2010s UI; Windows Metro, iOS 7 in 2013).
- **Skeuomorphism** (UI that imitates real materials: early iOS).
- **Neumorphism** (2020s).
- Apple's **"Liquid Glass"** design language (2025).

9. Famous Logos & Brands (frequently tested in MCQ years)

Logo	Hidden or notable feature	Designer / year (where known)
FedEx	Hidden arrow between E and x	Lindon Leader, Landor (1994)
Amazon	Arrow from **a to z** = a smile	Turner Duckworth (2000)
Apple	Bitten apple; "bite"/"byte" pun; rainbow version 1977–1998	Rob Janoff
Nike	Swoosh = the wing of Nike, Greek goddess of victory	Carolyn Davidson (1971)
Toyota	Three ovals; all letters of TOYOTA can be traced in the emblem	1989
Baskin-Robbins	Pink "31" in "BR" (31 flavours)	—
Tour de France	A cyclist hidden in the letter R	—
Toblerone	A bear hidden in the Matterhorn (the symbol of Bern)	—

Logo	Hidden or notable feature	Designer / year (where known)
WWF	Panda; **closure** principle	Peter Scott (1961); refined by Landor (1986)
Olympic rings	Five interlocking rings (blue, yellow, black, green, red on white). At least one of these colours appears in every nation's flag.	Pierre de Coubertin (1913)
Mercedes-Benz	Three-pointed star = land, sea, air	—
Audi	Four rings = the four merged companies (Auto Union)	—
BMW	Roundel in Bavarian blue and white (often wrongly said to be a propeller)	—
Starbucks	Twin-tailed siren (mermaid)	—
Shell	Scallop shell	Raymond Loewy redesign (1971)
McDonald's	Golden Arches	—
Pepsi	Globe	—
Google	Multicoloured wordmark; sans-serif since 2015 (Product Sans)	—
Unilever	The "U" is made of 25 icons (sun, hand, DNA, etc.)	Wolff Olins (2004)
Wikipedia	An incomplete puzzle globe	—
Indian brands	**Amul girl** (mascot, 1966, by Sylvester daCunha's agency; Eustace Fernandes drew her); **Air India Maharaja** (Bobby Kooka & Umesh Rao, 1946); **Tata** "T" (a well of knowledge); **LIC** (two hands protecting a lamp; motto "Yogakshemam Vahamyaham"); **Asian Paints Gattu** (R.K. Laxman); **India Post** (red, with a flying-envelope bird)	—
Indian govt symbols	**National emblem** (adapted from the **Lion Capital of Ashoka, Sarnath**; motto *Satyameva Jayate* from the Mundaka Upanishad); **₹ symbol** (Devanagari "र" + Roman "R" + a double line suggesting the tricolour / equality sign); **Swachh Bharat** logo (Gandhi's spectacles); **Make in India** lion (made of cogs); **Incredible India** (the "!" as the letter i)	—

10. Iconic Products (often asked: "who designed / which brand")

Product	Designer / company	Year	Why iconic
Coca-Cola contour bottle	Earl R. Dean (Root Glass Co.)	1915	Recognisable by touch in the dark
Thonet No. 14 chair	Michael Thonet	1859	Mass production; flat-pack
Swiss Army Knife	Karl Elsener (Victorinox)	1891	Multi-function
Zippo lighter	George Blaisdell	1933	Windproof
Anglepoise lamp	George Carwardine	1932	Spring-balanced arm
Moka pot	Alfonso Bialetti	1933	Octagonal aluminium
Vespa	Corradino D'Ascanio / Piaggio	1946	Step-through frame

Product	Designer / company	Year	Why iconic
Polaroid SX-70	Edwin Land / Polaroid	1972	Instant photography
Sony Walkman	Sony (Akio Morita, Masaru Ibuka)	1979	Personal music
iPhone	Apple (Jony Ive's team)	2007	Multi-touch smartphone
Monobloc plastic chair	(Many; Henry Massonnet's "Fauteuil 300", 1972)	1970s	The world's most common chair
Indian icons	**Ambassador car** (Hindustan Motors; based on the Morris Oxford); **Bajaj Chetak** scooter; **Royal Enfield Bullet**; **Hero/Atlas bicycles**; **Prestige/Hawkins pressure cooker**; **Tata Nano** (2008, "₹1 lakh car"); **Godrej Storwel almirah**; **steel tiffin / dabba**; **Bajaj fan**; the **kulhad**; **Jaipur Foot** (prosthetic limb, Dr P.K. Sethi & Ram Chandra Sharma, 1969)		

11. Museums, Institutions & Awards

Museums (India):

- **National Museum** (Delhi)
- **NGMA** (National Gallery of Modern Art: Delhi, Mumbai, Bengaluru)
- **Chhatrapati Shivaji Maharaj Vastu Sangrahalaya** (Mumbai; formerly the Prince of Wales Museum)
- **Indian Museum** (Kolkata, 1814, the oldest in India)
- **Salar Jung Museum** (Hyderabad)
- **Calico Museum of Textiles** (Ahmedabad)
- **Crafts Museum** (National Handicrafts and Handlooms Museum, Delhi; designed by Charles Correa)
- **Kiran Nadar Museum of Art** (Delhi)
- **Dr. Bhau Daji Lad Museum** (Mumbai)
- **Pradhanmantri Sangrahalaya** (Delhi, 2022)
- **Museum of Art & Photography (MAP)** (Bengaluru, 2023)

Museums (world):

- **Louvre** (Paris; Mona Lisa)
- **MoMA** (New York; its design collection)
- **Metropolitan Museum of Art** (New York)
- **Victoria and Albert Museum (V&A)** (London; art and design)
- **Design Museum** (London)
- **Vitra Design Museum** (Weil am Rhein, Germany; building by Gehry)
- **Cooper Hewitt, Smithsonian Design Museum** (New York)
- **Guggenheim** (New York, Bilbao)
- **Tate Modern** (London)
- **Uffizi** (Florence)
- **Rijksmuseum** and **Van Gogh Museum** (Amsterdam)
- **Prado** (Madrid)
- **Hermitage** (St Petersburg)
- **Louvre Abu Dhabi** (Jean Nouvel)
- **Grand Egyptian Museum** (Giza; fully opened Nov 2025)

Design awards & events:

- **Red Dot** (Germany)

- **iF Design Award** (Germany)
- **Compasso d'Oro** (Italy, since 1954, an idea of Gio Ponti; the "Oscar of design")
- **Good Design Award** (Japan's G Mark; also Chicago)
- **IDSA IDEA** (USA)
- **D&AD Pencils** (UK)
- **Cannes Lions** (advertising)
- **India Design Mark** (the I Mark, by the India Design Council)
- **CII Design Excellence Awards**
- **Kyoorius Design Awards** (India)

Design weeks & fairs:

- **Milan Design Week / Salone del Mobile** (April)
- **India Design ID** (New Delhi, February)
- **London Design Festival** (September)
- **Dutch Design Week** (Eindhoven, October)

Biennales:

- **Venice Biennale** (art and architecture in alternate years)
- **Kochi-Muziris Biennale** (India's largest; started 2012)
- **India Art Fair** (Delhi)
- **Serendipity Arts Festival** (Goa)

World Design Organization (WDO) names the **World Design Capital**: Torino 2008 (the first), Seoul 2010, Helsinki 2012, Cape Town 2014, Taipei 2016, Mexico City 2018, Lille Metropole 2020, Valencia 2022, San Diego–Tijuana 2024, **Frankfurt Rhine-Main 2026** (the first in Germany; theme "Design for Democracy").

India Design Council was set up under the **National Design Policy (2007)**.

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12. Sustainability, Technology & AI in Design (concepts)

- **Circular design**, **biomaterials** (mycelium packaging, banana-fibre textiles, Piñatex from pineapple leaves), **upcycling**, **zero-waste fashion**, **slow fashion** vs fast fashion.
- **Universal design / accessibility**: the Accessible India Campaign (Sugamya Bharat Abhiyan, 2015).
- **Generative AI in design**: text-to-image tools, AI-assisted layouts, the ethics of training data and authorship. **Human-in-the-loop** design keeps humans in charge of judgement, empathy and context.
- **UI/UX**: user interface vs user experience; wireframes; prototypes; user testing.
- **3D printing / additive manufacturing**; **CNC**; **parametric design**; **digital twins**.
- **Smart cities**: India's Smart Cities Mission (2015).
- **Electric mobility**: EV design, charging infrastructure.

NEXT VOLUME: Volume 3 — Drawing & Creative Preparation (`03-VOL3-DRAWING-CREATIVE.md`)

VOLUME 3 — DRAWING & CREATIVE PREPARATION

> **The single most important truth:** NID does **not** reward photorealism. It rewards **clear thinking made visible**: observation, ideas, storytelling, empathy, composition and communication. A confident, well-observed, slightly rough drawing with a strong idea beats a polished drawing of a weak idea. [INFERRED from the nature of NID questions and consistent with NID's stated aims]

Contents

1. What examiners look for
2. Tools & materials
3. Core drawing skills, with exercises for each
4. Perspective scenes
5. Viewpoints & imagination
6. Human figures, faces, hands, expressions, gestures
7. Light, shadow, texture & materials
8. Product sketching & redesign
9. Visual communication (posters, logos, signage, packaging)
10. Sequential drawing & visual storytelling
11. Memory drawing
12. Studio Test (Mains) and In-Person Test preparation
13. **The 110-prompt subjective bank** (with guidance and evaluation criteria)

1. What NID examiners are likely looking for [INFERRED]

Criterion	What it means	Self-check question
Observation	Real-life details that make a drawing believable	"Would someone from this place recognise it?"
Idea quality	A thoughtful, relevant interpretation	"Is my idea better than the first thing everyone thinks of?"
Originality	A personal, unexpected angle	"Would 100 other students draw this?"
Communication	The message is clear in about 5 seconds	"Can someone explain my drawing back to me?"
Composition	The frame is used well; there is a focal point	"Where does the eye land first?"
Proportion	Correct relative sizes	"Is the person taller than the door? (They shouldn't be.)"
Clarity	Clean lines; labels where useful	"Is anything confusing?"
Storytelling	Action, emotion, cause and effect	"What is happening, and why?"
Problem-solving	A solution that fits the user and context	"Does it actually solve the problem for this user?"

2. Tools & materials (exam kit)

- **Pencils:** HB (construction), 2B (outlines), 4B–6B (dark accents).
- **Eraser:** a plastic eraser plus a kneaded eraser (to lift highlights).
- **Black fine-liners:** 0.3 and 0.5 mm, for confident final lines.

- **Colour:** 24-shade colour pencils (layer them; don't press hard at first) and oil pastels (for big areas and posters). Use sketch pens for posters only, sparingly.
- **Other:** a 30 cm scale, a sharpener, and a watch (if permitted).
- **Practice paper:** an A4 sketchbook (80–100 sheets) plus newsprint for warm-ups.

3. Core drawing skills

3.1 Line quality

- **What:** confident, varied lines. Thick lines for outlines and near objects; thin for details and distance.
- **Rule:** draw from the shoulder or elbow for long lines and from the wrist for details. Use "ghosting": move the hand over the line path 2–3 times, then draw it.
- **Exercises:**
    1. Fill a page with parallel straight lines (horizontal, vertical, diagonal), 5 minutes daily for 2 weeks.
    2. Draw 50 ellipses of different sizes in boxes.
    3. Line-weight drill: redraw a simple cup, varying the weight (heavy on the shadow side and contact points).

3.2 Shapes & forms

- Everything can be built from **cubes, cylinders, spheres and cones**. A pressure cooker = cylinder + dome + a handle (an elongated box). A bus = a long box. A person = an oval head + a box torso + cylinder limbs.
- **Exercises:**
    1. Draw 10 household objects, starting from basic forms.
    2. Rotate a cube in 6 positions.
    3. Combine 2 forms into a new object (e.g. a cylinder + a cone becomes a lamp).

3.3 Proportion

- **Sight-size and measuring:** use a pencil at arm's length to compare heights and widths.
- **Human proportions:** an adult is about **7.5–8 heads tall**; a child of 5–6 years is about 5–6 heads. The halfway point of an adult is the hips/crotch. With arms down, the fingertips reach mid-thigh.
- **Indian context references:**

Object	Approximate size
Door	2.1 m
Auto-rickshaw	~1.7 m tall, 2.6 m long
Two-wheeler	~1.1 m tall
Cycle-rickshaw seat	~0.9 m
Bus	~3 m tall
Chair seat	~0.45 m
Table	~0.75 m
Kitchen platform	~0.85 m

- **Exercise:** draw a person standing beside a door, an auto, a scooter and a bus (4 small studies). Check the ratios against the table.

3.4 Light & shadow (more in §7)

- Identify the **light source**. Shade the planes facing away. Add the **cast shadow** on the ground, opposite the light.
- Elements: highlight, light, halftone, core shadow, reflected light, cast shadow.
- **Exercise:** draw a sphere, a cube and a cylinder lit from the top-left; render them with a 5-value scale.

4. Perspective scenes (VERY HIGH priority)

The 6-step scene method (15–20 minutes)

1. **Decide the story** (30 sec): "Evening vegetable market; a vendor weighs tomatoes for an old woman."
2. **Draw the horizon line** at eye level. Place a VP (1-point) or 2 VPs (2-point).
3. **Block in big shapes:** buildings, road, stalls, as boxes in perspective.
4. **Add people:** heads on the HL for standing adults; place them at different depths (foreground, mid-ground, background). Overlap them.
5. **Add details that tell the story:** weighing scale, tomatoes, cloth bag, price board in Hindi/English, stray dog, bicycle, wires, light bulbs on the stalls.
6. **Finish:** a darker line weight in front, light tone or colour, shadows consistent with the light source.

Scene checklist: focal action ✓ · 3 depth layers ✓ · 5+ people with different actions ✓ · context details ✓ · consistent light ✓.

Exercises (do 1 per day, rotating):

#	Scene	Perspective
1	A railway platform with a train arriving	1-point
2	Your classroom from the last bench	1-point
3	A street corner tea stall	2-point
4	A three-storey building cut open (a section view showing life on each floor)	1-point frontal, stacked
5	A busy bus stop during rain	2-point
6	A crowded temple or mosque entrance during a festival	1-point
7	A hospital waiting area	2-point
8	A rooftop on Makar Sankranti (kite festival)	3-point (bird's-eye)
9	A view looking up at a metro bridge	3-point (worm's-eye)
10	A narrow lane in an old city (like Varanasi)	1-point, steep

The "three-storey scene" (reported in DAT 2026): draw a building in **section** (front wall removed, like a doll's house). Show a different activity on each floor: a shop on the ground floor, a family dinner on the 1st, a student studying on the 2nd, and a terrace with clothes drying and water tanks. Connect the floors with a staircase showing a person moving between levels: that creates the story. Keep the floor heights consistent (about 3 m, so about 1.7 people tall per floor).

5. Viewpoints & imagination

Viewpoint	How to draw	Example prompt
Bird's-eye	High horizon (or off the page); 3rd VP below; we see tops of heads and roofs	"Your street as seen by a kite"

Viewpoint	How to draw	Example prompt
Worm's / ant's-eye	Very low horizon; 3rd VP above; huge foreground objects	"A cricket match seen by an ant on the pitch"
Crow's-view (DAT 2025 poster)	Elevated, slightly angled; a crow's perspective	"Poster for an orange juice corner from a crow's view"
Through an object	A frame within the frame	"The world seen through a keyhole / a spectacle lens"
Microscopic	Exaggerated scale; textures become landscapes	"A day in the life of a dust particle on your desk"
Inside an object	A cutaway	"Inside a pressure cooker"
Emotional viewpoint	Distortion that conveys feeling	"The exam hall as seen by a nervous student"

Imagination drills:

- "What if" prompts: what if buildings could walk?
- Combine 2 objects into 1 creature.
- Show a familiar place 100 years in the future.
- Personify objects: the life of a forgotten umbrella.

6. Human figures, faces, hands, expressions, gestures

6.1 Gesture & stick figures (most important for scenes)

- Start with the **line of action**, a single curve showing the pose's energy.
- Then draw a **bean** torso (chest + pelvis with a twist), oval head, cylinder limbs, and simple hands and feet.
- **Exercise:** 30-second gestures × 20 daily (people in videos, family members, or photos of crowds).

6.2 Body proportions & poses

- Draw standing, sitting (cross-legged on the floor, a common Indian pose), squatting, walking, running, bending, lifting, carrying a load on the head, cycling, and riding pillion.
- **Balance rule:** for a standing figure, the head is roughly above the weight-bearing foot.

6.3 Faces & expressions

- Face proportions: the eyes sit halfway down the head; the nose halfway between the eyes and the chin; the mouth a third of the way from nose to chin.
- **Expressions via eyebrows + mouth:**

Emotion	Eyebrows	Mouth
Happy	relaxed	upturned
Sad	inner ends raised	down
Angry	inner ends down (V-shape)	tight or showing teeth
Surprised	high	open "O"
Afraid	raised and tense	stretched
Disgust	lowered	upper lip raised, nose wrinkled

- **Exercise:** a 6-emotion grid with the same character, daily for a week.

6.4 Hands

- Treat the hand as a **mitten** shape first, then split the fingers. The palm is about as long as the middle finger.
- **Exercise:** draw your non-drawing hand in 10 poses (gripping a cup, pointing, holding a phone, fist, writing).

6.5 Indian people & clothing details

Saree pleats, kurta, dhoti, salwar, school uniforms, turbans (regional styles), a gamchha on the shoulder, chappals, a police uniform, a vendor's apron, a construction worker's helmet, a doctor's coat.

7. Light, shadow, texture & material representation

Material	How to show it
Glass	Strong, sharp highlights; see-through background with slight distortion; reflections at the edges
Polished metal (steel tiffin)	High contrast; reflected surroundings as bands; bright highlights next to dark
Matte plastic	Soft gradients; small soft highlights
Wood	Grain lines following the form; warm tones
Fabric	Folds radiating from points of tension (shoulders, knees); soft shadows
Water	Horizontal reflections; ripples; transparency
Terracotta / clay	Matte, warm; rough stippled texture
Concrete / stone	Speckle texture; cracks
Rubber	Dark, matte; soft highlights
Leaves / foliage	Clusters rather than single leaves; light on top, dark beneath

Exercise: draw the same cylinder in 6 materials: glass, steel, wood, clay, fabric-covered and rubber.

8. Product sketching & redesign (VERY HIGH priority)

8.1 The 5-step redesign method ("PUCIS")

1. **P**roblem: list 4–6 problems with the current product (observe users).
2. **U**ser: who is the user? Age, abilities, context (e.g. an elderly user on a train).
3. **C**onstraints: cost, material, manufacturing, space, electricity, water, culture.
4. **I**deas: 3 thumbnail concepts, each 1–2 minutes.
5. **S**ketch the selected concept: a big main view (3/4 perspective) + 1–2 detail views + labels + a small "in use" scene + a 1–2 line justification.

8.2 Product sketch layout for an answer box

```
+-------------------------------------------------------+
|  TITLE: "EasyPour Pressure Cooker for Elderly Users"   |
|  [Problems: 1. heavy  2. hot handle  3. whistle fear]  |
|                                                        |
|        [ BIG 3/4 VIEW OF PRODUCT ]      [detail 1]     |
|          labels with arrows →           [detail 2]     |
|                                                        |
|  [small in-use scene: user lifting with two handles]   |
|  Materials: ...    How it helps: ...                   |
+-------------------------------------------------------+
```

8.3 Product sketching technique

- Use 3/4 view (two-point) for most products; add a front or side elevation if helpful.
- Ghost the construction lines lightly, then darken the final silhouette.
- Show **mechanisms** with arrows (rotate, slide, fold) and **exploded views** for assembly.
- Show **scale** by adding a hand holding the product.

8.4 Products to practise redesigning (common Indian context)

pressure cooker · bucket and mug · water bottle for school kids · school bag · auto-rickshaw seat · bus stop · street-vendor cart · tiffin · steel almirah · ceiling fan cleaning · chappal · umbrella · gas cylinder trolley · clothes-drying stand for balconies · walking stick · pill organiser · broom · dustbin with segregation · railway berth ladder · hospital stretcher · temple shoe stand · public water tap · mobile charging kiosk · tea kettle for a chaiwala · kite spool (charkha) · rangoli tool · bicycle carrier.

9. Visual communication

9.1 Poster process (20–35 min)

1. **Message** in one sentence ("Drink fresh orange juice at Sunny Corner").
2. **Audience & tone** (kids? commuters? fun or serious?).
3. **Big idea:** a visual metaphor or twist (the orange as the sun, seen by a crow from above).
4. **Thumbnails:** 3 layouts at 3×4 cm.
5. **Final:** headline (big, 3–6 words), one dominant image, a supporting line, a call to action and logo, a limited palette (3–4 colours), and clear hierarchy.

Visual metaphor bank:

- Earth as a melting ice cream (climate)
- A tree with lungs as foliage (air)
- A tap dripping hourglass sand (water scarcity)
- A pencil as a rocket (education)
- A seat belt as a hug (road safety)

9.2 Logo / symbol process

1. Brainstorm keywords (the name, values, activity).
2. Look for **double meanings and negative space**.
3. Draw 10 quick thumbnails.
4. Select and refine using geometric construction.
5. Test it small (1 cm) and in black and white.
6. Show the final in colour plus one application (signboard, cup, app icon).

9.3 Signage set (e.g. "5 signboards for narrow streets," DAT 2026)

- Use a **consistent system**: the same shape family, colour code, icon style and font.
- Show the **sign in context**: one small drawing of a sign mounted in the lane, at a believable height.
- Prioritise icons with minimal bilingual text.

9.4 Packaging

- Show a dieline (net) + the 3D form + the front panel design + the opening mechanism.
- Consider protection, stacking, the user's grip, reuse after use, and material (paper, cloth, leaf).

10. Sequential drawing & visual storytelling (VERY HIGH priority)

Rules

1. **Plan the beats first:** setup → problem → action → resolution (4 frames), or add build-up frames for 6.
2. **Continuity:** the same character design (clothes, hair, props) and setting in every frame.
3. **Vary the shots:** wide (establishing), medium (action), close-up (emotion or detail).
4. **Use the 180° rule:** keep the camera on one side of the action, so characters don't swap sides.
5. **Show emotion** through faces and body language.
6. **Minimal text:** captions of 3–6 words if allowed; speech bubbles are sparing.
7. **Number the frames** and keep the reading order left→right, top→bottom.

Worked plan: "Helping an elderly person who falls during a morning walk" (DAT 2026, reported)

Frame	Shot	Content
1	Wide	A park at dawn, joggers, an elderly man walking with a stick; a loose paving stone visible (foreshadowing)
2	Medium	He trips on the stone; the stick flies; a young girl notices
3	Close-up	His pained face; the girl kneels and asks "Are you okay?"; her phone in hand
4	Medium	She helps him sit on a bench; another walker brings water; she calls his family / 108 ambulance
5	Medium	Family arrives; gratitude; the girl smiles
6	Wide / epilogue	Next morning, the paving stone is fixed and a "Mind your step" sign is placed: the systemic solution. This shows design thinking.

Exercises: 2 stories per week from the prompt bank (§13, category C).

11. Memory drawing

- **What:** drawing objects, places or events from memory, accurately and with characteristic details.
- **Method:** recall the overall form → key proportions → the defining details (logo position, buttons, joints) → the context of use.
- **Weekly drill:** draw 5 objects from memory, then compare them with reality and circle the errors in red. Common subjects:
  - pressure cooker
  - ceiling fan
  - auto-rickshaw
  - bicycle (chain on the right; pedals; seat; stand)
  - school bench
  - railway coach interior
  - steel glass

    - matchbox
    - water cooler
    - post box (red, cylindrical)
    - traffic signal
    - gas stove
    - street lamp
    - your own house's entrance
- **Food illustration (DAT 2026: shikanji / thandai):** show the vessel (steel glass or kulhad), the ingredients around it (lemon slices, mint, ice, black salt; for thandai: almonds, saffron, rose petals, fennel), condensation droplets, the serving context (a street cart, a Holi setting), and labels. Colour helps a lot here.

12. Mains preparation

12.1 Studio Sensitivity Test

Skills:

- clay modelling (pinch, coil, slab)
- paper engineering (folding, pop-up, scoring, curling)
- wire forms
- found-object assemblage
- material textures

Weekly practice (start right after Prelims, or earlier on weekends):

1. Clay: model 3 animals in different expressions (30 min).
2. Paper: create a structure that holds a 500 g weight using only 2 A4 sheets.
3. Wire: make a figure showing movement (dancing, running).
4. Theme models: "Peace", "Speed", "Friendship", "Growth", "Balance". Interpret abstractly and literally.
5. Explain each model in 3 lines (concept, material choice, what it shows).

Evaluation logic [INFERRED]: concept + craft + material sensitivity + use of time.

12.2 In-Person Sensitivity Test (interview-style)

Prepare answers (spoken, 1–2 min each) for:

1. Why design? Why NID?
2. Which discipline and why?
3. A product you love or hate, and why. Have a redesign idea ready.
4. A craft from your state: its process, the community, its challenges.
5. A recent design or architecture news item (Vol 11).
6. Your hobbies and what you observe through them.
7. A problem in your city you would like to solve.
8. Your Studio Test model: explain it.

Tips: be honest; think aloud; give specific examples; it's fine to say "I don't know, but I think...".

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13. THE 110-PROMPT SUBJECTIVE BANK

Each prompt lists: **Consider** (what to think about) · **Composition** · **Idea strategy** · **Common mistakes** · **Evaluation criteria**. Suggested time: 15–25 min each unless stated. The "Evaluation" line names what an examiner would look for. Use it to self-score out of 10.

A. Draw from Memory / Observation (A1–A12)

A1. Draw a pressure cooker from memory, showing how the whistle works.

- Consider: body, lid with gasket, handle(s), whistle weight, safety valve, base.
- Composition: a large 3/4 view + a cutaway of the whistle/vent.
- Idea: a cutaway with steam arrows.
- Mistakes: forgetting the gasket and safety valve; the handle on the wrong side.
- Evaluation: accuracy, mechanism clarity, labels, proportion.

A2. Draw the inside of a railway sleeper coach from the aisle.

- Consider: berths (upper, middle, lower, side), ladder, windows with bars, fans, chains, luggage, passengers.
- Composition: 1-point perspective down the aisle.
- Idea: show a story, e.g. a family sharing food.
- Mistakes: incorrect berth arrangement; a flat drawing without perspective.
- Evaluation: perspective, observation, life.

A3. Draw an auto-rickshaw from a 3/4 front view.

- Consider: the three wheels, single headlight (older models) or twin, canopy, meter, driver seat, passenger bench.
- Composition: 2-point perspective.
- Idea: add a driver and a passenger to show scale.
- Mistakes: four wheels; wrong canopy shape.
- Evaluation: form, proportion, detail.

A4. Draw your school's main gate at 8 am.

- Consider: guard, students, parents, vehicles, vendors.
- Composition: a frontal or angled perspective.
- Idea: show the morning rush.
- Mistakes: an empty scene.
- Evaluation: observation, activity, depth.

A5. Draw a vegetable vendor's cart with at least 8 identifiable vegetables.

- Consider: the weighing scale, umbrella, cart wheels, vendor, customer.
- Composition: 2-point perspective.
- Idea: show bargaining.
- Mistakes: vegetables that can't be identified.
- Evaluation: identifiability, colour, story.

A6. Draw a bicycle accurately, side view.

- Consider: the diamond frame, chain on the right, pedals, brakes, carrier, bell, stand.
- Composition: side elevation with accurate proportion.
- Idea: add a milk can on the carrier for context.

- Mistakes: the chain on the wrong side; a missing front fork; wrong frame triangles.
- Evaluation: accuracy.

A7. Draw a ceiling fan as seen from directly below it.

- Consider: the hub, 3 blades with a radial arrangement, the canopy, the down-rod foreshortened.
- Idea: show motion blur in one version.
- Mistakes: drawing a side view instead.
- Evaluation: viewpoint accuracy.

A8. Draw the dining table of your house after dinner.

- Consider: utensils, leftovers, cloth, objects that tell a story about the family.
- Idea: details that reveal the people.
- Mistakes: a sterile, empty table.
- Evaluation: storytelling through objects.

A9. Draw a "product you love" and write 200 words about it (DAT 2026, reported).

- Consider: the object's form, material, function, emotion, sustainability, and one improvement.
- Composition: a big 3/4 view + a small in-use frame; the text in a neat block.
- Idea: choose a humble, meaningful product (tiffin, grandfather's radio, a pencil box) over a flashy phone.
- Mistakes: an iPhone with generic text; going over or under the word count.
- Evaluation: drawing quality, depth of observation, writing structure (see Vol 2B §3.5).

A10. Illustrate a glass of shikanji or thandai with its ingredients (DAT 2026, reported).

- Consider: the vessel, condensation, ingredients arranged around it, the context (street cart or festival), labels.
- Composition: the vessel at a thirds point; ingredients in an arc.
- Idea: an "ingredients journey" with arrows into the glass.
- Mistakes: no colour; ingredients unlabelled.
- Evaluation: appetite appeal, accuracy, layout.

A11. Draw a market scene during a power cut at night.

- Consider: light sources (candles, phone torches, petromax lamps, an inverter-lit shop), long shadows.
- Composition: strong value contrast.
- Idea: one shop with an inverter glowing, the others dark.
- Mistakes: lighting the whole scene evenly.
- Evaluation: handling of light and shadow, mood.

A12. Draw your study table as it is right now, without tidying.

- Consider: honesty, clutter, the materials.
- Idea: a revealing self-portrait through objects.
- Evaluation: observation, composition.

B. Scenes & Perspective (B1–B14)

B1. A three-storey building with life on every floor (DAT 2026 theme, reported).

- Consider: a section view; different activities on each floor (ground: a shop; 1st: a family; 2nd: a student; terrace: water tanks and clothes lines); a staircase linking them.

- Composition: frontal 1-point; stack the floors.
- Idea: one character moving between floors (e.g. a cat chasing a ball down the stairs) ties it together.
- Mistakes: inconsistent floor heights; tiny people.
- Evaluation: perspective, proportion, storytelling.

B2. A railway platform as a train arrives, from the footbridge.

- Consider: bird's-eye (3-point downward), the train roof, the crowd surging, porters, the tea vendor.
- Mistakes: wrong VP placement.
- Evaluation: viewpoint, crowd handling.

B3. A flooded street during the monsoon.

- Consider: water levels on people (ankle/knee), reflections, umbrellas, rickshaws, plastic sheets over shops, kids playing with paper boats.
- Evaluation: atmosphere, observation.

B4. A government hospital OPD waiting area.

- Consider: queues, a token display, a fan, posters, a wheelchair, anxious faces.
- Idea: highlight one design problem (e.g. no seating for the elderly).
- Evaluation: empathy, observation.

B5. A wedding procession (baraat) in a narrow lane.

- Consider: the band, the groom on a horse, lights carried by workers, onlookers on balconies.
- Composition: 1-point down the lane.
- Evaluation: energy, depth.

B6. A view from inside a moving bus window.

- Consider: a frame within a frame, motion blur outside, the passengers' reflections.
- Evaluation: viewpoint creativity.

B7. A rooftop on Uttarayan / Makar Sankranti (kite festival).

- Consider: bird's-eye or rooftop level; many kites; people with spools; the neighbouring rooftops.
- Evaluation: perspective, crowd, colour.

B8. A school sports day seen from the top of the stands.

- Consider: track lanes in perspective, crowds, flags.
- Evaluation: perspective of curved lanes.

B9. A street-food lane at 9 pm.

- Consider: light from stalls, steam, signage, crowds.
- Evaluation: lighting, density.

B10. The view an ant sees on a picnic mat.

- Consider: worm's-eye; giant food items; legs like towers.
- Evaluation: scale manipulation, imagination.

B11. The interior of a traditional Indian kitchen vs a modular kitchen (split frame).

- Consider: a chulha / tawa vs a chimney and hob; floor-seated vs standing work.
- Evaluation: observation, comparison.

B12. A post office queue (or bank queue) and its problems.

- Consider: annotate the pain points; propose a fix in an inset.
- Evaluation: problem identification.

B13. A village well or handpump in the evening.

- Consider: women with pots, children, animals, the setting sun, long shadows.
- Evaluation: composition, light.

B14. A metro station entrance seen from below, at street level.

- Consider: worm's-eye; 3-point upward perspective.
- Evaluation: technical perspective.

C. Sequential & Storytelling (C1–C14)

C1. Six frames: helping an elderly person who falls on a morning walk (DAT 2026, reported).

- See the worked plan in §10.
- Evaluation: continuity, emotion, an empathy-driven resolution.

C2. Four frames: the life cycle of a plastic bottle.

- Consider: manufacture → use → discard → (a) ocean or (b) recycled into a T-shirt; an alternative ending.
- Evaluation: clarity, message.

C3. Five frames: making chai at a roadside stall.

- Consider: a process drawing, ingredients, hands close up, steam, serving.
- Evaluation: process accuracy, shot variety.

C4. Four frames: a lost child at a mela finds their parents.

- Evaluation: emotion, the solution (an announcement booth, a wristband).

C5. Six frames: a day in the life of a school bag.

- Consider: personification, the bag's viewpoint.
- Evaluation: imagination, continuity.

C6. Four frames: a seed becoming a tree that shades a bus stop.

- Consider: the passage of time (seasons).
- Evaluation: time depiction.

C7. Six frames: preparing for and celebrating Diwali, with a twist about safety or pollution.

- Evaluation: a story arc and its message.

C8. Four frames: how a letter or parcel reaches your home.

- Consider: the system: post box → sorting → transport → postman.
- Evaluation: process understanding.

C9. Three frames: before, during and after a cyclone in a coastal village.

- Evaluation: atmosphere, consequence, recovery.

C10. Five frames: an origami crane being folded (process drawing).

- Consider: clear steps with arrows (instructional).
- Evaluation: instructional clarity.

C11. Four frames: a stray dog's rainy night.

- Evaluation: empathy, mood.

C12. Six frames: a wordless story titled "The Umbrella".

- Idea: sharing an umbrella with a stranger.
- Evaluation: storytelling without words.

C13. Four frames: grandparents learning to use a smartphone.

- Consider: humour, frustration, success; the design insight (bigger icons).
- Evaluation: emotion, insight.

C14. Four frames: how a potter makes a matka.

- Consider: the craft process (wedging, throwing on the wheel, drying, firing).
- Evaluation: knowledge of the craft, process.

D. Product Design / Redesign (D1–D16)

D1. Items to carry while travelling with your grandmother; sketch and organise them (DAT 2026, reported).

- Consider: medicines, spectacles, a shawl, water, a cushion, documents, a phone with large numbers, snacks, a walking stick, a pill timer; plus a carry solution.
- Composition: a flat-lay "knolling" (items arranged in a neat grid) + one organiser product.
- Idea: a colour-coded travel pouch with large-print labels.
- Mistakes: listing items without design thought.
- Evaluation: empathy, completeness, organisation.

D2. Redesign a bucket and mug for an elderly person with arthritis.

- Consider: grip diameter, weight, a pouring spout, a non-slip base, wheels.
- Evaluation: ergonomics, feasibility.

D3. Design a school bag that reduces back strain.

- Consider: weight distribution, a waist belt, compartments, a trolley mode for stairs.
- Evaluation: user insight.

D4. Design a dustbin that encourages waste segregation in apartments.

- Consider: colour coding (green = wet, blue = dry, red/black = hazardous), foot pedal, odour control.
- Evaluation: behaviour design.

D5. Redesign a street vendor's cart for a pani-puri seller.

- Consider: hygiene, water storage, waste, shade, mobility, lighting, a payment QR.
- Evaluation: context, multi-function.

D6. Design a seat for a crowded local train that holds more people standing comfortably.

- Consider: lean bars, foldable seats, handles at different heights.
- Evaluation: space efficiency.

D7. Solve the neem-branch problem: a tree branch overhangs between two neighbours and its leaves fall into both homes (DAT 2025, reported).

- Consider: both neighbours' interests, the tree's value (shade, medicine), possible solutions: a shared leaf-catching net that funnels into a compost pit; a schedule; pruning without killing the tree; a shared seating area beneath it.
- Composition: plan view + elevation + a happy-ending frame.
- Mistakes: cutting the tree as the only solution.
- Evaluation: empathy for both sides, sustainability, creativity.

D8. Design a kit for a first-time train traveller who can't read.

- Consider: icons, a colour-coded ticket holder, a seat-finding guide.
- Evaluation: inclusive design.

D9. Design a wash-basin solution for a train coach that prevents water wastage and spillage.

- Consider: the problems in a train sink (spills while the train moves, clogging, low pressure, height, hygiene): a push tap with metered flow, a deep basin with splash guard, a foot-operated pump.
- Evaluation: problem identification (the 2025 paper reportedly had a train-based problem question), solution feasibility.

D10. Redesign a hospital drip stand for patients who need to walk.

- Evaluation: mobility, stability.

D11. Design a lunchbox that keeps food warm without electricity.

- Consider: vacuum insulation, a hay-box principle, a thermal sleeve.
- Evaluation: technical insight.

D12. Design an umbrella for two-wheeler riders.

- Consider: safety (no visibility blockage), wind, mounting, legal concerns.
- Evaluation: critical thinking (an alternative may be a better raincoat).

D13. Design a toy for a visually impaired child.

- Consider: sound, texture, shape, safety.
- Evaluation: inclusivity, multisensory approach.

D14. Design a footwear stand for a temple that prevents mix-ups.

- Consider: a token system, cubbies, colour or number coding.
- Evaluation: systems thinking.

D15. Redesign a ration-shop (PDS) queue experience.

- Consider: token numbers, shade, seating, an SMS alert.
- Evaluation: service design.

D16. Design a water bottle for a construction worker.

- Consider: durability, large capacity, a strap, refill ease, labelling.
- Evaluation: user specificity.

E. Visual Communication: Posters (E1–E10)

E1. Poster for an orange-juice corner from a crow's point of view (DAT 2025, reported).

- Consider: elevated perspective; the circular orange glass seen from above like a sun; a crow on a wire looking down; bold headline.
- Mistakes: ignoring the viewpoint constraint.
- Evaluation: viewpoint execution, appetite appeal, hierarchy.

E2. Poster: "Save Water" without showing a tap or a drop.

- Idea: avoid the clichés, e.g. a cracked earth map shaped like India.
- Evaluation: originality.

E3. Poster for a classical-music concert (Hindustani) at a heritage venue.

- Consider: tanpura curves as visual rhythm; elegant type.
- Evaluation: mood, typography.

E4. Road safety poster for helmet use, aimed at teenagers.

- Consider: humour or shock; youth language.
- Evaluation: audience fit.

E5. Poster announcing a local craft fair featuring Warli art.

- Consider: Warli style (circle, triangle, square, white on terracotta).
- Evaluation: cultural accuracy.

E6. Poster: "Digital Detox Weekend."

- Evaluation: concept, metaphor.

E7. Poster for a blood-donation camp.

- Consider: red as accent only; white space.
- Evaluation: hierarchy, restraint.

E8. Poster for a lost-and-found campaign at a railway station.

- Evaluation: information design.

E9. Poster celebrating your city's food (one dish as a hero).

- Evaluation: appetite appeal, place identity.

E10. Poster: "Stop Single-Use Plastic" with a maximum of 3 colours.

- Evaluation: constraint handling.

F. Symbols, Logos & Pictograms (F1–F10)

F1. A logo for a waste-pickers' cooperative.

- Consider: dignity, circularity; a hand + a recycle-arrow combination.
- Evaluation: meaning, simplicity.

F2. A symbol for "quiet zone" near hospitals that works without text.

- Evaluation: universality.

F3. A set of 6 pictograms for a railway station (toilet, drinking water, ticket counter, enquiry, waiting room, exit).

- Consider: consistent style, grid, line weight.
- Evaluation: system consistency.

F4. A logo for a bamboo-furniture startup from Assam.

- Evaluation: material and region references.

F5. A logo using negative space for a bookstore named "Owl".

- Evaluation: clever use of negative space.

F6. A symbol representing "accessibility for all" for a museum.

- Evaluation: going beyond the wheelchair cliché.

F7. A monogram from your initials, in 3 styles (classical, modern, playful).

- Evaluation: typographic range.

F8. A logo for a monsoon festival.

- Evaluation: mood, simplicity.

F9. A pictogram set for a school playground's rules.

- Evaluation: child readability.

F10. A symbol for "women-only compartment" that avoids stereotypes.

- Evaluation: sensitivity.

G. Signage & Wayfinding (G1–G6)

G1. Design 5 signboards for narrow streets (DAT 2026, reported).

- Consider:
  - one-way
  - no four-wheelers
  - dead end
  - low wires overhead
  - public tap / toilet
  - silence near a school
  - pedestrians first
  - shop directions
- Mounting: projecting blade signs at about 2.4 m, visible at night (reflective).
- Composition: a 5-sign system + 1 in-context drawing.
- Evaluation: consistency, clarity, contextual placement.

G2. A signage system for a large government hospital.

- Evaluation: colour coding by department, legibility.

G3. A bilingual (or trilingual) name sign for a heritage monument.

- Evaluation: typography balancing scripts.

G4. Wayfinding for a village weekly market (haat).

- Evaluation: low-cost materials, icon use.

G5. Emergency evacuation signage for a school.

- Evaluation: standard colour (green for exit), arrows.

G6. A bus-stop information panel for first-time visitors.

- Evaluation: information hierarchy.

H. Packaging (H1–H6)

H1. Eco-friendly packaging for 6 eggs for a village co-op.

- Consider: moulded pulp, a leaf, a cloth sling, protection.
- Evaluation: protection, sustainability.

H2. Packaging for a set of 4 handmade Channapatna toys.

- Consider: a window to show the toys, craft storytelling.
- Evaluation: brand story.

H3. A reusable packaging system for a food-delivery app.

- Evaluation: a return system, hygiene.

H4. Packaging for Darjeeling tea as a gift.

- Evaluation: premium feel, GI identity.

H5. Packaging for medicine for the elderly that prevents dosage mistakes.

- Evaluation: calendar-strip design, large print.

H6. Packaging for a single samosa that keeps it crisp.

- Evaluation: vent holes, grease resistance.

I. Imagination & Unusual Situations (I1–I12)

I1. What if gravity reversed for 10 minutes in your classroom?

- Evaluation: imagination, details, humour.

I2. Draw a creature that is half-elephant, half-kettle, and show where it lives.

- Evaluation: a believable hybrid.

I3. Your city in 2075 after sea levels rise by 1 m.

- Evaluation: speculative design, adaptation.

I4. If trees could talk, what would the neem tree outside your house say? Draw it as a comic.

- Evaluation: personality, message.

I5. A day when all colours disappeared except one.

- Evaluation: concept, colour use.

I6. A house for a family of birds designed by humans (or for humans designed by birds).

- Evaluation: role reversal.

I7. Draw "time" without drawing a clock.

- Evaluation: abstraction.

I8. Draw the sound of a pressure-cooker whistle.

- Evaluation: synaesthesia, visual sound.

I9. Draw "silence."

- Evaluation: abstraction, negative space.

I10. You shrink to 2 cm and land in your kitchen. Draw what you see.

- Evaluation: scale, perspective.

I11. A vehicle for a family of 5 in 2050 Indian cities.

- Evaluation: future thinking, user needs.

I12. Your favourite festival celebrated on Mars.

- Evaluation: adaptation, humour.

J. Social, Environmental & Sustainable Design Problems (J1–J10)

J1. Show 4 problems faced by a street vendor during summer, and solve one.

- Evaluation: problem identification and a solution.

J2. Design a solution for stray-cattle collisions on highways at night.

- Evaluation: low-cost systemic ideas (reflective collars).

J3. Reduce food waste at Indian weddings: visualise a system.

- Evaluation: systems thinking.

J4. Design shade and seating for traffic police at junctions.

- Evaluation: empathy, feasibility.

J5. Help a school with no electricity keep classrooms cool.

- Evaluation: passive cooling (jaali, khus curtains, white roofs).

J6. Design a system for rainwater harvesting in an apartment, explained visually.

- Evaluation: technical clarity.

J7. Make public toilets safer and cleaner for women: show your ideas.

- Evaluation: sensitivity, practical detail.

J8. Upcycle 50 used plastic bottles into something useful for a school.

- Evaluation: originality, construction detail.

J9. Design a way for migrant workers to send money and messages home easily.

- Evaluation: service design, low literacy.

J10. Reduce noise pollution during festivals without killing the joy.

- Evaluation: balanced, creative.

K. Colour, Pattern & Composition Exercises (K1–K6)

K1. Fill a 10×10 cm square with a pattern inspired by Ajrakh, using only 3 colours.

- Evaluation: rhythm, cultural reference.

K2. Show the same street in a warm palette (festival morning) and a cool palette (rainy night).

- Evaluation: colour mood.

K3. Create an abstract composition expressing "traffic jam" using only geometric shapes.

- Evaluation: abstraction.

K4. Design a border pattern for a sari inspired by your city's architecture.

- Evaluation: motif development.

K5. Make a composition with 5 circles showing tension.

- Evaluation: use of principles.

K6. Recolour a familiar logo in a monochrome scheme and explain the change in mood.

- Evaluation: colour theory.

L. Writing-plus-Drawing (L1–L4)

L1. Describe your favourite street in 150 words and draw its most interesting corner.
L2. Write a 6-word story and illustrate it.
L3. Write a product review of a public bench (100 words) with a redesign sketch.
L4. Write a letter from a discarded chair to its designer (120 words) with an illustration.

- For L1–L4, evaluation: concise descriptive writing, a coherent relationship between text and image, and creativity.

Total prompts: 12 + 14 + 14 + 16 + 10 + 10 + 6 + 6 + 12 + 10 + 6 + 4 = 120.

Self-scoring rubric for any prompt (out of 10)

Criterion	0	1	2
Followed instructions (count, viewpoint, constraints)	missed	partly	fully
Idea (relevance + originality)	cliché / off-topic	sensible	fresh and fitting

Criterion	0	1	2
Observation & detail	generic	some	rich, believable
Composition & proportion	poor	acceptable	strong
Communication (clarity, labels, writing)	confusing	readable	instantly clear

Score bands: 9–10 = excellent; 7–8 = strong; 5–6 = developing; below 5 = redo the prompt within 3 days. These are practice benchmarks, not NID's actual scoring.

NEXT VOLUME: Volume 4 — Massive Question Bank (`04A-…` through `04G-…`)

VOLUME 4A — QUESTION BANK: DESIGN APTITUDE (150 Questions)

Format of every item: Question → options → **Answer** → explanation → *Concept* · *Difficulty* (E = easy, M = medium, H = hard) · *Why NID may ask* · *Common mistake*.

Sections: A. Elements & principles (1–25) · B. Colour (26–45) · C. Typography & visual communication (46–65) · D. Perception & composition (66–80) · E. Human-centred design, usability & ergonomics (81–110) · F. Materials, sustainability & products (111–130) · G. Scenario-based design judgement (131–150)

A. Elements & Principles of Design (1–25)

DA-1. A designer wants a poster to feel calm and stable. Which line direction should dominate?
(a) Diagonal (b) Zig-zag (c) Horizontal (d) Spiral
Ans: (c). Horizontal lines echo the horizon and a body at rest, so they read as stability and calm. Diagonals suggest motion; zig-zags suggest tension. *Concept:* line psychology · *Diff:* E · *Why NID:* tests basic visual vocabulary · *Mistake:* choosing vertical, which signals strength or height rather than calm.

DA-2. Which element of design is defined as "the lightness or darkness of a colour or tone"?
(a) Hue (b) Value (c) Saturation (d) Texture
Ans: (b). Value is lightness or darkness. Hue is the colour family; saturation is intensity. *Concept:* elements · *Diff:* E · *Why NID:* terminology precision · *Mistake:* confusing value with saturation.

DA-3. The difference between "shape" and "form" is that:
(a) shape is 3D, form is 2D (b) shape is 2D, form is 3D (c) both are 3D (d) shape refers only to geometric figures
Ans: (b). A circle is a shape; a sphere is a form. *Concept:* elements · *Diff:* E · *Why NID:* the basis for product sketching · *Mistake:* using the words interchangeably.

DA-4. A temple corridor with pillars at equal intervals creates mainly:
(a) Radial balance (b) Regular rhythm (c) Emphasis (d) Variety
Ans: (b). Equal repetition at equal intervals produces regular rhythm. *Concept:* rhythm · *Diff:* E · *Why NID:* Indian architectural context · *Mistake:* answering "repetition", which is not offered here; rhythm is repetition with tempo.

DA-5. In a layout, a large pale-grey circle on the left is balanced by a small bright-red square on the right. This is:
(a) Symmetrical balance (b) Asymmetrical balance (c) Radial balance (d) Imbalance
Ans: (b). Different elements of equal visual weight balance each other: a high-saturation small shape against a low-saturation large one. *Concept:* visual weight · *Diff:* M · *Why NID:* composition judgement · *Mistake:* thinking that different sizes must mean imbalance.

DA-6. Which principle is MOST concerned with the order in which a viewer reads information?
(a) Unity (b) Hierarchy (c) Harmony (d) Proportion
Ans: (b). Hierarchy arranges elements by importance through size, weight and position. *Concept:* hierarchy · *Diff:* E · *Why NID:* poster and signage tasks · *Mistake:* choosing emphasis, which concerns a single focal point rather than an ordered sequence.

DA-7. A rangoli drawn around a central lamp, with motifs repeating outward, best demonstrates:
(a) Asymmetry (b) Radial balance (c) Progression (d) Contrast
Ans: (b). Elements radiate from a centre. *Concept:* balance types · *Diff:* E · *Why NID:* Indian visual culture · *Mistake:* choosing symmetry in general rather than its radial kind.

DA-8. A comic artist draws a hero much larger than surrounding buildings to show power. This uses:
(a) Proportion (b) Scale (exaggerated) (c) Texture (d) Rhythm
Ans: (b). Scale is size relative to surroundings or the human; exaggerating it creates drama. *Concept:* scale vs proportion · *Diff:* M · *Why NID:* imagination drawings · *Mistake:* choosing proportion, which compares parts within one object.

DA-9. Which pair BEST describes the risk of too much variety and too much unity respectively?
(a) Boredom, chaos (b) Chaos, boredom (c) Harmony, contrast (d) Emphasis, rhythm
Ans: (b). Too much variety creates chaos; too much unity creates monotony. *Concept:* unity–variety balance · *Diff:* M · *Why NID:* aesthetic judgement · *Mistake:* reversing the pair.

DA-10. Seurat's paintings are made entirely of tiny dots. Which element is the building block?
(a) Line (b) Point (c) Plane (d) Space
Ans: (b). Pointillism builds tone and colour from points. *Concept:* point · *Diff:* E · *Why NID:* links art history and elements · *Mistake:* answering "colour", which the dots carry but which is not the structural unit.

DA-11. "Visual texture" is best illustrated by:
(a) a jute bag you can touch (b) a drawing of tree bark on smooth paper (c) a ceramic tile's glaze (d) sandpaper
Ans: (b). The texture is implied visually, not felt. *Concept:* texture · *Diff:* E · *Why NID:* rendering skill · *Mistake:* picking a real material.

DA-12. An emphasis strategy that does NOT rely on colour is:
(a) red text on grey (b) isolation of one object in empty space (c) a yellow highlight (d) a complementary accent
Ans: (b). Isolation creates a focal point through space alone. *Concept:* emphasis methods · *Diff:* M · *Why NID:* monochrome tasks · *Mistake:* assuming emphasis always needs colour.

DA-13. Which arrangement shows *progressive* rhythm?
(a) ○ ○ ○ ○ (b) ○ □ ○ □ (c) ∘ ○ ◯ ● (increasing size) (d) random dots
Ans: (c). A gradual, step-by-step change is progressive rhythm. (a) is regular; (b) is alternating. *Concept:* rhythm types · *Diff:* E · *Why NID:* pattern logic · *Mistake:* choosing (b).

DA-14. Negative space refers to:
(a) dark areas (b) the empty area around and between subjects (c) shadows (d) unused paper only at margins
Ans: (b). *Concept:* space · *Diff:* E · *Why NID:* logo cleverness (FedEx, WWF) · *Mistake:* equating it with darkness.

DA-15. Which composition has a single dominant focal point?
(a) Two identical faces of the same size at the left and right edges (b) One red umbrella in a crowd of black umbrellas (c) A grid of 16 equal photos (d) A uniform wallpaper pattern
Ans: (b). Contrast in colour isolates one element. *Concept:* emphasis · *Diff:* E · *Why NID:* the classic visual example · *Mistake:* none common.

DA-16. A designer repeats the same blue, the same font, and aligns everything to a grid across 10 brochure pages. The main principle achieved is:
(a) Variety (b) Unity (c) Contrast (d) Movement

Ans: (b). Consistency creates unity. *Concept:* unity · *Diff:* E · *Why NID:* brand systems · *Mistake:* choosing harmony; unity is the better term for consistency across a system.

DA-17. In Indian miniature paintings, important figures are often painted larger regardless of distance. This convention is called:
(a) Linear perspective (b) Hierarchical (hieratic) scale (c) Foreshortening (d) Atmospheric perspective
Ans: (b). Size shows importance rather than distance. *Concept:* scale conventions · *Diff:* M · *Why NID:* Indian art awareness · *Mistake:* choosing foreshortening.

DA-18. Which line type best expresses "anger" in an abstract drawing?
(a) Thin wavy lines (b) Heavy jagged lines with sharp angles (c) Long horizontals (d) Gentle spirals
Ans: (b). *Concept:* expressive line · *Diff:* E · *Why NID:* "express an emotion using only lines" tasks · *Mistake:* choosing spirals, which suggest confusion or hypnosis.

DA-19. The golden ratio is approximately:
(a) 1 : 1.414 (b) 1 : 1.618 (c) 1 : 2 (d) 1 : 3.14
Ans: (b). 1.414 is the A-paper ratio (√2). *Concept:* proportion · *Diff:* E · *Why NID:* proportion knowledge · *Mistake:* choosing 1.414.

DA-20. Which BEST describes "movement" as a design principle?
(a) Objects that physically move (b) The path the viewer's eye follows through a composition (c) Animation only (d) Diagonal lines only
Ans: (b). *Concept:* movement · *Diff:* E · *Why NID:* reading posters · *Mistake:* taking it literally.

DA-21. A logo is made of only three vertical rectangles whose heights increase from left to right, like a rising bar chart. What does the arrangement most likely communicate?
(a) Decline (b) Growth or progress (c) Stability (d) Chaos
Ans: (b). For left-to-right readers, an ascending progression reads as growth. This is a common trope in finance and education logos. *Concept:* visual semantics · *Diff:* E · *Why NID:* symbol interpretation · *Mistake:* choosing "stability" because the shapes are rectangles.

DA-22. Which principle is violated when a poster has a headline, image, date and logo all of the same size and weight?
(a) Unity (b) Hierarchy (c) Repetition (d) Balance
Ans: (b). Nothing tells the viewer what to read first. *Concept:* hierarchy · *Diff:* E · *Why NID:* critique skills · *Mistake:* choosing balance.

DA-23. Which statement about "harmony" is correct?
(a) It requires identical elements (b) It arises when elements are compatible, e.g. analogous colours and similar shapes (c) It needs high contrast (d) It means symmetry
Ans: (b). *Concept:* harmony · *Diff:* E · *Why NID:* aesthetic vocabulary · *Mistake:* confusing it with symmetry.

DA-24. A designer places the horizon exactly through the middle of a landscape. A tutor would most likely suggest:
(a) Keep it centred for drama (b) Move it to a third line to give the sky or ground dominance (c) Remove the horizon (d) Add a second horizon
Ans: (b). A centred horizon splits the image into two competing halves; the rule of thirds creates dominance. *Concept:* composition · *Diff:* E · *Why NID:* scene drawing · *Mistake:* choosing (d), which breaks perspective.

DA-25. Which is an example of "contrast of texture"?
(a) Rough jute placemat beside a glossy steel plate (b) Two shades of blue (c) A big and a small circle (d) Black text on white

Ans: (a). The others are contrasts of hue, size and value. *Concept:* contrast types · *Diff:* E · *Why NID:* material awareness · *Mistake:* choosing (d).

B. Colour (26–45)

DA-26. On the RYB colour wheel, the complement of yellow is:
(a) Blue (b) Violet (c) Green (d) Orange
Ans: (b). *Concept:* complements · *Diff:* E · *Why NID:* standard item · *Mistake:* choosing blue (in RGB terms the complement of yellow is blue; NID-level questions assume the painter's RYB wheel).

DA-27. Mixing equal amounts of cyan and magenta ink gives approximately:
(a) Green (b) Blue / violet-blue (c) Red (d) Black
Ans: (b). In CMY subtractive mixing, C + M = blue, M + Y = red, C + Y = green. *Concept:* subtractive colour · *Diff:* M · *Why NID:* print awareness · *Mistake:* confusing it with RGB.

DA-28. Which colour model do mobile screens use?
(a) CMYK (b) RYB (c) RGB (d) Pantone
Ans: (c). Screens emit light (additive). *Concept:* colour models · *Diff:* E · *Why NID:* digital design awareness · *Mistake:* choosing CMYK.

DA-29. A "tone" is produced by adding to a hue:
(a) White (b) Black (c) Grey (d) Its complement only
Ans: (c). A tint adds white, a shade adds black, a tone adds grey. *Concept:* tint/shade/tone · *Diff:* E · *Why NID:* frequently confused trio · *Mistake:* choosing black.

DA-30. A palette of yellow-green, green and blue-green is:
(a) Complementary (b) Analogous (c) Triadic (d) Split-complementary
Ans: (b). These are neighbours on the wheel. *Concept:* schemes · *Diff:* E · *Why NID:* colour application · *Mistake:* choosing monochromatic, which uses only one hue.

DA-31. Which set is triadic on the RYB wheel?
(a) Red, orange, yellow (b) Orange, green, violet (c) Blue, orange, white (d) Red, red-violet, violet
Ans: (b). The secondaries are 120° apart. *Concept:* triadic · *Diff:* E · *Why NID:* scheme identification · *Mistake:* choosing (a), which is analogous.

DA-32. Why do many food brands use red and yellow?
(a) They are cool colours (b) Warm colours are associated with energy, appetite and quick attention (c) They are cheapest to print (d) They recede visually
Ans: (b). *Concept:* colour psychology · *Diff:* E · *Why NID:* branding awareness · *Mistake:* choosing (d); warm colours advance.

DA-33. In a landscape painting, which technique makes mountains appear far away?
(a) Brighter, warmer, more detailed (b) Paler, bluer, less detailed (c) Thicker outlines (d) Higher saturation
Ans: (b). This is atmospheric (aerial) perspective. *Concept:* depth through colour · *Diff:* E · *Why NID:* scene drawing · *Mistake:* choosing (d).

DA-34. Red text on a green background of equal brightness is a poor choice mainly because:
(a) the colours are analogous (b) the value contrast is low and red–green colour-blind viewers cannot separate them (c) red is too warm (d) green is unlucky
Ans: (b). *Concept:* accessibility, value contrast · *Diff:* M · *Why NID:* inclusive design · *Mistake:* choosing (a); red and green are complementary, not analogous.

DA-35. The "60-30-10 rule" refers to:
(a) Image : text : white-space ratio (b) Dominant : secondary : accent colour proportions (c) Font sizes (d) Margins
Ans: (b). *Concept:* colour proportion · *Diff:* E · *Why NID:* interior and poster colouring · *Mistake:* none.

DA-36. In India, which colour is traditionally associated with mourning in many communities (but with purity in others)?
(a) Red (b) White (c) Yellow (d) Green
Ans: (b). *Concept:* cultural colour meaning · *Diff:* E · *Why NID:* cultural sensitivity in design · *Mistake:* assuming the Western black-for-mourning convention applies.

DA-37. Which colour combination gives the highest legibility for a road sign seen at night with headlights?
(a) Yellow on white (b) White (retroreflective) on dark green or blue (c) Light grey on white (d) Pink on red
Ans: (b). High value contrast plus retroreflective sheeting. *Concept:* legibility · *Diff:* E · *Why NID:* signage tasks · *Mistake:* none.

DA-38. Adding white to blue produces:
(a) Navy (shade) (b) Sky blue (tint) (c) Grey-blue (tone) (d) Violet
Ans: (b). *Concept:* tint · *Diff:* E · *Why NID:* terminology · *Mistake:* answering "tone".

DA-39. Saturation describes:
(a) how light or dark a colour is (b) the purity or intensity of a colour (c) the colour family (d) its temperature
Ans: (b). *Concept:* colour attributes · *Diff:* E · *Why NID:* terminology · *Mistake:* answering value.

DA-40. A split-complementary scheme based on red uses red plus:
(a) green (b) yellow-green and blue-green (c) orange and violet (d) blue and yellow
Ans: (b). The complement of red is green; its two neighbours are yellow-green and blue-green. *Concept:* schemes · *Diff:* M · *Why NID:* applied colour · *Mistake:* choosing (d), which is triadic.

DA-41. Which colour is commonly used for "mandatory action" signs (e.g. "wear helmet") in the ISO safety system?
(a) Red (b) Yellow (c) Blue (d) Green
Ans: (c). Red = prohibition, yellow = warning, blue = mandatory, green = safe condition / exit. *Concept:* safety colours · *Diff:* M · *Why NID:* signage design · *Mistake:* choosing red.

DA-42. Mixing all three RYB paint primaries in equal amounts gives roughly:
(a) White (b) A dark brown or grey (c) Bright green (d) Pink
Ans: (b). Subtractive mixing absorbs most light. *Concept:* pigment mixing · *Diff:* E · *Why NID:* hands-on colouring · *Mistake:* choosing white, which is the result for light.

DA-43. A monochromatic poster using only shades of saffron would feel:
(a) chaotic (b) unified, warm and calm (c) cold (d) highly contrasted in hue
Ans: (b). One hue gives unity; saffron is warm. *Concept:* monochrome · *Diff:* E · *Why NID:* constraint tasks · *Mistake:* choosing (d).

DA-44. Which colour pair is complementary?
(a) Blue–green (b) Orange–blue (c) Yellow–orange (d) Red–violet
Ans: (b). *Concept:* complements · *Diff:* E · *Why NID:* basic · *Mistake:* none.

DA-45. A hospital ward is painted pale green or pale blue rather than bright red mainly because:
(a) red paint is expensive (b) cool, low-saturation colours are calming and reduce visual fatigue (c) green is a primary colour (d) it hides dirt
Ans: (b). *Concept:* colour psychology in spaces · *Diff:* E · *Why NID:* environment design · *Mistake:* (d) is a minor side benefit, not the main reason.

C. Typography & Visual Communication (46–65)

DA-46. Adjusting space uniformly across an entire word is called:
(a) Kerning (b) Tracking (c) Leading (d) Justifying
Ans: (b). *Concept:* type spacing · *Diff:* E · *Why NID:* terminology · *Mistake:* kerning (that adjusts one pair of letters).

DA-47. Which font category is most suitable for long paragraphs in a printed novel?
(a) Decorative display (b) Script (c) Text serif like Garamond (d) Blackletter
Ans: (c). Serifs guide the eye along lines in long-form print. *Concept:* type choice · *Diff:* E · *Why NID:* readability · *Mistake:* choosing script.

DA-48. Helvetica was designed in 1957 by:
(a) Paul Renner (b) Max Miedinger with Eduard Hoffmann (c) Eric Gill (d) Adrian Frutiger
Ans: (b). Renner designed Futura, Gill designed Gill Sans, and Frutiger designed Univers and Frutiger. *Concept:* type history · *Diff:* M · *Why NID:* design GK · *Mistake:* choosing Frutiger.

DA-49. "Legibility" refers to:
(a) how easily individual characters can be distinguished (b) how interesting a paragraph is (c) colour contrast only (d) the length of lines
Ans: (a). Readability is the property of whole text blocks. *Concept:* legibility vs readability · *Diff:* M · *Why NID:* signage · *Mistake:* swapping the two terms.

DA-50. To make the word "SHAKY" express its meaning, the best typographic treatment is:
(a) bold, perfectly aligned capitals (b) letters slightly rotated and offset, with vibration lines (c) a script font (d) all letters in red
Ans: (b). Expressive typography mirrors meaning through form. *Concept:* expressive type · *Diff:* E · *Why NID:* common drawing task · *Mistake:* relying on colour alone.

DA-51. Which logo type is the Amul girl?
(a) Wordmark (b) Lettermark (c) Mascot (d) Abstract mark
Ans: (c). *Concept:* logo types · *Diff:* E · *Why NID:* Indian branding · *Mistake:* none.

DA-52. A pictogram for "no smoking" should place:
(a) a green circle around a cigarette (b) a red circle with a diagonal bar over a black cigarette (c) a yellow triangle with a cigarette (d) a blue square
Ans: (b). This is the prohibition convention. *Concept:* signage standards · *Diff:* E · *Why NID:* pictogram tasks · *Mistake:* choosing yellow (warning).

DA-53. Highway signs in many countries use mixed-case (not all caps) destination names mainly because:
(a) capitals are expensive (b) word shapes (ascenders and descenders) aid quick recognition at speed (c) lowercase is more modern (d) law requires it
Ans: (b). *Concept:* legibility research · *Diff:* M · *Why NID:* wayfinding · *Mistake:* choosing (c).

DA-54. A good logo should work at 16 × 16 pixels. Which design attribute matters most for that?
(a) Gradients (b) Fine detail (c) Simplicity and bold shapes (d) Many colours
Ans: (c). *Concept:* scalability · *Diff:* E · *Why NID:* logo design tasks · *Mistake:* none.

DA-55. In a poster, the eye typically scans in which pattern for image-heavy layouts?
(a) A Z-pattern (b) Bottom to top (c) Right to left for English (d) Random
Ans: (a). Z-pattern for sparse visual layouts; F-pattern for text-heavy ones. *Concept:* eye movement · *Diff:* M · *Why NID:* layout · *Mistake:* choosing (d).

DA-56. Leading that is too tight in body text causes:
(a) lines to collide and reduce readability (b) words to be too far apart (c) letters to overlap horizontally (d) colours to clash
Ans: (a). *Concept:* leading · *Diff:* E · *Why NID:* terminology · *Mistake:* answering (c), which describes tight tracking or kerning.

DA-57. Which is the strongest reason to use icons in railway-station signage in India?
(a) They look modern (b) They communicate across languages and literacy levels (c) They are cheaper (d) Icons need no testing
Ans: (b). *Concept:* inclusive communication · *Diff:* E · *Why NID:* the Indian context · *Mistake:* none.

DA-58. The 1972 Munich Olympic pictograms, a landmark in pictogram design, were designed under:
(a) Paul Rand (b) Otl Aicher (c) Milton Glaser (d) Saul Bass
Ans: (b). *Concept:* design history · *Diff:* M · *Why NID:* pictogram heritage · *Mistake:* choosing Glaser.

DA-59. Which element should be largest on a concert poster aimed at fans of a famous singer?
(a) The venue address (b) The singer's name or image (c) The sponsors' logos (d) The ticket price
Ans: (b). *Concept:* hierarchy by audience priority · *Diff:* E · *Why NID:* poster tasks · *Mistake:* none.

DA-60. A packaging designer adds Braille dots to a medicine box. This primarily improves:
(a) aesthetics (b) accessibility (c) shelf impact (d) cost
Ans: (b). *Concept:* inclusive design · *Diff:* E · *Why NID:* social design · *Mistake:* none.

DA-61. A brand changes its logo from a detailed illustration to a flat, simple icon. The MOST likely reason is:
(a) to look old-fashioned (b) to work better on screens, apps and small sizes (c) to hide the brand (d) to increase printing cost
Ans: (b). *Concept:* flat design trend · *Diff:* E · *Why NID:* contemporary design awareness · *Mistake:* none.

DA-62. An infographic showing the % share of transport modes should most appropriately use:
(a) a line graph (b) a pie chart or a 100% stacked bar (c) a scatter plot (d) a map only
Ans: (b). These show parts of a whole. *Concept:* information design · *Diff:* M · *Why NID:* visual communication · *Mistake:* choosing a line graph, which is for trends over time.

DA-63. The space enclosed within letters like "o", "a" and "e" is called:
(a) Serif (b) Counter (c) Ascender (d) Kern
Ans: (b). *Concept:* type anatomy · *Diff:* M · *Why NID:* terminology · *Mistake:* none.

DA-64. Which of these is NOT a function of packaging?
(a) Protection (b) Communication (c) Convenience (d) Increasing the product's nutritional value
Ans: (d). *Concept:* packaging functions · *Diff:* E · *Why NID:* packaging tasks · *Mistake:* none.

DA-65. A sign for a narrow lane must be read by people walking toward it along the lane. The most effective mounting is:
(a) flat on the wall, parallel to the lane (b) projecting perpendicular from the wall at about 2.4 m height (c) on the ground (d) on a rooftop
Ans: (b). A blade sign faces oncoming pedestrians and clears heads and vehicles. *Concept:* wayfinding placement · *Diff:* M · *Why NID:* DAT 2026 signboard theme · *Mistake:* (a), which is invisible until you are beside it.

D. Perception & Composition (66–80)

DA-66. The WWF panda logo, with incomplete outlines, relies on:
(a) Proximity (b) Closure (c) Common fate (d) Continuity
Ans: (b). *Concept:* gestalt · *Diff:* E · *Why NID:* logo analysis · *Mistake:* choosing continuity.

DA-67. Buttons grouped together on a remote control appear related because of:
(a) Similarity (b) Proximity (c) Closure (d) Figure–ground
Ans: (b). *Concept:* gestalt · *Diff:* E · *Why NID:* product interface · *Mistake:* similarity, which would apply if the grouping came from looks rather than position.

DA-68. In the rule of thirds, the best place for the main subject is:
(a) dead centre (b) on an intersection of the third lines (c) touching the edge (d) the bottom corner
Ans: (b). *Concept:* composition · *Diff:* E · *Why NID:* every drawing · *Mistake:* none.

DA-69. A railway track drawn converging to a point on the horizon uses:
(a) Framing (b) Leading lines and one-point perspective (c) Radial symmetry (d) Atmospheric perspective
Ans: (b). *Concept:* leading lines · *Diff:* E · *Why NID:* scene drawing · *Mistake:* none.

DA-70. Rubin's vase illustrates:
(a) Closure (b) Figure–ground reversal (c) Similarity (d) Continuity
Ans: (b). *Concept:* perception · *Diff:* E · *Why NID:* negative space · *Mistake:* none.

DA-71. "Tangents" in a drawing (two edges just touching) are discouraged because they:
(a) flatten depth and confuse which object is in front (b) waste time (c) add colour (d) are too dark
Ans: (a). *Concept:* composition errors · *Diff:* M · *Why NID:* drawing evaluation · *Mistake:* none.

DA-72. Which visual element appears "heavier" in a composition, all else equal?
(a) A small light-grey square (b) A large dark textured square (c) A thin outline circle (d) Empty space
Ans: (b). Size, darkness and texture all add visual weight. *Concept:* visual weight · *Diff:* E · *Why NID:* balance · *Mistake:* none.

DA-73. A photographer frames a monument through a stone archway. This is:
(a) Framing (b) Leading lines (c) Symmetry only (d) Cropping
Ans: (a). *Concept:* composition device · *Diff:* E · *Why NID:* scene composition · *Mistake:* none.

DA-74. Odd numbers of subjects (3, 5) are often preferred in compositions because they:
(a) are easier to draw (b) avoid splitting attention into equal pairs and feel more natural (c) are luckier (d) need less colour
Ans: (b). *Concept:* the rule of odds · *Diff:* M · *Why NID:* composition sense · *Mistake:* none.

DA-75. In a drawing, overlapping two objects primarily communicates:
(a) colour harmony (b) depth (one is in front) (c) movement (d) texture
Ans: (b). *Concept:* depth cues · *Diff:* E · *Why NID:* scenes · *Mistake:* none.

DA-76. The "Müller-Lyer illusion" shows that:
(a) lines with outward-pointing fins look longer than equal lines with inward fins (b) circles look larger in red (c) parallel lines converge (d) colours change in the dark
Ans: (a). *Concept:* optical illusion · *Diff:* M · *Why NID:* perception awareness · *Mistake:* none.

DA-77. Which composition feels most dynamic?
(a) A centred, symmetrical figure (b) A figure running diagonally from bottom-left to top-right, with space in front (c) A horizontal line across the middle (d) A uniform grid
Ans: (b). Diagonals plus "lead room" in the direction of motion create dynamism. *Concept:* movement · *Diff:* M · *Why NID:* storytelling · *Mistake:* none.

DA-78. "Lead room" in a composition means:
(a) extra space in front of a moving or looking subject (b) a margin for binding (c) space for text (d) a darkroom
Ans: (a). *Concept:* composition · *Diff:* M · *Why NID:* sequential frames · *Mistake:* none.

DA-79. Two objects of the same real size: one is drawn smaller and higher on the page. Viewers perceive it as:
(a) closer (b) farther away (c) heavier (d) brighter
Ans: (b). Relative size and height in the picture plane are depth cues. *Concept:* depth cues · *Diff:* E · *Why NID:* perspective · *Mistake:* none.

DA-80. Which crop makes a portrait of a busy street vendor feel more intimate?
(a) A wide shot showing the whole street (b) A close crop on the vendor's hands and face (c) An aerial view (d) A distant silhouette
Ans: (b). *Concept:* framing and emotion · *Diff:* E · *Why NID:* storyboard shot choice · *Mistake:* none.

E. Human-Centred Design, Usability & Ergonomics (81–110)

DA-81. The correct sequence of the Stanford design thinking model is:
(a) Ideate, Empathise, Define, Test, Prototype (b) Empathise, Define, Ideate, Prototype, Test (c) Define, Empathise, Prototype, Ideate, Test (d) Test, Prototype, Ideate, Define, Empathise
Ans: (b). *Concept:* design process · *Diff:* E · *Why NID:* design thinking basics · *Mistake:* none.

DA-82. A "Norman door" is a door that:
(a) is very heavy (b) confuses users about whether to push or pull (c) opens automatically (d) is made in Normandy
Ans: (b). The term comes from Don Norman's *The Design of Everyday Things*. *Concept:* affordances / signifiers · *Diff:* E · *Why NID:* usability · *Mistake:* none.

DA-83. A flat plate on a door (with no handle) signifies:
(a) Pull (b) Push (c) Slide (d) Locked
Ans: (b). *Concept:* affordance / signifier · *Diff:* E · *Why NID:* usability · *Mistake:* none.

DA-84. Gas stove knobs arranged in the same spatial layout as the burners demonstrate good:
(a) Feedback (b) Mapping (c) Branding (d) Constraint
Ans: (b). *Concept:* mapping · *Diff:* E · *Why NID:* everyday product analysis · *Mistake:* none.

DA-85. The "click" you hear when a seatbelt locks is an example of:
(a) Affordance (b) Feedback (c) Mapping (d) Visibility
Ans: (b). *Concept:* feedback · *Diff:* E · *Why NID:* interaction design · *Mistake:* none.

DA-86. A SIM card tray with one cut corner that fits only one way is an example of:
(a) Aesthetics (b) Physical constraint / poka-yoke (c) Feedback (d) Mapping
Ans: (b). *Concept:* error prevention · *Diff:* M · *Why NID:* usability · *Mistake:* none.

DA-87. Which is the BEST problem statement for a design challenge?
(a) "Design an app." (b) "Elderly commuters need a way to board crowded buses safely because steps are high and buses don't wait." (c) "Make buses beautiful." (d) "Buses are bad."
Ans: (b). It names the user, the need and the insight. *Concept:* define stage · *Diff:* E · *Why NID:* problem-solving tasks · *Mistake:* jumping to a solution, as in (a).

DA-88. A low-floor bus with a kneeling feature most directly improves:
(a) Fuel economy (b) Accessibility for elderly, disabled and child passengers (c) Top speed (d) Advertising space
Ans: (b). *Concept:* universal design · *Diff:* E · *Why NID:* public design · *Mistake:* none.

DA-89. Designing for the 5th to the 95th percentile of users means:
(a) designing only for average users (b) accommodating 90% of the population's body dimensions (c) designing for the tallest users (d) ignoring children

Ans: (b). *Concept:* anthropometry · *Diff:* M · *Why NID:* ergonomics · *Mistake:* answering (a).

DA-90. A comfortable adult chair seat height is approximately:
(a) 20–25 cm (b) 40–45 cm (c) 60–65 cm (d) 75–80 cm
Ans: (b). *Concept:* ergonomic dimensions · *Diff:* E · *Why NID:* proportions in drawing · *Mistake:* choosing (d), which is table height.

DA-91. Kerb cuts (ramps at footpath edges), designed for wheelchair users, also help people with prams, luggage and bicycles. This is called:
(a) The curb-cut effect / universal design benefit (b) Planned obsolescence (c) Greenwashing (d) Skeuomorphism
Ans: (a). *Concept:* inclusive design · *Diff:* M · *Why NID:* social design awareness · *Mistake:* none.

DA-92. Tactile paving with raised parallel bars indicates:
(a) Stop / danger (b) Direction to walk (c) A shop (d) A seat
Ans: (b). Dots mean warning; bars mean direction. *Concept:* accessibility · *Diff:* M · *Why NID:* public space · *Mistake:* swapping the two.

DA-93. The design principle "recognition rather than recall" suggests:
(a) users should memorise commands (b) options should be visible so users needn't remember them (c) hide menus (d) use passwords
Ans: (b). *Concept:* usability heuristics · *Diff:* M · *Why NID:* UI awareness · *Mistake:* none.

DA-94. In a user interview, which is the BEST question?
(a) "Don't you think this bottle is great?" (b) "Tell me about the last time you carried water to school." (c) "Would you buy this?" (d) "Is blue better than red?"
Ans: (b). It is open and story-based, and avoids leading the user. *Concept:* empathy research · *Diff:* M · *Why NID:* HCD understanding · *Mistake:* (a) is a leading question.

DA-95. A persona is:
(a) the designer's own profile (b) a fictional but research-based representation of a target user (c) a celebrity endorser (d) a legal document
Ans: (b). *Concept:* HCD tools · *Diff:* E · *Why NID:* design process · *Mistake:* none.

DA-96. Dieter Rams' principle "Good design is as little design as possible" is closest to:
(a) Maximalism (b) "Less, but better" (c) Ornamentation (d) Kitsch
Ans: (b). *Concept:* design philosophy · *Diff:* E · *Why NID:* design GK · *Mistake:* none.

DA-97. An elderly user struggles to read medicine labels. The MOST effective design improvement is:
(a) adding a holographic sticker (b) larger high-contrast text plus colour- and shape-coded dosage icons (c) a smaller box (d) a glossy finish
Ans: (b). *Concept:* inclusive packaging · *Diff:* E · *Why NID:* empathy tasks · *Mistake:* (d) adds glare and makes things worse.

DA-98. In a crowded local train, the main ergonomic need of standing passengers is:
(a) Decorative interiors (b) Handholds at multiple heights for different statures (c) Tinted windows (d) Music
Ans: (b). *Concept:* anthropometric range · *Diff:* E · *Why NID:* public transport redesign · *Mistake:* none.

DA-99. "Prototype" in design thinking primarily means:
(a) the final mass-produced product (b) a quick, tangible model for testing ideas (c) a patent (d) a sales brochure
Ans: (b). *Concept:* prototyping · *Diff:* E · *Why NID:* process · *Mistake:* none.

DA-100. Why is iteration important in design?
(a) It increases cost (b) Testing reveals flaws, so repeated refinement improves the solution (c) It delays launch (d) It is optional
Ans: (b). *Concept:* iteration · *Diff:* E · *Why NID:* design mindset · *Mistake:* none.

DA-101. A pressure cooker's safety valve is an example of:
(a) Decoration (b) Fail-safe design (c) Mapping (d) Branding
Ans: (b). It releases pressure if the whistle is blocked. *Concept:* safety design · *Diff:* M · *Why NID:* everyday product knowledge · *Mistake:* none.

DA-102. A user journey map shows:
(a) GPS routes (b) the steps a user goes through with a service, including emotions and pain points (c) a factory layout (d) the brand history
Ans: (b). *Concept:* service design · *Diff:* E · *Why NID:* HCD · *Mistake:* none.

DA-103. Which is the MOST inclusive design for a public drinking-water tap?
(a) A single tap at 1.2 m height (b) Taps at two heights plus a push-button usable with an elbow (c) A tap requiring strong twisting (d) A coin-operated tap
Ans: (b). It serves children, wheelchair users and people with limited grip. *Concept:* universal design · *Diff:* M · *Why NID:* public product design · *Mistake:* none.

DA-104. "Form follows function" was coined by:
(a) Le Corbusier (b) Louis Sullivan (c) Mies van der Rohe (d) Frank Gehry
Ans: (b). Sullivan wrote "form ever follows function" in 1896. *Concept:* design history · *Diff:* M · *Why NID:* design GK · *Mistake:* choosing Mies ("less is more").

DA-105. The Double Diamond model alternates between:
(a) cost and profit (b) divergent and convergent thinking (c) 2D and 3D (d) art and science
Ans: (b). *Concept:* design process · *Diff:* M · *Why NID:* process awareness · *Mistake:* none.

DA-106. Handles of kitchen tools designed with a 30–40 mm diameter primarily provide:
(a) better looks (b) a comfortable power grip (c) cheaper production (d) faster heating
Ans: (b). *Concept:* ergonomics · *Diff:* M · *Why NID:* product redesign · *Mistake:* none.

DA-107. A railway ticket-vending machine is used by many first-time users. Which feature matters MOST?
(a) Animated advertisements (b) Clear step-by-step guidance in local languages with icons (c) A metallic finish (d) Hidden buttons
Ans: (b). *Concept:* usability · *Diff:* E · *Why NID:* public interfaces · *Mistake:* none.

DA-108. In empathy mapping, the four quadrants usually are:
(a) Says, Thinks, Does, Feels (b) Red, Blue, Green, Yellow (c) Past, Present, Future, Never (d) Cost, Time, Quality, Scope
Ans: (a). *Concept:* empathy tools · *Diff:* M · *Why NID:* HCD · *Mistake:* none.

DA-109. Jugaad (frugal improvisation) is valued as creative, but a designer should also consider:
(a) only cost (b) safety, durability and scalability (c) only looks (d) nothing else
Ans: (b). *Concept:* critical view of frugal innovation · *Diff:* M · *Why NID:* the Indian design context · *Mistake:* romanticising jugaad uncritically.

DA-110. A good way to test a signage prototype for a hospital is to:
(a) ask the hospital director only (b) watch first-time visitors try to find a department using the signs (c) print it in gold (d) skip testing
Ans: (b). Observing real users reveals real problems. *Concept:* testing · *Diff:* E · *Why NID:* design thinking · *Mistake:* none.

F. Materials, Sustainability & Products (111–130)

DA-111. Which material is MOST appropriate for a disposable plate that composts quickly?
(a) Polystyrene (b) Areca (supari) palm leaf (c) Melamine (d) Aluminium foil
Ans: (b). *Concept:* sustainable materials · *Diff:* E · *Why NID:* Indian sustainable traditions · *Mistake:* none.

DA-112. "Upcycling" means:
(a) burning waste for energy (b) converting waste into a product of higher value or quality (c) recycling into a lower-grade material (d) sending waste uphill
Ans: (b). Downcycling is the opposite case. *Concept:* sustainability terms · *Diff:* E · *Why NID:* sustainable design · *Mistake:* none.

DA-113. Using screws instead of adhesives in a chair primarily supports:
(a) design for disassembly and repair (b) lower strength (c) planned obsolescence (d) decoration
Ans: (a). *Concept:* circular design · *Diff:* M · *Why NID:* sustainability thinking · *Mistake:* none.

DA-114. A brand claims its bottle is "eco-friendly" because it is green-coloured, though it is ordinary plastic. This is:
(a) Biomimicry (b) Greenwashing (c) Upcycling (d) Life-cycle assessment
Ans: (b). *Concept:* ethics · *Diff:* E · *Why NID:* critical awareness · *Mistake:* none.

DA-115. Velcro was inspired by burrs sticking to a dog's fur. This is an example of:
(a) Biomimicry (b) Skeuomorphism (c) Brutalism (d) Kitsch
Ans: (a). The inventor was George de Mestral. *Concept:* biomimicry · *Diff:* E · *Why NID:* creative thinking · *Mistake:* none.

DA-116. The Eastgate Centre in Harare stays cool without conventional air conditioning by imitating:
(a) beehives (b) termite mounds (c) spider webs (d) lotus leaves
Ans: (b). The architect was Mick Pearce. *Concept:* biomimicry · *Diff:* M · *Why NID:* sustainability and architecture · *Mistake:* none.

DA-117. The "lotus effect" (self-cleaning surfaces) is based on:
(a) the colour of the lotus (b) microscopic surface structures that make water bead and roll off, carrying dirt (c) the lotus's scent (d) its petal shape
Ans: (b). *Concept:* biomimicry · *Diff:* M · *Why NID:* material innovation · *Mistake:* none.

DA-118. Which property makes terracotta (matka) pots cool water naturally?
(a) Glaze (b) Porosity, which allows evaporative cooling (c) Weight (d) Colour
Ans: (b). *Concept:* traditional material science · *Diff:* E · *Why NID:* Indian sustainable products · *Mistake:* none.

DA-119. A "cradle to cradle" product is one that:
(a) ends in a landfill (b) is designed so its materials re-enter biological or technical cycles indefinitely (c) is made for babies (d) is handmade
Ans: (b). *Concept:* circular economy · *Diff:* M · *Why NID:* sustainability vocabulary · *Mistake:* answering (c).

DA-120. Bamboo is considered a sustainable material mainly because it:
(a) is heavy (b) grows very fast, regenerates without replanting and has a high strength-to-weight ratio (c) is expensive (d) cannot be recycled
Ans: (b). *Concept:* materials · *Diff:* E · *Why NID:* NE India crafts and furniture · *Mistake:* none.

DA-121. Which is the BEST material for a lightweight, rust-free bicycle frame at moderate cost?
(a) Cast iron (b) Aluminium alloy (c) Glass (d) Lead
Ans: (b). *Concept:* material selection · *Diff:* E · *Why NID:* product design · *Mistake:* none.

DA-122. Mycelium (mushroom root) packaging is gaining attention because it:
(a) is compostable and can replace polystyrene foam (b) glows in the dark (c) is waterproof forever (d) is metallic
Ans: (a). *Concept:* biomaterials · *Diff:* M · *Why NID:* current trends · *Mistake:* none.

DA-123. A steel tiffin is more sustainable than single-use containers mainly because:
(a) it is shiny (b) it is durable and reusable for years, reducing waste (c) steel is light (d) it is cheap to throw away
Ans: (b). *Concept:* reuse · *Diff:* E · *Why NID:* everyday Indian design · *Mistake:* none.

DA-124. Planned obsolescence means:
(a) designing products to last forever (b) deliberately limiting a product's useful life to drive repurchase (c) designing for recycling (d) a historic style
Ans: (b). *Concept:* design ethics · *Diff:* E · *Why NID:* critical awareness · *Mistake:* none.

DA-125. A designer uses a single plastic type (mono-material) for a shampoo bottle and cap. The main benefit is:
(a) easier recycling (b) better smell (c) brighter colours (d) heavier weight
Ans: (a). *Concept:* design for recycling · *Diff:* M · *Why NID:* packaging · *Mistake:* none.

DA-126. Which traditional Indian building feature provides ventilation, light and privacy at once?
(a) A plain glass wall (b) A jaali (perforated screen) (c) A tin roof (d) A concrete slab
Ans: (b). *Concept:* vernacular design · *Diff:* E · *Why NID:* architecture GK · *Mistake:* none.

DA-127. Khus (vetiver) curtains, used in North Indian summers, cool rooms through:
(a) Reflection (b) Evaporative cooling when water is sprinkled on them (c) Insulation only (d) Electricity
Ans: (b). *Concept:* passive cooling · *Diff:* E · *Why NID:* sustainable design · *Mistake:* none.

DA-128. Which product property matters MOST for a chair meant for stacking in a community hall?
(a) Carved decoration (b) Nesting geometry with a lightweight, durable material (c) Upholstery (d) Fixed armrests that clash
Ans: (b). *Concept:* functional design · *Diff:* E · *Why NID:* product logic · *Mistake:* none.

DA-129. Life-cycle assessment (LCA) evaluates:
(a) only a product's price (b) the environmental impact from raw material to disposal (c) consumer age groups (d) a brand's history
Ans: (b). *Concept:* sustainability tools · *Diff:* M · *Why NID:* sustainability awareness · *Mistake:* none.

DA-130. Eri silk is called "ahimsa silk" because:
(a) it is dyed with flowers (b) the fibre can be spun after the moth emerges, so the silkworm isn't killed (c) it is grown in temples (d) it is synthetic
Ans: (b). *Concept:* ethical textiles · *Diff:* M · *Why NID:* textile awareness · *Mistake:* none.

G. Scenario-Based Design Judgement (131–150)

DA-131. A vegetable vendor's tomatoes spoil quickly in summer. Which solution is MOST appropriate for a low-income vendor without electricity?
(a) A solar refrigerator costing ₹80,000 (b) A shaded cart with a wet jute-cloth evaporative cover (c) Selling online (d) Buying fewer tomatoes forever
Ans: (b). It is low-cost, needs no power and uses evaporative cooling. *Concept:* context-appropriate design · *Diff:* M · *Why NID:* the 2025–26 trend for everyday problems · *Mistake:* choosing high-tech solutions that ignore constraints.

DA-132. Leaves from a neighbour's tree fall into two houses, causing disputes. Which response shows the best design thinking?
(a) Cut down the tree (b) Build a high wall (c) A shared leaf-collecting net that funnels leaves into a common compost pit, benefiting both homes (d) Complain to the municipality
Ans: (c). It balances both stakeholders, preserves the tree and turns waste into value. *Concept:* stakeholder-balanced, sustainable solution · *Diff:* M · *Why NID:* echoes the 2025 neem-branch question · *Mistake:* destructive or one-sided solutions.

DA-133. Passengers in trains splash water on the floor while using wash basins when the train sways. The best design fix is:
(a) Remove basins (b) A deeper basin with a splash rim, a sensor or push-metered tap, and a non-slip drained floor (c) A larger mirror (d) Brighter lights
Ans: (b). It addresses the root causes: sway, flow control and drainage. *Concept:* problem analysis · *Diff:* M · *Why NID:* the 2025 train-sink theme · *Mistake:* answering cosmetically.

DA-134. You must design 5 signboards for a narrow old-city lane. What should come FIRST?
(a) Choosing fonts (b) Observing what lane users actually need to know (directions, hazards, rules) (c) Picking colours (d) Drawing decorative borders
Ans: (b). Content and need come before styling. *Concept:* the HCD sequence · *Diff:* E · *Why NID:* the DAT 2026 signboard question · *Mistake:* starting with aesthetics.

DA-135. An elderly woman travels by train with her grandson. Which item MOST improves her independence?
(a) A fashionable handbag (b) A large-print, colour-coded pill organiser with an alarm (c) A tablet with games (d) Perfume
Ans: (b). It addresses a critical need (medication timing) and her abilities (eyesight). *Concept:* user-centred prioritisation · *Diff:* E · *Why NID:* the DAT 2026 grandmother theme · *Mistake:* none.

DA-136. A juice-stall poster must be designed "from a crow's point of view". The key visual decision is:
(a) Using only text (b) An elevated, top-down perspective, perhaps showing the round glass from above like a sun (c) Drawing the crow very large in the foreground only (d) Ignoring the viewpoint
Ans: (b). The viewpoint constraint is central to the question. *Concept:* following constraints creatively · *Diff:* M · *Why NID:* echoes DAT 2025 · *Mistake:* drawing a crow without the crow's viewpoint.

DA-137. A school wants children to wash hands before lunch. Which intervention is MOST likely to change behaviour?
(a) A notice in small text (b) Footprint stickers leading from the classroom to the wash basin, plus a fun timer song (c) A fine (d) A lecture once a year
Ans: (b). Nudges are fun, visible cues placed at the moment of action. *Concept:* behaviour design · *Diff:* M · *Why NID:* social design · *Mistake:* answering with punishment.

DA-138. Construction workers lose their water bottles easily on site. The BEST redesign is:
(a) A glass bottle (b) A brightly coloured, rugged bottle with a carabiner clip and a name tag area (c) A smaller bottle (d) A clear plastic disposable bottle
Ans: (b). Visibility, attachment and durability match the context. *Concept:* context-specific design · *Diff:* E · *Why NID:* user specificity · *Mistake:* none.

DA-139. Two products solve the same problem. One is cheaper but breaks in 6 months; the other costs 30% more and lasts 5 years. From a sustainability view:
(a) The cheaper one is better (b) The durable one is better: less waste and a lower cost per year of use (c) Both are equal (d) Neither matters
Ans: (b). *Concept:* life-cycle cost · *Diff:* M · *Why NID:* critical judgement · *Mistake:* considering only the upfront price.

DA-140. A poster for an adult-literacy drive in a village should rely MOST on:
(a) dense text (b) clear pictures and minimal text in the local language (c) English slogans (d) abstract art
Ans: (b). The audience may have limited literacy. *Concept:* audience-appropriate communication · *Diff:* E · *Why NID:* inclusive communication · *Mistake:* none.

DA-141. A ceiling fan collects dust and cleaning it is risky. Which idea is MOST user-friendly?
(a) Remove fans (b) Detachable blades with a quick-release clip that can be cleaned on the floor (c) Paint the fan black (d) Make the fan rotate faster
Ans: (b). *Concept:* maintenance-friendly design · *Diff:* E · *Why NID:* product redesign · *Mistake:* none.

DA-142. Which is the strongest reason to sketch 3 thumbnail ideas before a final drawing in the exam?
(a) To waste time (b) To explore alternatives and choose the best idea quickly (c) Examiners demand exactly 3 (d) To fill space
Ans: (b). *Concept:* divergent then convergent thinking · *Diff:* E · *Why NID:* exam strategy · *Mistake:* answering (c).

DA-143. A mobile app for farmers shows weather in °C, wind speed in m/s and complex graphs. Feedback says farmers find it confusing. The best redesign:
(a) Add more graphs (b) Simple icons and voice output in the local language, with actionable advice ("Don't spray pesticide today: rain expected") (c) Use English only (d) Smaller fonts
Ans: (b). Translate the data into decisions for the user. *Concept:* information design for users · *Diff:* M · *Why NID:* tech + HCD · *Mistake:* none.

DA-144. At a crowded temple, devotees' footwear gets mixed up. The BEST system-level solution is:
(a) Ban footwear (b) A numbered cubby and token system with colour-coded zones (c) A bigger pile (d) A guard shouting
Ans: (b). *Concept:* service and system design · *Diff:* E · *Why NID:* everyday Indian problems · *Mistake:* none.

DA-145. Which is the BEST evaluation question for any design solution you sketch in the exam?
(a) "Is it high-tech?" (b) "Does it solve the defined problem for the specific user within the given context?" (c) "Is it colourful?" (d) "Is it expensive?"
Ans: (b). *Concept:* evaluation criteria · *Diff:* E · *Why NID:* self-evaluation · *Mistake:* none.

DA-146. A designer proposes an app to solve street dogs biting people at night. A tutor says, "Think physical first." Which is a better first idea?
(a) A blockchain dog registry (b) Well-lit feeding zones away from homes, with community vaccination and sterilisation drives (c) A dog-themed game (d) A social media page
Ans: (b). It addresses the root causes (hunger, fear, territory) in the physical context. *Concept:* systems thinking · *Diff:* M · *Why NID:* social design · *Mistake:* reaching for an app by default.

DA-147. A bus stop is used by schoolchildren in the morning and vendors in the afternoon. A good design would:
(a) serve only one group (b) include adaptable features (seating and shade for children; foldable counters for vendors) (c) remove seating (d) add advertisements only
Ans: (b). *Concept:* multi-user design · *Diff:* M · *Why NID:* public space design · *Mistake:* none.

DA-148. Which of these sketches communicates "a product for one-handed use" MOST clearly?
(a) A product with no hand shown (b) The product drawn with a hand gripping and operating it, plus arrows (c) A product drawn in a box (d) Only text
Ans: (b). Showing the interaction communicates use and scale. *Concept:* visual communication of use · *Diff:* E · *Why NID:* product sketching · *Mistake:* none.

DA-149. A festival creates lots of single-use plastic plates. Which design approach is MOST sustainable and culturally appropriate?
(a) Thicker plastic plates (b) Leaf plates (pattal) or a steel-plate washing station for community reuse (c) Colourful plastic (d) Paper plates with plastic lining
Ans: (b). *Concept:* sustainable cultural practice · *Diff:* E · *Why NID:* Indian sustainability · *Mistake:* answering (d), which is not compostable.

DA-150. An examiner sees two answers to "redesign a water bottle for school children". Answer X: beautiful rendering, no explanation. Answer Y: decent sketch, labelled features (leak-proof lid, wide mouth for cleaning, name strip, strap) and a small in-use frame. Which likely scores higher?
(a) X (b) Y (c) Equal (d) Neither
Ans: (b). NID values the communication of ideas and problem-solving over rendering alone. *Concept:* evaluation priorities · *Diff:* E · *Why NID:* meta-understanding of the exam · *Mistake:* over-valuing rendering.

NEXT: Volume 4B — Visual Reasoning (150 questions) (`04B-VOL4-QB-VISUAL-REASONING.md`)

VOLUME 4B — QUESTION BANK: VISUAL REASONING (150 Questions)

> All items are **text/ASCII-solvable**, so you can practise without images. Items that would normally be shown as a picture are marked **[VISUAL QUESTION — IMAGE REQUIRED]** and include a precise description, so you can still solve them (or draw the figure yourself first: that is excellent practice). Conventions: coordinates are (x, y), with x measured from the left edge and y from the bottom edge. "Vertical mirror" = a mirror standing upright beside the object (left–right flip). "Water image" = a mirror lying below the object (top–bottom flip).

Sections: A. Mirror & water images (1–20) · B. Rotation & symmetry (21–40) · C. Paper folding & cutting (41–60) · D. Cubes, dice, nets (61–85) · E. Counting figures (86–100) · F. Series, matrices & analogies (101–120) · G. Orthographic views, shadows, perspective (121–135) · H. Spatial relations, gears, maps & mixed (136–150)

Format: Question → **Answer** → explanation → *Concept · Diff · Why NID · Mistake*.

A. Mirror & Water Images (1–20)

VR-1. How many of these capital letters look identical in a vertical mirror: A, B, C, M, T, Y, R, W?
(a) 4 (b) 5 (c) 6 (d) 3
Ans: (b). A, M, T, Y, W are symmetric about a vertical axis; B, C, R are not. *Concept:* vertical symmetry · *Diff:* E · *Why NID:* the basic mirror skill · *Mistake:* counting B (which is symmetric about a horizontal axis, not a vertical one).

VR-2. Which word reads exactly the same in a vertical mirror?
(a) TOOT (b) BOOB (c) DEED (d) NOON
Ans: (a). Two conditions are needed: the letters must be vertically symmetric, and the word must read the same backwards. TOOT meets both. B, D, E and N are not vertically symmetric. *Concept:* mirror words · *Diff:* M · *Why NID:* combines two rules · *Mistake:* checking only the palindrome condition.

VR-3. Which word looks unchanged in a water image (top–bottom flip), letter positions unchanged?
(a) CHOICE (b) MOTHER (c) WATER (d) PAPER
Ans: (a). C, H, O, I and E are all symmetric about a horizontal axis. In a water image the order of letters along the line is NOT reversed. *Concept:* water image · *Diff:* M · *Why NID:* the classic item · *Mistake:* reversing the letter order.

VR-4. A clock shows 4:10. What time does its image in a vertical mirror show?
(a) 7:50 (b) 8:50 (c) 7:10 (d) 8:10
Ans: (a). 11:60 − 4:10 = 7:50. *Concept:* mirror clock · *Diff:* E · *Why NID:* spatial transformation · *Mistake:* answering 8:50 by subtracting the hours and minutes separately from 12:60.

VR-5. In a vertical mirror a clock appears to show 2:45. What is the actual time?
(a) 9:15 (b) 10:15 (c) 9:45 (d) 3:15
Ans: (a). The relation is symmetric: 11:60 − 2:45 = 9:15. *Concept:* mirror clock (inverse) · *Diff:* E · *Why NID:* reverse reasoning · *Mistake:* none.

VR-6. Which digit(s) look unchanged in a vertical mirror (simple digital-style digits)?
(a) 0, 1, 8 (b) 2, 5 (c) 3, 8 (d) 6, 9
Ans: (a). In seven-segment form, 0, 8 and the centred "1" are vertically symmetric. 3 is symmetric only about a horizontal axis. *Concept:* digit symmetry · *Diff:* M · *Why NID:* attention to form · *Mistake:* choosing 3.

VR-7. Which digits look unchanged in a water image (seven-segment style)?
(a) 0, 1, 3, 8 (b) 2, 5, 6 (c) 4, 7 (d) 6, 9
Ans: (a). These are symmetric about a horizontal axis. 1 is a vertical stroke, so it stays the same when flipped top to bottom. *Concept:* horizontal symmetry · *Diff:* M · *Why NID:* precision · *Mistake:* choosing 6/9 (that pair swaps under a 180° rotation, not under a water image).

VR-8. The word "NID" is held in front of a vertical mirror. In the mirror, the first letter you read from the left is:
(a) N (b) mirrored D (c) I (d) mirrored N
Ans: (b). The order reverses, so D (now flipped) appears on the left. *Concept:* order reversal · *Diff:* E · *Why NID:* the standard trap · *Mistake:* keeping the original order.

VR-9. An arrow points to the upper right (north-east). In a vertical mirror it points to:
(a) the upper right (b) the upper left (c) the lower right (d) the lower left
Ans: (b). Left–right flips; up stays up. *Concept:* mirror of direction · *Diff:* E · *Why NID:* transformation logic · *Mistake:* answering (d), which is a 180° rotation.

VR-10. The same north-east arrow in a water image points to:
(a) the upper left (b) the lower right (c) the lower left (d) unchanged
Ans: (b). Up flips to down; right stays right. *Concept:* water image · *Diff:* E · *Why NID:* paired with VR-9 · *Mistake:* none.

VR-11. A figure is reflected in a vertical mirror and then reflected again in a horizontal mirror. The net result equals:
(a) the original (b) a 90° rotation (c) a 180° rotation (d) a vertical-mirror image
Ans: (c). Two perpendicular reflections compose to a 180° rotation. *Concept:* composition of transformations · *Diff:* H · *Why NID:* deeper spatial reasoning · *Mistake:* answering (a); two reflections across the *same* axis would give the original.

VR-12. How many letters of the word "EXHIBIT" look the same in a water image?
(a) 5 (b) 6 (c) 7 (d) 4
Ans: (b). E, X, H, I, B, I are symmetric about a horizontal axis; T is not (its bar would move to the bottom). *Concept:* horizontal symmetry · *Diff:* M · *Why NID:* counting with care · *Mistake:* counting T.

VR-13. Ambulances often have the word written as "ƎƆИA⅃UƎMA" on the bonnet. Why?
(a) For style (b) So drivers ahead read it correctly in their rear-view mirrors (c) It's a printing error (d) It's another language
Ans: (b). *Concept:* applied mirror design · *Diff:* E · *Why NID:* links reasoning and real-world design · *Mistake:* none.

VR-14. In a vertical mirror, a person raises their right hand. The mirror image appears to raise:
(a) its right hand (b) its left hand (c) both hands (d) no hand
Ans: (b). From the mirror figure's own point of view, it is raising its left hand. *Concept:* handedness reversal · *Diff:* M · *Why NID:* spatial empathy · *Mistake:* none.

VR-15. A clock shows 12:20. Its vertical-mirror image shows:
(a) 11:40 (b) 12:40 (c) 11:20 (d) 10:40
Ans: (a). For times between 11:00 and 12:59, use 23:60 − 12:20 = 11:40. Check: the minute hand at "4" (20

min) mirrors to "8" (40 min), and the hour hand just past 12 mirrors to just before 12, which gives 11:40 ✓. *Concept:* mirror clock edge case · *Diff:* H · *Why NID:* precision · *Mistake:* computing 11:60 − 12:20 as a negative number.

VR-16. Which letter looks identical in both a vertical mirror and a water image?
(a) A (b) B (c) X (d) E
Ans: (c). X is symmetric about both axes. *Concept:* double symmetry · *Diff:* E · *Why NID:* basic · *Mistake:* none.

VR-17. A car's number plate reads "MH 01". In a vertical mirror, which characters look the same as themselves (ignoring order)?
(a) M, H, 0, 1 (b) M, H only (c) H, 0 only (d) none
Ans: (a). M, H, 0 and 1 are all vertically symmetric. The string order still reverses. *Concept:* character symmetry · *Diff:* M · *Why NID:* observation · *Mistake:* none.

VR-18. Which pair of letters are mirror images of each other (vertical mirror) in standard lowercase?
(a) b and d (b) b and p (c) m and w (d) n and u
Ans: (a). b and p are water images (top–bottom); n and u are 180° rotations; m and w are related by a 180° rotation or water image. *Concept:* letter transformations · *Diff:* M · *Why NID:* fine discrimination · *Mistake:* choosing b/p.

VR-19. Which lowercase pair are water images of each other?
(a) b and d (b) b and p (c) p and q (d) d and q
Ans: (b). b flipped top to bottom becomes p; d flipped becomes q. *Concept:* water image · *Diff:* M · *Why NID:* pairs with VR-18 · *Mistake:* choosing p/q (those are vertical mirrors).

VR-20. A triangle points to the right (▶). Which transformation makes it point left (◀)? Choose ALL that work: (i) vertical mirror (ii) water image (iii) 180° rotation (iv) 90° rotation
(a) i only (b) i and iii (c) ii and iv (d) iii only
Ans: (b). Both a left–right flip and a half turn reverse the direction of a symmetric arrow-triangle. *Concept:* equivalent transformations for symmetric shapes · *Diff:* M · *Why NID:* flexible thinking · *Mistake:* stopping at one answer.

B. Rotation & Symmetry (21–40)

VR-21. An arrow points North. It is rotated 90° clockwise three times, then 45° anticlockwise. Where does it point?
(a) North-West (b) West (c) South-West (d) North
Ans: (c). 270° clockwise puts it at West. Turning 45° anticlockwise from West gives South-West. *Concept:* rotation arithmetic · *Diff:* M · *Why NID:* mental rotation · *Mistake:* turning 45° in the wrong direction.

VR-22. Which of these words reads the same when the page is rotated 180°? (Use simple sans-serif capitals.)
(a) SWIMS (b) MOM (c) DAD (d) WOW
Ans: (a). A 180° rotation reverses the letter order AND turns each letter upside down. S→S, W→M, I→I, M→W, so S-W-I-M-S becomes S-W-I-M-S ✓. MOM becomes WOW ×, WOW becomes MOM ×, and D upside-down is not D ×. *Concept:* 180° ambigrams · *Diff:* H · *Why NID:* careful verification · *Mistake:* only checking whether the word is a palindrome.

VR-23. The number of lines of symmetry of a regular hexagon is:
(a) 3 (b) 4 (c) 6 (d) 12
Ans: (c). Three lines pass through opposite vertices and three through midpoints of opposite sides. *Concept:* symmetry · *Diff:* E · *Why NID:* geometry sense · *Mistake:* answering 3.

VR-24. How many lines of symmetry does a rectangle (not a square) have?
(a) 1 (b) 2 (c) 4 (d) 0
Ans: (b). The diagonals are NOT lines of symmetry for a non-square rectangle. *Concept:* symmetry · *Diff:* E · *Why NID:* the classic trap · *Mistake:* answering 4.

VR-25. A parallelogram (not a rectangle or rhombus) has how many lines of symmetry and what order of rotational symmetry?
(a) 2 lines, order 2 (b) 0 lines, order 2 (c) 0 lines, order 1 (d) 1 line, order 2
Ans: (b). It has no line symmetry, but a half turn maps it onto itself. *Concept:* line vs rotational symmetry · *Diff:* M · *Why NID:* distinguishing the two kinds · *Mistake:* answering (a).

VR-26. Which letter has rotational symmetry of order 2 but NO line of symmetry?
(a) H (b) S (c) A (d) E
Ans: (b). S looks the same after a half turn and has no mirror axis. H has both kinds of symmetry. *Concept:* symmetry types · *Diff:* M · *Why NID:* discrimination · *Mistake:* choosing H.

VR-27. A five-pointed regular star has how many lines of symmetry?
(a) 5 (b) 10 (c) 2 (d) 1
Ans: (a). *Concept:* symmetry · *Diff:* E · *Why NID:* the star motif in logos · *Mistake:* answering 10.

VR-28. The Ashoka Chakra (24 equally spaced spokes) has rotational symmetry of order:
(a) 12 (b) 24 (c) 6 (d) 48
Ans: (b). *Concept:* rotational symmetry · *Diff:* E · *Why NID:* national symbol + geometry · *Mistake:* none.

VR-29. A square tile shows a single arrow pointing up. The tile is rotated 90° anticlockwise. Where does the arrow point?
(a) Right (b) Left (c) Down (d) Up
Ans: (b). Turning anticlockwise, up goes to left. *Concept:* rotation direction · *Diff:* E · *Why NID:* basics · *Mistake:* rotating clockwise by habit.

VR-30. The capital letter "F" is rotated 90° clockwise. What does it look like?
(a) A horizontal bar along the top, with two short strokes hanging down: one from the RIGHT end and one from the middle
(b) A horizontal bar along the top, with two short strokes hanging down: one from the LEFT end and one from the middle
(c) A horizontal bar along the bottom, with two short strokes pointing up from the right end and the middle
(d) It looks like the letter E
Ans: (a). In F the stem is on the left and the arms point right from its top and middle. Turning 90° clockwise sends the top of the stem to the right, so the stem lies horizontal and the arms now point downward. The top arm hangs from the right end; the middle arm hangs from the middle. (b) is the mirror image of the correct answer, and (c) is the result of an anticlockwise turn followed by a flip. *Concept:* mental rotation · *Diff:* H · *Why NID:* precision · *Mistake:* choosing (b). A rotation can never produce a mirror image. (Check: write F on a sticky note and turn it.)

VR-31. A wheel with 8 equally spaced spokes is rotated by 135°. Does it look identical to the start?
(a) Yes (b) No
Ans: (a). Order-8 symmetry means rotations by multiples of 45° look identical, and 135° = 3 × 45°. *Concept:* rotational symmetry · *Diff:* M · *Why NID:* reasoning with angles · *Mistake:* none.

VR-32. Which shape tiles a floor alone with no gaps?
(a) Regular pentagon (b) Regular octagon (c) Regular hexagon (d) Circle
Ans: (c). Its interior angle of 120° divides 360° exactly. *Concept:* tessellation · *Diff:* E · *Why NID:* pattern design · *Mistake:* choosing the octagon (octagons need extra squares to fill the gaps).

VR-33. A clock's minute hand moves from 12 to 9 going clockwise. Through how many degrees has it turned?
(a) 90° (b) 180° (c) 270° (d) 360°
Ans: (c). *Concept:* angle measurement · *Diff:* E · *Why NID:* rotation · *Mistake:* answering 90° (that is the anticlockwise route).

VR-34. A design is made by rotating a single petal around a centre 6 times at equal intervals. The angle between petals is:
(a) 30° (b) 45° (c) 60° (d) 72°
Ans: (c). 360° ÷ 6 = 60°. *Concept:* radial design · *Diff:* E · *Why NID:* rangoli / mandala construction · *Mistake:* none.

VR-35. Which letter has exactly one horizontal line of symmetry and no vertical one?
(a) A (b) K (c) H (d) T
Ans: (b). A and T have a vertical axis; H has both. *Concept:* symmetry · *Diff:* M · *Why NID:* discrimination · *Mistake:* none.

VR-36. A semicircle has how many lines of symmetry?
(a) 0 (b) 1 (c) 2 (d) infinite
Ans: (b). *Concept:* symmetry · *Diff:* E · *Why NID:* basics · *Mistake:* answering "infinite" by confusing it with a full circle.

VR-37. Which of these figures has rotational symmetry of order 3?
(a) Square (b) Equilateral triangle (c) Rectangle (d) Isosceles triangle
Ans: (b). *Concept:* rotational symmetry · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-38. A 3×3 grid has only its top-left cell shaded. The grid is rotated 90° clockwise, then mirrored left–right. Which cell is shaded now?
(a) top-left (b) top-right (c) bottom-left (d) bottom-right
Ans: (a). A 90° clockwise rotation moves top-left to top-right. The left–right mirror moves top-right back to top-left. *Concept:* composed transformations · *Diff:* M · *Why NID:* multi-step tracking · *Mistake:* doing the steps in the wrong order.

VR-39. In the same 3×3 grid, the shaded cell is top-middle. After a 180° rotation it is:
(a) top-middle (b) bottom-middle (c) middle-left (d) centre
Ans: (b). *Concept:* rotation · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-40. A paper strip has a half-twist and its ends are joined (a Möbius strip). How many sides and edges does it have?
(a) 2 sides, 2 edges (b) 1 side, 1 edge (c) 1 side, 2 edges (d) 2 sides, 1 edge
Ans: (b). *Concept:* topology · *Diff:* M · *Why NID:* curiosity / paper models (the recycling symbol is based on it) · *Mistake:* answering (a).

C. Paper Folding & Cutting (41–60)

VR-41. A square sheet, 8 cm × 8 cm, is folded in half with the left half going onto the right half (fold line x = 4). A hole is punched at (6, 6). Unfolded, where are the holes?
(a) (6,6) only (b) (2,6) and (6,6) (c) (6,2) and (6,6) (d) (2,2) and (6,6)
Ans: (b). Reflecting across x = 4 gives 4 − (6 − 4) = 2, with y unchanged. *Concept:* reflection across a fold · *Diff:* E · *Why NID:* the basic fold · *Mistake:* reflecting y as well.

VR-42. The same 8×8 sheet is folded left onto right (x = 4), then the top onto the bottom (y = 4). One hole is punched at (5, 3). How many holes appear, and where?
(a) 2 (b) 4: (5,3), (3,3), (5,5), (3,5) (c) 4: (5,3), (3,3), (5,1), (3,1) (d) 8

Ans: (b). Unfold in reverse order. First undo the top-onto-bottom fold: (5,3) reflects across y = 4 to (5,5). Then undo the left-onto-right fold: (5,3)→(3,3) and (5,5)→(3,5). *Concept:* reverse unfolding · *Diff:* M · *Why NID:* the core skill · *Mistake:* reflecting across the wrong axis.

VR-43. A sheet is folded in half 4 times, and a hole is punched through all layers once (not on a fold). How many holes appear when unfolded?
(a) 4 (b) 8 (c) 16 (d) 32
Ans: (c). $2^4 = 16$. *Concept:* layer doubling · *Diff:* E · *Why NID:* the quick rule · *Mistake:* answering 8.

VR-44. A square sheet is folded along its diagonal from bottom-left to top-right (the top-left triangle goes onto the bottom-right). A hole is punched at (6, 2) on an 8×8 sheet. Where is the second hole after unfolding?
(a) (2,6) (b) (6,6) (c) (2,2) (d) (6,2) only
Ans: (a). Reflection across the line y = x swaps the coordinates: (6,2) → (2,6). *Concept:* diagonal fold · *Diff:* M · *Why NID:* the harder fold type · *Mistake:* treating it as a vertical fold.

VR-45. A rectangle is folded in half once, and a semicircle is cut out with its flat side on the fold line. Unfolded, the hole is:
(a) a semicircle (b) a full circle (c) two semicircles at opposite edges (d) an ellipse
Ans: (b). The cut mirrors across the fold to complete a circle. *Concept:* cut on a fold · *Diff:* E · *Why NID:* paper cutting · *Mistake:* answering (c).

VR-46. A paper is folded in half twice (making 4 layers, folds meeting at one corner). A quarter-circle is cut from the corner where both folds meet. Unfolded, the result is:
(a) a quarter-circle notch (b) a full circular hole in the centre (c) four circles (d) a semicircle on an edge
Ans: (b). The corner where the folds meet is the centre of the original sheet, so four quarter-circles join into a full circle. *Concept:* the snowflake principle · *Diff:* M · *Why NID:* the classic item · *Mistake:* answering (c).

VR-47. A long strip of paper is folded in half 3 times, each fold parallel to the previous one (making 8 layers). It is then cut with ONE straight cut across all layers, parallel to the folds but not on a fold. How many pieces result?
(a) 2 (b) 5 (c) 8 (d) 9
Ans: (d). Each cut through the folded packet separates the strip at every layer, giving 8 cut lines on the unfolded strip and therefore 8 + 1 = 9 pieces. Test with small cases: 1 fold (2 layers) gives 3 pieces, and 2 folds (4 layers) give 5. The pattern is $2^n + 1$. *Concept:* fold-and-cut reasoning · *Diff:* H · *Why NID:* deeper reasoning · *Mistake:* answering 8 (the layers, not the pieces).

VR-48. A paper is folded like a fan (accordion) into 4 equal vertical strips. A small triangle is cut out of the CENTRE of the folded stack (not touching any edge or fold). How many triangular holes appear when unfolded?
(a) 1 (b) 2 (c) 4 (d) 3
Ans: (c). The accordion makes 4 layers, so the cut passes through 4 layers and gives 4 holes. Neighbouring holes are mirror images of each other across the folds. *Concept:* accordion fold · *Diff:* M · *Why NID:* garland and paper-chain designs · *Mistake:* answering 1.

VR-49. Paper doll chains (people holding hands) are made by accordion-folding and cutting a figure whose hands touch the folded edges. Why must the hands touch the folds?
(a) For decoration (b) Uncut fold segments keep the figures connected (c) To save paper (d) There is no reason
Ans: (b). *Concept:* the fold as a hinge or connection · *Diff:* E · *Why NID:* craft logic · *Mistake:* none.

VR-50. An 8×8 sheet is folded so that the top edge meets the bottom edge (fold at y = 4), then the right edge meets the left edge (fold at x = 4). A hole is punched at (1, 1) in the folded packet, which spans x 0–4 and y 0–4. Unfolded holes:

(a) (1,1), (7,1), (1,7), (7,7) (b) (1,1), (3,1), (1,3), (3,3) (c) (1,1) only (d) (1,1), (1,7)
Ans: (a). Undo the right-onto-left fold: (1,1)→(7,1). Undo the top-onto-bottom fold: each point reflects across y = 4, so (1,1)→(1,7) and (7,1)→(7,7). *Concept:* reverse unfolding · *Diff:* M · *Why NID:* practice · *Mistake:* reflecting across x = 2.

VR-51. Which fold produces a mirror-symmetric shape with a vertical axis when cutting a heart?
(a) Fold vertically and cut half a heart along the fold (b) Fold horizontally (c) Fold diagonally (d) No fold
Ans: (a). *Concept:* symmetric cutting · *Diff:* E · *Why NID:* craft · *Mistake:* none.

VR-52. A sheet is folded in half. A square punch is placed so that it straddles the folded edge (half of the punch over the paper, half hanging off the fold). Unfolded, you see:
(a) two separate square holes (b) one rectangular hole twice as wide as the part of the punch that bit into the paper, centred on the fold line (c) a notch on the outer edge (d) no hole
Ans: (b). The punch cuts a notch into the folded edge. Unfolding mirrors the notch across the fold, joining the two halves into one hole that straddles the fold line. *Concept:* punch on a fold · *Diff:* M · *Why NID:* the typical trap · *Mistake:* answering (a).

VR-53. How many layers result from folding a sheet in half, then in thirds (accordion, 3 panels)?
(a) 5 (b) 6 (c) 8 (d) 9
Ans: (b). 2 × 3 = 6. *Concept:* layer multiplication · *Diff:* E · *Why NID:* reasoning · *Mistake:* adding instead of multiplying.

VR-54. A square paper is folded in half twice to form a smaller square (4 layers). One small circle is punched at the centre of this small square (not on any fold). How many holes appear after unfolding, and how are they arranged?
(a) 4, one at the centre of each quadrant (b) 4, clustered at the centre (c) 1 at the centre (d) 2
Ans: (a). The centre of the folded square maps to the centre of each quarter of the original sheet. *Concept:* mapping folded positions · *Diff:* M · *Why NID:* practice · *Mistake:* answering (b).

VR-55. Origami is the Japanese art of paper folding. Kirigami differs because it:
(a) uses glue (b) involves cutting as well as folding (c) uses cloth (d) uses no folds
Ans: (b). *Concept:* paper craft vocabulary · *Diff:* E · *Why NID:* the studio test · *Mistake:* none.

VR-56. A strip of paper is folded in half 5 times along its length, then unfolded. How many crease lines are there?
(a) 5 (b) 16 (c) 31 (d) 32
Ans: (c). Folding 5 times creates 2^5 = 32 segments, separated by 31 creases. *Concept:* creases = segments − 1 · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 32.

VR-57. An 8×8 square is folded diagonally from the top-left corner to the bottom-right corner (the bottom-left triangle folds onto the top-right triangle). A hole is punched at (6, 6). The second hole after unfolding is at:
(a) (2,2) (b) (6,6) only (c) (2,6) (d) (6,2)
Ans: (a). The fold line runs from (0,8) to (8,0), i.e. x + y = 8. Reflecting (6,6) across it gives (8 − 6, 8 − 6) = (2,2). *Concept:* reflection across the anti-diagonal · *Diff:* H · *Why NID:* advanced folding · *Mistake:* swapping coordinates as if the fold were y = x.

VR-58. A circular paper (a coffee filter) is folded in half, then in half again, then once more, making 8 layers in a narrow wedge. One small hole is punched in the interior of the wedge, not touching any edge or fold. How many holes appear after unfolding?
(a) 4 (b) 8 (c) 16 (d) 2
Ans: (b). 8 layers give 8 holes, arranged in a ring with radial symmetry. *Concept:* radial cutting (snowflakes) · *Diff:* M · *Why NID:* radial design · *Mistake:* answering 4.

VR-59. To make a 6-pointed paper snowflake, the circle or square should be folded so that the wedge angle is:
(a) 90° (b) 60° (c) 30° (d) 45°
Ans: (c). Six-fold symmetry with mirror lines needs 12 identical mirrored wedges, and 360° ÷ 12 = 30°. (A 60° wedge gives 6 identical parts, but the mirror symmetry comes from folding the wedge in half.) *Concept:* radial symmetry construction · *Diff:* H · *Why NID:* deeper craft geometry · *Mistake:* answering 60°.

VR-60. A sheet folded in half twice, at right angles, has a hole punched through all layers. One of the unfolded holes is at the top-left quadrant's centre. Where are the others?
(a) The centres of the other three quadrants (b) Along the top edge only (c) The centre of the sheet only (d) Random
Ans: (a). *Concept:* symmetry about both fold lines · *Diff:* E · *Why NID:* reinforcement · *Mistake:* none.

D. Cubes, Dice & Nets (61–85)

VR-61. A cube has how many edges?
(a) 6 (b) 8 (c) 12 (d) 24
Ans: (c). *Concept:* polyhedra · *Diff:* E · *Why NID:* basics · *Mistake:* answering 8, which is the number of vertices.

VR-62. On a standard die, which number is opposite 3?
(a) 4 (b) 5 (c) 6 (d) 1
Ans: (a). Opposite faces sum to 7. *Concept:* the dice rule · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-63. In the net below, which face is opposite face C?

```
   [A]
[B][C][D][E]
   [F]
```

(a) A (b) E (c) F (d) B
Ans: (b). In a row of 4, faces with one face between them are opposite: B–D and C–E. A and F (above and below the same face C) are opposite each other. *Concept:* the net rule · *Diff:* M · *Why NID:* standard cube item · *Mistake:* answering D.

VR-64. In the same net, which face is opposite A?
(a) F (b) C (c) E (d) D
Ans: (a). Both attach to face C on opposite sides of it. *Concept:* net · *Diff:* M · *Why NID:* the same net, a new face · *Mistake:* answering C.

VR-65. Can this arrangement fold into a cube?

```
[1][2][3]
[4][5][6]
```

(a) Yes (b) No
Ans: (b). A 2×3 block has faces that would overlap; no face could close the cube. *Concept:* invalid nets · *Diff:* M · *Why NID:* net validity · *Mistake:* thinking 6 squares are always enough.

VR-66. Can this fold into a cube?

```
[1][2][3][4]
[5]      [6]
```

(1 is above 5; 4 is above 6)
(a) Yes (b) No
Ans: (b). With 4 in a row and the other two squares both hanging on the SAME side, at the two ends, faces 5 and 6 would both fold to the same side of the cube and collide; nothing would close the face opposite their side. Check: in the row 1-2-3-4, opposite pairs are 1–3 and 2–4. Face 5 (under 1) and face 6 (under 4) fold in from the same direction and land on the same face position. *Concept:* invalid net · *Diff:* H · *Why NID:* deeper visualisation · *Mistake:* assuming any 4 + 2 arrangement works.

VR-67. Can this fold into a cube? A row of four squares 1-2-3-4; square 5 is attached ABOVE square 2; square 6 is attached BELOW square 4.

```
   [5]
[1][2][3][4]
         [6]
```

(a) Yes (b) No
Ans: (a). Four in a row with one square above and one below (at any positions along the row) is always a valid net, and there are 6 such variants. *Concept:* the 1-4-1 net family · *Diff:* M · *Why NID:* net validity · *Mistake:* none.

VR-68. For the valid net in VR-67, which face is opposite face 5?
(a) 6 (b) 1 (c) 3 (d) 4
Ans: (a). In 1-4-1 nets, the square above the row and the square below the row are always opposite each other. *Concept:* net rule · *Diff:* M · *Why NID:* follow-up · *Mistake:* none.

VR-69. A 3×3×3 cube painted on all faces is cut into 27 small cubes. How many small cubes have exactly 2 painted faces?
(a) 8 (b) 12 (c) 6 (d) 1
Ans: (b). $12(n - 2) = 12 \times 1 = 12$, one on each edge. *Concept:* painted cube · *Diff:* E · *Why NID:* standard · *Mistake:* answering 8, which is the number of corners.

VR-70. A 5×5×5 painted cube is cut into 125 unit cubes. How many have exactly 1 painted face?
(a) 54 (b) 36 (c) 27 (d) 72
Ans: (a). $6(n - 2)^2 = 6 \times 9 = 54$. *Concept:* painted cube · *Diff:* M · *Why NID:* formula use · *Mistake:* answering 27, which is the unpainted count.

VR-71. A 6×6×6 painted cube is cut into unit cubes. How many have NO paint?
(a) 64 (b) 125 (c) 216 (d) 96
Ans: (a). $(6 - 2)^3 = 64$. *Concept:* painted cube · *Diff:* M · *Why NID:* formula · *Mistake:* none.

VR-72. A cuboid 3×4×5 is painted on all faces and cut into unit cubes. How many unit cubes have no paint?
(a) 6 (b) 12 (c) 24 (d) 60
Ans: (a). The interior is $(3-2)(4-2)(5-2) = 1 \times 2 \times 3 = 6$. *Concept:* painted cuboid · *Diff:* M · *Why NID:* the extension · *Mistake:* using a cube formula.

VR-73. In the same 3×4×5 cuboid, how many unit cubes have exactly 2 painted faces?
(a) 24 (b) 12 (c) 20 (d) 36
Ans: (a). Edge cubes excluding corners: $4[(3-2) + (4-2) + (5-2)] = 4 \times 6 = 24$. *Concept:* painted cuboid · *Diff:* H · *Why NID:* advanced · *Mistake:* none.

VR-74. What is the minimum number of straight cuts needed to divide a cube into 27 equal smaller cubes (pieces may not be rearranged)?
(a) 6 (b) 9 (c) 27 (d) 3
Ans: (a). Two cuts in each of the three directions. *Concept:* spatial division · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

VR-75. Two views of the same die: View 1 shows 1 on top, 2 in front. View 2 shows 1 on top, 3 in front. Which statement must be true?
(a) 2 is opposite 3 (b) 1 is opposite 6 (on a standard die) (c) 2 is opposite 4 (d) 3 is opposite 1
Ans: (b). On a standard die, opposite faces sum to 7. The two views alone show only that 2 and 3 are both adjacent to 1, so (a) cannot be concluded. *Concept:* dice logic · *Diff:* M · *Why NID:* inference discipline · *Mistake:* concluding (a).

VR-76. A die is rolled one quarter turn forward (away from you) on a table, starting with 1 on top and 2 facing you. After the roll, what is on top? (Standard die.)
(a) 2 (b) 5 (c) 6 (d) 3
Ans: (a). Rolling forward brings the front face up to the top. *Concept:* rolling dice · *Diff:* M · *Why NID:* mental simulation · *Mistake:* answering 5.

VR-77. A cube has faces coloured Red, Blue, Green, Yellow, Black, White. Red is opposite Blue, Green is opposite Yellow. Which colour is opposite White?
(a) Red (b) Black (c) Green (d) Yellow
Ans: (b). The remaining pair must be opposite. *Concept:* elimination · *Diff:* E · *Why NID:* logic · *Mistake:* none.

VR-78. How many different nets (unfolded shapes) does a cube have?
(a) 6 (b) 8 (c) 11 (d) 12
Ans: (c). *Concept:* nets · *Diff:* M · *Why NID:* the classic fact · *Mistake:* none.

VR-79. Which solid has a net made ONLY of squares?
(a) Cube (b) Square pyramid (c) Triangular prism (d) Cylinder
Ans: (a). A square pyramid needs 4 triangles plus a square, a triangular prism needs 2 triangles plus 3 rectangles, and a cylinder needs 2 circles plus a rectangle. *Concept:* nets of solids · *Diff:* E · *Why NID:* packaging dielines · *Mistake:* choosing the square pyramid because of its square base.

VR-80. The net of a cylinder consists of:
(a) 2 circles and 1 rectangle (b) 3 circles (c) 2 rectangles and 1 circle (d) 1 circle and 1 triangle
Ans: (a). *Concept:* nets · *Diff:* E · *Why NID:* packaging · *Mistake:* none.

VR-81. The net of a triangular prism contains:
(a) 2 triangles and 3 rectangles (b) 3 triangles and 2 rectangles (c) 5 triangles (d) 6 squares
Ans: (a). *Concept:* nets · *Diff:* E · *Why NID:* the Toblerone-style packaging · *Mistake:* none.

VR-82. A tetrahedron has how many faces, edges and vertices?
(a) 4, 6, 4 (b) 4, 4, 4 (c) 6, 12, 8 (d) 5, 8, 5
Ans: (a). Check with Euler's formula $V - E + F = 4 - 6 + 4 = 2$ ✓. *Concept:* polyhedra · *Diff:* M · *Why NID:* form vocabulary · *Mistake:* none.

VR-83. Using Euler's formula ($V - E + F = 2$), a polyhedron with 12 vertices and 30 edges has how many faces?
(a) 18 (b) 20 (c) 22 (d) 16
Ans: (b). $12 - 30 + F = 2$, so $F = 20$. (This is an icosahedron.) *Concept:* Euler · *Diff:* M · *Why NID:* logical geometry · *Mistake:* none.

VR-84. A stack of cubes has 3 layers: the bottom is 3×3 = 9 cubes, the middle 2×2 = 4, and the top is 1. How many cubes are in total, and how many are completely hidden from view (from all sides and the top, assuming the bottom isn't visible)?
(a) 14 total; 1 hidden (b) 14 total; 0 hidden (c) 13 total; 1 hidden (d) 14 total; 2 hidden

Ans: (a). 9 + 4 + 1 = 14. The bottom-centre cube is covered by the middle layer above and surrounded on all four sides by bottom-layer cubes, so it is hidden. In the middle layer every cube is on the edge, so all are visible. *Concept:* stacked solids · *Diff:* M · *Why NID:* spatial counting · *Mistake:* answering 0 hidden.

VR-85. A 2×2×2 cube made of 8 unit cubes is painted red on the outside. Then one unit cube is removed. How many unit faces of the remaining structure show red on the outside?
(a) 21 (b) 24 (c) 18 (d) 20
Ans: (a). Originally there are 24 red unit faces (6 faces × 4). The removed cube was a corner cube carrying 3 red faces, so 24 − 3 = 21 red faces remain. The removal exposes 3 new, unpainted inner faces of its neighbours. *Concept:* surface reasoning · *Diff:* H · *Why NID:* deeper spatial thinking · *Mistake:* forgetting that the exposed faces are unpainted.

E. Counting Figures (86–100)

VR-86. How many squares (of all sizes) are in a 4×4 grid of unit squares?
(a) 16 (b) 25 (c) 30 (d) 20
Ans: (c). 16 + 9 + 4 + 1 = 30. *Concept:* counting squares · *Diff:* M · *Why NID:* observation · *Mistake:* answering 16.

VR-87. How many rectangles (including squares) are in a 3×3 grid?
(a) 14 (b) 36 (c) 9 (d) 18
Ans: (b). C(4,2) × C(4,2) = 6 × 6 = 36. *Concept:* rectangle counting · *Diff:* M · *Why NID:* combinatorial reasoning · *Mistake:* answering 14, which counts squares only.

VR-88. A square with both diagonals drawn contains how many triangles?
(a) 4 (b) 6 (c) 8 (d) 10
Ans: (c). 4 small triangles plus 4 half-squares, each made of 2 small triangles. *Concept:* triangle counting · *Diff:* E · *Why NID:* standard · *Mistake:* answering 4.

VR-89. A square with both diagonals AND both midlines (the lines joining midpoints of opposite sides) drawn: how many triangles?
(a) 8 (b) 12 (c) 16 (d) 20
Ans: (c). There are 8 smallest triangles. Two adjacent smallest triangles meeting across a midline make 4 medium triangles (each with the centre and two adjacent corners). The diagonals split the square into halves, giving 4 large triangles. 8 + 4 + 4 = 16. *Concept:* systematic counting · *Diff:* H · *Why NID:* the classic tough item · *Mistake:* missing the medium triangles.

VR-90. An equilateral triangle is divided into 9 small equilateral triangles (3 per side). How many triangles in total?
(a) 9 (b) 10 (c) 13 (d) 16
Ans: (c). Upward triangles: 6 of size 1, 3 of size 2 and 1 of size 3 make 10. There are 3 downward triangles of size 1. Total 13. *Concept:* triangular grids · *Diff:* H · *Why NID:* systematic counting · *Mistake:* forgetting the downward triangles.

VR-91. A triangle has 5 lines from its apex to its base (including the two sides). How many triangles?
(a) 5 (b) 10 (c) 15 (d) 20
Ans: (b). Any two of the 5 lines, together with the base, form a triangle: C(5,2) = 10. *Concept:* combinations · *Diff:* M · *Why NID:* the formula approach · *Mistake:* none.

VR-92. The same figure as VR-91 with one horizontal line crossing all 5 lines (parallel to the base). How many triangles?
(a) 10 (b) 15 (c) 20 (d) 25

Ans: (c). Each horizontal "base" (the actual base plus the crossing line) gives 10 triangles, so 2 × 10 = 20. *Concept:* extending the formula · *Diff:* H · *Why NID:* deeper · *Mistake:* answering 15.

VR-93. How many squares are in a 2×3 grid?
(a) 6 (b) 8 (c) 9 (d) 11
Ans: (b). 6 unit squares plus 2 squares of size 2×2. *Concept:* counting in non-square grids · *Diff:* M · *Why NID:* care · *Mistake:* answering 6.

VR-94. How many straight lines are needed to draw a 5-pointed star (pentagram) without lifting the pencil?
(a) 4 (b) 5 (c) 6 (d) 10
Ans: (b). *Concept:* figure construction · *Diff:* E · *Why NID:* observation · *Mistake:* answering 10, which counts the outer edges.

VR-95. How many triangles are in a pentagram (5-pointed star drawn with 5 straight lines)?
(a) 5 (b) 10 (c) 15 (d) 35
Ans: (b). Every set of 3 of the 5 lines meets in pairs at points that lie on the drawn segments, so each set of 3 lines bounds a triangle: C(5,3) = 10. These are 5 small "tip" triangles and 5 larger triangles (each made of one tip plus part of the inner pentagon). *Concept:* systematic counting · *Diff:* H · *Why NID:* the classic puzzle · *Mistake:* answering 5 (only the tips).

VR-96. [VISUAL QUESTION — IMAGE REQUIRED] *Figure description:* a large rectangle divided by one vertical line into two equal halves; in the left half, one diagonal from top-left to bottom-right. How many triangles?
(a) 1 (b) 2 (c) 3 (d) 4
Ans: (b). The diagonal splits the left half into 2 triangles. The right half has no diagonal. *Concept:* careful counting · *Diff:* E · *Why NID:* attention · *Mistake:* imagining extra lines.

VR-97. How many cubes are in a staircase made of columns of heights 1, 2, 3, 4 (in a single row)?
(a) 8 (b) 10 (c) 12 (d) 16
Ans: (b). 1 + 2 + 3 + 4 = 10. *Concept:* counting solids · *Diff:* E · *Why NID:* spatial counting · *Mistake:* none.

VR-98. How many diagonals does an octagon have?
(a) 16 (b) 20 (c) 24 (d) 28
Ans: (b). n(n − 3)/2 = 8 × 5 / 2 = 20. *Concept:* diagonals · *Diff:* M · *Why NID:* geometry · *Mistake:* none.

VR-99. How many circles of diameter 2 cm fit in a single row inside a rectangle 10 cm long and 2 cm wide (without overlapping)?
(a) 4 (b) 5 (c) 10 (d) 20
Ans: (b). *Concept:* packing · *Diff:* E · *Why NID:* layout logic · *Mistake:* none.

VR-100. How many squares are in a 5×5 grid?
(a) 25 (b) 50 (c) 55 (d) 60
Ans: (c). 25 + 16 + 9 + 4 + 1 = 55. *Concept:* counting · *Diff:* M · *Why NID:* the formula · *Mistake:* none.

F. Series, Matrices & Visual Analogies (101–120)

VR-101. Series of arrows: ↑ → ↓ ← ↑ → ? What comes next?
(a) ↑ (b) ↓ (c) ← (d) →
Ans: (b). The arrow rotates 90° clockwise each step. *Concept:* rotation series · *Diff:* E · *Why NID:* pattern recognition · *Mistake:* none.

VR-102. Series: ● ○○ ●●● ○○○○ ? What comes next?
(a) ●●●●● (b) ○○○○○ (c) ●●●● (d) ○●○●○
Ans: (a). The count increases by 1 each step and the fill alternates between filled and empty. *Concept:* two-

rule series · *Diff:* E · *Why NID:* tracking multiple rules · *Mistake:* none.

VR-103. A shaded quadrant in a circle moves: top-left, top-right, bottom-right, bottom-left, ? Next:
(a) top-left (b) bottom-right (c) centre (d) top-right
Ans: (a). It moves clockwise and cycles. *Concept:* cyclic series · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-104. Sequence of polygons: triangle, square, pentagon, ?, heptagon. Missing:
(a) circle (b) hexagon (c) octagon (d) rhombus
Ans: (b). The number of sides increases by 1. *Concept:* series · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-105. Matrix (number of dots in each cell):

```
1  2  3
2  4  6
3  6  ?
```

(a) 8 (b) 9 (c) 12 (d) 7
Ans: (b). Each cell is row × column, so 3 × 3 = 9. *Concept:* matrix rule · *Diff:* E · *Why NID:* matrix reasoning · *Mistake:* answering 8.

VR-106. Matrix of shapes; each row contains one circle, one square and one triangle, with fill varying:

```
○  ■  △
■  △  ○
△  ○  ?
```

(Row 1: empty circle, filled square, empty triangle. Row 2: filled square, empty triangle, empty circle.)
Missing shape:
(a) a square (b) a circle (c) a triangle (d) a star
Ans: (a). Each shape appears once in each row and column (a Latin square), so row 3 needs the square. *Concept:* Latin-square logic · *Diff:* M · *Why NID:* matrix · *Mistake:* none.

VR-107. Analogy: a square is to a cube as a circle is to a:
(a) cylinder (b) sphere (c) cone (d) ring
Ans: (b). The 2D shape maps to its 3D counterpart, which has the same outline from every direction. *Concept:* 2D→3D analogy · *Diff:* E · *Why NID:* form vocabulary · *Mistake:* answering cylinder (which comes from extruding a circle).

VR-108. Analogy: a small circle inside a big square is to a big circle inside a small square as a small triangle inside a big circle is to:
(a) a small circle inside a big triangle (b) a big circle with a small triangle inside it (c) a small triangle inside a big square (d) two triangles
Ans: (a). The rule swaps which shape is outside and which is inside. Inner small triangle with outer big circle becomes inner small circle with outer big triangle. *Concept:* visual analogy (swap) · *Diff:* M · *Why NID:* analogical thinking · *Mistake:* choosing (b), which is just the original figure.

VR-109. Odd one out: cube, cylinder, sphere, square.
(a) cube (b) cylinder (c) sphere (d) square
Ans: (d). It is the only 2D shape. *Concept:* classification · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-110. Odd one out: ⬠ (pentagon), ⬡ (hexagon), △ (triangle), ◯ (circle).
(a) pentagon (b) hexagon (c) triangle (d) circle
Ans: (d). It is the only shape with no straight edges or vertices. *Concept:* classification · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-111. Series: a clock face shows 12:00, 12:15, 12:45, 1:30, ? What comes next?
(a) 2:15 (b) 2:30 (c) 2:00 (d) 3:00
Ans: (b). The intervals are +15, +30, +45, so the next is +60 min: 1:30 + 1:00 = 2:30. *Concept:* increasing intervals · *Diff:* M · *Why NID:* visual series · *Mistake:* answering 2:15.

VR-112. Series of figures with sides: 3, 4, 6, 9, ? (triangle, square, hexagon, nonagon…). Next number of sides:
(a) 12 (b) 13 (c) 14 (d) 11
Ans: (b). The differences are 1, 2, 3, so the next is +4: 9 + 4 = 13. *Concept:* series · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 12.

VR-113. Matrix: each row's third figure is formed by overlaying the first two and removing the lines they share (an XOR rule). Row: Figure 1 = a square with a vertical midline; Figure 2 = a square with a horizontal midline. Figure 3 = ?
(a) an empty square (b) a square with both midlines (a plus sign inside) (c) only a plus sign (d) a square with diagonals
Ans: (c). The outer square is common to both figures and is removed. The midlines are not shared, so they remain. *Concept:* superimposition rules · *Diff:* H · *Why NID:* matrix reasoning · *Mistake:* answering (b), which is the plain-overlay (OR) rule.

VR-114. Using the plain overlay rule (OR) instead, Figure 3 in VR-113 would be:
(a) an empty square (b) a square with both midlines (c) a plus sign only (d) diagonals
Ans: (b). *Concept:* rule variants · *Diff:* M · *Why NID:* rule precision · *Mistake:* none.

VR-115. Series: a dot moves around the 8 outer cells of a 3×3 grid, skipping one cell each time (moving 2 positions clockwise). Starting at the top-left corner, where is it after 3 moves?
(a) bottom-left (b) bottom-right (c) top-right (d) middle-left
Ans: (a). Number the outer ring clockwise from top-left: 1 TL, 2 TM, 3 TR, 4 MR, 5 BR, 6 BM, 7 BL, 8 ML. Starting at 1, each move adds 2: 1 → 3 → 5 → 7 = BL. *Concept:* positional series · *Diff:* M · *Why NID:* tracking · *Mistake:* counting only the corners.

VR-116. Analogy: a line is to a square as a curve is to a:
(a) circle (b) cube (c) point (d) triangle
Ans: (a). Closing straight lines makes a square; closing a curve makes a circle. *Concept:* element-to-shape analogy · *Diff:* E · *Why NID:* design vocabulary · *Mistake:* none.

VR-117. Series of stick-figure frames: standing → crouching → mid-air with arms up → ? Most logical next frame:
(a) lying down asleep (b) landing with bent knees (c) standing with arms crossed (d) sitting on a chair
Ans: (b). The sequence shows a jump, and landing completes the action. *Concept:* sequential logic · *Diff:* E · *Why NID:* storyboard thinking · *Mistake:* none.

VR-118. Odd one out among logos: Olympic rings, Audi rings, MasterCard circles, Nike swoosh.
(a) Olympic (b) Audi (c) MasterCard (d) Nike
Ans: (d). The other three are built from overlapping or interlocking circles. *Concept:* visual classification · *Diff:* E · *Why NID:* logo awareness · *Mistake:* none.

VR-119. Series of growing squares: 1×1, 2×2, 3×3, 4×4 grids of unit tiles. How many unit tiles are added to go from the 4×4 to the 5×5?
(a) 5 (b) 9 (c) 10 (d) 25
Ans: (b). 25 − 16 = 9 (an L-shaped band). *Concept:* visual series (odd numbers) · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 25.

VR-120. Analogy: footprint : foot :: shadow : ?
(a) light (b) object (c) sun (d) darkness
Ans: (b). Each is an impression or projection of the thing. *Concept:* relationship · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering "sun", which is the cause rather than the source shape.

G. Orthographic Views, Shadows & Perspective (121–135)

VR-121. The top view (plan) of a cone standing on its base is:
(a) a triangle (b) a circle with a dot at the centre (c) a square (d) an ellipse
Ans: (b). *Concept:* orthographic views · *Diff:* E · *Why NID:* 3D→2D · *Mistake:* answering triangle, which is the front view.

VR-122. The front view of a cylinder lying horizontally (its axis left–right) is:
(a) a circle (b) a rectangle (c) a triangle (d) an ellipse
Ans: (b). *Concept:* views · *Diff:* E · *Why NID:* basics · *Mistake:* answering circle, which is the end (side) view.

VR-123. The top view of a square pyramid is:
(a) a triangle (b) a square with both diagonals drawn (c) a circle (d) a pentagon
Ans: (b). The four sloping edges meet at the apex above the centre. *Concept:* views · *Diff:* M · *Why NID:* standard · *Mistake:* answering square only.

VR-124. A cube has a cylindrical hole drilled straight through from its front face to its back face. In technical drawing, the top view shows:
(a) a square with a circle (b) a square with two parallel dashed lines (c) a plain square (d) a circle only
Ans: (b). The hole is hidden inside, so its edges appear as dashed lines. *Concept:* hidden lines · *Diff:* H · *Why NID:* technical visualisation · *Mistake:* answering (a), which is the front view.

VR-125. The front view of a hemisphere resting on its flat face is:
(a) a circle (b) a semicircle (c) a square (d) a triangle
Ans: (b). *Concept:* views · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-126. At 8 am in India, a person faces their shadow. Which direction are they facing?
(a) East (b) West (c) North (d) South
Ans: (b). The morning sun is in the east, so the shadow points west. *Concept:* shadows and directions · *Diff:* E · *Why NID:* the classic item · *Mistake:* answering East.

VR-127. In a drawing, the light source is at the top-left. Where should the cast shadow of a ball on the ground fall?
(a) top-left (b) bottom-right of the ball (c) directly above (d) left
Ans: (b). Shadows fall away from the light. *Concept:* light logic · *Diff:* E · *Why NID:* rendering · *Mistake:* none.

VR-128. At what time of day are outdoor shadows shortest?
(a) Sunrise (b) Around solar noon (c) Sunset (d) Midnight
Ans: (b). The sun is highest then. *Concept:* shadow length · *Diff:* E · *Why NID:* scene realism · *Mistake:* none.

VR-129. In one-point perspective, lines that recede from the viewer meet at:
(a) two points on the horizon (b) one vanishing point on the horizon line (c) the bottom of the page (d) they never meet
Ans: (b). *Concept:* perspective · *Diff:* E · *Why NID:* scene drawing · *Mistake:* none.

VR-130. In a street scene at standing eye level, a distant standing adult's head should be drawn:
(a) above the horizon (b) on the horizon line (c) below the horizon (d) at the vanishing point only
Ans: (b). *Concept:* figures in perspective · *Diff:* M · *Why NID:* crowd scenes · *Mistake:* drawing far-off people as tiny figures below the horizon.

VR-131. You look down from the top of a tall building. Which perspective fits best?
(a) One-point at eye level (b) Three-point with the third vanishing point below (c) Isometric (d) Two-point with a low horizon
Ans: (b). *Concept:* bird's-eye view · *Diff:* M · *Why NID:* viewpoint tasks · *Mistake:* none.

VR-132. Isometric drawings differ from perspective drawings because:
(a) parallel edges remain parallel and nothing converges (b) they use 3 vanishing points (c) far objects look smaller (d) they are always 2D
Ans: (a). *Concept:* projection types · *Diff:* M · *Why NID:* product sketching · *Mistake:* none.

VR-133. In isometric projection, the three axes are drawn at:
(a) 90° to each other (b) 120° to each other on paper (with the receding axes at 30° from horizontal) (c) 45° (d) 60°
Ans: (b). *Concept:* isometric · *Diff:* M · *Why NID:* technical sketching · *Mistake:* answering 45°, which belongs to oblique / cabinet projection.

VR-134. [VISUAL QUESTION — IMAGE REQUIRED] *Description:* a silhouette shows a rounded body, a spout on the right curving upward, a looped handle on the left and a small knob on top. Which object is it?
(a) Watering can (b) Teapot / kettle (c) Pressure cooker (d) Vase
Ans: (b). A spout, a handle and a lid knob together identify a teapot or kettle. A watering can usually has a long straight spout and a top handle. *Concept:* silhouette recognition · *Diff:* E · *Why NID:* observation · *Mistake:* answering watering can.

VR-135. The side view of a standard mug (handle on the right, seen from the side) is roughly:
(a) a circle (b) a rectangle with a C-shape attached on the right (c) a triangle (d) an ellipse only
Ans: (b). *Concept:* orthographic views of everyday objects · *Diff:* E · *Why NID:* product drawing · *Mistake:* none.

H. Spatial Relations, Gears, Maps & Mixed (136–150)

VR-136. Gear A (turning clockwise) meshes with Gear B, which meshes with Gear C, in a line. Which way does Gear C turn?
(a) Clockwise (b) Anticlockwise (c) It doesn't move (d) It alternates
Ans: (a). Meshing gears alternate direction: A clockwise, B anticlockwise, C clockwise. *Concept:* mechanical reasoning · *Diff:* E · *Why NID:* product mechanisms · *Mistake:* none.

VR-137. Three gears are arranged in a closed triangle, each meshing with the other two. What happens when one is turned?
(a) All turn clockwise (b) They jam / cannot turn (c) They alternate (d) Only one turns
Ans: (b). An odd closed loop forces a direction conflict. *Concept:* mechanical logic · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

VR-138. Gear A has 20 teeth and drives Gear B with 60 teeth. If A turns 6 times, how many times does B turn?
(a) 18 (b) 2 (c) 3 (d) 6
Ans: (b). Teeth passed = 20 × 6 = 120, and 120 ÷ 60 = 2. *Concept:* gear ratio · *Diff:* M · *Why NID:* mechanism thinking · *Mistake:* answering 18.

VR-139. Two pulleys are connected by an open (uncrossed) belt. If the first turns clockwise, the second turns:
(a) clockwise (b) anticlockwise (c) it stays still (d) randomly
Ans: (a). An open belt keeps the same direction; a crossed belt reverses it. *Concept:* pulleys · *Diff:* E · *Why NID:* mechanisms · *Mistake:* none.

VR-140. You face North, turn right, turn right again, then turn left. Which direction do you face?
(a) North (b) East (c) South (d) West
Ans: (b). N → E → S → E. *Concept:* direction sense · *Diff:* E · *Why NID:* spatial orientation · *Mistake:* none.

VR-141. On a map, North is up. A temple is 3 km east and 4 km north of a school. What is the straight-line distance?
(a) 5 km (b) 7 km (c) 1 km (d) 12 km
Ans: (a). A 3-4-5 right triangle. *Concept:* map distance · *Diff:* E · *Why NID:* spatial reasoning · *Mistake:* answering 7.

VR-142. A person walks 5 m forward, turns 90° left and walks 5 m, turns 90° left and walks 5 m, then turns 90° left and walks 5 m. Where are they?
(a) 5 m from the start (b) Back at the start (c) 10 m away (d) 20 m away
Ans: (b). They have walked a closed square. *Concept:* path tracing · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-143. In a row of 5 objects, a cup is 2nd from the left and a vase is 2nd from the right. How many objects lie between them?
(a) 0 (b) 1 (c) 2 (d) 3
Ans: (b). The cup is at position 2 and the vase at position 4, so position 3 lies between. *Concept:* positional reasoning · *Diff:* E · *Why NID:* logic · *Mistake:* none.

VR-144. A ladder leans against a wall with its top at 4 m and its foot 3 m from the wall. How long is the ladder?
(a) 5 m (b) 7 m (c) 6 m (d) 4.5 m
Ans: (a). *Concept:* spatial geometry · *Diff:* E · *Why NID:* proportion in drawings · *Mistake:* none.

VR-145. A lever: a 10 kg load sits 1 m from the fulcrum. What force is needed at 2 m on the other side to balance it?
(a) 5 kg-force (b) 10 kg-force (c) 20 kg-force (d) 2 kg-force
Ans: (a). Moments balance: 10 × 1 = F × 2, so F = 5. *Concept:* levers · *Diff:* M · *Why NID:* product mechanisms (bottle openers, seesaws) · *Mistake:* none.

VR-146. A bicycle's front gear (chainring) has 48 teeth and the rear sprocket has 16. For each pedal turn, the rear wheel turns:
(a) 1/3 time (b) 3 times (c) 16 times (d) 48 times
Ans: (b). 48 ÷ 16 = 3. *Concept:* gear ratio · *Diff:* M · *Why NID:* product understanding · *Mistake:* inverting the ratio.

VR-147. Which knot or join holds best when two sticks must be tied at right angles for a bamboo frame?
(a) A reef knot (b) Square lashing (c) A bow (d) A slip knot
Ans: (b). Square lashing is designed for right-angle joins in pole structures. *Concept:* structural making (Studio Test) · *Diff:* M · *Why NID:* hands-on making · *Mistake:* none.

VR-148. Which structure is strongest for a paper bridge using the same amount of paper?
(a) A flat sheet (b) A sheet folded into a corrugated (zig-zag) pattern (c) A crumpled ball (d) A loosely rolled tube with gaps

Ans: (b). Corrugation increases stiffness by moving material away from the bending axis. *Concept:* structural form · *Diff:* M · *Why NID:* Studio Test challenges · *Mistake:* answering (a).

VR-149. Triangles are used in bridge trusses and roof frames because:
(a) they look nice (b) a triangle cannot change shape without changing a side length, so it is rigid (c) they use more material (d) squares are illegal
Ans: (b). *Concept:* structural rigidity · *Diff:* E · *Why NID:* structures and models · *Mistake:* none.

VR-150. [VISUAL QUESTION — IMAGE REQUIRED] *Description:* four identical circles have their centres at the corners of a square. Neighbouring circles (along the square's sides) overlap. The two diagonal pairs of circles just touch at the centre of the square, and all four circles pass through that centre point. How many distinct enclosed regions are formed (not counting the outside)?
(a) 8 (b) 9 (c) 12 (d) 13
Ans: (a). There are 4 lens-shaped overlaps, one between each pair of neighbouring circles. No point lies inside three circles at once, because the diagonal circles only touch. Each circle also keeps one non-overlapping region, giving 4 more. All four lenses meet at the centre point, so there is no separate central region. Total 4 + 4 = 8. *Concept:* region counting · *Diff:* H · *Why NID:* careful visual analysis · *Mistake:* inventing a central region (there would be one only if the circles overlapped more). (Draw it to confirm.)

Notes on items with special answers (please read)

- All items have exactly one correct answer. Items VR-22, 30, 47, 48, 52, 58, 79, 95, 108 and 150 were tightened during the answer-key audit to remove ambiguity.
- For fold, cut and rotation items, physically checking with a sheet of paper is the fastest way to build accuracy.

NEXT: Volume 4C — Logical Reasoning (100 questions) (`04C-VOL4-QB-LOGICAL-REASONING.md`)

VOLUME 4C — QUESTION BANK: LOGICAL REASONING (100 Questions)

Priority: LOW–MEDIUM for DAT 2027. Many items use design or craft contexts, so practising them also revises GK.
Sections: A. Number & letter series (1–15) · B. Analogies (16–30) · C. Classification (31–42) · D. Coding–decoding (43–52) · E. Syllogisms & deduction (53–65) · F. Statements, assumptions, cause–effect (66–78) · G. Arrangements, ranking, directions & puzzles (79–100)

Format: Question → **Answer** → explanation → *Concept* · *Diff* · *Why NID* · *Mistake*.

A. Number & Letter Series (1–15)

LR-1. 5, 11, 23, 47, 95, ?
(a) 190 (b) 191 (c) 189 (d) 193
Ans: (b). Each term is ×2 + 1: 95 × 2 + 1 = 191. *Concept:* multiplicative series · *Diff:* E · *Why NID:* logic warm-up · *Mistake:* answering 190 by forgetting the +1.

LR-2. 1, 4, 9, 16, 25, ?
(a) 30 (b) 36 (c) 49 (d) 35
Ans: (b). Squares: $6^2 = 36$. *Concept:* squares · *Diff:* E · *Why NID:* pattern recognition · *Mistake:* none.

LR-3. 2, 3, 5, 7, 11, 13, ?
(a) 15 (b) 17 (c) 19 (d) 16
Ans: (b). Primes. *Concept:* primes · *Diff:* E · *Why NID:* pattern · *Mistake:* answering 15 (15 = 3 × 5).

LR-4. 3, 6, 18, 72, ?
(a) 144 (b) 288 (c) 360 (d) 216
Ans: (c). The multipliers are ×2, ×3, ×4, so the next is ×5: 72 × 5 = 360. *Concept:* increasing multiplier · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 288 (×4 again).

LR-5. 1, 8, 27, 64, ?
(a) 100 (b) 125 (c) 81 (d) 216
Ans: (b). Cubes: $5^3 = 125$. *Concept:* cubes · *Diff:* E · *Why NID:* volume intuition · *Mistake:* none.

LR-6. 10, 9, 7, 4, 0, ?
(a) −4 (b) −5 (c) −6 (d) −3
Ans: (b). The differences are −1, −2, −3, −4, so the next is −5: 0 − 5 = −5. *Concept:* difference series · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-7. 2, 5, 10, 17, 26, ?
(a) 35 (b) 36 (c) 37 (d) 39
Ans: (c). Each term is $n^2 + 1$: $6^2 + 1 = 37$. *Concept:* squares + 1 · *Diff:* M · *Why NID:* pattern · *Mistake:* none.

LR-8. A, C, F, J, O, ?
(a) T (b) U (c) S (d) V
Ans: (b). The gaps grow by one: +2, +3, +4, +5, then +6. O(15) + 6 = 21 = U. *Concept:* letter series · *Diff:* M · *Why NID:* pattern · *Mistake:* miscounting the letter positions.

LR-9. Z, X, V, T, R, ?
(a) Q (b) P (c) O (d) S
Ans: (b). −2 each step: R(18) − 2 = 16 = P. *Concept:* backward letters · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-10. AZ, BY, CX, DW, ?
(a) EV (b) EU (c) FV (d) EW
Ans: (a). Opposite letters (each pair sums to 27). *Concept:* opposite letters · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-11. 1, 2, 6, 24, 120, ?
(a) 600 (b) 720 (c) 240 (d) 840
Ans: (b). Factorials: 6! = 720. *Concept:* factorial series · *Diff:* M · *Why NID:* pattern · *Mistake:* none.

LR-12. 4, 6, 9, 13.5, ?
(a) 18 (b) 20.25 (c) 20 (d) 17
Ans: (b). ×1.5 each step: 13.5 × 1.5 = 20.25. *Concept:* ratio series · *Diff:* M · *Why NID:* proportional thinking · *Mistake:* none.

LR-13. 2, 4, 3, 9, 4, 16, 5, ?
(a) 20 (b) 25 (c) 36 (d) 6
Ans: (b). The terms come in pairs (n, n^2), so 5 is followed by 25. *Concept:* paired series · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 6.

LR-14. 0, 1, 1, 2, 3, 5, 8, 13, ?
(a) 18 (b) 20 (c) 21 (d) 26
Ans: (c). Fibonacci: 8 + 13 = 21. (Fibonacci numbers relate to the golden ratio and to spirals in sunflowers.) *Concept:* Fibonacci · *Diff:* E · *Why NID:* design and nature link · *Mistake:* none.

LR-15. Which term is wrong in the series: 3, 7, 15, 31, 63, 128?
(a) 15 (b) 31 (c) 63 (d) 128
Ans: (d). The rule is ×2 + 1, so 63 × 2 + 1 = 127, not 128. *Concept:* the wrong-term type · *Diff:* M · *Why NID:* error-spotting · *Mistake:* none.

B. Analogies (16–30)

LR-16. Loom : Weaver :: Wheel : ?
(a) Driver (b) Potter (c) Tailor (d) Smith
Ans: (b). Tool : craftsperson. *Concept:* tool–user · *Diff:* E · *Why NID:* the craft context · *Mistake:* answering "driver" (a car wheel is not that person's main tool of making).

LR-17. Madhubani : Bihar :: Warli : ?
(a) Gujarat (b) Maharashtra (c) Odisha (d) Rajasthan
Ans: (b). Art form : state. *Concept:* region analogy · *Diff:* E · *Why NID:* Indian art GK · *Mistake:* none.

LR-18. Pashmina : Goat :: Silk : ?
(a) Sheep (b) Silkworm (c) Cotton plant (d) Spider
Ans: (b). Fibre : source. *Concept:* material source · *Diff:* E · *Why NID:* textile awareness · *Mistake:* none.

LR-19. Typeface : Font :: Family : ?
(a) Member (b) House (c) Tree (d) Name
Ans: (a). A font is one member of a typeface family. *Concept:* whole–part · *Diff:* M · *Why NID:* typography concepts · *Mistake:* none.

LR-20. Architect : Blueprint :: Composer : ?
(a) Orchestra (b) Score (c) Song (d) Stage
Ans: (b). Creator : planning document. *Concept:* creator–document · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering "song" (the product rather than the plan).

LR-21. Taj Mahal : Shah Jahan :: Qutub Minar : ?
(a) Akbar (b) Qutb-ud-din Aibak (c) Humayun (d) Ashoka
Ans: (b). Monument : the ruler who commissioned or began it. *Concept:* history · *Diff:* E · *Why NID:* architecture GK · *Mistake:* none.

LR-22. Pixel : Image :: Brick : ?
(a) Cement (b) Wall (c) Clay (d) Kiln
Ans: (b). Unit : whole. *Concept:* part–whole · *Diff:* E · *Why NID:* digital + material thinking · *Mistake:* answering "clay" (the material, not the whole).

LR-23. Bauhaus : Gropius :: De Stijl : ?
(a) Picasso (b) Mondrian (c) Dalí (d) Warhol
Ans: (b). Movement : key figure. (Theo van Doesburg was a co-founder of De Stijl; Mondrian is its most famous member.) *Concept:* design history · *Diff:* M · *Why NID:* design GK · *Mistake:* none.

LR-24. Kiln : Pottery :: Oven : ?
(a) Bread (b) Fire (c) Flour (d) Chef
Ans: (a). The heating device and what it bakes or fires. *Concept:* process–product · *Diff:* E · *Why NID:* reasoning · *Mistake:* none.

LR-25. Sculptor : Chisel :: Painter : ?
(a) Canvas (b) Brush (c) Easel (d) Colour
Ans: (b). Maker : main active tool. *Concept:* tool–user · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering canvas (the surface).

LR-26. Red : Stop :: Green : ?
(a) Wait (b) Go (c) Slow (d) Danger
Ans: (b). Signal colour : meaning. *Concept:* colour semantics · *Diff:* E · *Why NID:* signage · *Mistake:* none.

LR-27. Muga : Assam :: Paithani : ?
(a) Gujarat (b) Maharashtra (c) Karnataka (d) Tamil Nadu
Ans: (b). *Concept:* textile region · *Diff:* E · *Why NID:* textile GK · *Mistake:* none.

LR-28. Sketch : Drawing :: Draft : ?
(a) Essay (b) Pen (c) Paper (d) Reader
Ans: (a). Preliminary version : final work. *Concept:* stage relationship · *Diff:* M · *Why NID:* process thinking · *Mistake:* none.

LR-29. Cube : 6 :: Tetrahedron : ?
(a) 3 (b) 4 (c) 5 (d) 8
Ans: (b). Solid : number of faces. *Concept:* geometry · *Diff:* E · *Why NID:* form knowledge · *Mistake:* none.

LR-30. Helvetica : Sans serif :: Garamond : ?
(a) Script (b) Serif (c) Display (d) Monospace
Ans: (b). *Concept:* type classification · *Diff:* E · *Why NID:* typography · *Mistake:* none.

C. Classification / Odd One Out (31–42)

LR-31. Odd one out: Kanjeevaram, Banarasi, Paithani, Madhubani
Ans: Madhubani. It is a painting tradition; the others are woven saris. *Concept:* category · *Diff:* E · *Why NID:* craft GK · *Mistake:* none.

LR-32. Odd one out: Le Corbusier, Louis Kahn, Charles Correa, Raja Ravi Varma
Ans: Raja Ravi Varma. He was a painter; the others were architects. *Concept:* profession · *Diff:* E · *Why NID:* GK · *Mistake:* none.

LR-33. Odd one out: Futura, Helvetica, Times New Roman, Arial
Ans: Times New Roman. It is a serif; the others are sans serif. *Concept:* typography · *Diff:* E · *Why NID:* type · *Mistake:* none.

LR-34. Odd one out: Konark, Khajuraho, Lingaraja, Meenakshi
Ans: Meenakshi. It is Dravida style; the other three are Nagara-style temples. *Concept:* temple styles · *Diff:* M · *Why NID:* architecture · *Mistake:* none.

LR-35. Odd one out: Red, Yellow, Blue, Green (RYB painting primaries)
Ans: Green. It is a secondary colour in the RYB model. *Concept:* colour theory · *Diff:* E · *Why NID:* colour · *Mistake:* thinking in RGB, where green is a primary.

LR-36. Odd one out: Bandhani, Leheriya, Ikat, Chikankari
Ans: Chikankari. It is embroidery; the others are resist-dyeing techniques. *Concept:* textile process · *Diff:* M · *Why NID:* textile depth · *Mistake:* none.

LR-37. Odd one out: Picasso, Braque, Dalí, Gris
Ans: Dalí. He was a Surrealist; the others were Cubists. *Concept:* art movements · *Diff:* M · *Why NID:* art history · *Mistake:* none.

LR-38. Odd one out: Square, Rhombus, Trapezium, Cylinder
Ans: Cylinder. It is 3D; the others are 2D quadrilaterals. *Concept:* geometry · *Diff:* E · *Why NID:* form · *Mistake:* none.

LR-39. Odd one out: Dhokra, Bidri, Pietra dura, Kalamkari
Ans: Kalamkari. It is a textile art; the others are metal or stone crafts. *Concept:* craft materials · *Diff:* M · *Why NID:* craft GK · *Mistake:* none.

LR-40. Odd one out: Jony Ive, Dieter Rams, Philippe Starck, Milton Glaser
Ans: Milton Glaser. He was a graphic designer; the others are product or industrial designers. *Concept:* design fields · *Diff:* M · *Why NID:* designer GK · *Mistake:* none.

LR-41. Odd one out: 121, 144, 169, 196, 210
Ans: 210. It is not a perfect square (the others are 11^2, 12^2, 13^2, 14^2). *Concept:* number property · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-42. Odd one out: Sanchi Stupa, Ajanta Caves, Hawa Mahal, Ellora Caves
Ans: Hawa Mahal. It is Rajput and secular (1799); the others are ancient rock-cut or Buddhist-era monuments. *Concept:* period / type · *Diff:* M · *Why NID:* architecture GK · *Mistake:* none.

D. Coding–Decoding (43–52)

LR-43. If COLOUR is coded as DPMPVS, how is SHADE coded?
(a) TIBEF (b) TJBEF (c) RGZCD (d) TIBFE
Ans: (a). +1 per letter: S→T, H→I, A→B, D→E, E→F. *Concept:* shift code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-44. If DESIGN is written as NGISED, how is CRAFT written?
(a) TFARC (b) TFACR (c) CRAFT (d) FTARC
Ans: (a). Reversal. *Concept:* reverse code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-45. In a code, A = 1, B = 2 ... Z = 26. What is the sum of the letters in "NID"?
(a) 27 (b) 31 (c) 33 (d) 29
Ans: (a). N = 14, I = 9, D = 4; 14 + 9 + 4 = 27. *Concept:* letter values · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-46. If each letter is replaced by its opposite (A↔Z, B↔Y...), what does "ART" become?
(a) ZIG (b) ZJG (c) ZIH (d) YIG
Ans: (a). A→Z, R(18)→27 − 18 = 9 = I, T(20)→7 = G. *Concept:* opposite-letter code · *Diff:* M · *Why NID:* logic · *Mistake:* miscounting.

LR-47. If PAINT = 5 and SKETCH = 6, then CANVAS = ?
(a) 5 (b) 6 (c) 7 (d) 4
Ans: (b). The code is the number of letters. *Concept:* simple property code · *Diff:* E · *Why NID:* logic · *Mistake:* overthinking.

LR-48. If "red blue green" means "sun sky leaf", "blue yellow" means "sky light" and "green pink" means "leaf petal", what does "blue" mean?
(a) sun (b) sky (c) leaf (d) light
Ans: (b). "blue" appears in the first two sentences, and the only common word in their meanings is "sky". *Concept:* common-word decoding · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-49. In LR-48, what does "red" mean?
(a) sun (b) sky (c) leaf (d) light
Ans: (a). blue = sky (LR-48), and green = leaf (common to the first and third sentences), so red = sun. *Concept:* elimination · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-50. If FORM is coded as 6-15-18-13, what is SPACE?
(a) 19-16-1-3-5 (b) 19-15-1-3-5 (c) 18-16-1-3-5 (d) 19-16-2-3-5
Ans: (a). Letter positions: S19 P16 A1 C3 E5. *Concept:* position code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-51. If LINE is coded as KHMD (each letter −1), how is DOT coded?
(a) CNS (b) EPU (c) CNT (d) DNS
Ans: (a). *Concept:* shift code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-52. In a code, each symbol stands for a digit: ○ = 1, △ = 3, □ = 4. A row of symbols is read as the number formed by those digits in order (so □○ = 41). What number is △□○?
(a) 8 (b) 341 (c) 12 (d) 314
Ans: (b). △ = 3, □ = 4, ○ = 1, which gives 341. *Concept:* symbol substitution · *Diff:* E · *Why NID:* reading the rule carefully · *Mistake:* adding the digits (8) when the rule says to place them side by side.

E. Syllogisms & Deduction (53–65)

For each: decide which conclusions *definitely* follow.

LR-53. All potters are artists. All artists are observers. Conclusion: All potters are observers.
(a) Follows (b) Does not follow
Ans: (a). A chain of "all" statements is valid. *Concept:* syllogism · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-54. Some designers are painters. Some painters are poets. Conclusion: Some designers are poets.
(a) Follows (b) Does not follow
Ans: (b). Two "some" premises give no certain conclusion. *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* assuming the overlap.

LR-55. All chairs are furniture. No furniture is edible. Conclusion: No chair is edible.
(a) Follows (b) Does not follow
Ans: (a). *Concept:* syllogism · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-56. No sculptor is lazy. Some students are sculptors. Conclusion: Some students are not lazy.
(a) Follows (b) Does not follow
Ans: (a). The students who are sculptors are not lazy. *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-57. All logos are symbols. Conclusion I: Some symbols are logos. Conclusion II: All symbols are logos.
(a) Only I (b) Only II (c) Both (d) Neither
Ans: (a). "All A are B" implies "some B are A", but not "all B are A". *Concept:* conversion · *Diff:* M · *Why NID:* logic · *Mistake:* answering (c).

LR-58. Some fabrics are silk. All silk is natural. Conclusion: Some fabrics are natural.
(a) Follows (b) Does not follow
Ans: (a). *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-59. All cubes are prisms. Some prisms are hollow. Conclusion: Some cubes are hollow.
(a) Follows (b) Does not follow
Ans: (b). The hollow prisms may not include any cubes. *Concept:* the undistributed middle · *Diff:* M · *Why NID:* logic · *Mistake:* assuming it follows.

LR-60. If it rains, the street floods. The street is not flooded. Conclusion: It did not rain.
(a) Follows (b) Does not follow
Ans: (a). Modus tollens: denying the consequent is valid. *Concept:* conditional logic · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

LR-61. If it rains, the street floods. The street is flooded. Conclusion: It rained.
(a) Follows (b) Does not follow
Ans: (b). Affirming the consequent is invalid; a burst pipe could have flooded the street. *Concept:* logical fallacy · *Diff:* M · *Why NID:* critical thinking · *Mistake:* assuming it follows.

LR-62. Every NID student sketches daily. Riya does not sketch daily. Conclusion: Riya is not an NID student.
(a) Follows (b) Does not follow
Ans: (a). *Concept:* contrapositive · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-63. No square is a circle. All circles are round. Conclusion: No square is round.
(a) Follows (b) Does not follow
Ans: (b). Other things (e.g. ovals) can be round, so the premises do not rule out round squares. (Real

squares aren't round, but that fact is not deducible from these premises.) *Concept:* validity vs truth · *Diff:* H · *Why NID:* rigour · *Mistake:* using outside knowledge.

LR-64. All A are B. No B is C. Conclusion I: No A is C. Conclusion II: No C is A.
(a) Only I (b) Only II (c) Both (d) Neither
Ans: (c). "No A is C" is equivalent to "no C is A". *Concept:* E-statement conversion · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-65. Some lamps are brass. No brass is plastic. Conclusion: Some lamps are not plastic.
(a) Follows (b) Does not follow
Ans: (a). The brass lamps are not plastic. *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* none.

F. Statements, Assumptions, Inferences, Cause–Effect (66–78)

LR-66. Statement: "Install ramps at all metro stations." Implicit assumption:
(a) Some passengers need step-free access (b) Everyone uses wheelchairs (c) Stairs are illegal (d) Ramps are cheap
Ans: (a). *Concept:* assumption · *Diff:* E · *Why NID:* inclusive-design reasoning · *Mistake:* choosing an extreme option like (b).

LR-67. Statement: "Our new bottle uses 30% less plastic." Which assumption is implicit?
(a) Less plastic is desirable (b) The bottle is free (c) Plastic is banned (d) Customers hate glass
Ans: (a). *Concept:* assumption · *Diff:* E · *Why NID:* sustainability reasoning · *Mistake:* none.

LR-68. Cause: heavy rains for a week. Effect most likely:
(a) Vegetable prices rise (b) Ice-cream sales rise (c) Fewer umbrellas sold (d) Dust storms
Ans: (a). Crop damage and transport disruption push prices up. *Concept:* cause–effect · *Diff:* E · *Why NID:* everyday reasoning · *Mistake:* none.

LR-69. Statement: "Visitors at the museum doubled after audio guides in 5 Indian languages were added." Best inference:
(a) Language accessibility may have increased engagement (b) Audio guides cause all visits (c) Museums are useless without guides (d) English is not spoken
Ans: (a). This is the cautious inference. *Concept:* inference vs overgeneralisation · *Diff:* M · *Why NID:* critical thinking · *Mistake:* choosing (b).

LR-70. Statement: "Use steel tumblers instead of paper cups in the canteen." Which argument is STRONG?
(a) Steel is shiny (b) It reduces daily waste and long-term cost (c) Paper cups are colourful (d) Everyone likes steel
Ans: (b). It is relevant and substantive. *Concept:* argument strength · *Diff:* E · *Why NID:* reasoning · *Mistake:* none.

LR-71. Two events: (A) many people started cycling to work; (B) the city built protected cycle lanes. Most likely relation:
(a) A caused B (b) B caused A (c) Both from a common cause (d) Unrelated
Ans: (b). Infrastructure typically enables behaviour. (Some argue the reverse feedback exists, but the "most likely" reading is B → A.) *Concept:* causal direction · *Diff:* M · *Why NID:* urban-design reasoning · *Mistake:* none.

LR-72. Statement: "The handloom sector employs millions but weavers' incomes are low." Which course of action is MOST useful?
(a) Ban power looms entirely (b) Connect weavers directly to markets and design support (c) Stop

handloom (d) Ignore it
Ans: (b). *Concept:* course of action · *Diff:* M · *Why NID:* craft and design policy · *Mistake:* choosing an extreme action.

LR-73. Statement: "All entries must be submitted on A3 sheets." Inference:
(a) A4 entries may be rejected (b) A3 is the largest paper (c) Everyone has A3 (d) A3 is cheap
Ans: (a). *Concept:* inference from rules · *Diff:* E · *Why NID:* reading instructions (a key exam skill!) · *Mistake:* none.

LR-74. Statement: "Since the new signage was installed, fewer visitors asked for directions." The best conclusion:
(a) The signage probably improved wayfinding (b) Visitors stopped coming (c) Signs always work (d) Guards were fired
Ans: (a). *Concept:* evaluating evidence · *Diff:* E · *Why NID:* design evaluation · *Mistake:* none.

LR-75. Which statement is a FACT rather than an opinion?
(a) The Taj Mahal is the most beautiful building (b) The Taj Mahal is made mainly of white marble (c) Mughal art is better than Rajput art (d) Agra is boring
Ans: (b). *Concept:* fact vs opinion · *Diff:* E · *Why NID:* critical reading · *Mistake:* none.

LR-76. Statement: "Children in this school carry 8 kg bags." Which conclusion is NOT justified?
(a) Bags may be heavy for young children (b) All children have back pain (c) Lockers might help (d) A timetable change might reduce load
Ans: (b). It is an over-generalisation. *Concept:* conclusion validity · *Diff:* E · *Why NID:* problem framing · *Mistake:* none.

LR-77. Argument: "We should design for the average user only." Which is the strongest counter-argument?
(a) Average users don't exist in reality; designing for diverse users (5th–95th percentile and beyond) improves usability for all (b) Averages are mathematical (c) Designers like averages (d) None
Ans: (a). *Concept:* inclusive-design argument · *Diff:* M · *Why NID:* design philosophy · *Mistake:* none.

LR-78. Statement: "The poster had too much text, so few people read it." Implicit assumption:
(a) People are less likely to read text-heavy posters at a glance (b) Posters should have no text (c) Nobody reads (d) Text is illegal
Ans: (a). *Concept:* assumption · *Diff:* E · *Why NID:* visual-communication reasoning · *Mistake:* choosing the extreme (b).

G. Arrangements, Ranking, Directions & Puzzles (79–100)

LR-79. Five designers P, Q, R, S, T sit in a row facing north. Q is to the immediate right of P. R is at the left end. T is between R and P. Where is S?
(a) Right end (b) Second from left (c) Middle (d) Left end
Ans: (a). Seats 1–5 from left: R = 1. T lies between R and P, and Q is immediately right of P, so the order begins R, T, P, Q. That leaves S at seat 5, the right end. *Concept:* linear arrangement · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-80. In a class of 40, Asha is 12th from the top. What is her rank from the bottom?
(a) 28 (b) 29 (c) 30 (d) 27
Ans: (b). 40 − 12 + 1 = 29. *Concept:* ranking · *Diff:* E · *Why NID:* logic · *Mistake:* answering 28.

LR-81. Six people A–F sit around a circular table facing the centre. A sits opposite D. B is to the immediate left of A. C sits opposite B. F is to the immediate right of D. Who sits opposite E?
(a) F (b) B (c) A (d) C
Ans: (a). Number the seats 1–6 clockwise (seen from above). For someone facing the centre, their

immediate left is the next seat clockwise and their immediate right is the next seat anticlockwise. A = 1, D = 4. B is immediately left of A, so B = 2. C is opposite B, so C = 5. F is immediately right of D, so F = 3. That leaves E = 6, and seat 3 (F) is opposite seat 6. *Concept:* circular arrangement · *Diff:* H · *Why NID:* rigorous spatial logic · *Mistake:* mixing up left and right for people facing the centre.

LR-82. A walks 10 m South, turns left and walks 6 m, turns left and walks 10 m. How far and in which direction is A from the start?
(a) 6 m East (b) 6 m West (c) 10 m North (d) 16 m East
Ans: (a). Turning left from South faces East, so A walks 6 m east. Turning left from East faces North, and 10 m north returns A to the starting latitude. A ends 6 m east of the start. *Concept:* directions · *Diff:* E · *Why NID:* spatial logic · *Mistake:* turning in the wrong direction.

LR-83. If today is Wednesday, what day will it be 100 days from now?
(a) Thursday (b) Friday (c) Saturday (d) Sunday
Ans: (b). 100 mod 7 = 2, so Wednesday + 2 = Friday. *Concept:* calendar · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-84. Four friends (Ana, Ben, Cara, Dev) each like a different craft: pottery, weaving, carving, painting. Ana doesn't like pottery or painting. Ben likes weaving. Cara doesn't like carving. Who likes carving?
(a) Ana (b) Cara (c) Dev (d) Ben
Ans: (a). Ben has weaving. Ana cannot have pottery or painting, so Ana has carving. *Concept:* grid puzzle · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-85. In LR-84, Cara likes:
(a) Pottery or painting, undetermined (b) Pottery (c) Painting (d) Carving
Ans: (a). Pottery and painting remain for Cara and Dev, and no clue separates them. *Concept:* determinacy · *Diff:* M · *Why NID:* rigour · *Mistake:* forcing a unique answer.

LR-86. A clock gains 5 minutes every hour. Set correctly at 9 am, what does it show at 3 pm?
(a) 3:25 pm (b) 3:30 pm (c) 3:35 pm (d) 3:00 pm
Ans: (b). 6 hours × 5 min = 30 min ahead. *Concept:* clock logic · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-87. Books on a shelf: Red is left of Blue; Green is right of Blue; Yellow is left of Red. Which is rightmost?
(a) Red (b) Blue (c) Green (d) Yellow
Ans: (c). Order: Yellow, Red, Blue, Green. *Concept:* ordering · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-88. A, B, C, D, E are of different heights. B is taller than D but shorter than E. A is taller than E. C is the shortest. Who is the tallest?
(a) A (b) E (c) B (d) D
Ans: (a). A > E > B > D > C. *Concept:* comparison · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-89. Pointing to a photograph, Ravi says, "She is the daughter of my grandfather's only son." How is she related to Ravi?
(a) Sister (b) Cousin (c) Mother (d) Aunt
Ans: (a). The grandfather's only son is Ravi's father, so his daughter is Ravi's sister. *Concept:* relations · *Diff:* M · *Why NID:* logic · *Mistake:* answering cousin.

LR-90. Three boxes are labelled "Pens", "Pencils" and "Mixed". All labels are wrong. You may draw one item from one box. Which box should you pick from to relabel all correctly?
(a) Pens (b) Pencils (c) Mixed (d) Any
Ans: (c). The "Mixed" box must contain only pens or only pencils, so one item reveals its contents. The other two labels then follow by elimination. *Concept:* classic logic puzzle · *Diff:* H · *Why NID:* problem-solving · *Mistake:* none.

LR-91. A designer must arrange 3 posters (A, B, C) in a row. How many arrangements are possible?
(a) 3 (b) 6 (c) 9 (d) 27
Ans: (b). 3! = 6. *Concept:* permutations · *Diff:* E · *Why NID:* layout options · *Mistake:* none.

LR-92. With 4 colours, how many ways can you paint 2 different stripes so that the stripes are different colours (the order matters: top/bottom)?
(a) 6 (b) 8 (c) 12 (d) 16
Ans: (c). 4 × 3 = 12. *Concept:* counting · *Diff:* M · *Why NID:* colour combinations · *Mistake:* answering 16 (repetition allowed) or 6 (order ignored).

LR-93. A snail climbs 3 m up a 10 m wall each day and slides back 2 m each night. On which day does it reach the top?
(a) 10th (b) 8th (c) 7th (d) 9th
Ans: (b). After 7 days and nights it is at 7 m. On day 8 it climbs 3 m, reaching 10 m, and stops before sliding back. *Concept:* classic puzzle · *Diff:* M · *Why NID:* lateral thinking · *Mistake:* answering 10.

LR-94. You have 8 identical-looking clay beads; one is heavier. What is the minimum number of weighings on a balance scale guaranteed to find it?
(a) 1 (b) 2 (c) 3 (d) 4
Ans: (b). Weigh 3 against 3. If they balance, weigh the remaining 2 against each other. If they don't, take the heavier 3 and weigh 1 against 1; if those balance, the third is heavy. *Concept:* the divide-by-three strategy · *Diff:* H · *Why NID:* problem-solving · *Mistake:* answering 3.

LR-95. How many times do a clock's hands overlap in 12 hours?
(a) 12 (b) 11 (c) 24 (d) 22
Ans: (b). *Concept:* clock logic · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 12.

LR-96. A team of 4 has to be selected from 6 students. How many ways?
(a) 15 (b) 24 (c) 30 (d) 360
Ans: (a). C(6,4) = 15. *Concept:* combinations · *Diff:* M · *Why NID:* logic · *Mistake:* using permutations.

LR-97. Lateral thinking: a man pushes his car to a hotel and tells the owner he's bankrupt. What's happening?
(a) He's playing Monopoly (b) His car broke down (c) He's lying (d) He's a hotelier
Ans: (a). In the board game, landing on a property with a hotel can bankrupt a player. *Concept:* lateral thinking · *Diff:* M · *Why NID:* creative reasoning · *Mistake:* reading it literally.

LR-98. Which is the next figure in the pattern of matchstick squares: 1 square = 4 sticks, 2 squares in a row = 7, 3 in a row = 10? How many sticks for 10 squares in a row?
(a) 30 (b) 31 (c) 40 (d) 33
Ans: (b). Sticks = 3n + 1 = 31. *Concept:* visual-numeric pattern · *Diff:* M · *Why NID:* pattern generalisation · *Mistake:* answering 40.

LR-99. At a party, everyone shakes hands with everyone else exactly once, and there are 28 handshakes. How many people attended?
(a) 7 (b) 8 (c) 9 (d) 14
Ans: (b). n(n − 1)/2 = 28, so n = 8. *Concept:* combinations · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

LR-100. Five tiles numbered 1–5 are placed in a row so that no two consecutive numbers are adjacent. Which is a valid arrangement?
(a) 1 2 3 4 5 (b) 2 4 1 3 5 (c) 1 3 2 4 5 (d) 3 4 1 2 5
Ans: (b). Check the adjacent pairs: 2-4, 4-1, 1-3, 3-5 differ by 2, 3, 2 and 2, so none are consecutive ✓. (c) contains 3-2 and 4-5; (d) contains 3-4 and 1-2. *Concept:* constraint arrangement · *Diff:* M · *Why NID:* logic · *Mistake:* not checking every pair.

NEXT: Volume 4D — Quantitative / Mathematical Aptitude (100 questions) (`04D-VOL4-QB-QUANT.md`)

VOLUME 4D — QUESTION BANK: QUANTITATIVE / MATHEMATICAL APTITUDE (100 Questions)

> **Reality check:** NID needs Class 8–10 maths in design contexts (scale, area, ratio, percentages). This bank is deliberately set in those contexts. Priority: LOW. Aim for speed and accuracy, then stop. Sections: A. Percentages, ratio, proportion (1–20) · B. Averages, statistics & data (21–32) · C. Units, paper & digital measures (33–45) · D. Time, speed, distance, work (46–58) · E. Geometry & mensuration (59–85) · F. Design-context word problems & puzzles (86–100)

Format: Question → **Answer** → working → *Concept* · *Diff* · *Why NID* · *Mistake*.

A. Percentages, Ratio, Proportion (1–20)

QA-1. What is 25% of 480?
(a) 100 (b) 120 (c) 140 (d) 96
Ans: (b). 480 ÷ 4 = 120. *Concept:* percentage · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-2. A ₹1,200 table lamp's price rises by 15%. What is the new price?
(a) ₹1,350 (b) ₹1,380 (c) ₹1,400 (d) ₹1,320
Ans: (b). 1200 × 1.15 = 1380. *Concept:* % increase · *Diff:* E · *Why NID:* everyday · *Mistake:* none.

QA-3. A ₹2,000 item is discounted 10%, and the discounted price is then raised by 10%. What is the final price?
(a) ₹2,000 (b) ₹1,980 (c) ₹2,020 (d) ₹1,800
Ans: (b). 2000 × 0.9 × 1.1 = 1980, a net change of −1%. *Concept:* successive % · *Diff:* M · *Why NID:* the classic trap · *Mistake:* assuming the two changes cancel out.

QA-4. Successive discounts of 20% and 10% equal a single discount of:
(a) 30% (b) 28% (c) 25% (d) 32%
Ans: (b). 1 − 0.8 × 0.9 = 0.28. *Concept:* successive % · *Diff:* M · *Why NID:* reasoning · *Mistake:* adding them to get 30%.

QA-5. By what percentage does a price rising from ₹80 to ₹100 increase?
(a) 20% (b) 25% (c) 80% (d) 15%
Ans: (b). 20 ÷ 80 = 25%. *Concept:* % change · *Diff:* E · *Why NID:* data reading · *Mistake:* dividing by the new value (20 ÷ 100 = 20%).

QA-6. By what percentage does a price falling from ₹100 to ₹80 decrease?
(a) 20% (b) 25% (c) 80% (d) 15%
Ans: (a). 20 ÷ 100 = 20%. *Concept:* % change · *Diff:* E · *Why NID:* contrast with QA-5 · *Mistake:* none.

QA-7. 12 is what percentage of 48?
(a) 20% (b) 25% (c) 30% (d) 40%
Ans: (b). *Concept:* % · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-8. Red and white paint are mixed in the ratio 2 : 3 to make 10 L of pink. How much red paint is used?
(a) 3 L (b) 4 L (c) 5 L (d) 6 L
Ans: (b). 10 × 2/5 = 4. *Concept:* ratio · *Diff:* E · *Why NID:* colour mixing context · *Mistake:* none.

QA-9. ₹64 is divided in the ratio 3 : 5. What is the larger share?
(a) ₹24 (b) ₹40 (c) ₹38 (d) ₹32
Ans: (b). 64 × 5/8 = 40. *Concept:* ratio · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-10. If a : b = 2 : 3 and b : c = 4 : 5, find a : b : c.
(a) 8 : 12 : 15 (b) 2 : 3 : 5 (c) 4 : 6 : 5 (d) 8 : 12 : 10
Ans: (a). Make b equal in both ratios: a : b = 8 : 12 and b : c = 12 : 15. *Concept:* compound ratio · *Diff:* M · *Why NID:* proportion · *Mistake:* answering (b).

QA-11. On a 1 : 200 site plan, a wall measures 4.5 cm. What is its real length?
(a) 4.5 m (b) 9 m (c) 90 m (d) 0.9 m
Ans: (b). 4.5 × 200 = 900 cm = 9 m. *Concept:* scale · *Diff:* E · *Why NID:* architecture drawings · *Mistake:* converting the units wrongly.

QA-12. On a map where 1 cm = 5 km, two towns are 12.4 cm apart. What is the real distance?
(a) 60 km (b) 62 km (c) 64 km (d) 6.2 km
Ans: (b). *Concept:* map scale · *Diff:* E · *Why NID:* scale · *Mistake:* none.

QA-13. A 1 : 10 model of a chair is made. The real seat height is 45 cm. What is the model seat height?
(a) 45 mm (b) 4.5 mm (c) 450 mm (d) 4.5 m
Ans: (a). 45 cm ÷ 10 = 4.5 cm = 45 mm. *Concept:* model scale · *Diff:* E · *Why NID:* Studio Test models · *Mistake:* mixing up cm and mm.

QA-14. A table top measures 50 cm² on a 1 : 20 model. What is the real area?
(a) 1,000 cm² (b) 2 m² (c) 0.2 m² (d) 20 m²
Ans: (b). Areas scale by 20² = 400: 50 × 400 = 20,000 cm² = 2 m². *Concept:* area scaling · *Diff:* M · *Why NID:* scale reasoning · *Mistake:* multiplying by 20 only.

QA-15. If every length of a box is doubled, its volume becomes:
(a) 2× (b) 4× (c) 6× (d) 8×
Ans: (d). $2^3 = 8$. *Concept:* volume scaling · *Diff:* E · *Why NID:* packaging · *Mistake:* answering 2×.

QA-16. Five artisans make a set of baskets in 12 days. How long would 6 artisans take at the same rate?
(a) 10 days (b) 14 days (c) 11 days (d) 9 days
Ans: (a). Total work = 5 × 12 = 60 artisan-days; 60 ÷ 6 = 10. *Concept:* inverse proportion · *Diff:* E · *Why NID:* planning · *Mistake:* using direct proportion.

QA-17. 3 pencils cost ₹12. How much do 15 pencils cost?
(a) ₹45 (b) ₹60 (c) ₹50 (d) ₹36
Ans: (b). *Concept:* direct proportion · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-18. A stall buys a mug for ₹400 and sells it for ₹500. What is the profit percentage?
(a) 20% (b) 25% (c) 10% (d) 100%
Ans: (b). Profit % is calculated on the cost price: 100 ÷ 400 = 25%. *Concept:* profit · *Diff:* E · *Why NID:* basics · *Mistake:* calculating on the selling price (20%).

QA-19. A rectangle's length increases by 20% and its width by 10%. By what percentage does its area increase?
(a) 30% (b) 32% (c) 28% (d) 22%
Ans: (b). 1.2 × 1.1 = 1.32. *Concept:* combined % · *Diff:* M · *Why NID:* resizing artwork · *Mistake:* adding to 30%.

QA-20. A poster's width : height is 3 : 4. If the width is 45 cm, what is the height?
(a) 55 cm (b) 60 cm (c) 50 cm (d) 64 cm
Ans: (b). 45 × 4/3 = 60. *Concept:* aspect ratio · *Diff:* E · *Why NID:* layout · *Mistake:* none.

B. Averages, Statistics & Data Interpretation (21–32)

QA-21. Find the average of 12, 15, 18, 21, 24.
(a) 17 (b) 18 (c) 19 (d) 20
Ans: (b). The numbers are evenly spaced, so the average is the middle value, 18. *Concept:* average · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-22. Six numbers average 20. When one number is removed, the remaining five average 18. What number was removed?
(a) 28 (b) 30 (c) 32 (d) 20
Ans: (b). 6 × 20 − 5 × 18 = 120 − 90 = 30. *Concept:* average manipulation · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

QA-23. A student scored 70 and 80 in two tests. What score in the third test gives an average of 80?
(a) 85 (b) 90 (c) 95 (d) 80
Ans: (b). 240 − 150 = 90. *Concept:* target average · *Diff:* E · *Why NID:* planning · *Mistake:* none.

QA-24. For the data 2, 3, 3, 5, 7, find the mean, median and mode.
(a) 4, 3, 3 (b) 4, 5, 3 (c) 3, 3, 4 (d) 5, 3, 3
Ans: (a). Mean = 20 ÷ 5 = 4; median = the middle value = 3; mode = the most frequent = 3. *Concept:* central tendency · *Diff:* E · *Why NID:* survey analysis · *Mistake:* none.

QA-25. Find the range of 4, 9, 2, 15.
(a) 11 (b) 13 (c) 15 (d) 6
Ans: (b). 15 − 2 = 13. *Concept:* range · *Diff:* E · *Why NID:* data · *Mistake:* none.

QA-26. In a pie chart, a category is 25% of the total. What angle does its slice make?
(a) 25° (b) 90° (c) 45° (d) 100°
Ans: (b). 0.25 × 360 = 90°. *Concept:* pie chart · *Diff:* E · *Why NID:* infographics · *Mistake:* none.

QA-27. A pie-chart slice measures 72°. What percentage does it represent?
(a) 18% (b) 20% (c) 25% (d) 72%
Ans: (b). 72 ÷ 360 = 0.2. *Concept:* pie chart · *Diff:* E · *Why NID:* infographics · *Mistake:* none.

QA-28. In a survey of 50 students, 60% like drawing. How many like drawing?
(a) 25 (b) 30 (c) 35 (d) 20
Ans: (b). *Concept:* % of a total · *Diff:* E · *Why NID:* research · *Mistake:* none.

QA-29. A bar chart shows the visitors to a craft fair: Mon 200, Tue 300, Wed 250, Thu 450, Fri 300. On which day did visitors increase the most compared with the previous day?
(a) Tue (b) Wed (c) Thu (d) Fri
Ans: (c). The day-on-day changes are +100, −50, +200, −150. *Concept:* data reading · *Diff:* E · *Why NID:* data visualisation · *Mistake:* choosing the highest absolute value instead of the biggest change.

QA-30. In QA-29, what is the average number of visitors per day?
(a) 280 (b) 300 (c) 250 (d) 320
Ans: (b). Total 1500 ÷ 5 = 300. *Concept:* average from a chart · *Diff:* E · *Why NID:* data · *Mistake:* none.

QA-31. A designer estimated 95 units but the actual number was 100. What is the percentage error, relative to the actual?
(a) 5% (b) 5.26% (c) 95% (d) 10%
Ans: (a). 5 ÷ 100 = 5%. *Concept:* error % · *Diff:* E · *Why NID:* estimation · *Mistake:* dividing by the estimate.

QA-32. A class's marks are: 10 students scored 60 and 30 students scored 80. What is the class average?
(a) 70 (b) 75 (c) 72 (d) 78
Ans: (b). (600 + 2400) ÷ 40 = 75. *Concept:* weighted average · *Diff:* M · *Why NID:* reasoning · *Mistake:* taking the simple average of 60 and 80 (70).

C. Units, Paper & Digital Measures (33–45)

QA-33. How many centimetres are in 2.5 m?
(a) 25 (b) 250 (c) 2,500 (d) 0.25
Ans: (b). *Concept:* units · *Diff:* E · *Why NID:* measurement · *Mistake:* none.

QA-34. 10 inches is approximately:
(a) 2.54 cm (b) 25.4 cm (c) 254 cm (d) 30.5 cm
Ans: (b). *Concept:* imperial–metric · *Diff:* E · *Why NID:* product sizes (TV screens, for example) · *Mistake:* none.

QA-35. A person is 6 feet tall. Approximately how tall is that in cm?
(a) 172 (b) 183 (c) 190 (d) 160
Ans: (b). 6 × 30.48 = 182.88. *Concept:* units · *Diff:* E · *Why NID:* anthropometry · *Mistake:* none.

QA-36. What is the area of an A0 sheet?
(a) 0.5 m² (b) 1 m² (c) 2 m² (d) 0.25 m²
Ans: (b). A0 is defined as 1 m² with sides in the ratio 1 : √2 (841 × 1189 mm). *Concept:* ISO paper · *Diff:* M · *Why NID:* design materials · *Mistake:* none.

QA-37. How many A4 sheets have the same area as one A1 sheet?
(a) 4 (b) 6 (c) 8 (d) 16
Ans: (c). A1 = 2 A2 = 4 A3 = 8 A4. *Concept:* paper sizes · *Diff:* E · *Why NID:* design practice · *Mistake:* answering 4.

QA-38. The long side ÷ short side ratio of any A-series sheet is about:
(a) 1.5 (b) 1.414 (c) 1.618 (d) 1.333
Ans: (b). √2. This ratio means that halving a sheet keeps the same proportions. *Concept:* the √2 ratio · *Diff:* M · *Why NID:* proportion · *Mistake:* answering 1.618 (the golden ratio).

QA-39. How many pixels does a screen of 1920 × 1080 have?
(a) about 2.07 million (b) about 1 million (c) about 20.7 million (d) 3,000
Ans: (a). 1920 × 1080 = 2,073,600. *Concept:* digital measure · *Diff:* E · *Why NID:* digital design · *Mistake:* none.

QA-40. What is the aspect ratio of 1920 × 1080?
(a) 4 : 3 (b) 16 : 9 (c) 3 : 2 (d) 21 : 9
Ans: (b). Divide both by 120. *Concept:* aspect ratio · *Diff:* E · *Why NID:* screen and video design · *Mistake:* none.

QA-41. A 4 × 6 inch photo is printed at 300 dpi. What pixel size is needed?
(a) 400 × 600 (b) 1200 × 1800 (c) 300 × 450 (d) 2400 × 3600
Ans: (b). Multiply each side by 300. *Concept:* print resolution · *Diff:* M · *Why NID:* graphic production · *Mistake:* none.

QA-42. In typography 72 points ≈ 1 inch. What is the height of 36 pt type?
(a) 0.25 in (b) 0.5 in (c) 1 in (d) 2 in
Ans: (b). *Concept:* type measure · *Diff:* E · *Why NID:* typography · *Mistake:* none.

QA-43. How many litres does a tank measuring 1 m × 1 m × 1 m hold?
(a) 100 (b) 1,000 (c) 10,000 (d) 10
Ans: (b). *Concept:* volume–capacity · *Diff:* E · *Why NID:* product sizing · *Mistake:* none.

QA-44. How many square metres are in 1 hectare?
(a) 1,000 (b) 10,000 (c) 100,000 (d) 100
Ans: (b). *Concept:* area units · *Diff:* E · *Why NID:* landscape/campus scale · *Mistake:* none.

QA-45. India Standard Time is UTC + 5:30. What is the IST time at 12:00 noon UTC?
(a) 6:30 pm (b) 5:30 pm (c) 6:30 am (d) 5:30 am
Ans: (b). *Concept:* time zones · *Diff:* E · *Why NID:* general awareness · *Mistake:* none.

D. Time, Speed, Distance, Work (46–58)

QA-46. How long does it take to cover 150 km at 60 km/h?
(a) 2 h (b) 2.5 h (c) 3 h (d) 1.5 h
Ans: (b). *Concept:* d = st · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-47. What is 36 km/h in m/s?
(a) 10 (b) 36 (c) 3.6 (d) 100
Ans: (a). 36 × 5/18 = 10. *Concept:* unit conversion · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-48. A 200 m train moving at 72 km/h passes a pole. How long does it take?
(a) 10 s (b) 20 s (c) 2.8 s (d) 15 s
Ans: (a). 72 km/h = 20 m/s, and 200 ÷ 20 = 10 s. *Concept:* train problems · *Diff:* M · *Why NID:* reasoning · *Mistake:* forgetting to convert the units.

QA-49. The same train crosses a 300 m platform. How long does it take?
(a) 15 s (b) 25 s (c) 10 s (d) 30 s
Ans: (b). It must cover its own length plus the platform: 500 ÷ 20 = 25 s. *Concept:* train + platform · *Diff:* M · *Why NID:* reasoning · *Mistake:* using only 300 m.

QA-50. A cyclist goes 60 km at 30 km/h and returns at 60 km/h. What is the average speed for the whole trip?
(a) 45 km/h (b) 40 km/h (c) 50 km/h (d) 42 km/h
Ans: (b). Total time = 2 h + 1 h = 3 h for 120 km, so 40 km/h. *Concept:* average speed (a harmonic mean) · *Diff:* M · *Why NID:* the classic trap · *Mistake:* averaging the two speeds (45).

QA-51. Walking at 5 km/h, how long does it take to cover 2.5 km?
(a) 20 min (b) 30 min (c) 45 min (d) 50 min
Ans: (b). *Concept:* time · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-52. A can paint a mural in 6 days and B can paint it in 12 days. How long do they take working together?
(a) 9 days (b) 4 days (c) 3 days (d) 18 days
Ans: (b). 1/6 + 1/12 = 3/12 = 1/4 of the job per day. *Concept:* work rates · *Diff:* M · *Why NID:* collaboration problems · *Mistake:* averaging the times.

QA-53. A tap fills a tank in 4 h and a drain empties it in 6 h. With both open, how long does the tank take to fill?
(a) 10 h (b) 12 h (c) 5 h (d) 2.4 h
Ans: (b). 1/4 − 1/6 = 1/12. *Concept:* rates · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

QA-54. What is the angle between the clock hands at 4:00?
(a) 90° (b) 120° (c) 150° (d) 100°
Ans: (b). 4 × 30° = 120°. *Concept:* clock angles · *Diff:* E · *Why NID:* reasoning · *Mistake:* none.

QA-55. What is the smaller angle between the clock hands at 9:15?
(a) 172.5° (b) 180° (c) 187.5° (d) 165°
Ans: (a). $|30 \times 9 - 5.5 \times 15| = |270 - 82.5| = 187.5°$, so the smaller angle is 360° − 187.5° = 172.5°. *Concept:* clock angles · *Diff:* M · *Why NID:* precision · *Mistake:* reporting the larger angle (187.5°).

QA-56. What is the angle between the clock hands at 3:30?
(a) 90° (b) 75° (c) 60° (d) 105°
Ans: (b). $|90 - 165| = 75°$. *Concept:* clock angles · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 90°.

QA-57. A bus leaves every 15 minutes from 6:00 am. How many buses leave between 6:00 am and 8:00 am, inclusive of both times?
(a) 8 (b) 9 (c) 10 (d) 7
Ans: (b). 120 ÷ 15 = 8 intervals, so there are 9 departures. *Concept:* fence-post counting · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 8.

QA-58. A father is 3 times as old as his son. In 10 years he will be twice as old. How old is the son now?
(a) 8 (b) 10 (c) 12 (d) 15
Ans: (b). 3x + 10 = 2(x + 10), so x = 10. *Concept:* linear equations · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

E. Geometry & Mensuration (59–85)

(Use π = 22/7 where convenient.)

QA-59. Find the area and perimeter of a 12 × 8 rectangle.
(a) 96, 40 (b) 96, 20 (c) 80, 40 (d) 48, 40
Ans: (a). *Concept:* rectangle · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-60. What is the area of a circle of radius 7 cm?
(a) 44 cm² (b) 154 cm² (c) 144 cm² (d) 49 cm²
Ans: (b). 22/7 × 49 = 154. *Concept:* circle area · *Diff:* E · *Why NID:* basics · *Mistake:* computing the circumference (44).

QA-61. What is the circumference of a circular table of radius 14 cm?
(a) 44 (b) 88 (c) 616 (d) 28
Ans: (b). 2 × 22/7 × 14 = 88. *Concept:* circumference · *Diff:* E · *Why NID:* edge-banding a table, for example · *Mistake:* none.

QA-62. Find the volume and surface area of a cube of side 5.
(a) 125, 150 (b) 125, 100 (c) 25, 150 (d) 150, 125
Ans: (a). *Concept:* cube · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-63. Find the volume and total surface area of a 10 × 5 × 4 cuboid.
(a) 200, 220 (b) 200, 110 (c) 190, 220 (d) 220, 200
Ans: (a). Surface area = 2(50 + 20 + 40) = 220. *Concept:* cuboid · *Diff:* E · *Why NID:* packaging · *Mistake:* none.

QA-64. A cylindrical bottle has radius 7 cm and height 10 cm. What is its capacity?
(a) 1.54 L (b) 15.4 L (c) 0.154 L (d) 154 L
Ans: (a). 22/7 × 49 × 10 = 1540 cm³ = 1.54 L. *Concept:* cylinder · *Diff:* M · *Why NID:* product design ·

Mistake: converting the units wrongly.

QA-65. A cone has radius 3 and height 4. What is its slant height?
(a) 5 (b) 7 (c) 25 (d) 12
Ans: (a). √(9 + 16) = 5. *Concept:* Pythagoras in 3D · *Diff:* E · *Why NID:* packaging (cones) · *Mistake:* none.

QA-66. What is the curved surface area of that cone (r = 3, l = 5)?
(a) 15π (b) 12π (c) 9π (d) 30π
Ans: (a). πrl = 15π ≈ 47.1. *Concept:* cone surface · *Diff:* M · *Why NID:* paper cone cups · *Mistake:* using the height instead of the slant height.

QA-67. What is the volume of a sphere of radius 3?
(a) 36π (b) 27π (c) 12π (d) 108π
Ans: (a). (4/3)π × 27 = 36π. *Concept:* sphere · *Diff:* M · *Why NID:* form · *Mistake:* none.

QA-68. What is the area of a triangle with base 10 and height 6?
(a) 60 (b) 30 (c) 16 (d) 36
Ans: (b). *Concept:* triangle · *Diff:* E · *Why NID:* basics · *Mistake:* forgetting the ½.

QA-69. A 5 m × 4 m room is tiled with 25 cm × 25 cm tiles. How many tiles are needed?
(a) 200 (b) 320 (c) 400 (d) 80
Ans: (b). 500 × 400 ÷ 625 = 320. *Concept:* area division · *Diff:* E · *Why NID:* interior design · *Mistake:* mixing m and cm.

QA-70. A 3 m × 4 m wall needs two coats of paint, and 1 L covers 6 m² per coat. How much paint is needed?
(a) 2 L (b) 4 L (c) 6 L (d) 3 L
Ans: (b). 12 m² × 2 coats = 24 m², and 24 ÷ 6 = 4 L. *Concept:* applied area · *Diff:* E · *Why NID:* practical estimation · *Mistake:* forgetting the second coat.

QA-71. What is the interior angle of a regular pentagon?
(a) 100° (b) 108° (c) 120° (d) 72°
Ans: (b). (5 − 2) × 180 ÷ 5 = 108. *Concept:* polygon angles · *Diff:* E · *Why NID:* pattern geometry · *Mistake:* giving the exterior angle (72°).

QA-72. What is the sum of the interior angles of a hexagon?
(a) 540° (b) 720° (c) 900° (d) 360°
Ans: (b). *Concept:* angle sum · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-73. What is each exterior angle of a regular octagon?
(a) 30° (b) 45° (c) 135° (d) 60°
Ans: (b). 360 ÷ 8. *Concept:* exterior angles · *Diff:* E · *Why NID:* geometry · *Mistake:* giving the interior angle (135°).

QA-74. What is the diagonal of a square of side 10?
(a) 10 (b) 14.14 (c) 20 (d) 12
Ans: (b). 10√2 ≈ 14.14. *Concept:* Pythagoras · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-75. What is the diagonal of a 6 × 8 rectangle?
(a) 10 (b) 14 (c) 12 (d) 9
Ans: (a). *Concept:* Pythagoras · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-76. Circles have radii in the ratio 2 : 3. What is the ratio of their areas?
(a) 2 : 3 (b) 4 : 9 (c) 8 : 27 (d) 4 : 6
Ans: (b). *Concept:* area scaling · *Diff:* E · *Why NID:* scale · *Mistake:* none.

QA-77. What is the area of a trapezium with parallel sides 8 and 12 and height 5?
(a) 40 (b) 50 (c) 60 (d) 100
Ans: (b). ½ × 20 × 5 = 50. *Concept:* trapezium · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-78. A ring (annulus) has outer radius 7 and inner radius 3.5. What is its area?
(a) 115.5 (b) 154 (c) 38.5 (d) 77
Ans: (a). π(49 − 12.25) = 22/7 × 36.75 = 115.5. *Concept:* annulus · *Diff:* M · *Why NID:* product forms (washers, bangles) · *Mistake:* subtracting the radii before squaring.

QA-79. What is the perimeter of a semicircle of radius 7?
(a) 22 (b) 36 (c) 44 (d) 29
Ans: (b). Curved part + diameter: πr + 2r = 22 + 14 = 36. *Concept:* composite perimeter · *Diff:* M · *Why NID:* geometry · *Mistake:* forgetting the diameter.

QA-80. If a sphere's radius doubles, its volume becomes:
(a) 2× (b) 4× (c) 8× (d) 6×
Ans: (c). *Concept:* scaling · *Diff:* E · *Why NID:* scale · *Mistake:* answering 4×.

QA-81. What is the area of an equilateral triangle of side 6?
(a) 9√3 ≈ 15.6 (b) 18 (c) 36 (d) 12√3
Ans: (a). (√3/4) × 36 = 9√3. *Concept:* special triangles · *Diff:* M · *Why NID:* geometry · *Mistake:* none.

QA-82. What is the area of a regular hexagon of side 2?
(a) 6√3 ≈ 10.4 (b) 12 (c) 4√3 (d) 24
Ans: (a). It is made of 6 equilateral triangles: 6 × (√3/4) × 4 = 6√3. *Concept:* hexagon area · *Diff:* M · *Why NID:* tiling · *Mistake:* none.

QA-83. An open box is made from a 20 cm × 20 cm sheet by cutting a 5 cm square from each corner and folding up the sides. What is its volume?
(a) 500 cm³ (b) 1,000 cm³ (c) 2,000 cm³ (d) 400 cm³
Ans: (a). The box is 10 × 10 × 5. *Concept:* nets to volume · *Diff:* M · *Why NID:* packaging · *Mistake:* using 20 − 5 instead of 20 − 2 × 5 for the base.

QA-84. What is the largest square tile that can exactly cover a 48 cm × 36 cm board, and how many such tiles are needed?
(a) 12 cm, 12 tiles (b) 6 cm, 48 tiles (c) 12 cm, 16 tiles (d) 4 cm, 108 tiles
Ans: (a). HCF(48, 36) = 12, giving a 4 × 3 grid of 12 tiles. *Concept:* HCF · *Diff:* M · *Why NID:* tiling layouts · *Mistake:* none.

QA-85. One border motif repeats every 6 cm and another every 8 cm. After how many cm do they line up again?
(a) 14 (b) 24 (c) 48 (d) 12
Ans: (b). LCM(6, 8) = 24. *Concept:* LCM · *Diff:* E · *Why NID:* textile pattern repeats · *Mistake:* none.

F. Design-Context Word Problems & Puzzles (86–100)

QA-86. Twelve 50 × 50 cm squares are needed for six cushion covers (front and back). Fabric comes 2 m wide. What length of fabric is needed, ignoring seams and waste?
(a) 1 m (b) 1.5 m (c) 2 m (d) 3 m
Ans: (b). 4 squares fit across the 2 m width, so 12 squares need 3 rows × 0.5 m = 1.5 m. *Concept:* layout optimisation · *Diff:* M · *Why NID:* textile planning · *Mistake:* dividing the area (3 m² ÷ 2 m) without checking that the squares actually fit, which happens to give the same answer here.

QA-87. A rectangle has width 10. What length makes it a golden rectangle?
(a) 14.1 (b) 16.18 (c) 15 (d) 20
Ans: (b). 10 × 1.618. *Concept:* golden ratio · *Diff:* E · *Why NID:* proportion · *Mistake:* using √2.

QA-88. Fencing a 20 m straight side with posts every 2 m, including both ends, needs how many posts?
(a) 10 (b) 11 (c) 12 (d) 9
Ans: (b). *Concept:* fence-post problem · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering 10.

QA-89. How many cuts are needed to cut a 10 m rope into 1 m pieces?
(a) 10 (b) 9 (c) 11 (d) 5
Ans: (b). *Concept:* cuts = pieces − 1 · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering 10.

QA-90. A staircase rises 3 m with risers of 15 cm. How many risers are needed?
(a) 15 (b) 20 (c) 25 (d) 30
Ans: (b). 300 ÷ 15. *Concept:* architectural maths · *Diff:* E · *Why NID:* ergonomics · *Mistake:* none.

QA-91. A wheelchair ramp must have a slope of 1 : 12. How long must it run horizontally to rise 60 cm?
(a) 5 m (b) 7.2 m (c) 6 m (d) 12 m
Ans: (b). 60 × 12 = 720 cm. *Concept:* slope ratio · *Diff:* E · *Why NID:* accessible design · *Mistake:* none.

QA-92. Find 3/4 of 2/3 of 120.
(a) 60 (b) 80 (c) 90 (d) 40
Ans: (a). 120 × 2/3 = 80, and 80 × 3/4 = 60. *Concept:* fractions · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-93. Write 0.125 as a fraction.
(a) 1/4 (b) 1/8 (c) 1/6 (d) 1/5
Ans: (b). *Concept:* decimals · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-94. A die is rolled. What is the probability of an even number?
(a) 1/3 (b) 1/2 (c) 1/6 (d) 2/3
Ans: (b). *Concept:* probability · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-95. A bag holds 3 red and 5 blue beads. What is the probability of drawing red?
(a) 3/5 (b) 3/8 (c) 5/8 (d) 1/3
Ans: (b). *Concept:* probability · *Diff:* E · *Why NID:* basics · *Mistake:* answering 3/5 (red : blue).

QA-96. What is the sum of the first 10 natural numbers?
(a) 45 (b) 55 (c) 50 (d) 100
Ans: (b). n(n + 1)/2 = 55. *Concept:* series sum · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

QA-97. What is the sum of the first 10 odd numbers?
(a) 50 (b) 100 (c) 99 (d) 121
Ans: (b). The sum of the first n odd numbers is n^2 (picture growing square layers). *Concept:* visual number pattern · *Diff:* M · *Why NID:* links to visual series · *Mistake:* none.

QA-98. A Prelims paper has 3 hours. If 25 minutes go to 22 MCQs, 10 minutes to reading and 10 minutes to a buffer, how many minutes per mark remain for a 70-mark subjective section?
(a) about 1.5 (b) about 1.93 (c) about 2.5 (d) about 3
Ans: (b). 180 − 45 = 135 min, and 135 ÷ 70 ≈ 1.93 min per mark, so a 20-mark question gets about 38–39 min. *Concept:* time allocation · *Diff:* M · *Why NID:* exam strategy · *Mistake:* none.

QA-99. 37 °C equals how many °F?
(a) 96.8 (b) 98.6 (c) 100.4 (d) 99.5
Ans: (b). 37 × 9/5 + 32 = 98.6. *Concept:* temperature conversion · *Diff:* E · *Why NID:* general awareness · *Mistake:* none.

QA-100. What is the ratio of consecutive Fibonacci numbers 21 and 13 (21/13), to two decimals, and what does it approach?
(a) 1.62, the golden ratio (b) 1.41, √2 (c) 1.50, 3/2 (d) 1.73, √3
Ans: (a). 21 ÷ 13 ≈ 1.615, and the ratio approaches φ ≈ 1.618. *Concept:* Fibonacci and the golden ratio · *Diff:* M · *Why NID:* design and maths link · *Mistake:* none.

NEXT: Volume 4E — Verbal Ability (100 questions) (`04E-VOL4-QB-VERBAL.md`)

VOLUME 4E — QUESTION BANK: VERBAL ABILITY (100 Questions)

> Priority: LOW as MCQs, but HIGH as a support skill. Good vocabulary and clear sentences improve the 100–200-word writing tasks and your interview answers.
> Sections: A. Synonyms (1–15) · B. Antonyms (16–30) · C. Words in context & one-word substitution (31–45) · D. Idioms & phrases (46–55) · E. Grammar & sentence correction (56–70) · F. Fill in the blanks (71–80) · G. Reading comprehension: 4 passages (81–100)

Format: Question → **Answer** → explanation → *Concept* · *Diff* · *Why NID* · *Mistake*.

A. Synonyms (1–15)

VA-1. AESTHETIC: (a) useful (b) artistic/beautiful (c) cheap (d) ancient
Ans: (b). *Concept:* design vocabulary · *Diff:* E · *Why NID:* core term · *Mistake:* confusing it with "ascetic" (self-denying).

VA-2. UBIQUITOUS: (a) rare (b) omnipresent (c) unique (d) hidden
Ans: (b). It means found everywhere. *Diff:* E · *Concept:* vocab · *Why NID:* describing everyday objects · *Mistake:* none.

VA-3. EPHEMERAL: (a) permanent (b) transient (c) heavy (d) loud
Ans: (b). It means short-lived; a rangoli is ephemeral art. *Diff:* M · *Concept:* vocab · *Why NID:* art writing · *Mistake:* none.

VA-4. FRUGAL: (a) wasteful (b) economical (c) fruity (d) fragile
Ans: (b). *Diff:* E · *Concept:* vocab (frugal innovation) · *Why NID:* design for India · *Mistake:* confusing it with "fragile".

VA-5. INTRICATE: (a) simple (b) complex/detailed (c) broken (d) intimate
Ans: (b). *Diff:* E · *Concept:* vocab · *Why NID:* describing craft · *Mistake:* none.

VA-6. TACTILE: (a) strategic (b) related to touch (c) tasteful (d) silent
Ans: (b). *Diff:* E · *Concept:* sensory vocab · *Why NID:* material design · *Mistake:* confusing it with "tactical".

VA-7. VERNACULAR: (a) foreign (b) local/native (c) vertical (d) formal
Ans: (b). Vernacular architecture is built by local people with local materials. *Diff:* M · *Concept:* vocab · *Why NID:* architecture · *Mistake:* none.

VA-8. INNOVATIVE: (a) traditional (b) novel/inventive (c) careless (d) indoor
Ans: (b). *Diff:* E · *Concept:* vocab · *Why NID:* the design core · *Mistake:* none.

VA-9. METICULOUS: (a) careless (b) extremely careful (c) metallic (d) quick
Ans: (b). *Diff:* E · *Concept:* vocab · *Why NID:* craft description · *Mistake:* none.

VA-10. DURABLE: (a) fragile (b) long-lasting (c) heavy (d) costly
Ans: (b). *Diff:* E · *Concept:* product vocab · *Why NID:* product writing · *Mistake:* none.

VA-11. VIBRANT: (a) dull (b) lively/bright (c) vibrating (d) violent
Ans: (b). *Diff:* E · *Concept:* colour vocab · *Why NID:* describing art · *Mistake:* reading it literally as "vibrating".

VA-12. ORNATE: (a) plain (b) elaborately decorated (c) orange (d) organic
Ans: (b). *Diff:* E · *Concept:* style vocab · *Why NID:* architecture description · *Mistake:* none.

VA-13. PRAGMATIC: (a) idealistic (b) practical (c) dramatic (d) magical
Ans: (b). *Diff:* M · *Concept:* vocab · *Why NID:* design attitude · *Mistake:* none.

VA-14. CONCISE: (a) lengthy (b) brief and clear (c) confused (d) precise
Ans: (b). "Precise" means exact, which is close but not the same. *Diff:* M · *Concept:* vocab nuance · *Why NID:* communication · *Mistake:* choosing "precise".

VA-15. JUXTAPOSE: (a) separate widely (b) place side by side for contrast (c) justify (d) join permanently
Ans: (b). *Diff:* M · *Concept:* art vocab · *Why NID:* composition writing · *Mistake:* none.

B. Antonyms (16–30)

VA-16. MINIMAL: (a) sparse (b) excessive (c) small (d) simple
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* design styles · *Mistake:* none.

VA-17. TRANSPARENT: (a) clear (b) opaque (c) glassy (d) obvious
Ans: (b). *Diff:* E · *Concept:* material vocab · *Why NID:* rendering materials · *Mistake:* none.

VA-18. ABUNDANT: (a) plentiful (b) scarce (c) huge (d) wealthy
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* sustainability writing · *Mistake:* none.

VA-19. RIGID: (a) stiff (b) flexible (c) strong (d) hard
Ans: (b). *Diff:* E · *Concept:* material vocab · *Why NID:* product · *Mistake:* none.

VA-20. SYMMETRICAL: (a) balanced (b) asymmetrical (c) even (d) mirrored
Ans: (b). *Diff:* E · *Concept:* design vocab · *Why NID:* composition · *Mistake:* none.

VA-21. OBSCURE: (a) hidden (b) evident (c) dark (d) vague
Ans: (b). *Diff:* M · *Concept:* antonym · *Why NID:* reading · *Mistake:* none.

VA-22. ARTIFICIAL: (a) synthetic (b) natural (c) fake (d) manmade
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* materials · *Mistake:* none.

VA-23. CONVERGENT (thinking): (a) focused (b) divergent (c) logical (d) narrow
Ans: (b). *Diff:* E · *Concept:* creativity terms · *Why NID:* design thinking · *Mistake:* none.

VA-24. MONOTONOUS: (a) boring (b) varied (c) single (d) quiet
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* composition critique · *Mistake:* none.

VA-25. OPTIMISTIC: (a) hopeful (b) pessimistic (c) realistic (d) optical
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* basics · *Mistake:* none.

VA-26. GENEROUS: (a) kind (b) stingy (c) generic (d) large
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* basics · *Mistake:* none.

VA-27. ANCIENT: (a) archaic (b) modern (c) old (d) historic
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* basics · *Mistake:* none.

VA-28. COHERENT: (a) logical (b) disjointed (c) clear (d) united
Ans: (b). *Diff:* M · *Concept:* antonym · *Why NID:* writing quality · *Mistake:* none.

VA-29. DILIGENT: (a) hardworking (b) lazy (c) clever (d) dull
Ans: (b). *Diff:* E · *Concept:* antonym · *Why NID:* basics · *Mistake:* none.

VA-30. CONSPICUOUS: (a) noticeable (b) inconspicuous (c) suspicious (d) bright
Ans: (b). *Diff:* M · *Concept:* antonym · *Why NID:* signage vocab · *Mistake:* none.

C. Words in Context & One-Word Substitution (31–45)

VA-31. A person who designs buildings: (a) engineer (b) architect (c) mason (d) surveyor
Ans: (b). *Diff:* E · *Concept:* one-word substitution · *Why NID:* basics · *Mistake:* none.

VA-32. A study of the measurement of the human body for design: (a) ergonomics (b) anthropometry (c) anthropology (d) aesthetics
Ans: (b). Ergonomics applies anthropometric data; anthropology studies human societies. *Diff:* M · *Concept:* precise terms · *Why NID:* design vocab · *Mistake:* choosing ergonomics.

VA-33. Art that imitates the style of another work, often with respect or humour: (a) parody/pastiche (b) plagiarism (c) original (d) collage
Ans: (a). *Diff:* M · *Concept:* art terms · *Why NID:* art vocab · *Mistake:* plagiarism, which is copying without credit.

VA-34. A picture made by sticking pieces of paper, cloth etc. onto a surface: (a) mosaic (b) collage (c) fresco (d) mural
Ans: (b). *Diff:* E · *Concept:* art terms · *Why NID:* media knowledge · *Mistake:* choosing mosaic, which uses small tiles or stones.

VA-35. A painting on wet plaster: (a) fresco (b) tempera (c) gouache (d) etching
Ans: (a). *Diff:* M · *Concept:* art techniques · *Why NID:* art GK · *Mistake:* none.

VA-36. The study of the meaning of symbols and signs: (a) semantics (b) semiotics (c) syntax (d) symmetry
Ans: (b). *Diff:* M · *Concept:* communication theory · *Why NID:* visual communication · *Mistake:* choosing semantics (the meaning of words).

VA-37. "The museum's new wing is a *pastiche* of Mughal styles." Here, pastiche means:
(a) original invention (b) an imitation mixing elements of a style (c) a ruin (d) a painting
Ans: (b). *Diff:* M · *Concept:* context meaning · *Why NID:* reading architecture reviews · *Mistake:* none.

VA-38. A place where pottery is fired: (a) loom (b) kiln (c) forge (d) mill
Ans: (b). *Diff:* E · *Concept:* craft vocab · *Why NID:* craft · *Mistake:* none.

VA-39. A small preliminary sketch, usually tiny: (a) thumbnail (b) blueprint (c) mural (d) poster
Ans: (a). *Diff:* E · *Concept:* design vocab · *Why NID:* process · *Mistake:* none.

VA-40. A word or design that reads the same upside-down or in another orientation: (a) palindrome (b) ambigram (c) acronym (d) anagram
Ans: (b). A palindrome reads the same forwards and backwards. *Diff:* M · *Concept:* typographic vocab · *Why NID:* logos · *Mistake:* choosing palindrome.

VA-41. "The designer chose a *muted* palette." Muted means:
(a) silent (b) soft, subdued colours (c) bright neon (d) black only
Ans: (b). *Diff:* E · *Concept:* colour vocab · *Why NID:* colour writing · *Mistake:* reading it literally as "silent".

VA-42. Someone who collects and studies coins: (a) philatelist (b) numismatist (c) cartographer (d) calligrapher
Ans: (b). *Diff:* M · *Concept:* one-word · *Why NID:* GK vocab · *Mistake:* choosing philatelist (stamps).

VA-43. The art of beautiful handwriting: (a) typography (b) calligraphy (c) lithography (d) cartography
Ans: (b). *Diff:* E · *Concept:* one-word · *Why NID:* lettering · *Mistake:* none.

VA-44. A map-maker: (a) cartographer (b) geologist (c) surveyor (d) navigator
Ans: (a). *Diff:* E · *Concept:* one-word · *Why NID:* information design · *Mistake:* none.

VA-45. "The product's design is *intuitive*." It means the product:
(a) is expensive (b) can be used without instructions (c) is hand-made (d) is large
Ans: (b). *Diff:* E · *Concept:* UX vocab · *Why NID:* product writing · *Mistake:* none.

D. Idioms & Phrases (46–55)

VA-46. "Back to the drawing board" means:
(a) start drawing class (b) start again after a failed attempt (c) buy a new board (d) finish a project
Ans: (b). *Diff:* E · *Concept:* idioms · *Why NID:* design vocabulary · *Mistake:* none.

VA-47. "Think outside the box" means:
(a) think in an unconventional, creative way (b) leave the room (c) think about packaging (d) think slowly
Ans: (a). *Diff:* E · *Concept:* idioms · *Why NID:* creativity · *Mistake:* none.

VA-48. "The devil is in the details" means:
(a) details are evil (b) small details can cause problems or make the difference (c) ignore details (d) details are unimportant
Ans: (b). *Diff:* E · *Concept:* idioms · *Why NID:* craft attitude · *Mistake:* none.

VA-49. "Reinvent the wheel" means:
(a) innovate radically (b) waste time creating something that already exists (c) repair a car (d) design a vehicle
Ans: (b). *Diff:* M · *Concept:* idioms · *Why NID:* design research · *Mistake:* choosing (a).

VA-50. "A picture is worth a thousand words" emphasises:
(a) pictures are expensive (b) visuals communicate complex ideas efficiently (c) words are useless (d) count words in pictures
Ans: (b). *Diff:* E · *Concept:* idioms · *Why NID:* visual communication · *Mistake:* none.

VA-51. "Cut corners" means:
(a) design rounded edges (b) do something poorly or cheaply to save time or money (c) take a shortcut road (d) trim paper
Ans: (b). *Diff:* E · *Concept:* idioms · *Why NID:* craft ethics · *Mistake:* reading it literally.

VA-52. "Form follows function" is a principle associated with:
(a) Louis Sullivan (b) Picasso (c) Shakespeare (d) Gandhi
Ans: (a). *Diff:* M · *Concept:* design maxims · *Why NID:* design GK · *Mistake:* none.

VA-53. "Less is more" is associated with:
(a) Mies van der Rohe (b) Frank Gehry (c) Andy Warhol (d) Coco Chanel
Ans: (a). *Diff:* M · *Concept:* design maxims · *Why NID:* design GK · *Mistake:* none.

VA-54. "At the eleventh hour" means:
(a) at 11 o'clock (b) at the last possible moment (c) very early (d) at night
Ans: (b). *Diff:* E · *Concept:* idioms · *Why NID:* basics · *Mistake:* none.

VA-55. "To see eye to eye" means:
(a) to stare (b) to agree fully (c) to be the same height (d) to argue
Ans: (b). *Diff:* E · *Concept:* idioms · *Why NID:* basics · *Mistake:* none.

E. Grammar & Sentence Correction (56–70)

VA-56. Choose the correct sentence:
(a) Each of the students have a sketchbook. (b) Each of the students has a sketchbook.
Ans: (b). "Each" is singular. *Diff:* E · *Concept:* subject–verb agreement · *Why NID:* writing accuracy · *Mistake:* letting "students" decide the verb.

VA-57. Choose the correct form: "The team of designers ___ presenting today."
(a) are (b) is
Ans: (b). "Team" is a singular collective noun, which is standard in Indian/British exam convention when the group acts as one. *Diff:* M · *Concept:* collective nouns · *Why NID:* writing · *Mistake:* none.

VA-58. Correct the error: "Neither the pencils nor the eraser were in the box."
(a) "was" instead of "were" (b) no error (c) "is" (d) "are"
Ans: (a). With neither/nor, the verb agrees with the nearer subject ("eraser", singular). *Diff:* M · *Concept:* agreement · *Why NID:* writing · *Mistake:* none.

VA-59. Choose the correct word: "The lamp's design had a big ___ on the market."
(a) affect (b) effect
Ans: (b). "Effect" is the noun; "affect" is usually the verb. *Diff:* E · *Concept:* commonly confused words · *Why NID:* writing · *Mistake:* none.

VA-60. Choose the correct word: "___ studio is on the second floor."
(a) Their (b) There (c) They're
Ans: (a). *Diff:* E · *Concept:* homophones · *Why NID:* writing · *Mistake:* none.

VA-61. Choose the correct word: "This chair is more comfortable ___ that one."
(a) then (b) than
Ans: (b). *Diff:* E · *Concept:* comparison · *Why NID:* writing · *Mistake:* none.

VA-62. Identify the correctly punctuated sentence:
(a) Its colours are vivid. (b) It's colours are vivid.
Ans: (a). "Its" is possessive; "it's" means "it is". *Diff:* E · *Concept:* apostrophes · *Why NID:* writing · *Mistake:* none.

VA-63. Choose the correct option: "She has been sketching ___ two hours."
(a) since (b) for
Ans: (b). "For" goes with a duration; "since" goes with a starting point. *Diff:* E · *Concept:* prepositions · *Why NID:* writing · *Mistake:* none.

VA-64. Choose the correct option: "If I ___ a designer, I would redesign bus stops."
(a) was (b) were
Ans: (b). This is the subjunctive for a hypothetical. *Diff:* M · *Concept:* conditionals · *Why NID:* writing · *Mistake:* none.

VA-65. Identify the error: "Fewer people uses public transport when buses are late."
(a) "Fewer" (b) "uses" should be "use" (c) "when" (d) no error
Ans: (b). "People" is plural. *Diff:* E · *Concept:* agreement · *Why NID:* writing · *Mistake:* none.

VA-66. Choose the correct option: "There were ___ visitors than expected."
(a) less (b) fewer
Ans: (b). "Fewer" is for countable nouns. *Diff:* M · *Concept:* fewer/less · *Why NID:* writing · *Mistake:* none.

VA-67. Choose the correct sentence:
(a) The poster, which was designed by Asha, won. (b) The poster which was designed by Asha, won.
Ans: (a). A non-restrictive clause needs a pair of commas. *Diff:* M · *Concept:* punctuation · *Why NID:* writing · *Mistake:* none.

VA-68. Passive voice of "The artisan wove the shawl":
(a) The shawl was woven by the artisan. (b) The shawl is weaved by the artisan. (c) The shawl woven the artisan. (d) The artisan was woven.
Ans: (a). *Diff:* E · *Concept:* voice · *Why NID:* writing · *Mistake:* writing "weaved" (the past participle of "weave" in this sense is "woven").

VA-69. Choose the correct article: "She is ___ honest designer."
(a) a (b) an
Ans: (b). "Honest" begins with a vowel sound because the h is silent. *Diff:* E · *Concept:* articles · *Why NID:* writing · *Mistake:* none.

VA-70. Which sentence is the MOST concise and clear?
(a) Due to the fact that it was raining, the event was postponed. (b) Because it rained, the event was postponed. (c) The event, owing to the fact of rain, was postponed due to rain. (d) It was raining and due to that the event got postponed because of it.
Ans: (b). *Diff:* E · *Concept:* concise writing · *Why NID:* writing under time pressure · *Mistake:* none.

F. Fill in the Blanks (71–80)

VA-71. A good logo must be simple enough to be ___ at a glance.
(a) recognised (b) ignored (c) complicated (d) hidden
Ans: (a). *Diff:* E · *Concept:* context vocab · *Why NID:* logo design · *Mistake:* none.

VA-72. The designer interviewed users to ___ with their problems.
(a) empathise (b) emphasise (c) emulate (d) emancipate
Ans: (a). "Emphasise" is a confusable word. *Diff:* M · *Concept:* confusables · *Why NID:* HCD vocab · *Mistake:* choosing "emphasise".

VA-73. The old haveli's frescoes have ___ with time; restoration is urgent.
(a) flourished (b) deteriorated (c) improved (d) multiplied
Ans: (b). *Diff:* E · *Concept:* context · *Why NID:* heritage · *Mistake:* none.

VA-74. Bamboo is ___ than steel by weight for some uses, making it a smart material.
(a) stronger (b) heavier (c) rustier (d) costlier
Ans: (a). Bamboo has a high strength-to-weight ratio. *Diff:* M · *Concept:* context + knowledge · *Why NID:* materials · *Mistake:* none.

VA-75. The exhibition was so ___ that visitors queued for hours.
(a) popular (b) dull (c) empty (d) closed
Ans: (a). *Diff:* E · *Concept:* context · *Why NID:* basics · *Mistake:* none.

VA-76. A prototype allows designers to ___ ideas before mass production.
(a) test (b) sell (c) hide (d) frame
Ans: (a). *Diff:* E · *Concept:* design process · *Why NID:* process vocab · *Mistake:* none.

VA-77. The signage was ___: visitors kept getting lost.
(a) ambiguous (b) clear (c) helpful (d) legible
Ans: (a). *Diff:* E · *Concept:* context · *Why NID:* wayfinding · *Mistake:* none.

VA-78. Handloom weavers often face ___ from cheaper machine-made imitations.
(a) competition (b) cooperation (c) celebration (d) completion
Ans: (a). *Diff:* E · *Concept:* context · *Why NID:* craft economics · *Mistake:* none.

VA-79. The minimalist room had ___ furniture and lots of empty space.
(a) sparse (b) cluttered (c) abundant (d) ornate
Ans: (a). *Diff:* E · *Concept:* context · *Why NID:* interior vocab · *Mistake:* none.

VA-80. Good design should be ___ to people of all abilities.
(a) accessible (b) excessive (c) exclusive (d) expensive
Ans: (a). *Diff:* E · *Concept:* inclusive vocab · *Why NID:* inclusive design · *Mistake:* none.

G. Reading Comprehension (81–100)

Passage 1 (questions 81–85)

> *The humble kulhad, an unglazed clay cup, has made a comeback at Indian railway stations. Unlike plastic cups, kulhads are made from local clay by potters, are used once and can return to the earth. Tea drinkers often say the earthy aroma improves the taste. Critics point out that kulhads use topsoil and need firing, which consumes fuel; they also break easily in transport. Still, many see the kulhad as a symbol of design that is local, low-tech and circular, provided production is managed responsibly.*

VA-81. The main idea of the passage is:
(a) Kulhads are better than everything (b) Kulhads are returning as a local, circular alternative to plastic, with some environmental trade-offs (c) Plastic cups are banned (d) Tea tastes bad in kulhads
Ans: (b). It captures both the benefits and the criticism. *Diff:* M · *Concept:* main idea · *Why NID:* critical reading · *Mistake:* choosing one-sided (a).

VA-82. According to the passage, one criticism of kulhads is that they:
(a) taste of plastic (b) consume topsoil and fuel (c) are imported (d) are expensive
Ans: (b). *Diff:* E · *Concept:* detail retrieval · *Why NID:* reading · *Mistake:* none.

VA-83. "Circular" in the passage most nearly means:
(a) round in shape (b) returning materials to natural or production cycles (c) moving in circles (d) repeated
Ans: (b). *Diff:* M · *Concept:* vocabulary in context · *Why NID:* sustainability vocab · *Mistake:* answering "round".

VA-84. The author's tone is best described as:
(a) balanced and appreciative (b) angry (c) sarcastic (d) indifferent
Ans: (a). *Diff:* M · *Concept:* tone · *Why NID:* reading · *Mistake:* none.

VA-85. Which statement would the author most likely AGREE with?
(a) Kulhads are perfect (b) Responsible production is key to the kulhad's sustainability (c) Plastic is better (d) Potters should stop working
Ans: (b). *Diff:* M · *Concept:* inference · *Why NID:* reading · *Mistake:* none.

Passage 2 (questions 86–90)

> *In 1958, Charles and Ray Eames travelled across India and wrote "The India Report", recommending an institute that would train designers to address the country's needs. They warned against the blind imitation of Western products and urged attention to India's own traditions, such as the lota, a simple brass water vessel that they admired for how perfectly it served its purpose. The report led to the founding of the National Institute of Design in Ahmedabad in 1961.*

VA-86. Which object did the Eameses praise as an example of good design?
(a) The charkha (b) The lota (c) The bullock cart (d) The tiffin
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* NID history (a frequent interview topic) · *Mistake:* none.

VA-87. The report warned against:
(a) Indian traditions (b) blind imitation of Western products (c) building institutes (d) using brass
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* history · *Mistake:* none.

VA-88. The passage implies that good design, for the Eameses, was:
(a) decorative (b) appropriate to purpose and context (c) imported (d) expensive
Ans: (b). *Diff:* M · *Concept:* inference · *Why NID:* design philosophy · *Mistake:* none.

VA-89. When was NID founded, according to the passage?
(a) 1958 (b) 1961 (c) 1947 (d) 1969
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* history · *Mistake:* choosing 1958, the year of the report.

VA-90. The best title for the passage is:
(a) "Brass Vessels of India" (b) "How a Report Shaped Indian Design Education" (c) "American Travel" (d) "Modern Products"
Ans: (b). *Diff:* M · *Concept:* title selection · *Why NID:* reading · *Mistake:* choosing the too-narrow (a).

Passage 3 (questions 91–95)

> *Wayfinding is more than putting up signs. A good wayfinding system considers how people think about space: landmarks, paths, decision points and destinations. At a decision point, such as a junction in a hospital corridor, people need a sign; along a straight path, too many signs create clutter. Colour coding can help, but must never be the only cue, because colour-blind visitors may miss it. Pictograms reduce dependence on language, an important consideration in multilingual countries.*

VA-91. According to the passage, signs are MOST needed at:
(a) every few metres (b) decision points (c) the exit only (d) on the ceiling
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* signage design · *Mistake:* none.

VA-92. Why shouldn't colour be the only cue?
(a) Colour is expensive (b) Colour-blind people may miss it (c) Colours fade (d) It is illegal
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* accessibility · *Mistake:* none.

VA-93. The passage suggests that too many signs:
(a) always help (b) create clutter (c) are required (d) reduce cost
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* design restraint · *Mistake:* none.

VA-94. "Pictograms reduce dependence on language" is especially relevant in India because:
(a) India is multilingual and literacy levels vary (b) Indians dislike text (c) Pictograms are free (d) English is banned
Ans: (a). *Diff:* E · *Concept:* application · *Why NID:* the Indian context · *Mistake:* none.

VA-95. The passage is mainly about:
(a) hospital management (b) principles of effective wayfinding design (c) colour blindness (d) language policy
Ans: (b). *Diff:* E · *Concept:* main idea · *Why NID:* reading · *Mistake:* none.

Passage 4 (questions 96–100)

> *Generative AI tools can now produce images from text prompts in seconds. Some designers fear that such tools will replace them; others see them as powerful assistants for exploring ideas quickly. However, AI-generated images raise questions about authorship, the use of artists' work in training data, and cultural accuracy: a prompt for "Indian wedding" may return clichés. Designers who understand context, empathy and ethics, and who can judge and refine outputs, are likely to remain essential.*

VA-96. According to the passage, which concern is raised about AI images?
(a) They are too slow (b) Authorship and training-data ethics (c) They cost too much (d) They are black and white
Ans: (b). *Diff:* E · *Concept:* detail · *Why NID:* current AI-and-design debates · *Mistake:* none.

VA-97. The "Indian wedding" example illustrates:
(a) speed (b) the risk of cultural cliché or inaccuracy (c) high resolution (d) cost
Ans: (b). *Diff:* M · *Concept:* example function · *Why NID:* cultural sensitivity · *Mistake:* none.

VA-98. The author's view is that designers:
(a) will disappear (b) will remain essential if they bring judgement, context and ethics (c) should ban AI (d) should only use AI
Ans: (b). *Diff:* M · *Concept:* author's view · *Why NID:* interview-relevant topic · *Mistake:* none.

VA-99. Which word best describes the passage's stance?
(a) alarmist (b) balanced (c) celebratory (d) dismissive
Ans: (b). *Diff:* M · *Concept:* tone · *Why NID:* reading · *Mistake:* none.

VA-100. Which is an ASSUMPTION underlying the last sentence?
(a) Human judgement adds value that AI alone cannot provide (b) AI is illegal (c) Designers dislike technology (d) Clients don't care about quality
Ans: (a). *Diff:* M · *Concept:* assumption · *Why NID:* critical reading · *Mistake:* none.

NEXT: Volume 4F — General Awareness (200 questions) (`04F-VOL4-QB-GENERAL-AWARENESS.md`)

VOLUME 4F — QUESTION BANK: GENERAL AWARENESS, DESIGN-FOCUSED (200 Questions)

> Facts here are **confirmed historical or institutional facts**. Recent (2025–26) events have their own bank in Volume 11.
> Compact format: **Q** (options) → **Ans** + explanation · *Concept* · *Diff* · *Why NID* · *Trap*.

Sections: A. National symbols & landmarks (1–20) · B. GI & regional products (21–45) · C. UNESCO, currency & performing arts (46–70) · D. Institutions, museums, awards (71–95) · E. Brands, logos & products (96–125) · F. Science, technology & digital design (126–150) · G. Environment & sustainability (151–175) · H. Indian culture, festivals & media (176–200)

A. National Symbols & Landmarks (1–20)

GA-1. India's national emblem is adapted from: (a) Ashoka Pillar at Vaishali (b) Lion Capital of Ashoka at Sarnath (c) Sanchi gateway (d) Konark wheel
Ans: (b). Adopted on 26 Jan 1950. *Concept:* national symbols · *Diff:* E · *Why NID:* symbol design and heritage · *Trap:* Sanchi.

GA-2. "Satyameva Jayate", inscribed below the emblem, is taken from the: (a) Rigveda (b) Mundaka Upanishad (c) Bhagavad Gita (d) Arthashastra
Ans: (b). *Concept:* emblem · *Diff:* M · *Why NID:* GK · *Trap:* the Gita.

GA-3. The ratio of width to length of the Indian national flag is: (a) 1:2 (b) 2:3 (c) 3:5 (d) 1:1
Ans: (b). *Concept:* proportion · *Diff:* E · *Why NID:* observation / drawing accuracy · *Trap:* 1:2.

GA-4. How many spokes does the Ashoka Chakra on the flag have? (a) 12 (b) 24 (c) 32 (d) 16
Ans: (b). It is navy blue. *Concept:* observation · *Diff:* E · *Why NID:* detail memory · *Trap:* none.

GA-5. The design of the national flag is based on a design by: (a) Pingali Venkayya (b) Nandalal Bose (c) Rabindranath Tagore (d) Raja Ravi Varma
Ans: (a). *Concept:* design history · *Diff:* E · *Why NID:* the designer behind a symbol · *Trap:* Nandalal Bose (who illustrated the Constitution).

GA-6. The ₹ symbol was designed by: (a) D. Udaya Kumar (b) Satyajit Ray (c) R.K. Laxman (d) Kirti Trivedi
Ans: (a). Adopted in 2010. Udaya Kumar later taught at IIT Guwahati. *Concept:* symbol design · *Diff:* E · *Why NID:* the most famous Indian graphic-design story · *Trap:* none.

GA-7. India's national tree is: (a) Neem (b) Banyan (c) Peepal (d) Ashoka
Ans: (b). *Concept:* national symbols · *Diff:* E · *Why NID:* GK · *Trap:* peepal.

GA-8. India's national aquatic animal is the: (a) Olive Ridley turtle (b) Gangetic river dolphin (c) Mahseer (d) Dugong
Ans: (b). *Concept:* national symbols · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-9. The original hand-written Constitution of India was illustrated by artists of Santiniketan led by: (a) Nandalal Bose (b) Jamini Roy (c) M.F. Husain (d) Abanindranath Tagore
Ans: (a). The calligraphy is by Prem Behari Narain Raizada. *Concept:* art + national history · *Diff:* M · *Why NID:* the art–state link · *Trap:* Abanindranath.

GA-10. The national calendar of India is based on the: (a) Vikram Samvat (b) Saka era (c) Hijri (d) Gregorian
Ans: (b). *Concept:* GK · *Diff:* M · *Why NID:* GK · *Trap:* Vikram Samvat.

GA-11. The Statue of Unity (182 m), which depicts Sardar Vallabhbhai Patel, was sculpted by: (a) Ram V. Sutar (b) Anish Kapoor (c) Ram Kinkar Baij (d) Satish Gujral
Ans: (a). Completed 2018 at Ekta Nagar (Kevadia), Gujarat. *Concept:* monuments · *Diff:* E · *Why NID:* scale and sculpture · *Trap:* Ram Kinkar Baij.

GA-12. The new Parliament building (inaugurated 2023) was designed by: (a) Hafeez Contractor (b) Bimal Patel / HCP Design (c) Charles Correa (d) B.V. Doshi
Ans: (b). It has a triangular plan. *Concept:* contemporary architecture · *Diff:* E · *Why NID:* architecture awareness · *Trap:* Correa (who died in 2015).

GA-13. India Gate in New Delhi was designed by: (a) Herbert Baker (b) Edwin Lutyens (c) George Wittet (d) Le Corbusier
Ans: (b). *Concept:* colonial architecture · *Diff:* E · *Why NID:* GK · *Trap:* Baker.

GA-14. The Gateway of India (Mumbai, 1924) was designed by: (a) George Wittet (b) F.W. Stevens (c) Lutyens (d) William Emerson
Ans: (a). *Concept:* Indo-Saracenic architecture · *Diff:* M · *Why NID:* GK · *Trap:* Stevens (who designed CSMT).

GA-15. Hawa Mahal ("Palace of Winds") is in: (a) Jodhpur (b) Jaipur (c) Udaipur (d) Agra
Ans: (b). Built 1799 under Sawai Pratap Singh; its many jharokhas let royal women watch the street unseen and allow air flow. *Concept:* Rajput architecture · *Diff:* E · *Why NID:* climate-responsive design · *Trap:* none.

GA-16. The Lotus Temple in Delhi is a House of Worship of the: (a) Jain (b) Bahá'í faith (c) Buddhist (d) Sikh
Ans: (b). Designed by Fariborz Sahba, 1986. *Concept:* architecture · *Diff:* E · *Why NID:* form inspired by nature · *Trap:* Buddhist.

GA-17. India's oldest museum is the: (a) National Museum, Delhi (b) Indian Museum, Kolkata (c) Salar Jung Museum (d) CSMVS, Mumbai
Ans: (b). Founded 1814. *Concept:* museums · *Diff:* M · *Why NID:* GK · *Trap:* National Museum.

GA-18. The Charminar (1591) was built by: (a) Muhammad Quli Qutb Shah (b) Akbar (c) Tipu Sultan (d) Aurangzeb
Ans: (a). *Concept:* Deccan architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-19. Victoria Memorial, Kolkata, was designed by: (a) William Emerson (b) Lutyens (c) Wittet (d) Baker
Ans: (a). *Concept:* colonial architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-20. Chhatrapati Shivaji Maharaj Terminus (Mumbai) is a UNESCO site built in which style? (a) Art Deco (b) Victorian Gothic Revival blended with Indian elements (c) Mughal (d) Modernist
Ans: (b). Designed by F.W. Stevens, 1888. *Concept:* architecture styles · *Diff:* M · *Why NID:* style identification · *Trap:* Art Deco (Marine Drive).

B. GI Tags & Regional Products (21–45)

GA-21. India's first GI-tagged product (2004) was: (a) Mysore Silk (b) Darjeeling Tea (c) Basmati (d) Banarasi Saree
Ans: (b). *Concept:* GI · *Diff:* E · *Why NID:* craft and IP awareness · *Trap:* Mysore Silk.

GA-22. The Geographical Indications Registry of India is in: (a) Delhi (b) Chennai (c) Mumbai (d) Kolkata
Ans: (b). *Concept:* GI · *Diff:* M · *Why NID:* GK · *Trap:* Delhi.

GA-23. India's GI law, the Geographical Indications of Goods (Registration and Protection) Act, was passed in: (a) 1995 (b) 1999 (c) 2003 (d) 2010
Ans: (b). It came into force in 2003. *Concept:* GI law · *Diff:* M · *Why NID:* GK · *Trap:* 2003.

GA-24. Channapatna toys come from: (a) Andhra Pradesh (b) Karnataka (c) Tamil Nadu (d) Kerala
Ans: (b). *Concept:* crafts · *Diff:* E · *Why NID:* craft geography · *Trap:* AP (Kondapalli).

GA-25. Pochampally Ikat is from: (a) Odisha (b) Telangana (c) Gujarat (d) West Bengal
Ans: (b). *Concept:* textiles · *Diff:* E · *Why NID:* textile geography · *Trap:* Odisha (which has its own ikat).

GA-26. The naturally golden Muga silk is from: (a) Assam (b) Karnataka (c) Bihar (d) J&K
Ans: (a). *Concept:* textiles · *Diff:* E · *Why NID:* textile GK · *Trap:* none.

GA-27. Blue Pottery is a famous craft of: (a) Khurja (b) Jaipur (c) Lucknow (d) Kutch
Ans: (b). It is quartz-based, not clay. *Concept:* crafts · *Diff:* E · *Why NID:* material awareness · *Trap:* Khurja (glazed ceramics).

GA-28. Bidriware, metal inlaid with silver, originates from: (a) Bidar, Karnataka (b) Bikaner (c) Bhopal (d) Bastar
Ans: (a). *Concept:* crafts · *Diff:* E · *Why NID:* name–place link · *Trap:* Bikaner.

GA-29. Aranmula Kannadi from Kerala is a mirror made of: (a) glass (b) metal alloy (c) mica (d) silver-coated wood
Ans: (b). *Concept:* materials · *Diff:* M · *Why NID:* material awareness · *Trap:* glass.

GA-30. Which of these is NOT a silk sari? (a) Kanjeevaram (b) Mysore Silk (c) Baluchari (d) Kota Doria
Ans: (d). Kota Doria is a light cotton (or cotton-silk) fabric with small checks. *Concept:* textiles · *Diff:* M · *Why NID:* discrimination · *Trap:* none.

GA-31. Chanderi fabric is from: (a) Madhya Pradesh (b) Rajasthan (c) UP (d) Gujarat
Ans: (a). *Concept:* textiles · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-32. Banarasi brocade is woven in: (a) Varanasi, UP (b) Patan (c) Kanchipuram (d) Surat
Ans: (a). *Concept:* textiles · *Diff:* E · *Why NID:* GK · *Trap:* Surat (a zari and synthetics centre).

GA-33. Kolhapuri chappals hold a GI tag registered for districts in: (a) Maharashtra only (b) Maharashtra and Karnataka (c) Gujarat (d) Goa
Ans: (b). *Concept:* GI · *Diff:* M · *Why NID:* craft IP (see Vol 11 for the 2025 global fashion controversy) · *Trap:* (a).

GA-34. Rogan art, painted with castor-oil-based paint, is practised in: (a) Nirona, Kutch, Gujarat (b) Raghurajpur, Odisha (c) Madhubani (d) Kalighat
Ans: (a). *Concept:* crafts · *Diff:* M · *Why NID:* rare-craft GK · *Trap:* Raghurajpur (Pattachitra).

GA-35. The Nagpur Orange is a GI product of: (a) Maharashtra (b) MP (c) Punjab (d) Assam
Ans: (a). *Concept:* GI · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-36. Both West Bengal and Odisha hold separate GI tags for varieties of: (a) Sandesh (b) Rasgulla/Rasagola (c) Mishti doi (d) Chhena poda
Ans: (b). *Concept:* GI · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-37. Dhokra metal craft uses which technique? (a) Forging (b) Lost-wax casting (c) Stamping (d) Electroplating
Ans: (b). *Concept:* process · *Diff:* E · *Why NID:* craft processes · *Trap:* forging.

GA-38. Sohrai and Khovar paintings are from: (a) Jharkhand (b) Odisha (c) Chhattisgarh (d) Assam
Ans: (a). *Concept:* art geography · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-39. The Kani shawl is from: (a) Kashmir (b) Himachal (c) Sikkim (d) Punjab
Ans: (a). It is woven with small wooden bobbins (kanis). *Concept:* textiles · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-40. Tirupati Laddu holds a GI tag. It belongs to: (a) Andhra Pradesh (b) Tamil Nadu (c) Karnataka (d) Telangana
Ans: (a). *Concept:* GI · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-41. Kutch embroidery belongs to: (a) Gujarat (b) Rajasthan (c) Punjab (d) Sindh only
Ans: (a). *Concept:* textiles · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-42. Longpi black pottery is made without a potter's wheel in: (a) Manipur (b) Mizoram (c) Nagaland (d) Tripura
Ans: (a). *Concept:* crafts · *Diff:* M · *Why NID:* NE crafts · *Trap:* Nagaland.

GA-43. Pattachitra cloth painting is strongly associated with: (a) Raghurajpur, Odisha (b) Kutch (c) Bhuj (d) Jaipur
Ans: (a). *Concept:* art geography · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-44. Which craft is made from golden grass (sikki) in Bihar? (a) Basketry and figurines (b) Shawls (c) Pottery (d) Metalware
Ans: (a). *Concept:* crafts · *Diff:* M · *Why NID:* sustainable materials · *Trap:* none.

GA-45. Mysore Silk is from: (a) Karnataka (b) Kerala (c) TN (d) Goa
Ans: (a). *Concept:* textiles · *Diff:* E · *Why NID:* GK · *Trap:* none.

C. UNESCO, Currency Motifs & Performing Arts (46–70)

GA-46. India's 44th World Heritage Site (2025), "Maratha Military Landscapes", includes how many forts? (a) 10 (b) 12 (c) 15 (d) 8
Ans: (b). Eleven are in Maharashtra and one (Gingee) is in Tamil Nadu. *Concept:* UNESCO · *Diff:* M · *Why NID:* recent heritage · *Trap:* none.

GA-47. The Moidams of Charaideo (UNESCO 2024) are mound-burials of which dynasty? (a) Ahom (b) Chola (c) Kakatiya (d) Pala
Ans: (a). In Assam. *Concept:* UNESCO · *Diff:* M · *Why NID:* recent heritage · *Trap:* none.

GA-48. The "Sacred Ensembles of the Hoysalas" (UNESCO 2023) include: (a) Belur, Halebidu, Somanathapura (b) Hampi, Badami (c) Khajuraho (d) Thanjavur
Ans: (a). *Concept:* UNESCO · *Diff:* M · *Why NID:* architecture · *Trap:* Hampi.

GA-49. Santiniketan was inscribed as a UNESCO site in: (a) 2021 (b) 2023 (c) 2019 (d) 2016
Ans: (b). *Concept:* UNESCO · *Diff:* M · *Why NID:* the art-education heritage · *Trap:* none.

GA-50. Dholavira, a Harappan city inscribed in 2021, is in: (a) Rajasthan (b) Gujarat (c) Haryana (d) Punjab
Ans: (b). *Concept:* UNESCO · *Diff:* E · *Why NID:* town planning · *Trap:* Haryana (Rakhigarhi).

GA-51. The Ramappa (Rudreswara) Temple, famous for its "floating bricks", is in: (a) Telangana (b) Karnataka (c) Odisha (d) AP
Ans: (a). Kakatiya dynasty; UNESCO 2021. *Concept:* UNESCO · *Diff:* M · *Why NID:* materials · *Trap:* AP.

GA-52. India's first UNESCO World Heritage City (2017) is: (a) Jaipur (b) Ahmedabad (c) Varanasi (d) Hyderabad
Ans: (b). *Concept:* UNESCO · *Diff:* E · *Why NID:* NID's home city · *Trap:* Jaipur (2019).

GA-53. India's only "mixed" (natural + cultural) World Heritage Site is: (a) Kaziranga (b) Khangchendzonga National Park (c) Sundarbans (d) Western Ghats
Ans: (b). *Concept:* UNESCO · *Diff:* M · *Why NID:* GK · *Trap:* Kaziranga.

GA-54. The current ₹100 note (lavender) shows which heritage site? (a) Rani ki Vav (b) Hampi (c) Sanchi (d) Red Fort
Ans: (a). *Concept:* currency design · *Diff:* M · *Why NID:* observation + heritage · *Trap:* none.

GA-55. The current ₹10 note (chocolate brown) shows: (a) Konark Sun Temple (b) Ellora (c) Hampi (d) Mangalyaan
Ans: (a). *Concept:* currency design · *Diff:* M · *Why NID:* observation · *Trap:* none.

GA-56. The ₹50 note (fluorescent blue) shows: (a) the stone chariot of Hampi (b) Sanchi (c) Rani ki Vav (d) Red Fort
Ans: (a). *Concept:* currency design · *Diff:* M · *Why NID:* observation · *Trap:* none.

GA-57. The ₹20 note (greenish-yellow) shows: (a) Ellora Caves (b) Konark (c) Hampi (d) Sanchi
Ans: (a). *Concept:* currency · *Diff:* M · *Why NID:* observation · *Trap:* none.

GA-58. The ₹200 note shows: (a) Sanchi Stupa (b) Red Fort (c) Hampi (d) Ellora
Ans: (a). *Concept:* currency · *Diff:* M · *Why NID:* observation · *Trap:* none.

GA-59. The ₹500 note shows: (a) Red Fort (b) Sanchi (c) Rani ki Vav (d) India Gate
Ans: (a). *Concept:* currency · *Diff:* M · *Why NID:* observation · *Trap:* none.

GA-60. Which festival was inscribed on UNESCO's Intangible Cultural Heritage list in 2023? (a) Garba of Gujarat (b) Holi (c) Diwali (d) Pongal
Ans: (a). *Concept:* ICH · *Diff:* M · *Why NID:* culture GK · *Trap:* none.

GA-61. Durga Puja in Kolkata was inscribed as ICH in: (a) 2019 (b) 2021 (c) 2023 (d) 2017
Ans: (b). Pandal design is a major design industry. *Concept:* ICH · *Diff:* M · *Why NID:* temporary architecture · *Trap:* none.

GA-62. Kumbh Mela was inscribed as ICH in: (a) 2010 (b) 2017 (c) 2020 (d) 2013
Ans: (b). *Concept:* ICH · *Diff:* M · *Why NID:* the temporary-city design challenge · *Trap:* none.

GA-63. Chhau dance is performed in: (a) Odisha, Jharkhand, West Bengal (b) Kerala (c) Manipur (d) Gujarat
Ans: (a). Masks feature especially in Purulia and Seraikela. *Concept:* performing arts · *Diff:* M · *Why NID:* mask and costume design · *Trap:* none.

GA-64. Kathakali, known for elaborate face make-up and costume, is from: (a) Kerala (b) Tamil Nadu (c) Karnataka (d) Odisha
Ans: (a). *Concept:* performing arts · *Diff:* E · *Why NID:* costume colour coding · *Trap:* none.

GA-65. Kuchipudi is a classical dance of: (a) Andhra Pradesh (b) Assam (c) Odisha (d) Kerala
Ans: (a). *Concept:* dance · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-66. Sattriya is a classical dance from: (a) Assam (b) Manipur (c) Bengal (d) Tripura
Ans: (a). *Concept:* dance · *Diff:* E · *Why NID:* GK · *Trap:* Manipur (Manipuri dance).

GA-67. The Mountain Railways of India (UNESCO) include: (a) Darjeeling, Nilgiri, Kalka–Shimla (b) Konkan, Matheran (c) Vande Bharat (d) Rajdhani
Ans: (a). *Concept:* UNESCO · *Diff:* M · *Why NID:* engineering heritage · *Trap:* Matheran (not inscribed).

GA-68. Bhimbetka rock shelters (prehistoric paintings) are in: (a) Madhya Pradesh (b) Maharashtra (c) Karnataka (d) Rajasthan
Ans: (a). *Concept:* prehistoric art · *Diff:* E · *Why NID:* the origins of Indian visual art · *Trap:* none.

GA-69. The Ajanta Caves are famous mainly for: (a) Buddhist murals and paintings (b) Mughal miniatures (c) Jain bronzes (d) colonial architecture
Ans: (a). *Concept:* art heritage · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-70. The Kailasa Temple (Ellora Cave 16) is remarkable because it was: (a) built of bricks (b) carved top-down from a single rock (c) made of wood (d) built by the British
Ans: (b). *Concept:* monolithic architecture · *Diff:* E · *Why NID:* the subtractive process · *Trap:* none.

D. Institutions, Museums, Awards (71–95)

GA-71. NID was established in: (a) 1958 (b) 1961 (c) 1969 (d) 1986
Ans: (b). *Concept:* NID history · *Diff:* E · *Why NID:* the interview staple · *Trap:* 1958 (The India Report).

GA-72. NIFT (National Institute of Fashion Technology) was established in: (a) 1961 (b) 1986 (c) 1995 (d) 2000
Ans: (b). *Concept:* institutions · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-73. The IDC School of Design is part of: (a) IIT Bombay (b) IIT Delhi (c) IISc (d) IIT Kanpur
Ans: (a). Founded 1969. *Concept:* institutions · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-74. NID became an Institution of National Importance through an Act passed in: (a) 2010 (b) 2014 (c) 2019 (d) 1961
Ans: (b). *Concept:* NID status · *Diff:* M · *Why NID:* the interview · *Trap:* none.

GA-75. The World Design Organization (WDO) was formerly called: (a) ICSID (b) AIGA (c) ADI (d) RIBA
Ans: (a). The International Council of Societies of Industrial Design. *Concept:* organisations · *Diff:* M · *Why NID:* GK · *Trap:* AIGA (a US graphic design body).

GA-76. The first Pritzker Architecture Prize (1979) went to: (a) Philip Johnson (b) Frank Gehry (c) I.M. Pei (d) Oscar Niemeyer
Ans: (a). *Concept:* awards · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-77. The Red Dot Design Award is based in: (a) Germany (b) Italy (c) Japan (d) USA
Ans: (a). Essen. *Concept:* design awards · *Diff:* E · *Why NID:* GK · *Trap:* Japan (the G Mark).

GA-78. The Compasso d'Oro is awarded by ADI in: (a) Italy (b) France (c) Spain (d) Denmark
Ans: (a). Since 1954, an idea of Gio Ponti. *Concept:* design awards · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-79. Cannes Lions recognises excellence in: (a) architecture (b) advertising and creative communication (c) film acting (d) fashion
Ans: (b). *Concept:* awards · *Diff:* E · *Why NID:* communication design · *Trap:* film acting (that is the Cannes Film Festival).

GA-80. MoMA, famous for its design collection, is in: (a) London (b) New York (c) Paris (d) Berlin
Ans: (b). *Concept:* museums · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-81. The Victoria and Albert Museum (art and design) is in: (a) London (b) Edinburgh (c) Sydney (d) Mumbai
Ans: (a). *Concept:* museums · *Diff:* E · *Why NID:* GK · *Trap:* Mumbai (whose Dr. Bhau Daji Lad Museum was formerly the Victoria & Albert Museum, Bombay).

GA-82. The glass pyramid at the Louvre was designed by: (a) I.M. Pei (b) Norman Foster (c) Renzo Piano (d) Jean Nouvel
Ans: (a). 1989. *Concept:* architecture · *Diff:* E · *Why NID:* GK · *Trap:* Jean Nouvel (Louvre Abu Dhabi).

GA-83. The Calico Museum of Textiles is in: (a) Ahmedabad (b) Surat (c) Mumbai (d) Jaipur
Ans: (a). It was founded by the Sarabhai family. *Concept:* museums · *Diff:* M · *Why NID:* NID's neighbourhood · *Trap:* none.

GA-84. The Salar Jung Museum is in: (a) Hyderabad (b) Lucknow (c) Bhopal (d) Mysuru
Ans: (a). *Concept:* museums · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-85. India's largest contemporary art biennale, first held in 2012, is the: (a) Kochi-Muziris Biennale (b) India Art Fair (c) Serendipity Arts Festival (d) Jaipur Literature Festival
Ans: (a). *Concept:* art events · *Diff:* E · *Why NID:* the art scene · *Trap:* India Art Fair (which is a fair, not a biennale).

GA-86. The first Indian to win an Academy Award was costume designer: (a) Bhanu Athaiya (b) A.R. Rahman (c) Satyajit Ray (d) Resul Pookutty
Ans: (a). *Gandhi* (1983). *Concept:* design and film · *Diff:* M · *Why NID:* the fashion and costume field · *Trap:* Satyajit Ray (honorary Oscar, 1992).

GA-87. India's national academy of fine arts is the: (a) Lalit Kala Akademi (b) Sangeet Natak Akademi (c) Sahitya Akademi (d) NGMA
Ans: (a). 1954. *Concept:* institutions · *Diff:* M · *Why NID:* GK · *Trap:* Sangeet Natak Akademi (performing arts).

GA-88. The National Design Policy of India was approved in: (a) 2002 (b) 2007 (c) 2014 (d) 2020
Ans: (b). It led to the India Design Council. *Concept:* policy · *Diff:* M · *Why NID:* the design ecosystem · *Trap:* none.

GA-89. The Indian Institute of Crafts and Design (IICD) is in: (a) Jaipur (b) Bhopal (c) Kolkata (d) Pune
Ans: (a). *Concept:* institutions · *Diff:* M · *Why NID:* craft education · *Trap:* none.

GA-90. The Bauhaus school was founded in 1919 in: (a) Weimar (b) Berlin (c) Dessau (d) Munich
Ans: (a). It later moved to Dessau (1925) and Berlin (1932). *Concept:* design history · *Diff:* M · *Why NID:* the design-education roots · *Trap:* Dessau.

GA-91. The Stanford d.school is closely associated with popularising: (a) design thinking (b) Bauhaus (c) Brutalism (d) Art Deco
Ans: (a). *Concept:* design education · *Diff:* E · *Why NID:* process · *Trap:* none.

GA-92. The India Design Mark (I Mark) is awarded by the: (a) India Design Council (b) NID (c) NIFT (d) Lalit Kala Akademi
Ans: (a). *Concept:* awards · *Diff:* M · *Why NID:* GK · *Trap:* NID.

GA-93. Which organisation designates the World Design Capital? (a) WDO (b) UNESCO (c) UN-Habitat (d) ADI
Ans: (a). *Concept:* organisations · *Diff:* M · *Why NID:* GK · *Trap:* UNESCO (which runs the Creative Cities Network).

GA-94. Which Indian city is a member of the UNESCO Creative Cities Network in the "Crafts & Folk Art" category? (a) Jaipur (b) Mumbai (c) Hyderabad (d) Kozhikode
Ans: (a). Jaipur joined in 2015, and Srinagar is also listed for crafts (2021). Mumbai = Film, Hyderabad = Gastronomy, Kozhikode = Literature. Other Indian members: Varanasi, Chennai and Gwalior (Music), and Lucknow (Gastronomy, 2025). *Concept:* UNESCO creative cities · *Diff:* M · *Why NID:* the culture–city link · *Trap:* Hyderabad.

GA-95. Lucknow was designated a UNESCO Creative City in 2025 for: (a) Gastronomy (b) Music (c) Crafts (d) Film
Ans: (a). Designated on 30–31 Oct 2025 [REPORTED]. *Concept:* recent culture · *Diff:* E · *Why NID:* current affairs · *Trap:* crafts (chikankari).

E. Brands, Logos & Products (96–125)

GA-96. The FedEx logo hides: (a) a star (b) an arrow between E and x (c) a plane (d) a box
Ans: (b). *Concept:* negative space · *Diff:* E · *Why NID:* logo literacy · *Trap:* none.

GA-97. The arrow under "amazon" points from: (a) a to z (b) m to n (c) a to n (d) o to n
Ans: (a). It suggests "everything from A to Z" and a smile. *Concept:* logo semantics · *Diff:* E · *Why NID:* logos · *Trap:* none.

GA-98. The Apple logo (1977 rainbow version) was designed by: (a) Rob Janoff (b) Paul Rand (c) Susan Kare (d) Jony Ive
Ans: (a). *Concept:* logo history · *Diff:* M · *Why NID:* GK · *Trap:* Susan Kare (Mac icons).

GA-99. The Nike Swoosh was designed by: (a) Carolyn Davidson (b) Milton Glaser (c) Saul Bass (d) Paula Scher
Ans: (a). 1971. *Concept:* logo history · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-100. "I ❤ NY" was designed by: (a) Milton Glaser (b) Paul Rand (c) Massimo Vignelli (d) Herb Lubalin
Ans: (a). 1977. *Concept:* graphic design history · *Diff:* E · *Why NID:* GK · *Trap:* Vignelli.

GA-101. The Olympic rings logo consists of how many coloured rings on a white background? (a) 4 (b) 5 (c) 6 (d) 7
Ans: (b). Blue, yellow, black, green and red. *Concept:* symbols · *Diff:* E · *Why NID:* observation · *Trap:* none.

GA-102. The three-pointed star of Mercedes-Benz symbolises: (a) land, sea, air (b) three founders (c) three cities (d) speed
Ans: (a). *Concept:* logo meaning · *Diff:* M · *Why NID:* symbols · *Trap:* none.

GA-103. Audi's four rings represent: (a) four merged companies (Auto Union) (b) four wheels (c) four founders' families (d) the Olympics
Ans: (a). *Concept:* logo meaning · *Diff:* M · *Why NID:* symbols · *Trap:* four wheels.

GA-104. Which car logo is designed so that every letter of its name can be traced within it? (a) Toyota (b) Honda (c) Hyundai (d) Kia
Ans: (a). *Concept:* logo construction · *Diff:* M · *Why NID:* logo cleverness · *Trap:* none.

GA-105. Twitter was rebranded as "X" in: (a) 2021 (b) 2023 (c) 2024 (d) 2019
Ans: (b). *Concept:* rebranding · *Diff:* E · *Why NID:* brand-identity debates · *Trap:* none.

GA-106. Air India's 2023 logo, inspired by a window frame, is called: (a) The Vista (b) The Maharaja (c) The Wheel (d) The Swan
Ans: (a). *Concept:* Indian rebranding · *Diff:* M · *Why NID:* Indian identity design · *Trap:* Maharaja (the mascot).

GA-107. The Amul girl mascot first appeared in: (a) 1946 (b) 1966 (c) 1980 (d) 1991
Ans: (b). *Concept:* advertising history · *Diff:* M · *Why NID:* Indian visual culture · *Trap:* 1946 (the Amul cooperative).

GA-108. The Tata Nano, marketed as the "₹1 lakh car", was unveiled in: (a) 2004 (b) 2008 (c) 2012 (d) 2015
Ans: (b). *Concept:* product history · *Diff:* M · *Why NID:* frugal-design debate · *Trap:* none.

GA-109. The Ambassador car was manufactured by: (a) Hindustan Motors (b) Maruti (c) Tata (d) Premier
Ans: (a). It was based on the Morris Oxford. *Concept:* Indian design icons · *Diff:* E · *Why NID:* product history · *Trap:* Premier (Padmini).

GA-110. The Coca-Cola contour bottle (1915) was designed so it could be recognised: (a) only by colour (b) even in the dark or when broken (c) by smell (d) by price
Ans: (b). *Concept:* form as brand · *Diff:* M · *Why NID:* product identity · *Trap:* none.

GA-111. The Sony Walkman was launched in: (a) 1969 (b) 1979 (c) 1989 (d) 1999
Ans: (b). *Concept:* product history · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-112. The first iPhone was released in: (a) 2005 (b) 2007 (c) 2009 (d) 2010
Ans: (b). *Concept:* product history · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-113. The Vespa scooter was created by which Italian company? (a) Piaggio (b) Fiat (c) Ferrari (d) Ducati
Ans: (a). 1946, by Corradino D'Ascanio. *Concept:* product design · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-114. The Jaipur Foot is a famous low-cost: (a) prosthetic limb (b) shoe brand (c) dance (d) craft
Ans: (a). Developed in 1969 by Dr P.K. Sethi and Ram Chandra Sharma. *Concept:* social design · *Diff:* E · *Why NID:* Indian frugal innovation · *Trap:* shoe brand.

GA-115. LEGO comes from: (a) Denmark (b) Sweden (c) Germany (d) USA
Ans: (a). "Leg godt" means "play well". *Concept:* brands · *Diff:* E · *Why NID:* toy design · *Trap:* Sweden.

GA-116. IKEA, known for flat-pack furniture, is from: (a) Sweden (b) Norway (c) Finland (d) Denmark
Ans: (a). *Concept:* brands · *Diff:* E · *Why NID:* furniture design · *Trap:* Denmark.

GA-117. MUJI (Japan) is known for: (a) logo-heavy luxury (b) "no-brand", minimal, functional products (c) sports cars (d) fast fashion only
Ans: (b). *Concept:* design philosophy · *Diff:* M · *Why NID:* minimalism · *Trap:* none.

GA-118. The Unilever "U" logo is made of how many small icons? (a) 15 (b) 25 (c) 35 (d) 10
Ans: (b). *Concept:* logo construction · *Diff:* H · *Why NID:* logo literacy · *Trap:* none.

GA-119. The WWF panda logo uses which Gestalt principle most clearly? (a) Closure (b) Proximity (c) Common fate (d) Similarity
Ans: (a). *Concept:* gestalt · *Diff:* E · *Why NID:* logo analysis · *Trap:* none.

GA-120. Google changed its logo to a sans-serif typeface in: (a) 2010 (b) 2015 (c) 2018 (d) 2020
Ans: (b). *Concept:* rebranding · *Diff:* M · *Why NID:* digital branding · *Trap:* none.

GA-121. The BMW roundel's blue and white come from the flag of: (a) Bavaria (b) Bavaria's propeller company (c) Germany (d) Austria
Ans: (a). The "propeller" story is a later marketing myth. *Concept:* logo history · *Diff:* M · *Why NID:* the critical-fact habit · *Trap:* (b).

GA-122. "Swachh Bharat" uses which iconic object in its logo? (a) Gandhi's spectacles (b) a broom only (c) a charkha (d) a lotus
Ans: (a). *Concept:* government branding · *Diff:* E · *Why NID:* Indian public design · *Trap:* broom.

GA-123. The "Make in India" logo is a lion made of: (a) cogs/gears (b) flowers (c) circuits (d) flags
Ans: (a). *Concept:* government branding · *Diff:* E · *Why NID:* symbol design · *Trap:* none.

GA-124. In the "Incredible India" logo, the "!" replaces the letter: (a) i (b) l (c) t (d) e
Ans: (a). *Concept:* typographic logos · *Diff:* E · *Why NID:* logos · *Trap:* none.

GA-125. The "Gattu" mascot of Asian Paints was created by: (a) R.K. Laxman (b) Mario Miranda (c) Anant Pai (d) Sudhir Dar
Ans: (a). *Concept:* Indian illustration · *Diff:* M · *Why NID:* Indian visual culture · *Trap:* Mario Miranda.

F. Science, Technology & Digital Design (126–150)

GA-126. The movable-type printing press in Europe (c. 1440) is credited to: (a) Gutenberg (b) Caxton (c) Edison (d) Da Vinci
Ans: (a). *Concept:* communication history · *Diff:* E · *Why NID:* typography roots · *Trap:* none.

GA-127. The World Wide Web was invented by: (a) Tim Berners-Lee (b) Bill Gates (c) Steve Jobs (d) Vint Cerf
Ans: (a). 1989, at CERN. *Concept:* technology history · *Diff:* E · *Why NID:* digital design · *Trap:* Vint Cerf (internet protocols).

GA-128. The computer mouse was invented by: (a) Douglas Engelbart (b) Alan Turing (c) Steve Wozniak (d) Tim Cook
Ans: (a). *Concept:* interaction design history · *Diff:* M · *Why NID:* HCI history · *Trap:* none.

GA-129. The QR code was invented in 1994 by: (a) Denso Wave, Japan (b) Google (c) IBM (d) Samsung
Ans: (a). Masahiro Hara. *Concept:* information design · *Diff:* M · *Why NID:* UPI-era India · *Trap:* none.

GA-130. UPI (Unified Payments Interface) was launched in 2016 by: (a) NPCI (b) RBI only (c) SBI (d) Paytm
Ans: (a). *Concept:* digital public infrastructure · *Diff:* E · *Why NID:* the design of India's fintech · *Trap:* Paytm.

GA-131. Chandrayaan-3 landed near the Moon's south pole on: (a) 23 Aug 2023 (b) 14 Jul 2023 (c) 15 Aug 2023 (d) 2 Sep 2023
Ans: (a). 23 August is now India's National Space Day. *Concept:* science · *Diff:* E · *Why NID:* GK · *Trap:* 14 July (the launch date).

GA-132. Aditya-L1 is India's mission to study the: (a) Sun (b) Mars (c) Venus (d) Moon
Ans: (a). *Concept:* science · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-133. Gaganyaan is India's: (a) human spaceflight programme (b) Mars rover (c) navigation satellite (d) fighter jet
Ans: (a). *Concept:* science · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-134. In 2025, Shubhanshu Shukla became the first Indian to visit the: (a) International Space Station (b) Moon (c) Tiangong (d) Mir
Ans: (a). Axiom Mission 4, June–July 2025. He was the second Indian in space after Rakesh Sharma (1984). *Concept:* recent science · *Diff:* E · *Why NID:* current affairs · *Trap:* Moon.

GA-135. Velcro was inspired by: (a) burrs sticking to dog fur (b) gecko feet (c) spider silk (d) lotus leaves
Ans: (a). *Concept:* biomimicry · *Diff:* E · *Why NID:* creative thinking · *Trap:* gecko feet (which inspired dry adhesives).

GA-136. 3D printing by stereolithography was patented in the 1980s by: (a) Chuck Hull (b) Elon Musk (c) Dean Kamen (d) James Dyson
Ans: (a). *Concept:* making technology · *Diff:* M · *Why NID:* prototyping · *Trap:* none.

GA-137. "UI" and "UX" stand for: (a) User Interface and User Experience (b) Unique Idea and User Extension (c) Universal Input and Universal Exit (d) Utility Index and User Index
Ans: (a). *Concept:* digital design · *Diff:* E · *Why NID:* interaction design · *Trap:* none.

GA-138. Augmented reality (AR) differs from virtual reality (VR) because AR: (a) overlays digital content on the real world (b) fully replaces the real world (c) is audio only (d) needs no device
Ans: (a). *Concept:* emerging tech · *Diff:* E · *Why NID:* tech and design · *Trap:* none.

GA-139. Vector graphics differ from raster graphics because vectors: (a) are made of pixels (b) are defined by mathematical paths and scale without losing quality (c) are always photos (d) can't be coloured
Ans: (b). *Concept:* digital graphics · *Diff:* E · *Why NID:* logo production · *Trap:* (a).

GA-140. A "pixel" is: (a) the smallest controllable unit of a digital image (b) a printing ink (c) a vector path (d) a font
Ans: (a). *Concept:* digital basics · *Diff:* E · *Why NID:* basics · *Trap:* none.

GA-141. Haptic feedback refers to: (a) feedback through touch, such as vibration (b) sound (c) colour (d) smell
Ans: (a). *Concept:* interaction design · *Diff:* M · *Why NID:* product UX · *Trap:* none.

GA-142. "Internet of Things" (IoT) means: (a) everyday objects connected to the internet and exchanging data (b) internet shops (c) social media (d) online games
Ans: (a). *Concept:* tech · *Diff:* E · *Why NID:* smart products · *Trap:* none.

GA-143. E-ink displays (as in e-readers) are valued because they: (a) use very little power and mimic paper (b) show fast video (c) are very bright (d) are 3D
Ans: (a). *Concept:* display tech · *Diff:* M · *Why NID:* product design · *Trap:* none.

GA-144. Generative AI image tools create images from: (a) text prompts, using models trained on large datasets (b) cameras only (c) hand-drawing only (d) printers
Ans: (a). *Concept:* AI and design · *Diff:* E · *Why NID:* current debates · *Trap:* none.

GA-145. A "digital twin" is: (a) a virtual replica of a physical object or system used for simulation (b) a sibling's photo (c) a duplicate account (d) a backup drive
Ans: (a). *Concept:* tech · *Diff:* M · *Why NID:* emerging design tools · *Trap:* none.

GA-146. Which display technology does not need a backlight because each pixel emits light? (a) OLED (b) LCD (c) CRT (d) Projector
Ans: (a). *Concept:* display tech · *Diff:* M · *Why NID:* product · *Trap:* LCD.

GA-147. "Skeuomorphism" in UI design means: (a) imitating real-world materials and objects (e.g. a notepad with a leather texture) (b) flat colours (c) 3D printing (d) a font
Ans: (a). *Concept:* UI styles · *Diff:* M · *Why NID:* digital design history · *Trap:* none.

GA-148. Apple's design language introduced in 2025 across its platforms is called: (a) Liquid Glass (b) Material You (c) Aqua (d) Metro
Ans: (a). Material You is Google's (2021); Aqua was the original Mac OS X look; Metro was Microsoft's. *Concept:* current UI design · *Diff:* M · *Why NID:* current affairs · *Trap:* Material You.

GA-149. The DigiLocker, Aadhaar and UPI ecosystem is often described as India's: (a) Digital Public Infrastructure (b) Smart City (c) Make in India (d) Startup India
Ans: (a). *Concept:* service design at national scale · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-150. The barcode (linear) was invented by: (a) Norman Woodland and Bernard Silver (b) Edison (c) Tesla (d) Bell
Ans: (a). *Concept:* information design history · *Diff:* H · *Why NID:* GK · *Trap:* none.

G. Environment & Sustainability (151–175)

GA-151. The Paris Agreement (2015) aims to limit global warming to well below 2 °C, pursuing efforts to limit it to: (a) 1 °C (b) 1.5 °C (c) 2.5 °C (d) 3 °C
Ans: (b). *Concept:* climate · *Diff:* E · *Why NID:* sustainability context · *Trap:* none.

GA-152. COP30 (November 2025) was held in: (a) Belém, Brazil (b) Dubai (c) Baku (d) Glasgow
Ans: (a). *Concept:* recent climate events · *Diff:* M · *Why NID:* current affairs · *Trap:* Baku (COP29).

GA-153. India's "Mission LiFE" stands for: (a) Lifestyle for Environment (b) Life Insurance for Everyone (c) Light Fixtures Efficiency (d) Living in Forests
Ans: (a). *Concept:* policy · *Diff:* E · *Why NID:* sustainable behaviour design · *Trap:* none.

GA-154. India banned identified single-use plastic items from: (a) 1 July 2022 (b) 2 Oct 2019 (c) 1 Jan 2020 (d) 15 Aug 2023
Ans: (a). *Concept:* policy · *Diff:* M · *Why NID:* packaging context · *Trap:* 2 Oct 2019 (the campaign announcement).

GA-155. The Swachh Bharat Mission was launched on: (a) 2 Oct 2014 (b) 15 Aug 2014 (c) 26 Jan 2015 (d) 5 Jun 2014
Ans: (a). Gandhi Jayanti. *Concept:* policy · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-156. The Chipko movement (1973), in which villagers hugged trees to prevent felling, began in present-day: (a) Uttarakhand (b) Kerala (c) Rajasthan (d) Assam
Ans: (a). *Concept:* environmental history · *Diff:* E · *Why NID:* social design stories · *Trap:* Rajasthan (the Bishnoi story).

GA-157. The 1730 Khejarli sacrifice to protect trees is associated with the: (a) Bishnoi community (b) Warli tribe (c) Toda tribe (d) Bhil tribe
Ans: (a). Led by Amrita Devi. *Concept:* conservation history · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-158. The Montreal Protocol (1987) addresses: (a) ozone-depleting substances (b) plastics (c) whaling (d) noise
Ans: (a). *Concept:* environment · *Diff:* M · *Why NID:* materials (CFCs in fridges and sprays) · *Trap:* none.

GA-159. "Compostable" differs from "biodegradable" because compostable materials: (a) break down into non-toxic compost within a defined time under composting conditions (b) never break down (c) are always plastic (d) are recyclable metals
Ans: (a). *Concept:* material claims · *Diff:* M · *Why NID:* packaging honesty · *Trap:* treating the two as the same.

GA-160. The International Solar Alliance is headquartered in: (a) Gurugram, India (b) Paris (c) Geneva (d) New York
Ans: (a). *Concept:* organisations · *Diff:* M · *Why NID:* GK · *Trap:* Paris.

GA-161. GRIHA is India's: (a) green building rating system (b) housing loan scheme (c) craft fair (d) design award
Ans: (a). *Concept:* sustainable architecture · *Diff:* M · *Why NID:* architecture GK · *Trap:* none.

GA-162. LEED is a green building certification from: (a) USA (b) UK (c) India (d) Japan
Ans: (a). US Green Building Council. *Concept:* green certification · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-163. World Environment Day is observed on: (a) 5 June (b) 22 April (c) 16 September (d) 21 March
Ans: (a). 22 April is Earth Day. *Concept:* days · *Diff:* E · *Why NID:* poster topics · *Trap:* 22 April.

GA-164. "Urban heat island" means: (a) cities are warmer than surrounding rural areas due to built surfaces and fewer trees (b) islands in cities (c) hot springs (d) a type of building
Ans: (a). *Concept:* urban design · *Diff:* E · *Why NID:* design for heat · *Trap:* none.

GA-165. Green roofs help by: (a) insulating buildings, managing rainwater and cooling cities (b) adding weight only (c) making roofs slippery (d) increasing heat
Ans: (a). *Concept:* sustainable architecture · *Diff:* E · *Why NID:* design solutions · *Trap:* none.

GA-166. "Fast fashion" is criticised mainly for: (a) high waste, pollution and poor labour conditions (b) being too slow (c) being handmade (d) using natural dyes
Ans: (a). *Concept:* fashion sustainability · *Diff:* E · *Why NID:* textile and fashion design · *Trap:* none.

GA-167. Microplastics are: (a) plastic fragments smaller than 5 mm (b) tiny 3D printers (c) a type of fabric (d) micro-chips
Ans: (a). *Concept:* pollution · *Diff:* E · *Why NID:* material design · *Trap:* none.

GA-168. The Sundarbans, the world's largest mangrove forest, lies across India and: (a) Bangladesh (b) Myanmar (c) Sri Lanka (d) Nepal
Ans: (a). *Concept:* geography · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-169. Project Tiger was launched in India in: (a) 1973 (b) 1986 (c) 1999 (d) 2006
Ans: (a). *Concept:* conservation · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-170. Which traditional practice is an example of rainwater conservation in Rajasthan? (a) Johads, kunds and stepwells (b) Paddy terraces (c) Snow harvesting (d) Desalination
Ans: (a). *Concept:* vernacular water design · *Diff:* E · *Why NID:* sustainable design · *Trap:* none.

GA-171. "Carbon footprint" measures: (a) the total greenhouse gases caused by an activity or product (b) shoe size (c) soot on walls (d) coal reserves
Ans: (a). *Concept:* sustainability metric · *Diff:* E · *Why NID:* design impact · *Trap:* none.

GA-172. E-waste refers to: (a) discarded electrical and electronic devices (b) email spam (c) energy loss (d) eco-friendly waste
Ans: (a). *Concept:* waste · *Diff:* E · *Why NID:* product end-of-life · *Trap:* none.

GA-173. The "Great Indian Bustard", a critically endangered bird, is threatened notably by: (a) collisions with power lines (b) skyscrapers (c) cold weather (d) fishing nets
Ans: (a). *Concept:* conservation and infrastructure design · *Diff:* M · *Why NID:* systemic design thinking · *Trap:* none.

GA-174. Earth Hour encourages people to: (a) switch off non-essential lights for one hour (b) plant trees for an hour (c) walk for an hour (d) fast
Ans: (a). Organised by WWF. *Concept:* awareness campaigns · *Diff:* E · *Why NID:* campaign design · *Trap:* none.

GA-175. A circular economy aims to: (a) keep materials in use and eliminate waste (b) produce round products (c) increase disposal (d) use only paper
Ans: (a). *Concept:* sustainability · *Diff:* E · *Why NID:* design thinking · *Trap:* none.

H. Indian Culture, Festivals & Media (176–200)

GA-176. Uttarayan, the international kite festival, is famously celebrated in: (a) Gujarat (b) Kerala (c) Assam (d) Goa
Ans: (a). *Concept:* festivals · *Diff:* E · *Why NID:* scene-drawing prompts · *Trap:* none.

GA-177. The Hornbill Festival is held in: (a) Nagaland (b) Mizoram (c) Meghalaya (d) Arunachal
Ans: (a). *Concept:* festivals · *Diff:* E · *Why NID:* NE culture · *Trap:* none.

GA-178. The floral carpet called "pookalam" is made during: (a) Onam (b) Pongal (c) Bihu (d) Lohri
Ans: (a). Kerala. *Concept:* festival design · *Diff:* E · *Why NID:* ephemeral design · *Trap:* none.

GA-179. Kolam, drawn with rice flour at doorsteps, is a tradition mainly of: (a) Tamil Nadu (b) Punjab (c) Gujarat (d) Kashmir
Ans: (a). *Concept:* floor art · *Diff:* E · *Why NID:* pattern design · *Trap:* none.

GA-180. Bihu is the major festival of: (a) Assam (b) Bihar (c) Odisha (d) Bengal
Ans: (a). *Concept:* festivals · *Diff:* E · *Why NID:* GK · *Trap:* Bihar.

GA-181. "Alpana" is the floor-art tradition of: (a) Bengal (b) Rajasthan (c) Maharashtra (d) Karnataka
Ans: (a). *Concept:* floor art · *Diff:* M · *Why NID:* GK · *Trap:* Rajasthan (Mandana).

GA-182. "Mandana" floor and wall art belongs to: (a) Rajasthan and MP (b) Kerala (c) Assam (d) Punjab
Ans: (a). *Concept:* floor art · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-183. Eco-friendly Ganesh idols are encouraged instead of Plaster of Paris because PoP: (a) does not dissolve easily and pollutes water bodies (b) is too heavy (c) is expensive (d) breaks easily
Ans: (a). *Concept:* sustainable festival design · *Diff:* E · *Why NID:* social design · *Trap:* none.

GA-184. Kathputli, the string-puppet tradition, is from: (a) Rajasthan (b) Kerala (c) Bengal (d) Punjab
Ans: (a). *Concept:* puppetry · *Diff:* E · *Why NID:* craft and performance · *Trap:* none.

GA-185. Tholu Bommalata (leather shadow puppetry) is from: (a) Andhra Pradesh (b) Rajasthan (c) Assam (d) Gujarat
Ans: (a). *Concept:* puppetry · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-186. Bismillah Khan is famous for playing the: (a) shehnai (b) sitar (c) sarod (d) tabla
Ans: (a). *Concept:* culture · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-187. Satyajit Ray, besides films, designed typefaces such as: (a) Ray Roman and Ray Bizarre (b) Helvetica (c) Futura (d) Devanagari MT
Ans: (a). He also worked as a commercial artist at the D.J. Keymer agency and designed book covers. *Concept:* Indian graphic-design history · *Diff:* H · *Why NID:* the designer–filmmaker link · *Trap:* none.

GA-188. The "father of Indian cinema", who made *Raja Harishchandra* (1913), is: (a) Dadasaheb Phalke (b) Raj Kapoor (c) V. Shantaram (d) Guru Dutt
Ans: (a). *Concept:* cinema history · *Diff:* E · *Why NID:* film design discipline · *Trap:* none.

GA-189. R.K. Laxman's iconic cartoon character is: (a) The Common Man (b) Chacha Chaudhary (c) Suppandi (d) Motu Patlu
Ans: (a). *Concept:* illustration · *Diff:* E · *Why NID:* Indian illustration · *Trap:* Chacha Chaudhary (Pran).

GA-190. Amar Chitra Katha comics were founded by: (a) Anant Pai (b) Pran (c) R.K. Laxman (d) Mario Miranda
Ans: (a). *Concept:* illustration · *Diff:* M · *Why NID:* sequential art · *Trap:* Pran.

GA-191. Chacha Chaudhary was created by: (a) Pran Kumar Sharma (b) Anant Pai (c) Abid Surti (d) Uncle Pai
Ans: (a). *Concept:* comics · *Diff:* M · *Why NID:* sequential art · *Trap:* none.

GA-192. The Mumbai dabbawalas are famed for: (a) a highly accurate lunchbox delivery system using colour and symbol codes (b) dance (c) textiles (d) architecture
Ans: (a). *Concept:* service and information design · *Diff:* E · *Why NID:* a systems-design case study · *Trap:* none.

GA-193. The Maha Kumbh 2025 was held at: (a) Prayagraj (b) Haridwar (c) Ujjain (d) Nashik
Ans: (a). Jan–Feb 2025. *Concept:* recent culture · *Diff:* E · *Why NID:* the temporary-city design challenge · *Trap:* none.

GA-194. Mario Miranda was a famous cartoonist from: (a) Goa (b) Kerala (c) Bengal (d) Punjab
Ans: (a). *Concept:* illustration · *Diff:* E · *Why NID:* crowd-scene drawing inspiration · *Trap:* none.

GA-195. The tabla is a: (a) percussion instrument (b) string instrument (c) wind instrument (d) keyboard
Ans: (a). *Concept:* culture · *Diff:* E · *Why NID:* object drawing · *Trap:* none.

GA-196. The famous wall paintings of Shekhawati are found on: (a) havelis in Rajasthan (b) temples in Kerala (c) caves in MP (d) forts in Punjab
Ans: (a). *Concept:* architecture and art · *Diff:* M · *Why NID:* GK · *Trap:* none.

GA-197. Lohri is mainly celebrated in: (a) Punjab (b) TN (c) Kerala (d) Assam
Ans: (a). *Concept:* festivals · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-198. A "diya" is traditionally made of: (a) terracotta clay (b) glass (c) steel (d) wax
Ans: (a). *Concept:* craft materials · *Diff:* E · *Why NID:* sustainable products · *Trap:* none.

GA-199. Which Indian city is known for "Chikankari" embroidery? (a) Lucknow (b) Jaipur (c) Kolkata (d) Surat
Ans: (a). *Concept:* textiles · *Diff:* E · *Why NID:* GK · *Trap:* none.

GA-200. Which Indian art tradition uses mainly three basic shapes (circle, triangle, square) in white on a mud background? (a) Warli (b) Gond (c) Madhubani (d) Kalighat
Ans: (a). *Concept:* folk art · *Diff:* E · *Why NID:* styles for poster and illustration tasks · *Trap:* Gond.

NEXT: Volume 4G — Art, Design, Architecture & Culture (200 questions) (`04G-VOL4-QB-ART-DESIGN-ARCH-CULTURE.md`)

VOLUME 4G — QUESTION BANK: ART, DESIGN, ARCHITECTURE & CULTURE (200 Questions)

> These questions go deeper than Vol 4F, with no overlap. Use them with Vol 2C (theory) and Vol 12 (flashcards).
> Compact format: **Q** (options) → **Ans** + explanation · *Concept* · *Diff* · *Why NID* · *Trap*.

Sections: A. Indian painting & artists (1–40) · B. Indian textiles & crafts (41–70) · C. Indian architecture (71–100) · D. World architecture (101–130) · E. Art movements & world artists (131–160) · F. Designers: graphic, type, industrial, fashion (161–200)

A. Indian Painting Traditions & Artists (1–40)

ADC-1. Madhubani painting traditionally originated on: (a) cloth scrolls (b) walls and floors of homes in the Mithila region (c) palm leaves (d) glass
Ans: (b). It moved to paper in the 1960s, encouraged during the Bihar drought relief efforts. *Concept:* the origin of a folk art · *Diff:* M · *Why NID:* craft evolution · *Trap:* cloth scrolls (that is Pattachitra or Phad).

ADC-2. A distinctive feature of Madhubani is: (a) empty backgrounds (b) double outlines and every space filled with patterns (c) only black ink (d) realistic shading
Ans: (b). *Concept:* style identification · *Diff:* E · *Why NID:* drawing in a style · *Trap:* none.

ADC-3. Warli paintings depict the Tarpa dance with figures arranged in a: (a) straight line (b) spiral or circle (c) grid (d) pyramid
Ans: (b). *Concept:* Warli composition · *Diff:* M · *Why NID:* style recognition · *Trap:* none.

ADC-4. Jivya Soma Mashe is celebrated for bringing which art form to global attention? (a) Warli (b) Gond (c) Madhubani (d) Kalighat
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* Gond.

ADC-5. Jangarh Singh Shyam was a pioneering artist of: (a) Gond art (b) Pattachitra (c) Phad (d) Tanjore
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-6. Bhajju Shyam's illustrated book, recording a Gond artist's impressions of London, is titled: (a) *The London Jungle Book* (b) *Gond London* (c) *The Night Life of Trees* (d) *Signature*
Ans: (a). (*The Night Life of Trees* is another Tara Books title featuring Gond artists.) *Concept:* contemporary folk illustration · *Diff:* H · *Why NID:* the illustration field · *Trap:* (c).

ADC-7. Gond art fills its forms using: (a) dots and dashes (b) gold leaf (c) gradients (d) cross-hatching only
Ans: (a). *Concept:* style · *Diff:* E · *Why NID:* pattern drawing · *Trap:* none.

ADC-8. Phad scroll paintings of Rajasthan narrate the stories of folk deities: (a) Pabuji and Devnarayan (b) Rama and Sita only (c) Krishna (d) Buddha
Ans: (a). *Concept:* narrative art · *Diff:* M · *Why NID:* sequential storytelling heritage · *Trap:* Krishna.

ADC-9. Phad scrolls are traditionally "performed" by: (a) Bhopas (priest-singers) (b) Kathakali dancers (c) Baul singers (d) Dhrupad singers
Ans: (a). *Concept:* art and performance · *Diff:* M · *Why NID:* GK · *Trap:* Baul.

ADC-10. Kalighat paintings developed in the 19th century near a temple in: (a) Kolkata (b) Puri (c) Varanasi (d) Madurai
Ans: (a). *Concept:* urban folk art · *Diff:* E · *Why NID:* GK · *Trap:* Puri.

ADC-11. Kalighat painting is known for: (a) bold, sweeping brush lines and social satire (b) gold leaf (c) dot patterns (d) tiny miniatures
Ans: (a). *Concept:* style · *Diff:* M · *Why NID:* style recognition · *Trap:* none.

ADC-12. Tanjore paintings are distinguished by: (a) gold foil and gem inlay with raised relief (b) dot patterns (c) white on mud (d) scrolls
Ans: (a). *Concept:* style · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-13. Srikalahasti-style Kalamkari is made: (a) freehand with a bamboo pen (kalam) (b) by block printing only (c) by embroidery (d) on walls
Ans: (a). The Machilipatnam style uses blocks. *Concept:* craft technique · *Diff:* M · *Why NID:* process knowledge · *Trap:* (b).

ADC-14. Cheriyal scroll paintings come from: (a) Telangana (b) Kerala (c) Odisha (d) Rajasthan
Ans: (a). *Concept:* folk art · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-15. Pithora paintings are made by the Rathwa community mainly in: (a) Gujarat (b) Assam (c) TN (d) Punjab
Ans: (a). Also in MP. Horses are a key motif. *Concept:* tribal art · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-16. Saura paintings (idital) are from: (a) Odisha (b) Bihar (c) Maharashtra (d) Rajasthan
Ans: (a). *Concept:* tribal art · *Diff:* M · *Why NID:* GK · *Trap:* Maharashtra (Warli, which looks similar).

ADC-17. Thangka paintings are associated with: (a) Tibetan Buddhism (Sikkim, Ladakh) (b) Jainism (c) Mughal courts (d) Bengal School
Ans: (a). *Concept:* religious art · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-18. The famous Kishangarh painting "Bani Thani", called the "Indian Mona Lisa", is by: (a) Nihal Chand (b) Mansur (c) Basawan (d) Raja Ravi Varma
Ans: (a). *Concept:* miniatures · *Diff:* M · *Why NID:* GK · *Trap:* Mansur.

ADC-19. Ustad Mansur was a Mughal painter at Jahangir's court famous for: (a) birds and animals (b) battle scenes (c) portraits of Akbar only (d) architecture
Ans: (a). *Concept:* Mughal art · *Diff:* M · *Why NID:* nature observation · *Trap:* none.

ADC-20. The *Hamzanama*, a huge illustrated manuscript project, was commissioned by: (a) Akbar (b) Babur (c) Shah Jahan (d) Aurangzeb
Ans: (a). *Concept:* Mughal art · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-21. The Kangra school of painting belongs to which group? (a) Pahari (b) Rajasthani (c) Deccan (d) Pala
Ans: (a). *Concept:* miniature schools · *Diff:* M · *Why NID:* GK · *Trap:* Rajasthani.

ADC-22. Pala-period paintings (8th–12th c.) were mainly: (a) Buddhist palm-leaf manuscript illustrations (b) Mughal portraits (c) colonial oils (d) murals of Ajanta
Ans: (a). *Concept:* early Indian painting · *Diff:* H · *Why NID:* GK · *Trap:* none.

ADC-23. Raja Ravi Varma made his paintings widely affordable by: (a) setting up a lithographic (oleograph) press (b) painting on walls (c) selling only originals (d) using photography
Ans: (a). *Concept:* the democratisation of art · *Diff:* M · *Why NID:* reproduction and print design · *Trap:* none.

ADC-24. Abanindranath Tagore's famous painting *Bharat Mata* is associated with the: (a) Bengal School / Swadeshi movement (b) Progressive Artists' Group (c) Company School (d) Kalighat
Ans: (a). *Concept:* modern Indian art · *Diff:* E · *Why NID:* GK · *Trap:* Progressive group.

ADC-25. The Progressive Artists' Group (1947) was founded in: (a) Bombay (b) Calcutta (c) Delhi (d) Madras
Ans: (a). Its members included F.N. Souza, S.H. Raza and M.F. Husain. *Concept:* modern art · *Diff:* M · *Why NID:* GK · *Trap:* Calcutta.

ADC-26. S.H. Raza is best known for his series centred on the: (a) Bindu (dot) (b) horse (c) Mother Teresa (d) bull
Ans: (a). *Concept:* artists · *Diff:* E · *Why NID:* GK · *Trap:* horse (Husain).

ADC-27. Amrita Sher-Gil's paintings include: (a) *Three Girls* (b) *Bharat Mata* (c) *Shakuntala* (d) *Santhal Family*
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-28. Jamini Roy drew inspiration mainly from: (a) Kalighat and Bengali folk art (b) European Cubism only (c) Mughal miniatures (d) Japanese prints
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-29. Ram Kinkar Baij's famous outdoor sculpture *Santhal Family* is at: (a) Santiniketan (b) Baroda (c) Delhi (d) Chennai
Ans: (a). *Concept:* sculpture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-30. Nandalal Bose's Haripura posters (1938) were made for the: (a) Indian National Congress session (b) Durga Puja (c) British exhibition (d) a film
Ans: (a). Gandhi commissioned them. *Concept:* the art and nation link · *Diff:* H · *Why NID:* poster history · *Trap:* none.

ADC-31. Anish Kapoor's *Cloud Gate* ("The Bean") is in: (a) Chicago (b) London (c) Mumbai (d) New York
Ans: (a). *Concept:* contemporary art · *Diff:* E · *Why NID:* GK · *Trap:* London (the ArcelorMittal Orbit).

ADC-32. Subodh Gupta is known for sculptures made of: (a) stainless-steel kitchen utensils (b) clay (c) paper (d) glass
Ans: (a). *Concept:* contemporary art · *Diff:* M · *Why NID:* everyday objects as art · *Trap:* none.

ADC-33. Bharti Kher is widely known for artworks using: (a) bindis (b) bangles only (c) pottery (d) stamps
Ans: (a). *Concept:* contemporary art · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-34. V.S. Gaitonde is associated with: (a) meditative abstract painting (b) realism (c) cartoons (d) photography
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-35. Tyeb Mehta's paintings often feature: (a) a diagonal dividing the canvas and figures like Kali and Mahishasura (b) dots (c) landscapes only (d) cartoons
Ans: (a). *Concept:* artists · *Diff:* H · *Why NID:* GK · *Trap:* none.

ADC-36. Manjusha art, featuring snake motifs, is from: (a) Bhagalpur, Bihar (b) Kutch (c) Kangra (d) Mysuru
Ans: (a). *Concept:* folk art · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-37. Baua Devi is a renowned artist of: (a) Madhubani / Mithila (b) Gond (c) Warli (d) Pithora
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-38. Bhuri Bai is a celebrated artist of: (a) Bhil art (b) Kalighat (c) Tanjore (d) Phad
Ans: (a). *Concept:* artists · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-39. The Company School of painting refers to: (a) Indian artists painting for East India Company officials in a hybrid style (b) corporate logos (c) Mughal court art (d) modern abstraction
Ans: (a). *Concept:* colonial art · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-40. Satish Gujral, painter and sculptor, also designed which building? (a) Belgian Embassy, New Delhi (b) Lotus Temple (c) India Habitat Centre (d) Parliament
Ans: (a). *Concept:* the art–architecture crossover · *Diff:* H · *Why NID:* GK · *Trap:* India Habitat Centre (Joseph Allen Stein).

B. Indian Textiles & Crafts (41–70)

ADC-41. Patola of Patan is special because it is: (a) double ikat, with both warp and weft tie-dyed before weaving (b) block printed (c) embroidered (d) painted
Ans: (a). *Concept:* textile process · *Diff:* M · *Why NID:* textile depth · *Trap:* single ikat.

ADC-42. Kanjeevaram saris often have borders woven separately and interlocked with the body. This technique is called: (a) korvai (b) jamdani (c) bandha (d) zari
Ans: (a). *Concept:* weaving technique · *Diff:* H · *Why NID:* textile depth · *Trap:* none.

ADC-43. Paithani saris are known for which motif? (a) peacocks and parrots (b) elephants (c) mango only (d) geometric checks
Ans: (a). *Concept:* motifs · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-44. Baluchari saris are famous for: (a) mythological narrative scenes woven on the pallu (b) tie-dye (c) mirror work (d) plain white
Ans: (a). *Concept:* textiles · *Diff:* M · *Why NID:* narrative textiles · *Trap:* none.

ADC-45. Jamdani weaving creates motifs using: (a) supplementary weft inserted by hand on fine muslin (b) printing blocks (c) embroidery (d) wax resist
Ans: (a). *Concept:* weaving technique · *Diff:* H · *Why NID:* textile depth · *Trap:* embroidery.

ADC-46. Ajrakh printing, practised in Kutch, traditionally uses which colours? (a) indigo blue and madder red (b) pink and green (c) gold and silver (d) black only
Ans: (a). *Concept:* dyeing · *Diff:* M · *Why NID:* natural dyes · *Trap:* none.

ADC-47. Dabu printing uses: (a) mud resist (b) wax resist (c) stencils (d) screen printing
Ans: (a). *Concept:* resist printing · *Diff:* M · *Why NID:* process · *Trap:* wax resist (batik).

ADC-48. Leheriya is a tie-dye technique from Rajasthan producing: (a) diagonal wave-like stripes (b) dots (c) checks (d) florals
Ans: (a). *Concept:* tie-dye · *Diff:* E · *Why NID:* pattern recognition · *Trap:* dots (Bandhani).

ADC-49. Bandhani produces patterns of: (a) tiny dots via tied knots (b) stripes via weaving (c) block prints (d) embroidery
Ans: (a). *Concept:* tie-dye · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-50. Phulkari literally means: (a) flower work (b) gold work (c) mirror work (d) thread work
Ans: (a). *Concept:* embroidery · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-51. Chikankari is traditionally done: (a) white thread on fine white cotton (b) gold on velvet (c) mirrors on cotton (d) beads on silk
Ans: (a). Today it is done in many colours, but white-on-white is the classic form. *Concept:* embroidery · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-52. Kantha embroidery uses which basic stitch? (a) running stitch (b) chain stitch (c) cross stitch (d) satin stitch only
Ans: (a). *Concept:* stitches · *Diff:* M · *Why NID:* textile basics · *Trap:* chain stitch (Kutch, Aari).

ADC-53. Kasuti embroidery, with geometric counted-thread patterns, is from: (a) Karnataka (b) Punjab (c) Bengal (d) Gujarat
Ans: (a). *Concept:* embroidery · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-54. Zardozi embroidery uses: (a) metallic gold and silver threads (b) mirrors (c) beads only (d) cotton only
Ans: (a). *Concept:* embroidery · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-55. Toda embroidery (Pukhoor), in red and black on white, is from the: (a) Nilgiris, Tamil Nadu (b) Kutch (c) Kashmir (d) Assam
Ans: (a). *Concept:* tribal textiles · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-56. Pipli in Odisha is famous for: (a) appliqué work (b) pottery (c) bell metal (d) carpets
Ans: (a). *Concept:* crafts · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-57. The Assamese "Mekhela Chador" is commonly woven in: (a) Muga or Pat silk (b) wool (c) jute (d) polyester only
Ans: (a). *Concept:* regional dress · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-58. Bidriware's base metal turns black through treatment with: (a) soil from the Bidar fort region mixed with ammonium chloride (b) paint (c) charcoal (d) oil
Ans: (a). *Concept:* craft process · *Diff:* H · *Why NID:* material process · *Trap:* paint.

ADC-59. Blue Pottery uses which main material instead of clay? (a) quartz stone powder (b) porcelain (c) terracotta (d) plastic
Ans: (a). *Concept:* materials · *Diff:* M · *Why NID:* material GK · *Trap:* porcelain.

ADC-60. Channapatna toys are coloured using: (a) natural lacquer and vegetable dyes (b) spray paint (c) enamel (d) oil paint
Ans: (a). *Concept:* safe toy materials · *Diff:* M · *Why NID:* product safety · *Trap:* none.

ADC-61. Bankura terracotta is famous for its stylised: (a) horse (b) elephant (c) fish (d) peacock
Ans: (a). *Concept:* crafts · *Diff:* E · *Why NID:* form stylisation · *Trap:* none.

ADC-62. Meenakari is the craft of: (a) enamelling on metal (b) wood inlay (c) embroidery (d) leatherwork
Ans: (a). *Concept:* crafts · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-63. Pietra dura (parchinkari) in Agra involves: (a) inlaying semi-precious stones in marble (b) painting on glass (c) wood carving (d) ceramic glazing
Ans: (a). *Concept:* crafts · *Diff:* E · *Why NID:* the Taj Mahal link · *Trap:* none.

ADC-64. Thewa jewellery (Pratapgarh, Rajasthan) fuses: (a) intricately worked gold sheet onto coloured glass (b) silver onto wood (c) shell onto metal (d) beads onto cloth
Ans: (a). *Concept:* crafts · *Diff:* H · *Why NID:* GK · *Trap:* none.

ADC-65. The Assamese "jaapi" is a: (a) traditional bamboo and palm-leaf hat (b) drum (c) shawl (d) boat
Ans: (a). *Concept:* NE crafts · *Diff:* M · *Why NID:* product and culture · *Trap:* none.

ADC-66. Kondapalli toys are made from: (a) soft light wood called tella poniki (b) clay (c) metal (d) paper
Ans: (a). *Concept:* materials · *Diff:* H · *Why NID:* GK · *Trap:* none.

ADC-67. Kashmiri papier-mâché begins with: (a) paper pulp moulded into shapes (b) wood (c) clay (d) metal
Ans: (a). *Concept:* process · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-68. Sankheda furniture from Gujarat is made of: (a) lacquered teak wood with painted patterns (b) cane (c) steel (d) plastic
Ans: (a). *Concept:* furniture crafts · *Diff:* M · *Why NID:* furniture design · *Trap:* cane.

ADC-69. Mata ni Pachedi is a sacred textile honouring: (a) the Mother Goddess (b) the Sun (c) Buddha (d) rivers
Ans: (a). From Gujarat. *Concept:* ritual textiles · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-70. The Thatheras of Jandiala Guru (Punjab), on UNESCO's ICH list, are known for: (a) brass and copper utensil making (b) weaving (c) pottery (d) painting
Ans: (a). *Concept:* ICH crafts · *Diff:* H · *Why NID:* GK · *Trap:* none.

C. Indian Architecture (71–100)

ADC-71. The curvilinear tower over a North Indian (Nagara) temple sanctum is called the: (a) shikhara (b) gopuram (c) vimana (d) mandapa
Ans: (a). *Concept:* temple terms · *Diff:* E · *Why NID:* architecture vocabulary · *Trap:* vimana.

ADC-72. The monumental gateway tower of South Indian temples is the: (a) gopuram (b) shikhara (c) torana (d) chhatri
Ans: (a). *Concept:* temple terms · *Diff:* E · *Why NID:* GK · *Trap:* torana.

ADC-73. The sanctum sanctorum of a Hindu temple is the: (a) garbhagriha (b) mandapa (c) antarala (d) prakara
Ans: (a). *Concept:* temple terms · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-74. The Brihadeeswara Temple at Thanjavur was built by: (a) Rajaraja Chola I (b) Krishnadeva Raya (c) Narasimhavarman (d) Rajendra Chola only
Ans: (a). Completed around 1010 CE. *Concept:* Chola architecture · *Diff:* M · *Why NID:* GK · *Trap:* (d), who built Gangaikonda Cholapuram.

ADC-75. The Konark Sun Temple is designed as a: (a) giant chariot with 24 wheels pulled by 7 horses (b) lotus (c) mountain (d) boat
Ans: (a). *Concept:* symbolic architecture · *Diff:* E · *Why NID:* form and meaning · *Trap:* none.

ADC-76. Khajuraho temples were built by the: (a) Chandela dynasty (b) Chola (c) Pallava (d) Hoysala
Ans: (a). *Concept:* temple architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-77. The Shore Temple and the Pancha Rathas at Mahabalipuram are works of the: (a) Pallavas (b) Cholas (c) Pandyas (d) Chalukyas
Ans: (a). *Concept:* temple architecture · *Diff:* M · *Why NID:* GK · *Trap:* Cholas.

ADC-78. Hoysala temples are known for: (a) star-shaped plans and intricate soapstone carving (b) huge gopurams (c) brick construction (d) domes
Ans: (a). *Concept:* the Vesara style · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-79. Hampi was the capital of the: (a) Vijayanagara Empire (b) Maratha Empire (c) Mughal Empire (d) Bahmani Sultanate
Ans: (a). *Concept:* architecture and history · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-80. The Great Bath was found at: (a) Mohenjo-daro (b) Harappa (c) Lothal (d) Dholavira
Ans: (a). *Concept:* Indus architecture · *Diff:* E · *Why NID:* GK · *Trap:* Harappa.

ADC-81. The Indus city of Lothal is famous for its: (a) dockyard (b) great bath (c) stupa (d) palace
Ans: (a). *Concept:* Indus architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-82. The four gateways around the Great Stupa at Sanchi are called: (a) toranas (b) gopurams (c) darwazas (d) chhatris
Ans: (a). *Concept:* Buddhist architecture · *Diff:* M · *Why NID:* GK · *Trap:* gopurams.

ADC-83. A chaitya is a Buddhist: (a) prayer hall (b) monastery (c) pillar (d) gateway
Ans: (a). A vihara is a monastery. *Concept:* Buddhist terms · *Diff:* M · *Why NID:* GK · *Trap:* monastery.

ADC-84. Humayun's Tomb is significant as: (a) the first garden tomb on the Indian subcontinent (b) the tallest minaret (c) a Hindu temple (d) a British building
Ans: (a). It inspired the Taj Mahal. *Concept:* Mughal architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-85. The Persian four-part garden layout used in Mughal tombs is called: (a) charbagh (b) baradari (c) haveli (d) diwan
Ans: (a). *Concept:* landscape design · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-86. The chief architect usually credited with the Taj Mahal is: (a) Ustad Ahmad Lahauri (b) Mirak Mirza Ghiyas (c) Lal Chand Ustad (d) Mir Abdul Karim
Ans: (a). *Concept:* architecture history · *Diff:* M · *Why NID:* GK · *Trap:* Mirak Mirza Ghiyas (Humayun's Tomb).

ADC-87. The Buland Darwaza is at: (a) Fatehpur Sikri (b) Agra Fort (c) Delhi (d) Lahore
Ans: (a). Built by Akbar. *Concept:* Mughal architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-88. Gol Gumbaz at Bijapur is famous for its: (a) whispering gallery under a huge dome (b) minarets (c) stepwell (d) garden
Ans: (a). *Concept:* acoustics and architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-89. Jantar Mantar in Jaipur is a(n): (a) astronomical observatory (b) palace (c) temple (d) fort
Ans: (a). Built by Sawai Jai Singh II; a UNESCO site. *Concept:* scientific architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-90. A "jharokha" is a(n): (a) enclosed overhanging balcony window (b) stepwell (c) dome (d) gate
Ans: (a). *Concept:* architectural elements · *Diff:* E · *Why NID:* drawing Rajasthani scenes · *Trap:* none.

ADC-91. A chhatri is a(n): (a) elevated dome-shaped pavilion (b) basement (c) courtyard (d) gateway
Ans: (a). *Concept:* architectural elements · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-92. Chand Baori, a deep stepwell with thousands of steps, is in: (a) Abhaneri, Rajasthan (b) Patan (c) Adalaj (d) Delhi
Ans: (a). *Concept:* stepwells · *Diff:* M · *Why NID:* geometry in architecture · *Trap:* Patan (Rani ki Vav).

ADC-93. The Capitol Complex of Chandigarh was designed by: (a) Le Corbusier (b) Louis Kahn (c) B.V. Doshi (d) Lutyens
Ans: (a). *Concept:* modernism in India · *Diff:* E · *Why NID:* GK · *Trap:* Kahn.

ADC-94. IIM Ahmedabad's iconic brick campus was designed by: (a) Louis Kahn (b) Le Corbusier (c) Charles Correa (d) Anant Raje
Ans: (a). Anant Raje later added buildings. *Concept:* modernism · *Diff:* E · *Why NID:* the Ahmedabad context · *Trap:* none.

ADC-95. B.V. Doshi's Aranya low-cost housing project is in: (a) Indore (b) Ahmedabad (c) Bengaluru (d) Pune
Ans: (a). *Concept:* social architecture · *Diff:* M · *Why NID:* GK · *Trap:* Ahmedabad (Sangath).

ADC-96. Charles Correa's Jawahar Kala Kendra in Jaipur is planned on the basis of: (a) the navagraha (nine-square mandala) (b) the golden ratio (c) a spiral (d) a lotus
Ans: (a). *Concept:* concept-driven architecture · *Diff:* H · *Why NID:* the idea behind a design · *Trap:* none.

ADC-97. Laurie Baker is known for: (a) low-cost, climate-sensitive brick architecture in Kerala (b) skyscrapers (c) Mughal restoration (d) steel bridges
Ans: (a). *Concept:* sustainable architecture · *Diff:* E · *Why NID:* frugal design · *Trap:* none.

ADC-98. The "rat-trap bond", popularised in India by Laurie Baker, saves: (a) bricks and gives better insulation by creating cavities (b) cement only (c) steel (d) glass
Ans: (a). *Concept:* building technique · *Diff:* H · *Why NID:* materials efficiency · *Trap:* none.

ADC-99. The Gandhi Smarak Sangrahalaya at Sabarmati Ashram was designed by: (a) Charles Correa (b) B.V. Doshi (c) Le Corbusier (d) Achyut Kanvinde
Ans: (a). *Concept:* modern Indian architecture · *Diff:* M · *Why NID:* GK · *Trap:* Doshi.

ADC-100. The Indo-Saracenic style combines: (a) Indian/Islamic elements with British Gothic/neoclassical planning (b) Chinese and Japanese elements (c) Greek and Roman elements (d) Art Deco and Bauhaus
Ans: (a). *Concept:* colonial architecture · *Diff:* M · *Why NID:* style identification · *Trap:* none.

D. World Architecture & Architects (101–130)

ADC-101. Fallingwater (1930s), a house built over a waterfall, was designed by: (a) Frank Lloyd Wright (b) Le Corbusier (c) Mies (d) Gropius
Ans: (a). *Concept:* organic architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-102. The Guggenheim Museum in New York has a: (a) spiral ramp (b) glass pyramid (c) titanium cladding (d) green roof
Ans: (a). By Frank Lloyd Wright. *Concept:* architecture · *Diff:* E · *Why NID:* form recognition · *Trap:* titanium (Bilbao).

ADC-103. The Guggenheim Bilbao (1997), clad in titanium, was designed by: (a) Frank Gehry (b) Zaha Hadid (c) Renzo Piano (d) Rem Koolhaas
Ans: (a). *Concept:* deconstructivism · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-104. The "Bilbao effect" refers to: (a) a city's economic revival through iconic architecture (b) a lighting effect (c) a type of glass (d) earthquake damage
Ans: (a). *Concept:* architecture and urbanism · *Diff:* M · *Why NID:* the design-impact concept · *Trap:* none.

ADC-105. Le Corbusier's "Five Points of Architecture" include all EXCEPT: (a) pilotis (b) roof garden (c) free façade (d) ornamental columns
Ans: (d). The five are pilotis, free plan, free façade, horizontal windows and roof garden. *Concept:* modernism · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-106. Le Corbusier's proportion system based on human height is the: (a) Modulor (b) Golden Mean (c) Vitruvian Man (d) Fibonacci
Ans: (a). *Concept:* proportion systems · *Diff:* M · *Why NID:* proportion · *Trap:* Vitruvian Man (Leonardo).

ADC-107. Villa Savoye (1931) near Paris is by: (a) Le Corbusier (b) Mies (c) Aalto (d) Wright
Ans: (a). *Concept:* modernism · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-108. The Barcelona Pavilion (1929) was designed by: (a) Mies van der Rohe (b) Gaudí (c) Gropius (d) Breuer
Ans: (a). *Concept:* modernism · *Diff:* M · *Why NID:* GK · *Trap:* Gaudí (who was from Barcelona).

ADC-109. The Sagrada Família is the work of: (a) Antoni Gaudí (b) Santiago Calatrava (c) Ricardo Bofill (d) Rafael Moneo
Ans: (a). *Concept:* architecture · *Diff:* E · *Why NID:* GK · *Trap:* Calatrava.

ADC-110. The Sydney Opera House was designed by: (a) Jørn Utzon (b) Norman Foster (c) Renzo Piano (d) Zaha Hadid
Ans: (a). *Concept:* architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-111. The Centre Pompidou, with its exposed pipes and services, was designed by: (a) Renzo Piano and Richard Rogers (b) Jean Nouvel (c) I.M. Pei (d) Koolhaas
Ans: (a). It is a landmark of high-tech architecture (1977). *Concept:* high-tech · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-112. "The Gherkin" (30 St Mary Axe) in London is by: (a) Norman Foster (b) Renzo Piano (c) Zaha Hadid (d) Richard Rogers
Ans: (a). *Concept:* architecture · *Diff:* E · *Why NID:* GK · *Trap:* Renzo Piano (The Shard).

ADC-113. The Shard in London was designed by: (a) Renzo Piano (b) Foster (c) Rogers (d) Hadid
Ans: (a). *Concept:* architecture · *Diff:* M · *Why NID:* GK · *Trap:* Foster.

ADC-114. The first woman to win the Pritzker Prize (2004) was: (a) Zaha Hadid (b) Kazuyo Sejima (c) Anne Lacaton (d) Yvonne Farrell
Ans: (a). *Concept:* awards · *Diff:* E · *Why NID:* GK · *Trap:* Sejima (2010).

ADC-115. The Heydar Aliyev Center in Baku is known for: (a) fluid curving forms, by Zaha Hadid Architects (b) brutalist concrete blocks (c) Gothic spires (d) a pyramid
Ans: (a). *Concept:* parametric architecture · *Diff:* M · *Why NID:* form recognition · *Trap:* none.

ADC-116. Tadao Ando's Church of the Light features: (a) a cross-shaped slit in a concrete wall letting light in (b) stained glass (c) a dome (d) a timber frame
Ans: (a). *Concept:* light in architecture · *Diff:* M · *Why NID:* light as a design element · *Trap:* none.

ADC-117. Shigeru Ban is famous for using which material in emergency shelters? (a) paper tubes (b) steel (c) glass (d) stone
Ans: (a). *Concept:* humanitarian architecture · *Diff:* E · *Why NID:* materials creativity · *Trap:* none.

ADC-118. Diébédo Francis Kéré, Pritzker 2022, is from: (a) Burkina Faso (b) Nigeria (c) Kenya (d) Ghana
Ans: (a). He was the first African laureate. *Concept:* awards · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-119. Alejandro Aravena's "half-houses" (Quinta Monroy, Chile) are an example of: (a) incremental social housing (b) luxury villas (c) temples (d) skyscrapers
Ans: (a). *Concept:* social design · *Diff:* M · *Why NID:* design for low-income users · *Trap:* none.

ADC-120. The geodesic dome is associated with: (a) Buckminster Fuller (b) Eero Saarinen (c) Gaudí (d) Kahn
Ans: (a). *Concept:* structural innovation · *Diff:* E · *Why NID:* structures · *Trap:* none.

ADC-121. The Gateway Arch in St. Louis was designed by: (a) Eero Saarinen (b) Frank Gehry (c) I.M. Pei (d) SOM
Ans: (a). *Concept:* architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-122. Burj Khalifa's Y-shaped plan was inspired by a desert flower and designed by: (a) SOM (Adrian Smith) (b) Zaha Hadid (c) Foster (d) Gensler
Ans: (a). At 828 m, it is the world's tallest building. *Concept:* biomorphic inspiration · *Diff:* M · *Why NID:* nature-inspired design · *Trap:* none.

ADC-123. Beijing National Stadium ("Bird's Nest") was designed by: (a) Herzog & de Meuron (with Ai Weiwei as consultant) (b) Zaha Hadid (c) Kengo Kuma (d) Foster
Ans: (a). *Concept:* architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-124. Kengo Kuma designed the stadium for which Olympics? (a) Tokyo 2020 (b) Paris 2024 (c) Beijing 2008 (d) London 2012
Ans: (a). *Concept:* architecture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-125. Oscar Niemeyer is famed for buildings in: (a) Brasília (b) Buenos Aires (c) Mexico City (d) Lima
Ans: (a). *Concept:* architecture · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-126. Habitat 67 in Montreal, a stacked modular housing complex, was designed by: (a) Moshe Safdie (b) Fuller (c) Kahn (d) Ingels
Ans: (a). Safdie also designed Marina Bay Sands. *Concept:* modular design · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-127. Bjarke Ingels' CopenHill in Copenhagen is a waste-to-energy plant with a: (a) ski slope on its roof (b) swimming pool (c) farm (d) museum
Ans: (a). *Concept:* hedonistic sustainability · *Diff:* M · *Why NID:* creative multi-use design · *Trap:* none.

ADC-128. The Eiffel Tower was built for the: (a) 1889 World's Fair (Exposition Universelle) (b) 1900 Olympics (c) French Revolution centenary only (d) WWI memorial
Ans: (a). (It also coincided with the Revolution's centenary, but the fair is the main answer.) *Concept:* engineering icons · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-129. The Pritzker Prize 2025 was awarded to: (a) Liu Jiakun (b) Riken Yamamoto (c) David Chipperfield (d) Kéré
Ans: (a). *Concept:* recent awards · *Diff:* M · *Why NID:* current affairs · *Trap:* Yamamoto (2024).

ADC-130. The Bosco Verticale ("Vertical Forest") towers in Milan, covered with trees, were designed by: (a) Stefano Boeri (b) Renzo Piano (c) Gae Aulenti (d) Aldo Rossi
Ans: (a). *Concept:* green architecture · *Diff:* M · *Why NID:* sustainability · *Trap:* none.

E. Art Movements & World Artists (131–160)

ADC-131. Impressionism got its name from Monet's painting: (a) *Impression, Sunrise* (b) *Water Lilies* (c) *Haystacks* (d) *Rouen Cathedral*
Ans: (a). *Concept:* art movements · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-132. Pointillism, painting with small dots of pure colour, is associated with: (a) Georges Seurat (b) Van Gogh (c) Monet (d) Matisse
Ans: (a). *Concept:* art techniques · *Diff:* E · *Why NID:* the point as an element · *Trap:* none.

ADC-133. *The Starry Night* was painted by: (a) Vincent van Gogh (b) Gauguin (c) Cézanne (d) Munch
Ans: (a). *Concept:* artists · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-134. *The Scream* is a key work of: (a) Expressionism, by Edvard Munch (b) Surrealism (c) Cubism (d) Pop Art
Ans: (a). *Concept:* movements · *Diff:* E · *Why NID:* GK · *Trap:* Surrealism.

ADC-135. Cubism was developed by: (a) Picasso and Braque (b) Dalí and Magritte (c) Warhol and Lichtenstein (d) Monet and Renoir
Ans: (a). *Concept:* movements · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-136. Picasso's *Guernica* (1937) depicts: (a) the bombing of a Spanish town during the Spanish Civil War (b) a bullfight (c) a garden (d) a portrait
Ans: (a). *Concept:* art and protest · *Diff:* M · *Why NID:* the power of visual communication · *Trap:* bullfight.

ADC-137. *The Persistence of Memory*, with melting clocks, is by: (a) Salvador Dalí (b) René Magritte (c) Max Ernst (d) Joan Miró
Ans: (a). *Concept:* Surrealism · *Diff:* E · *Why NID:* GK · *Trap:* Magritte.

ADC-138. "Ceci n'est pas une pipe" ("This is not a pipe") appears in a painting by: (a) René Magritte (b) Dalí (c) Duchamp (d) Picasso
Ans: (a). *The Treachery of Images*. *Concept:* image vs reality · *Diff:* M · *Why NID:* semiotics thinking · *Trap:* Duchamp.

ADC-139. Marcel Duchamp's *Fountain* (1917) was a: (a) readymade urinal signed "R. Mutt" (b) marble fountain (c) painting (d) photograph
Ans: (a). *Concept:* Dada · *Diff:* M · *Why NID:* "what is art/design?" debates · *Trap:* none.

ADC-140. Piet Mondrian's compositions use: (a) primary colours, black lines and white in a grid (b) dots (c) drips (d) soup cans
Ans: (a). *Concept:* De Stijl · *Diff:* E · *Why NID:* grid composition · *Trap:* none.

ADC-141. Andy Warhol is a leading figure of: (a) Pop Art (b) Cubism (c) Impressionism (d) Baroque
Ans: (a). *Campbell's Soup Cans*. *Concept:* movements · *Diff:* E · *Why NID:* consumer culture · *Trap:* none.

ADC-142. Roy Lichtenstein's paintings imitate: (a) comic strips with Ben-Day dots (b) Japanese prints (c) cave art (d) mosaics
Ans: (a). *Concept:* Pop Art · *Diff:* M · *Why NID:* printing techniques · *Trap:* none.

ADC-143. Jackson Pollock is famous for: (a) drip painting (Abstract Expressionism) (b) pointillism (c) photorealism (d) Cubism
Ans: (a). *Concept:* movements · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-144. Mark Rothko is known for: (a) large blocks of soft-edged colour (colour field) (b) portraits (c) sculpture (d) comics
Ans: (a). *Concept:* colour field painting · *Diff:* M · *Why NID:* colour emotion · *Trap:* none.

ADC-145. Op Art, creating optical illusions, is associated with: (a) Bridget Riley and Victor Vasarely (b) Monet (c) Dalí (d) Rembrandt
Ans: (a). *Concept:* movements · *Diff:* M · *Why NID:* perception · *Trap:* none.

ADC-146. Henri Matisse led which movement known for wild, non-naturalistic colour? (a) Fauvism (b) Futurism (c) Dada (d) Minimalism
Ans: (a). *Concept:* movements · *Diff:* M · *Why NID:* colour · *Trap:* Futurism.

ADC-147. Futurism (Italy, 1909) celebrated: (a) speed, machines and modern technology (b) nature (c) religion (d) the past
Ans: (a). *Concept:* movements · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-148. Chiaroscuro means: (a) strong contrast between light and dark (b) bright colours (c) dot painting (d) flat colour
Ans: (a). Associated with Caravaggio and Rembrandt. *Concept:* art terms · *Diff:* M · *Why NID:* shading ·

Trap: none.

ADC-149. Linear perspective in Renaissance art was first demonstrated by: (a) Filippo Brunelleschi (b) Michelangelo (c) Raphael (d) Botticelli
Ans: (a). *Concept:* perspective history · *Diff:* M · *Why NID:* perspective · *Trap:* none.

ADC-150. Leonardo da Vinci's drawing of a man in a circle and a square, illustrating ideal proportions, is called: (a) Vitruvian Man (b) Modulor (c) David (d) Golden Man
Ans: (a). *Concept:* proportion · *Diff:* E · *Why NID:* proportion · *Trap:* Modulor.

ADC-151. Michelangelo painted the ceiling of the: (a) Sistine Chapel (b) Louvre (c) St Paul's Cathedral (d) Notre-Dame
Ans: (a). *Concept:* Renaissance · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-152. Gustav Klimt's *The Kiss* is associated with: (a) Art Nouveau / Vienna Secession (b) Pop Art (c) Cubism (d) Baroque
Ans: (a). *Concept:* movements · *Diff:* M · *Why NID:* decorative art · *Trap:* none.

ADC-153. Alphonse Mucha is famous for: (a) Art Nouveau posters with flowing female figures (b) Cubist paintings (c) photographs (d) sculptures
Ans: (a). *Concept:* poster history · *Diff:* M · *Why NID:* poster design · *Trap:* none.

ADC-154. Japanese woodblock prints (ukiyo-e), such as Hokusai's *The Great Wave off Kanagawa*, influenced: (a) Impressionists and Post-Impressionists (b) Cave art (c) Renaissance (d) Gothic
Ans: (a). This influence is called "Japonisme". *Concept:* cross-cultural influence · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-155. Banksy is known for: (a) anonymous satirical street art (b) Renaissance portraits (c) ceramic design (d) typography
Ans: (a). *Concept:* contemporary art · *Diff:* E · *Why NID:* public art · *Trap:* none.

ADC-156. Frida Kahlo is known for: (a) self-portraits rich in Mexican folk imagery (b) abstract drips (c) pop art (d) architecture
Ans: (a). *Concept:* artists · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-157. The Bauhaus artists Wassily Kandinsky and Paul Klee were known for: (a) abstraction and colour theory teaching (b) Gothic architecture (c) photorealism (d) fashion
Ans: (a). *Concept:* Bauhaus · *Diff:* M · *Why NID:* colour theory roots · *Trap:* none.

ADC-158. Josef Albers' book *Interaction of Color* explores: (a) how colours change appearance depending on their neighbours (b) mixing paints (c) printing (d) dyes
Ans: (a). *Concept:* colour perception · *Diff:* H · *Why NID:* colour theory · *Trap:* none.

ADC-159. The Arts and Crafts movement was led by: (a) William Morris (b) Le Corbusier (c) Warhol (d) Gropius
Ans: (a). It reacted against industrial mass production. *Concept:* design history · *Diff:* E · *Why NID:* the craft–industry debate · *Trap:* Gropius.

ADC-160. The Memphis Group (1981), known for bold colours and playful geometry, was founded by: (a) Ettore Sottsass (b) Philippe Starck (c) Dieter Rams (d) Eames
Ans: (a). *Concept:* design movements · *Diff:* M · *Why NID:* GK · *Trap:* none.

F. Designers: Graphic, Typography, Industrial, Fashion (161–200)

ADC-161. Paul Rand designed logos for: (a) IBM, ABC, UPS (b) Nike, Apple (c) FedEx (d) Google
Ans: (a). *Concept:* graphic design · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-162. Saul Bass is renowned for: (a) film title sequences and logos (b) typefaces only (c) furniture (d) cars
Ans: (a). *Vertigo*, *Psycho*. *Concept:* motion graphics · *Diff:* M · *Why NID:* the film and communication field · *Trap:* none.

ADC-163. Massimo Vignelli designed the 1972: (a) New York City subway map (b) London Tube map (c) Paris Métro signs (d) Tokyo map
Ans: (a). *Concept:* information design · *Diff:* M · *Why NID:* maps · *Trap:* London (Harry Beck).

ADC-164. Harry Beck's London Underground map (1931/33) is famous for: (a) a diagrammatic layout ignoring true geography (b) photographic accuracy (c) 3D drawings (d) using no colours
Ans: (a). *Concept:* information design · *Diff:* M · *Why NID:* simplification · *Trap:* none.

ADC-165. Edward Johnston designed a typeface (1916) for: (a) the London Underground (b) *The Times* (c) Penguin Books (d) the BBC
Ans: (a). *Concept:* typography · *Diff:* M · *Why NID:* wayfinding type · *Trap: The Times* (Times New Roman).

ADC-166. Times New Roman (1932) was commissioned by *The Times* and supervised by: (a) Stanley Morison (b) Eric Gill (c) Jan Tschichold (d) Frutiger
Ans: (a). *Concept:* typography · *Diff:* M · *Why NID:* GK · *Trap:* Gill.

ADC-167. Futura (1927), a geometric sans serif, was designed by: (a) Paul Renner (b) Max Miedinger (c) Adrian Frutiger (d) Matthew Carter
Ans: (a). *Concept:* typography · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-168. Gill Sans was designed by: (a) Eric Gill (b) Edward Johnston (c) Morison (d) Renner
Ans: (a). *Concept:* typography · *Diff:* E · *Why NID:* GK · *Trap:* Johnston (Gill's teacher).

ADC-169. Verdana and Georgia were designed for screen readability by: (a) Matthew Carter (b) Erik Spiekermann (c) Hermann Zapf (d) Zuzana Licko
Ans: (a). *Concept:* screen typography · *Diff:* H · *Why NID:* digital type · *Trap:* none.

ADC-170. Adrian Frutiger designed the typeface Frutiger originally for signage at: (a) Charles de Gaulle Airport, Paris (b) Heathrow (c) JFK (d) Changi
Ans: (a). *Concept:* wayfinding type · *Diff:* H · *Why NID:* signage · *Trap:* none.

ADC-171. Jan Tschichold's *Die neue Typographie* (1928) promoted: (a) asymmetric layouts and sans-serif type (b) blackletter (c) ornament (d) centred classical layouts
Ans: (a). He later returned to classical typography. *Concept:* typography history · *Diff:* H · *Why NID:* GK · *Trap:* none.

ADC-172. Josef Müller-Brockmann is associated with: (a) the Swiss grid system and International Typographic Style (b) Art Nouveau (c) Pop Art (d) Memphis
Ans: (a). *Concept:* graphic design history · *Diff:* M · *Why NID:* grids · *Trap:* none.

ADC-173. Otl Aicher designed: (a) the 1972 Munich Olympics identity and pictograms (b) the Nike logo (c) the London Tube map (d) the Coca-Cola bottle
Ans: (a). *Concept:* identity systems · *Diff:* M · *Why NID:* pictograms · *Trap:* none.

ADC-174. Susan Kare designed: (a) the original Apple Macintosh icons (b) the Google logo (c) the Windows logo (d) the IBM logo
Ans: (a). *Concept:* UI design history · *Diff:* M · *Why NID:* icon design · *Trap:* none.

ADC-175. Dieter Rams worked for many years as chief designer at: (a) Braun (b) Apple (c) Sony (d) Philips
Ans: (a). *Concept:* industrial design · *Diff:* E · *Why NID:* GK · *Trap:* Apple (whom he influenced).

ADC-176. Raymond Loewy's "MAYA" principle stands for: (a) Most Advanced Yet Acceptable (b) Make All Your Art (c) Maximum Aesthetic, Yielding Access (d) Modern And Young Always
Ans: (a). *Concept:* design philosophy · *Diff:* M · *Why NID:* innovation acceptance · *Trap:* none.

ADC-177. The Wassily Chair (1925), made of tubular steel, was designed by: (a) Marcel Breuer (b) Mies (c) Le Corbusier (d) Aalto
Ans: (a). Bicycle handlebars inspired it. *Concept:* furniture design · *Diff:* M · *Why NID:* GK · *Trap:* Mies.

ADC-178. The Red and Blue Chair is by: (a) Gerrit Rietveld (b) Mondrian (c) Breuer (d) Eames
Ans: (a). It is associated with De Stijl. *Concept:* furniture design · *Diff:* M · *Why NID:* GK · *Trap:* Mondrian.

ADC-179. The Eames Lounge Chair (1956) combines: (a) moulded plywood and leather (b) tubular steel and canvas (c) plastic only (d) cane
Ans: (a). *Concept:* materials · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-180. Thonet's No. 14 chair (1859) was revolutionary because it used: (a) steam-bent wood and could be shipped flat-packed in parts (b) plastic (c) aluminium (d) carved stone
Ans: (a). *Concept:* manufacturing innovation · *Diff:* M · *Why NID:* mass production · *Trap:* none.

ADC-181. The Panton Chair is notable as: (a) the first single-piece moulded plastic cantilever chair (b) the first folding chair (c) a rocking chair (d) a bamboo chair
Ans: (a). *Concept:* materials innovation · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-182. Arne Jacobsen designed the: (a) Egg Chair (b) Tulip Chair (c) Wassily Chair (d) Barcelona Chair
Ans: (a). *Concept:* furniture · *Diff:* M · *Why NID:* GK · *Trap:* Tulip (Saarinen).

ADC-183. Philippe Starck's Juicy Salif (1990) is a citrus squeezer made for: (a) Alessi (b) IKEA (c) Braun (d) Muji
Ans: (a). It is famous as a sculptural object that some say sacrifices function. *Concept:* form vs function debate · *Diff:* M · *Why NID:* critique · *Trap:* none.

ADC-184. Ettore Sottsass designed the red portable typewriter: (a) Olivetti Valentine (b) Remington (c) IBM Selectric (d) Brother
Ans: (a). 1969. *Concept:* product design · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-185. Kenji Ekuan designed the iconic soy-sauce bottle for: (a) Kikkoman (b) Toyota (c) Sony (d) Yamaha
Ans: (a). 1961. *Concept:* product design · *Diff:* H · *Why NID:* simple everyday design · *Trap:* none.

ADC-186. Sori Yanagi is known for the: (a) Butterfly Stool (b) Egg Chair (c) Walkman (d) Vespa
Ans: (a). *Concept:* furniture · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-187. Isamu Noguchi's Akari lamps are made of: (a) washi paper and bamboo (b) steel (c) glass (d) plastic
Ans: (a). *Concept:* materials · *Diff:* M · *Why NID:* craft and modern design · *Trap:* none.

ADC-188. Jony Ive led the design of: (a) the iMac G3, iPod and iPhone (b) Braun radios (c) Sony Walkman (d) Nokia 3310
Ans: (a). *Concept:* industrial design · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-189. The Volkswagen Beetle is associated with: (a) Ferdinand Porsche (b) Enzo Ferrari (c) Henry Ford (d) Alec Issigonis
Ans: (a). *Concept:* automotive design · *Diff:* M · *Why NID:* GK · *Trap:* Issigonis (Mini).

ADC-190. The Fiat 500 (1957) was designed by: (a) Dante Giacosa (b) Giugiaro (c) Pininfarina (d) Bertone
Ans: (a). *Concept:* automotive design · *Diff:* H · *Why NID:* GK · *Trap:* none.

ADC-191. Coco Chanel popularised: (a) the little black dress (b) the New Look (c) punk fashion (d) pleats
Ans: (a). *Concept:* fashion history · *Diff:* E · *Why NID:* GK · *Trap:* the New Look (Dior).

ADC-192. Christian Dior's "New Look" (1947) featured: (a) cinched waists and full skirts (b) mini skirts (c) tuxedos for women (d) denim
Ans: (a). *Concept:* fashion history · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-193. Yves Saint Laurent's "Le Smoking" (1966) was a: (a) tuxedo for women (b) perfume (c) cigarette pack (d) dress
Ans: (a). *Concept:* fashion history · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-194. Issey Miyake is known for: (a) "Pleats Please" and technology-driven pleating (b) punk (c) little black dress (d) denim
Ans: (a). *Concept:* fashion · *Diff:* M · *Why NID:* textile innovation · *Trap:* none.

ADC-195. Rei Kawakubo founded: (a) Comme des Garçons (b) Chanel (c) Prada (d) Uniqlo
Ans: (a). *Concept:* fashion · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-196. Ritu Kumar is credited with: (a) reviving traditional Indian block printing and crafts in fashion (b) designing the ₹ symbol (c) architecture (d) car design
Ans: (a). *Concept:* Indian fashion · *Diff:* E · *Why NID:* craft–fashion link · *Trap:* none.

ADC-197. Rahul Mishra is known for: (a) intricate hand-embroidered couture shown at Paris Haute Couture Week (b) sports shoes (c) logos (d) furniture
Ans: (a). *Concept:* Indian fashion · *Diff:* M · *Why NID:* GK · *Trap:* none.

ADC-198. Sabyasachi Mukherjee is especially known for: (a) heritage-inspired bridal wear (b) streetwear (c) sportswear (d) footwear
Ans: (a). *Concept:* Indian fashion · *Diff:* E · *Why NID:* GK · *Trap:* none.

ADC-199. Charles and Ray Eames, besides furniture, made the influential short film: (a) *Powers of Ten* (b) *Metropolis* (c) *Koyaanisqatsi* (d) *Helvetica*
Ans: (a). *Concept:* the scale concept · *Diff:* M · *Why NID:* scale and imagination prompts · *Trap: Helvetica* (Gary Hustwit's 2007 documentary).

ADC-200. Shepard Fairey created the famous 2008 poster for Barack Obama titled: (a) "Hope" (b) "Change" (c) "Yes We Can" (d) "Unity"
Ans: (a). *Concept:* political poster design · *Diff:* M · *Why NID:* poster communication · *Trap:* "Yes We Can".

NEXT: Volume 5 — Most Expected Question Types for the Upcoming NID DAT (`05-VOL5-EXPECTED-QUESTIONS.md`)

VOLUME 5 — MOST EXPECTED QUESTION TYPES FOR THE UPCOMING NID DAT (20 Dec 2026)

> **Honesty statement:** nobody outside NID knows the actual questions. This volume ranks **question patterns** by probability, using (1) the official Handbook scheme, (2) the reported DAT 2024–2026 papers, (3) long-run repetition, (4) NID's education priorities (empathy, Indian context, sustainability, communication) and (5) the 2026 pattern change. **Every item below is [PREDICTION].**

1. Two scenarios for the paper

Scenario	Likelihood (my estimate)	Basis
S1: subjective-only, like 2026 (5–7 questions, multiple sub-parts)	~55–60%	The most recent actual paper; NID tends to repeat a new format for at least one more cycle
S2: Handbook scheme (22 objective + 4 subjective)	~35–40%	Printed in the Handbook text; used through DAT 2025
S3: something new (e.g. more writing, or a hybrid)	~5%	NID changes formats without notice

Implication: prepare drawing and ideation as the core (they matter in all scenarios), and keep the objective skills at about 85% accuracy as insurance.

2. Tier A — VERY HIGH probability patterns

A1. Scene drawing with people in a specific Indian setting or viewpoint

- **Why:** it appears almost every year; 2026 had a three-storey scene and perspective tasks; it tests observation, perspective, proportion and storytelling all at once.
- **Likely forms:**
  - "Draw a scene of ___ from the viewpoint of ___"
  - "a multi-level building with activities"
  - "a crowded place during a festival / monsoon / power cut"
  - "a view through a window / from a bridge"
- **High-yield practice prompts:**
  1. A railway station at 6 am from the footbridge
  2. A housing society during a water-supply hour
  3. A school canteen at lunch break
  4. A Diwali market lane at night
  5. A hospital waiting area seen by a child
  6. A bus stop in heavy rain

A2. Sequential drawing / storyboard with an empathy or social angle

- **Why:** 2026 (an elderly person falls on a morning walk) continues a long sequence-drawing tradition; NID increasingly emphasises empathy.

- **Likely forms:** a 4–6-frame story of help, an accident or rescue, a process (making something), a day in the life, a transformation.
- **High-yield prompts:**
    1. A child lost in a crowd and reunited
    2. A street dog injured in traffic being helped
    3. A blind person crossing a road with help
    4. The life of a discarded plastic bottle
    5. Grandparents learning UPI payments
    6. A farmer's day during a drought

A3. Product redesign / problem-solving for a specific user

- **Why:** 2025 (the neem branch, a train-related problem), 2026 (redesign, the grandmother's travel kit), and the long-run trend.
- **Likely forms:** "Identify problems faced by ___ while ___ and propose a solution"; "Redesign ___ for ___ (elderly, children, differently-abled, vendors)".
- **High-yield prompts:**
    1. A pressure cooker for elderly users
    2. A school bag for primary students
    3. A water bottle for construction workers
    4. A public tap for a slum cluster
    5. An auto-rickshaw for wheelchair users
    6. A clothes-drying solution for small flats
    7. A lunch-delivery box for hospital patients

A4. Visual communication set: signboards, posters, pictograms

- **Why:** 2026 asked for 5 signboards; 2025 a poster with a viewpoint constraint; posters and symbols recur.
- **Likely forms:** "Design N signs for ___"; "a poster for ___ using only ___ colours"; "a symbol for ___".
- **High-yield prompts:**
    1. 6 signs for a heritage site
    2. A poster for a water-conservation week, using only 2 colours
    3. Pictograms for a village health centre
    4. Safety signs for a school laboratory
    5. A poster for a local craft fair in a folk-art style

A5. Object from memory + written description (100–200 words)

- **Why:** 2026 asked for "a product you love + 200 words", plus food illustration; writing has risen as a component.
- **Likely forms:** "Draw your favourite ___ and describe its features"; "Draw an object you use daily and explain one improvement"; "Illustrate an Indian dish or drink with its ingredients".
- **High-yield prompts:**
    1. Your grandmother's kitchen tool
    2. A street food item with its cart
    3. A traditional toy
    4. A musical instrument
    5. Your school uniform's accessories

3. Tier B — HIGH probability patterns

B1. Unusual viewpoint / scale ("as seen by an ant / crow / fish / drone")

- **Why:** the 2025 crow's-viewpoint poster; imagination prompts recur.
- **Practice:**
    1. A cricket match seen by the ball
    2. Your classroom seen by a ceiling fan
    3. A pond seen by a fish looking up

B2. Imagination / "what if" scenarios

- **Why:** these test originality; they are classic NID prompts.
- **Practice:**
    1. What if all vehicles disappeared for a day?
    2. What if your shadow had its own life?
    3. What if plants could walk to water?

B3. Packaging / object-system design

- **Why:** a frequent product-plus-communication hybrid.
- **Practice:**
    1. Packaging for 4 kulhads for shipping
    2. Sustainable packaging for street-food takeaway

B4. Organising and listing visually (knolling, kits, flat-lays)

- **Why:** 2026's "items to carry with your grandmother" is a new pattern type that blends observation, empathy and layout.
- **Practice:**
    1. A first-aid kit for a trek
    2. A monsoon survival kit for a commuter
    3. A new-born baby's travel kit
    4. A kit for a first-time hostel student

B5. Colour and composition constraints inside drawings

- **Practice:** draw the same scene in a warm and in a cool palette; use only 3 colours; use complementary colours to highlight the focal point.

B6. Logo or symbol for a specific organisation or cause

- **Practice:**
    1. A logo for a women-run millet café
    2. A symbol for a city-wide cycling initiative

4. Tier C — MODERATE probability patterns (mainly if the objective Part I returns)

Pattern	Why it is still possible	How much to prepare
Visual reasoning MCQs (folding, cubes, mirror, counting, series)	2016–2025 papers had them; the Handbook still lists objective questions	Vol 4B to 85%+ accuracy

Pattern	Why it is still possible	How much to prepare
Logo identification / brand awareness	Reported in 2025	Vol 2C §9 and the flashcards
Indian craft / textile / art-form identification	Long-run GK items; also feeds the subjective work	Vol 2C §2–4
Observation detail questions (e.g. "how many spokes...")	Old GAT favourite	Vol 2A §3.1
Design-awareness MCQs (designers, movements, architecture)	Older papers	Vol 4G (skim)
Maths-lite / logic puzzles	Very occasional	Vol 4D (light)
Current affairs MCQs	Rare in recent Prelims; useful for the interview	Vol 11

5. Expected "twists" to be ready for

1. **Count constraints:** "exactly 5", "6 frames", "200 words". Missing the count costs marks.
2. **Viewpoint constraints:** a crow's view, a bird's-eye view, from inside an object.
3. **User constraints:** elderly, children, visually impaired, vendors, migrants.
4. **Material or colour constraints:** "using only lines", "monochrome", "only 3 colours".
5. **Mixed-mode answers:** drawing + labels + a written justification.
6. **Multi-part questions:** (a) identify problems, (b) sketch 3 ideas, (c) develop one in detail.

6. A high-yield 20-question drill set (do one a day for 20 days)

Time each at the given minutes and self-score using Vol 3 §13's rubric.

#	Prompt	Time	Tests
1	A three-storey building section showing four different family activities and one connecting event	25	Perspective, story
2	Six frames: helping a cyclist who has fallen after hitting a pothole, ending with a systemic fix	25	Sequence, empathy
3	Redesign a steel bucket for an elderly person with arthritis (3 ideas + 1 developed)	25	Product, ergonomics
4	5 signboards for a busy market lane (consistent system + 1 in context)	20	Signage
5	Draw your favourite kitchen object and write 150 words about it	20	Memory, writing
6	A poster for a lemonade stall from a pigeon's viewpoint	20	Viewpoint, poster
7	Items for a 2-day trip with a toddler, knolled + an organiser design	20	Kit, empathy
8	A narrow lane in an old city at dusk, 1-point perspective, with 6+ people	20	Perspective
9	A symbol for "free drinking water" that works without text	10	Pictogram
10	Illustrate a glass of masala chai with its ingredients and a tea-stall context	15	Food illustration
11	What if your school bag could talk? 4-frame comic	15	Imagination
12	Packaging for 6 handmade soaps with zero plastic	20	Packaging
13	Problems of a street cobbler during the monsoon; propose 1 solution	20	Problem-solving
14	A rooftop on kite-festival day, bird's-eye view	20	3-point perspective
15	A poster for a blood-donation camp using only red, black and white	15	Colour constraint
16	Redesign a hospital bed-side table for patients who can't sit up	20	Product, empathy
17	Draw a bicycle from memory; then label 10 parts	15	Memory accuracy
18	Six pictograms for a railway station, in a consistent style	20	Icon system

#	Prompt	Time	Tests
19	Your neighbourhood 50 years in the future	20	Speculative design
20	Two neighbours share one parking space; draw a fair solution	20	Conflict resolution

7. Pre-solved model approaches for 5 Tier-A questions

Q1. "Draw a three-storey building and show life on every floor." (20 marks, 30 min)

- *Plan (3 min):* a section view with the front wall removed. Ground floor: a kirana shop with a customer. 1st floor: a family dinner, with the TV on. 2nd floor: a student studying late under a lamp. Terrace: a grandmother drying papad, water tanks, a dish antenna. **Connecting thread:** a cat chasing a ball of wool down the staircase across all floors.
- *Execute (22 min):* a frontal 1-point layout with the VP at the 1st-floor eye level. Keep the floor heights consistent (about 1.7 adult heights). Draw people in gesture first, then add the details that show life (utensils, schoolbooks, clothes line).
- *Finish (5 min):* light colour, a dark outline for the building section, and a short title.
- *Scoring focus:* perspective, proportion, the variety of activities, storytelling (the cat), and Indian context details.

Q2. "Six frames: an elderly person falls during a morning walk." (20 marks, 25 min)

- Use the plan in Vol 3 §10. The key to a high score is **frame 6**, a systemic improvement (the fixed paving, a sign, a railing), which demonstrates design thinking beyond the story.

Q3. "Draw items you would carry when travelling with your grandmother." (15 marks, 20 min)

- *Layout:* a knolled grid of 12 items, each labelled:
  - medicines
  - pill organiser
  - spectacles
  - shawl
  - neck pillow
  - water bottle with a sipper
  - snacks
  - phone with big buttons
  - ID/ticket pouch
  - walking stick
  - torch
  - sanitiser
- *Add value:* a final "Gran's Travel Pouch" design. Colour-coded compartments, large-print tags, a cross-body strap so her hands stay free, a reflective strip, and a bright colour so it's easy to find.
- *Scoring focus:* completeness, empathy, and the organising design.

Q4. "Design 5 signboards for narrow streets." (15 marks, 20 min)

- *System:* all signs are blade signs. Rounded rectangles, a bold icon on top, a bilingual word below; blue for information, red for prohibition, yellow for warning.
- *The five signs:*
  1. One-way lane (arrow)
  2. No four-wheelers (crossed car)
  3. Low overhead wires (a person under a zig-zag line)
  4. Dead end ahead (a T-bar)
  5. Public tap / toilet → 50 m (icon + arrow)

- *Context drawing:* one lane in 1-point perspective with the signs mounted at about 2.4 m.

Q5. "Draw a product you love and write 200 words." (20 marks, 25 min)

- Use the template and model answer in Vol 2B §3.5. Choose a product whose form you know well. Draw a 3/4 view plus a usage inset; write within 180–220 words.

NEXT VOLUME: Volume 6 — Model Paper 1 (`06-VOL6-MODEL-PAPER-1.md`)

VOLUME 6 — MODEL PAPER 1 (Difficulty: Easy → Moderate)

NID DAT (B.Des) Prelims: Practice Model Paper 1
Format follows the scheme printed in the NID Admissions Handbook: Part I (22 objective) + Part II (4 subjective). This is a practice paper written for this package. It is not an official NID paper.

INSTRUCTIONS

1. Time: **3 hours (180 minutes)**. Total marks: **100**. No negative marking.
2. **Part I** (Q1–22): objective. Q1–14 carry 1 mark each; Q15–22 carry 2 marks each. Total 30.
3. **Part II** (Q23–26): subjective. Q23 = 15, Q24 = 20, Q25 = 15, Q26 = 20. Total 70.
4. Answer Part II in the space given (use an A4 sheet per question when practising at home).
5. Allowed: pencils, eraser, sharpener, scale, colour pencils, crayons or oil pastels, sketch pens.
6. Suggested timing: Part I = 25 min · Q23 = 28 min · Q24 = 38 min · Q25 = 28 min · Q26 = 38 min · buffer = 23 min.

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. On the painter's (RYB) colour wheel, the complementary colour of red is:
(a) Blue (b) Green (c) Yellow (d) Violet

Q2. How many letters of the word **TAXI** look exactly the same in a vertical mirror?
(a) 1 (b) 2 (c) 3 (d) 4

Q3. Warli painting is a tribal art form of:
(a) Bihar (b) Maharashtra (c) Odisha (d) Madhya Pradesh

Q4. How many faces does a triangular prism have?
(a) 4 (b) 5 (c) 6 (d) 9

Q5. Which company's logo hides an arrow in its negative space?
(a) Apple (b) Nike (c) FedEx (d) Pepsi

Q6. Odd one out: Chair, Table, Sofa, Spoon
(a) Chair (b) Table (c) Sofa (d) Spoon

Q7. A "tint" of a colour is produced by adding:
(a) black (b) grey (c) white (d) its complement

Q8. The Gateway of India is located in:
(a) Delhi (b) Kolkata (c) Mumbai (d) Chennai

Q9. Which regular shape can tile a floor by itself with no gaps?
(a) Circle (b) Regular pentagon (c) Regular hexagon (d) Regular octagon

Q10. 2, 4, 8, 16, ?
(a) 24 (b) 32 (c) 30 (d) 64

Q11. A straight railway track drawn going away to the horizon is best shown using:
(a) one-point perspective (b) two-point perspective (c) isometric view (d) plan view

Q12. A square sheet is folded in half and then in half again. One hole is punched through all layers, away from the folds. How many holes appear when it is unfolded?
(a) 2 (b) 3 (c) 4 (d) 8

Q13. Kanjeevaram silk saris come from:
(a) Karnataka (b) Tamil Nadu (c) Kerala (d) Andhra Pradesh

Q14. Which of these is a sans-serif typeface?
(a) Times New Roman (b) Garamond (c) Helvetica (d) Baskerville

Q15–Q22 (2 marks each)

Q15. A 4 × 4 × 4 cube is painted on all faces and cut into 64 unit cubes. How many unit cubes have exactly ONE face painted?
(a) 8 (b) 16 (c) 24 (d) 32

Q16. A wall clock shows 7:20. What time does its image in a vertical mirror show?
(a) 4:40 (b) 5:40 (c) 4:20 (d) 3:40

Q17. Five posters P, Q, R, S, T hang in a row. R is in the middle. P is at the left end. Q is immediately to the right of R. T is not next to P. Which poster is second from the left?
(a) Q (b) S (c) T (d) R

Q18. In the cube net below, which face is opposite face 2?

```
   [1]
[2][3][4][5]
   [6]
```

(a) 3 (b) 4 (c) 5 (d) 6

Q19. A room 3 m × 2.4 m is to be tiled with 30 cm × 30 cm square tiles. How many tiles are needed?
(a) 64 (b) 72 (c) 80 (d) 96

Q20. At a busy bus stop, elderly passengers cannot read the small route numbers on approaching buses. Which solution is MOST effective?
(a) Print a timetable booklet
(b) Put large, high-contrast route numbers on the bus front and sides, with an audio announcement at the stop
(c) Paint the buses in brighter colours
(d) Ask passengers to carry binoculars

Q21. A painting shows a single red apple among many green apples. The design principle MOST clearly used is:
(a) Rhythm (b) Emphasis through contrast (c) Symmetry (d) Proportion

Q22. Four gears A, B, C, D are in a line, each meshing with the next. If A turns clockwise, D turns:
(a) clockwise (b) anticlockwise (c) not at all (d) alternately

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PART II — SUBJECTIVE (70 marks)

Q23. Memory Drawing (15 marks)
Draw a roadside **tea stall (chai tapri)** from memory, with the tea-seller and three customers. Your drawing must include **at least eight objects** that make the place recognisable. Use pencil shading; colour is optional.

Q24. Sequential Drawing (20 marks)
In **four frames**, show how a small seed planted by a child grows over the years into a tree that gives shade to a bus stop. Show the passing of time clearly. Keep the same location in all frames.

Q25. Poster Design (15 marks)
Design a poster on **"Save Electricity"** for a school notice board. Use **not more than three colours** (black and white count as colours if used). Include a short headline of your own.

Q26. Problem Solving (20 marks)
Children in your housing colony play cricket in the parking area, and cars, windows and children often get hurt or damaged.
(a) List **three** problems in this situation (3 marks).
(b) Design **one** solution and present it as a labelled sketch (14 marks).
(c) Write **two lines** explaining how your solution helps everyone involved (3 marks).

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	c	13	b	19	c
2	d	8	c	14	c	20	b
3	b	9	c	15	c	21	b
4	b	10	b	16	a	22	b
5	c	11	a	17	b		
6	d	12	c	18	b		

DETAILED SOLUTIONS — PART I

Q1 → (b) Green. On the RYB wheel red sits opposite green. *Why not blue?* Blue is a primary colour 120° away (the red–yellow–blue triad). *Concept:* complements · *Diff:* E · *Skill:* colour theory.

Q2 → (d) 4. T, A, X and I are all symmetric about a vertical axis, so each looks the same in a vertical mirror. (The word as a whole reads "IXAT", but the question asks about the letters.) *Concept:* mirror symmetry · *Diff:* E.

Q3 → (b) Maharashtra. Warli comes from the Thane–Palghar region and is painted in white rice paste on mud walls. Bihar = Madhubani; Odisha = Saura and Pattachitra; MP = Gond. *Concept:* folk art · *Diff:* E.

Q4 → (b) 5. Two triangles + three rectangles = 5 faces. It has 9 edges (option d is the trap) and 6 vertices. *Concept:* solids · *Diff:* E.

Q5 → (c) FedEx. The arrow is formed between the "E" and the "x". *Concept:* negative space · *Diff:* E.

Q6 → (d) Spoon. It is cutlery; the rest are furniture. *Concept:* classification · *Diff:* E.

Q7 → (c) white. Tint = hue + white; shade = + black; tone = + grey. *Concept:* colour terms · *Diff:* E.

Q8 → (c) Mumbai. Designed by George Wittet; completed 1924. *Concept:* GK · *Diff:* E.

Q9 → (c) Regular hexagon. Its interior angle of 120° fits exactly 3 times around a point (360°). Pentagons (108°) and octagons (135°) leave gaps; circles always leave gaps. *Concept:* tessellation · *Diff:* E.

Q10 → (b) 32. Each term doubles. *Concept:* series · *Diff:* E.

Q11 → (a) one-point. The rails recede to a single vanishing point on the horizon. *Concept:* perspective · *Diff:* E.

Q12 → (c) 4. Two folds make $2^2 = 4$ layers, so 4 holes. *Concept:* paper folding · *Diff:* E.

Q13 → (b) Tamil Nadu (Kanchipuram). Karnataka = Mysore silk. *Concept:* textiles · *Diff:* E.

Q14 → (c) Helvetica. The other three are serif typefaces. *Concept:* typography · *Diff:* E.

Q15 → (c) 24. One-face cubes = $6(n-2)^2 = 6 \times 2^2 = 24$. Check: corners 8 + edges 12 × 2 = 24 + faces 24 + inside 8 = 64 ✓. *Concept:* painted cube · *Diff:* M.

Q16 → (a) 4:40. 11:60 − 7:20 = 4:40. *Check:* the minute hand at "4" (20 min) mirrors to "8" (40 min); the hour hand just past 7 mirrors to just before 5. So 4:40 ✓. *Concept:* mirror clock · *Diff:* M.

Q17 → (b) S. Positions 1–5: P = 1, R = 3, Q = 4. Positions 2 and 5 are left for S and T. T is not next to P, so T ≠ 2. Therefore T = 5 and S = 2. *Concept:* arrangement · *Diff:* M.

Q18 → (b) 4. In the row 2-3-4-5, faces with one face between them are opposite: 2–4 and 3–5. Faces 1 and 6, attached above and below face 3, are opposite each other. *Concept:* nets · *Diff:* M.

Q19 → (c) 80. 300 ÷ 30 = 10 tiles along the length; 240 ÷ 30 = 8 tiles along the width; 10 × 8 = 80. *Concept:* area · *Diff:* M.

Q20 → (b). It fixes the root cause (legibility at a distance) and adds a non-visual channel (audio) for people with poor eyesight. (a) does not help identify an arriving bus, (c) does not identify the route, and (d) is absurd. *Concept:* inclusive design · *Diff:* M.

Q21 → (b) Emphasis through contrast. The one different colour becomes the focal point. *Concept:* principles · *Diff:* E.

Q22 → (b) anticlockwise. Meshing gears alternate: A clockwise, B anticlockwise, C clockwise, D anticlockwise. *Concept:* mechanical reasoning · *Diff:* M.

DETAILED SOLUTIONS — PART II (model approaches + marking rubrics)

> These rubrics are **practice benchmarks** built from the inferred NID criteria (Vol 1 §4). They are not NID's official scheme.

Q23. Chai tapri from memory (15 marks)

What a strong answer includes

- **Setting:** a small wooden or metal stall on a footpath, sometimes under a tree, with a tarpaulin or tin roof.

- **The 8+ identifying objects:**
    1. kerosene/LPG stove or angithi
    2. aluminium or steel kettle (patila)
    3. strainer
    4. glass jars of rusk, biscuits and toast
    5. small glass tumblers or kulhads
    6. sugar and tea-leaf tins
    7. milk vessel
    8. ginger and grater
    9. bench or stools
    10. hanging packets of snacks/chips
    11. a small radio
    12. a handwritten price board ("Chai ₹10")
    13. dustbin
    14. bicycle parked nearby
- **People:** the vendor pouring tea from a height (a classic action); one customer sipping, one reading a newspaper, one waiting with coins.
- **Composition:** a 2-point view of the stall corner with the vendor as the focal point (at a thirds point), and steam rising from the kettle. Customers overlap to show depth.

Step-by-step approach (28 min)

1. 2 min: list the objects in the margin.
2. 4 min: horizon line, then the stall block in 2-point perspective.
3. 8 min: gesture figures in correct proportion against the stall (counter height about 0.9 m, roughly the waist of an adult).
4. 10 min: details and objects; label them lightly if there is any doubt.
5. 4 min: shading (light from one side), steam, ground shadow.

Marking rubric (15)

Criterion	Marks
Recognisability, 8+ authentic objects	4
People: proportion, actions, interaction	3
Perspective & composition	3
Observation detail and memory accuracy	3
Rendering: line quality, shading	2

Common mistakes: a generic shop with no tea-specific objects; people taller than the stall roof; leaving out the activity (an empty stall).

Q24. Seed to shade tree, 4 frames (20 marks)

Model plan (fixed camera position, same bus-stop location)

Frame	Time	Content
1	Year 0	A child (about 8) plants a sapling beside a bare bus-stop pole on a hot day. A parent waters it. The sun is harsh; commuters squint.
2	Year 3	A sapling with a tree guard. The child (now about 11) waters it with a school bottle. A small bench has been added.
3	Year 8	A young tree as tall as the stop sign. The teenager chats with friends under thin shade. Birds arrive.

Frame	Time	Content
4	Year 15	A large tree spreading shade over the whole stop. Commuters, a vendor and elderly people rest in the shade. The young adult (now about 23) passes by and smiles. A nest is in the tree.

Showing the passing of time:

- the child ages (height, clothes: school uniform → college bag → office wear)
- the tree grows
- seasonal cues
- the surroundings change (a newer bus model, a new building in the background)
- optionally, small captions: "Year 0, 3, 8, 15"

Marking rubric (20)

Criterion	Marks
Clear narrative and the passage of time	5
Continuity (same location and camera, recognisable character)	4
Drawing quality: proportions of people and tree	4
Composition of each frame	3
Emotional or social message (shade = care for community)	2
Completion & neatness	2

Common mistakes: changing the viewpoint so the location becomes unrecognisable; the tree growing but the child staying the same; empty frames without people.

Q25. "Save Electricity" poster, ≤ 3 colours (15 marks)

Idea options (choose one and execute it boldly)

1. **Visual metaphor:** a light bulb drawn as an hourglass, with the "sand" (light) running out. Headline: "**Light is running out. Switch off.**"
2. **Switch as hero:** a giant switch in the OFF position casts a shadow shaped like a planet Earth smiling. Headline: "**One click. One planet.**"
3. **Classroom cue:** an empty classroom with fans and lights on, labelled "Nobody's here... except the bill." Headline: "**Last one out? Lights out.**"

Execution (for idea 1)

- **Colours:** yellow (the light), black (outlines and text) and white (the background).
- **Hierarchy:** the headline at the top (largest), the bulb-hourglass in the middle (dominant visual), and a small tip line below: "Switch off fans & lights when you leave the room."
- **Composition:** central, symmetrical, with plenty of negative space for impact.

Marking rubric (15)

Criterion	Marks
Concept originality & clarity	5
Hierarchy & layout	3
Typography / lettering	3
Colour constraint followed effectively	2
Overall impact & neatness	2

Common mistakes: more than 3 colours (an instruction violation); too much text; the cliché of a crying Earth with nothing new added.

Q26. Cricket in the parking area (20 marks)

(a) Three problems (3 marks, 1 each): any three of:

- balls hit cars and windows, causing damage
- children are at risk from moving cars
- there is no designated play space
- noise disturbs residents
- conflict between parents and car owners
- the elderly face falling risks from flying balls

(b) Model solution (14 marks): "Net-Pitch Corner"

- Convert a little-used corner (or a rooftop or terrace) into a **netted cricket cage**: a frame of GI pipes with nylon netting on all sides and the top, a **concrete pitch strip** with a mat, a soft-rubber **ball-only rule**, and **timing slots** (5–7 pm) displayed on a signboard.
- Parking lines are moved to separate the play zone with **bollards and a painted pedestrian path**.
- A **rebound wall** inside lets solo players practise.
- **Labels on the sketch:**
  - 3 m high netting
  - entry gate with spring hinge
  - soft-ball rule board
  - bench for parents
  - LED light for evening
  - pitch mat
  - bollards
- **Present as:** an isometric or 2-point sketch of the cage in the parking area, plus a small plan view showing the zones (play / parking / walkway).

(c) Two lines (3 marks): "The netted cage keeps balls inside, so cars and windows are safe, and the bollards keep cars out of the play zone, so children are safe. Fixed play timings and soft balls reduce noise and conflict, so both residents and children win."

Marking rubric for (b)

Criterion	Marks
Solution addresses all 3 stated problems	4
Feasibility (cost, space, materials)	3
Clarity of sketch & labels	4
User consideration (children, parents, car owners, elderly)	2
Originality	1

Common mistakes: "ban cricket" (not design thinking); an app-based solution with no physical change; an unlabelled sketch.

SCORING INTERPRETATION — MODEL PAPER 1

Practice benchmarks only. These are NOT NID cutoffs (NID does not publish verifiable cutoffs, and Prelims performance only shortlists for Mains).

Total score	Band	Meaning	Next step
80–100	**Excellent**	Strong command of easy–moderate material	Move to Papers 3–5; push originality
65–79	**Strong**	Solid base; a few gaps	Review mistakes; do Paper 2 within 5 days
50–64	**Good**	Competitive, but drawing or reasoning needs work	Daily scene drawing + Vol 4B drills
35–49	**Needs improvement**	Clear weaknesses in one or more areas	Revisit Vol 2A & 3 basics; retake in 10 days
< 35	**Major improvement required**	Fundamentals missing	Follow the 90-day plan from Day 1 (Vol 13)

Section diagnostics: Part I below 20/30 means drill visual reasoning. Any subjective question below 50% of its marks means repeat a similar prompt from Vol 3 §13 within 3 days.

NEXT VOLUME: Volume 7 — Model Paper 2 (`07-VOL7-MODEL-PAPER-2.md`)

VOLUME 7 — MODEL PAPER 2 (Difficulty: Moderate)

NID DAT (B.Des) Prelims: Practice Model Paper 2
Handbook-scheme format. This practice paper was written for this package; it is not an official NID paper.
Focus mix: observation, spatial reasoning, Indian crafts, pictograms, elevated-viewpoint scene, memory + writing, sustainable toy design.

INSTRUCTIONS

1. Time: **3 hours**. Total: **100 marks**. No negative marking.
2. Part I: Q1–14 (1 mark each), Q15–22 (2 marks each) = 30 marks.
3. Part II: Q23 (15), Q24 (20), Q25 (15), Q26 (20) = 70 marks.
4. Suggested timing: Part I = 25 min · Q23 = 25 · Q24 = 40 · Q25 = 28 · Q26 = 40 · buffer = 22.

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. Which capital letter has rotational symmetry of order 2 but NO line of symmetry?
(a) A (b) Z (c) E (d) T

Q2. Pochampally Ikat is woven in:
(a) Odisha (b) Telangana (c) Gujarat (d) Assam

Q3. Which of these is NOT a primary colour of light (RGB)?
(a) Red (b) Green (c) Blue (d) Yellow

Q4. Which of these is a *lettermark* logo?
(a) Apple (b) IBM (c) Nike swoosh (d) Starbucks siren

Q5. An arrow points East. It is rotated 270° clockwise. Where does it point now?
(a) North (b) South (c) West (d) East

Q6. Which Indian craft uses quartz powder instead of clay?
(a) Khurja pottery (b) Jaipur Blue Pottery (c) Longpi pottery (d) Bankura terracotta

Q7. Loom : Weaver :: Kiln : ?
(a) Blacksmith (b) Potter (c) Carpenter (d) Tailor

Q8. How many lines of symmetry does a square have?
(a) 2 (b) 4 (c) 6 (d) 8

Q9. "Less is more" is a motto closely associated with:
(a) Baroque (b) Modernism (c) Rococo (d) Memphis Design

Q10. Odd one out: Gond, Warli, Madhubani, Chikankari
(a) Gond (b) Warli (c) Madhubani (d) Chikankari

Q11. How many A4 sheets have the same area as one A3 sheet?
(a) 1.5 (b) 2 (c) 3 (d) 4

Q12. Which word looks unchanged in a water image (flipped top to bottom)?
(a) BOX (b) CAT (c) TOP (d) MAN

Q13. At 5 pm, in which direction does your shadow point?
(a) North (b) South (c) East (d) West

Q14. The handle of a cup "inviting" you to hold it is an example of:
(a) Feedback (b) Affordance (c) Mapping (d) Constraint

Q15–Q22 (2 marks each)

Q15. How many squares of all sizes are there in a grid of 3 rows × 4 columns of unit squares?
(a) 12 (b) 18 (c) 20 (d) 24

Q16. Statements: All bowls are vessels. Some vessels are made of brass.
Conclusions: I. Some bowls are made of brass. II. Some vessels are bowls.
(a) Only I follows (b) Only II follows (c) Both follow (d) Neither follows

Q17. An 8 cm × 8 cm sheet is folded so the bottom half goes up onto the top half (fold at y = 4). Then the left half is folded onto the right half (fold at x = 4). A hole is punched at (6, 6). Where are the holes when the sheet is unfolded? (x is measured from the left edge, y from the bottom edge.)
(a) (6,6) and (2,6)
(b) (6,6), (2,6), (6,2), (2,2)
(c) (6,6), (6,2)
(d) (2,2) only

Q18. A standard die (opposite faces add up to 7) shows 1 on top, 2 in front and 3 on the right. Which number is on the left face?
(a) 4 (b) 5 (c) 6 (d) 2

Q19. On a 1 : 50 drawing, a bed measures 4 cm × 3.8 cm. What is its real floor area?
(a) 1.9 m² (b) 3.8 m² (c) 7.6 m² (d) 15.2 m²

Q20. A village school has no electricity, and its classrooms become very hot in summer. Which is the BEST low-cost design approach?
(a) Install air-conditioners powered by a diesel generator
(b) Use a white reflective roof coating, cross-ventilation openings, jaali screens and shade trees
(c) Close all windows to keep the heat out
(d) Paint the walls dark brown

Q21. Dots arranged as ●● ●● ●● (pairs with gaps between them) are seen as three groups. Which Gestalt principle explains this?
(a) Similarity (b) Proximity (c) Closure (d) Continuity

Q22. A person walks 4 km North, then 3 km East, then 4 km South. How far are they from the starting point, and in which direction?
(a) 3 km East (b) 3 km West (c) 5 km North-East (d) 11 km East

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PART II — SUBJECTIVE (70 marks)

Q23. Pictogram System (15 marks)
Design **four pictograms** for a public park: (i) drinking water, (ii) toilet, (iii) "do not pluck flowers", (iv) dustbin. They must look like **one family** (consistent style). Draw each inside a 6 cm × 6 cm square.

Q24. Scene in Perspective (20 marks)
Draw a **Sunday-morning vegetable market** as seen by a person **standing on the back of a parked truck** (a slightly elevated view). Include at least **eight people** doing different things.

Q25. Memory + Writing (15 marks)
(a) Draw **your school bag from memory** in a 3/4 view (8 marks).
(b) In about **120 words**, describe **one improvement** you would make to it, and why (7 marks).

Q26. Sustainable Toy Design (20 marks)
Design a **toy for a five-year-old child** using **only waste materials** commonly found at home. Show: (a) the toy (main drawing), (b) the materials used (labelled), (c) how a child plays with it (a small scene), and (d) one safety consideration.

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	b	13	c	19	b
2	b	8	b	14	b	20	b
3	d	9	b	15	c	21	b
4	b	10	d	16	b	22	a
5	a	11	b	17	b		
6	b	12	a	18	a		

DETAILED SOLUTIONS — PART I

Q1 → (b) Z. Z looks the same after a half turn but has no mirror axis. A and T have a vertical axis; E has a horizontal one. *Concept:* symmetry types · *Diff:* M.

Q2 → (b) Telangana. The weave is named after Pochampally (Bhoodan Pochampally) in Nalgonda/Yadadri district; it holds a GI tag. *Concept:* textiles · *Diff:* E.

Q3 → (d) Yellow. The light primaries are red, green and blue. Yellow is a painter's (RYB) primary, and it is also what red + green light makes. *Concept:* colour models · *Diff:* E.

Q4 → (b) IBM. It is made of initials (Paul Rand's striped version). Apple is pictorial; the swoosh is abstract; the siren is an emblem or mascot. *Concept:* logo types · *Diff:* E.

Q5 → (a) North. 90° clockwise from East = South, 180° = West, 270° = North. Equivalently, 90° anticlockwise from East is North. *Concept:* rotation · *Diff:* E.

Q6 → (b) Jaipur Blue Pottery. It uses quartz stone powder, powdered glass and Multani mitti, not clay. *Concept:* materials · *Diff:* M.

Q7 → (b) Potter. This is the main equipment–worker pair: a weaver uses a loom, a potter fires in a kiln. *Concept:* analogy · *Diff:* E.

Q8 → (b) 4. Two diagonals + two midlines. *Concept:* symmetry · *Diff:* E.

Q9 → (b) Modernism. The phrase is associated with Mies van der Rohe. *Concept:* design history · *Diff:* E.

Q10 → (d) Chikankari. It is embroidery (Lucknow); the others are painting traditions. *Concept:* classification · *Diff:* E.

Q11 → (b) 2. Each A-size is half of the size before it. *Concept:* paper sizes · *Diff:* E.

Q12 → (a) BOX. B, O and X are all symmetric about a horizontal axis. In CAT, the A and T are not. In TOP, the T and P are not. In MAN, the M, A and N are not. *Concept:* water image · *Diff:* M.

Q13 → (c) East. The evening sun is in the west, so shadows point east. *Concept:* shadows · *Diff:* E.

Q14 → (b) Affordance. The handle's form suggests the action of holding. *Concept:* HCD · *Diff:* E.

Q15 → (c) 20. 1×1 squares: 3 × 4 = 12. 2×2 squares: 2 × 3 = 6. 3×3 squares: 1 × 2 = 2. Total 20. *Concept:* counting · *Diff:* M.

Q16 → (b) Only II. "All bowls are vessels" guarantees that some vessels are bowls (II). The brass vessels may or may not include bowls, so I is only possible, not certain. *Concept:* syllogism · *Diff:* M.

Q17 → (b). Unfold in reverse order. First undo the left-onto-right fold (x = 4): (6,6) reflects to (2,6). Then undo the bottom-onto-top fold (y = 4): (6,6) reflects to (6,2) and (2,6) reflects to (2,2). Four holes: (6,6), (2,6), (6,2), (2,2). *Concept:* folding · *Diff:* M.

Q18 → (a) 4. The left face is opposite the right face (3), and 7 − 3 = 4. *Concept:* dice · *Diff:* E.

Q19 → (b) 3.8 m². Real size: 4 × 50 = 200 cm = 2 m, and 3.8 × 50 = 190 cm = 1.9 m. Area = 2 × 1.9 = 3.8 m². *Concept:* scale and area · *Diff:* M.

Q20 → (b). It needs no electricity and uses passive cooling: reflection, ventilation, shading and evaporative tree cover. (a) is costly and polluting, (c) traps heat, and (d) absorbs more heat. *Concept:* sustainable, context-appropriate design · *Diff:* M.

Q21 → (b) Proximity. Elements close together are seen as a group. *Concept:* gestalt · *Diff:* E.

Q22 → (a) 3 km East. The north and south legs cancel out, leaving the 3 km east leg. *Concept:* directions · *Diff:* E.

DETAILED SOLUTIONS — PART II

Q23. Park pictograms (15 marks)

Model system

- **Grid:** a 6 cm square with a 0.5 cm margin. All figures use the same stroke weight (filled silhouettes) and the same "human" figure style: a round head and a rounded-rectangle body.
- **Colour code:**
  - blue background + white symbol = information (water, toilet, dustbin)
  - red circle and slash on white = prohibition (do not pluck flowers)
- **Designs:**
  1. *Drinking water:* a tap with three droplets falling into a cup, or a person drinking from a fountain spout.
  2. *Toilet:* standard male and female figures side by side with a vertical divider (a widely understood convention); an accessible-toilet variant can be added.
  3. *Do not pluck flowers:* a hand reaching for a flower, with a red circle and diagonal slash over it.
  4. *Dustbin:* a figure dropping an item into a bin (more active, and teaches the behaviour).

Rubric (15)

Criterion	Marks
Instant recognisability of each icon	5
Consistency (stroke, figure style, grid, colour code)	4
Simplicity and no unnecessary text	3
Correct use of prohibition convention	2
Neat execution	1

Mistakes: a different figure style in each icon; using words instead of symbols; overly detailed drawings.

Q24. Sunday vegetable market from a truck (20 marks)

Approach

- **Viewpoint:** about 2.5–3 m high, so the **horizon line sits high in the frame**. We look down on the tops of umbrellas and baskets, and nearby people show the tops of their heads. Use 2-point perspective with slightly converging verticals (a hint of a 3rd VP below), or simplified 2-point.
- **Layout:** the foreground shows the edge of the truck bed (frames the view, explains the viewpoint). The middle ground has rows of stalls with tarpaulin umbrellas, mounds of tomatoes, onions and greens, weighing scales and baskets. The background has buildings, a temple top, and wires.
- **8+ people with different actions:**
  - a vendor weighing tomatoes
  - a woman bargaining
  - a child pointing at mangoes
  - an old man with a cloth bag
  - a porter carrying a sack on his head
  - a boy on a cycle
  - a customer paying with a phone (UPI QR code on the stall)
  - a cow eating discarded leaves
  - a delivery person
- **Colour:** fresh produce colours make a natural focal point. Use warm morning light, with shadows falling to one side.

Rubric (20)

Criterion	Marks
Correct elevated viewpoint (high horizon, top views)	5
People: number, variety of actions, proportion	5
Observation of authentic market details	4
Composition & depth layers	3
Rendering / colour	3

Mistakes: drawing an eye-level view (ignoring "on a truck"); people all facing the viewer; no produce detail.

Q25. School bag + 120 words (15 marks)

(a) Drawing (8): a 3/4 view showing the main compartment, front pocket, side bottle pocket, padded straps, top handle, zippers with pullers, a brand patch and reflective strip, and the bottom panel. Add wear marks for realism (an observation bonus).

(b) Model 120-word answer:

> My school bag's biggest problem is weight distribution. All my heavy textbooks sit at the back of a single deep compartment, so the bag pulls away from my shoulders and I lean forward while walking. I would add a padded internal divider with an elastic strap to hold heavy books close to my back, plus a detachable waist belt that transfers some weight to the hips. A firm back panel with air channels would reduce sweating on hot afternoons. The side bottle pocket should also be deeper, because my bottle falls out on the bus. These changes cost little but would protect my posture every day, which matters because I carry this bag for over 200 days a year.

(About 120 words.) Structure: problem → cause → solution(s) → benefit.

Rubric: drawing accuracy 4 + detail 2 + rendering 2; writing relevance 3 + clarity 2 + word-limit discipline 2.

Q26. Toy from waste (20 marks)

Model concept: "Bottle-Cap Caterpillar Pull-Toy"

- **Materials:**
  - 6 plastic bottle caps (washed)
  - an old shoelace or string
  - a cardboard tube (head)
  - a shoebox lid cut into wheels
  - an old sock (the caterpillar's body cover)
  - buttons (eyes)
  - bottle-cap wheels on toothpick axles
- **Main drawing:** a 3/4 view of the caterpillar made of linked segments on cap wheels, with a pull string.
- **Exploded labels:** segment = cardboard ring + sock cover; wheels = caps; axle = ice-cream stick; joints = shoelace threaded through holes.
- **Play scene:** a 5-year-old pulls it across the floor; the segments wiggle. A second use: stack the caps to learn colours and counting.
- **Safety:** no small detachable parts for children under 3 (secure the buttons by stitching); round the edges; use non-toxic glue; no sharp toothpick ends (cover them with cap bits).

Rubric (20)

Criterion	Marks
Uses only household waste; materials clearly labelled	4
Play value for the age group (fun, learning)	4
Clarity of main drawing + construction	5
Play scene communicates use	3
Safety consideration	2
Originality	2

Mistakes: materials that aren't household waste (e.g. bought motors); ignoring safety for young children; no play scene.

SCORING INTERPRETATION — MODEL PAPER 2

Practice benchmarks only, not NID cutoffs.

Score	Band
78–100	**Excellent**
62–77	**Strong**
48–61	**Good**
33–47	**Needs improvement**
< 33	**Major improvement required**

Diagnostic tip: if Q24 scored low because of the viewpoint, practise 5 elevated-view sketches (from a balcony, bridge or bus top) this week.

NEXT VOLUME: Volume 8 — Model Paper 3 (`08-VOL8-MODEL-PAPER-3.md`)

VOLUME 8 — MODEL PAPER 3 (Difficulty: Moderate → Difficult)

NID DAT (B.Des) Prelims: Practice Model Paper 3
Handbook-scheme format. This practice paper was written for this package; it is not an official NID paper.
Focus mix: art history, harder spatial reasoning, expressive typography + logo, bird's-eye storytelling, packaging with dieline, service/system design.

INSTRUCTIONS

1. Time: **3 hours**. Total: **100 marks**. No negative marking.
2. Part I: Q1–14 (1 mark each), Q15–22 (2 marks each) = 30.
3. Part II: Q23 (15), Q24 (20), Q25 (15), Q26 (20) = 70.
4. Suggested timing: Part I = 25 min · Q23 = 25 · Q24 = 40 · Q25 = 28 · Q26 = 42 · buffer = 20.

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. Which artist led Fauvism, known for wild, non-natural colour?
(a) Claude Monet (b) Henri Matisse (c) Salvador Dalí (d) Edvard Munch

Q2. Patola saris of Patan are unique because they are:
(a) block printed (b) double ikat (c) embroidered with mirrors (d) woven with zari only

Q3. What is the angle between the hour and minute hands of a clock at 3:40?
(a) 120° (b) 130° (c) 140° (d) 150°

Q4. Which is NOT one of Le Corbusier's "Five Points of Architecture"?
(a) Pilotis (b) Roof garden (c) Horizontal ribbon windows (d) Flying buttress

Q5. Which word reads exactly the same when the page is turned upside down (rotated 180°)?
(a) NOON (b) MOON (c) SOON (d) BOON

Q6. How many edges does a regular octahedron (8 triangular faces, 6 vertices) have?
(a) 8 (b) 10 (c) 12 (d) 14

Q7. Srikalahasti-style Kalamkari is made mainly by:
(a) block printing (b) freehand drawing with a bamboo pen (c) tie-dye (d) machine printing

Q8. Odd one out: Futura, Gill Sans, Bodoni, Helvetica
(a) Futura (b) Gill Sans (c) Bodoni (d) Helvetica

Q9. A palette of blue, red-orange and yellow-orange is a:
(a) triadic scheme (b) split-complementary scheme (c) analogous scheme (d) monochromatic scheme

Q10. The Konark Sun Temple was built by:
(a) Rajaraja Chola I (b) Narasimhadeva I (c) Krishnadeva Raya (d) Raja Bhoja

Q11. 1, 3, 6, 10, 15, ?
(a) 20 (b) 21 (c) 22 (d) 25

Q12. The striped IBM logo is still read as solid letters because of the Gestalt principle of:
(a) Proximity (b) Closure (c) Common fate (d) Figure–ground reversal

Q13. Bauhaus : Germany :: De Stijl : ?
(a) France (b) Netherlands (c) Italy (d) Russia

Q14. Aluminium is widely used for beverage cans mainly because it is:
(a) heavy and magnetic (b) lightweight, corrosion-resistant and highly recyclable (c) transparent (d) biodegradable

Q15–Q22 (2 marks each)

Q15. A triangle has 4 straight lines drawn from its top vertex to its base (the two sides plus two internal lines). Two horizontal lines are then drawn across the triangle, parallel to the base and crossing all 4 lines. How many triangles are there in total?
(a) 12 (b) 15 (c) 18 (d) 24

Q16. An 8 × 8 square sheet is folded along the diagonal joining the top-left corner (0, 8) to the bottom-right corner (8, 0), so that the bottom-left triangle lies on the top-right triangle. A hole is punched at (7, 5). After unfolding, where is the second hole?
(a) (5, 7) (b) (1, 3) (c) (3, 1) (d) (7, 3)

Q17. A 2 × 3 × 4 cuboid is painted on all faces and cut into 24 unit cubes. How many unit cubes have exactly ONE face painted?
(a) 0 (b) 4 (c) 8 (d) 12

Q18. In a design competition, A scored more than B but less than C. D scored less than B. E scored more than C. Who came third?
(a) C (b) A (c) B (d) E

Q19. A label wraps exactly once around a cylindrical tin of radius 7 cm and height 10 cm, with 1 cm extra for overlap. What size should the label be? (π = 22/7)
(a) 44 cm × 10 cm (b) 45 cm × 10 cm (c) 22 cm × 10 cm (d) 154 cm × 10 cm

Q20. In a large hospital, first-time visitors keep getting lost. Which solution is MOST effective?
(a) More security guards
(b) Colour-coded floor lines leading to departments, with consistent pictogram signs at every decision point
(c) A long text notice at the entrance
(d) Brighter wall paint everywhere

Q21. Which statement is correct?
(a) Kerning adjusts the space across a whole line; tracking adjusts one letter pair
(b) Kerning adjusts the space between a specific letter pair; tracking adjusts it uniformly across a word or line
(c) Both terms mean line spacing
(d) Leading adjusts the letter width

Q22. A 15-tooth gear drives a 45-tooth gear. How many turns must the small gear make for the large gear to turn 4 times?
(a) 4 (b) 9 (c) 12 (d) 15

PART II — SUBJECTIVE (70 marks)

Q23. Expressive Typography + Logo (15 marks)
(a) Write each of the words **MELT, FLY, CRACK** and **ECHO** so that its appearance expresses its meaning (8 marks).
(b) Design a logo for a monsoon festival called **"BAARISH"**. Show it in colour and in single-colour (black) versions (7 marks).

Q24. Bird's-Eye Storytelling (20 marks)
A pigeon sits on the roof girder of a busy railway station and looks down. Draw what it sees in **bird's-eye perspective**. Within the crowd, include **one small story** the pigeon notices (for example, someone helping, a reunion, a chase). Make the story findable but not obvious.

Q25. Packaging with Dieline (15 marks)
Design **eco-friendly packaging for six kulhads** (clay cups) to be sent by courier. Show: (a) the flat net (dieline) with fold and cut lines, (b) the assembled package, (c) how each kulhad is protected from breaking, and (d) a simple graphic identity on the box.

Q26. Service / System Design (20 marks)
At a very large festival gathering (like the Kumbh Mela), many people, especially children and the elderly, get separated from their families.
(a) Identify **three user groups** and one problem for each (4 marks).
(b) Design a **lost-and-found system**: include an identification idea (e.g. a wristband or token), a booth design and signage (10 marks).
(c) Show in a **3-frame storyboard** how a lost child is reunited using your system (6 marks).

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	b	13	b	19	b
2	b	8	c	14	b	20	b
3	b	9	b	15	c	21	b
4	d	10	b	16	c	22	c
5	a	11	b	17	b		
6	c	12	b	18	b		

DETAILED SOLUTIONS — PART I

Q1 → (b) Matisse. Monet = Impressionism; Dalí = Surrealism; Munch = Expressionism. *Diff:* E.

Q2 → (b) double ikat. Both the warp and weft threads are resist-dyed before weaving, so the pattern appears on both sides. *Diff:* M.

Q3 → (b) 130°. $\theta = |30H - 5.5M| = |90 - 220| = 130°$. *Diff:* M.

Q4 → (d) Flying buttress. That is a Gothic element. The five points are pilotis, free plan, free façade, horizontal windows and roof garden. *Diff:* M.

Q5 → (a) NOON. A 180° rotation reverses the letter order and turns each letter upside down. N and O both look the same upside down, and NOON is the same in reverse, so it reads NOON again. MOON becomes NOOW, because M turns into W. SOON becomes NOOS, and the B in BOON doesn't survive. *Diff:* H.

Q6 → (c) 12. Euler's formula V − E + F = 2 gives 6 − E + 8 = 2, so E = 12. *Diff:* M.

Q7 → (b) freehand pen. Machilipatnam Kalamkari is the block-printed style. *Diff:* M.

Q8 → (c) Bodoni. It is a high-contrast serif (Didone) typeface; the other three are sans serif. *Diff:* M.

Q9 → (b) split-complementary. The complement of blue is orange; its neighbours red-orange and yellow-orange are used instead. *Diff:* M.

Q10 → (b) Narasimhadeva I of the Eastern Ganga dynasty, 13th century. Rajaraja Chola I built Thanjavur's Brihadeeswara. *Diff:* M.

Q11 → (b) 21. These are triangular numbers, with differences +2, +3, +4, +5, +6. *Diff:* E.

Q12 → (b) Closure. The mind fills the gaps between the stripes. *Diff:* M.

Q13 → (b) Netherlands. *Diff:* E.

Q14 → (b). Aluminium is light, does not rust and can be recycled repeatedly with large energy savings. *Diff:* E.

Q15 → (c) 18. Each triangle uses 2 of the 4 apex lines plus one horizontal "base". There are C(4,2) = 6 pairs of lines and 3 bases (the 2 crossing lines + the actual base), so 6 × 3 = 18. *Diff:* H.

Q16 → (c) (3, 1). The fold line is x + y = 8. Reflecting (x, y) across it gives (8 − y, 8 − x) = (8 − 5, 8 − 7) = (3, 1). *Check:* the midpoint of (7,5) and (3,1) is (5,3), which lies on x + y = 8 ✓. The trap is (5, 7), which comes from swapping coordinates as if the fold were y = x. *Diff:* H.

Q17 → (b) 4. One-face cubes = 2[(a−2)(b−2) + (b−2)(c−2) + (a−2)(c−2)] with a = 2, b = 3, c = 4. That is 2[0 × 1 + 1 × 2 + 0 × 2] = 2 × 2 = 4. *Check:* corners 8 + two-face 4[(0) + (1) + (2)] = 12 + one-face 4 + inside 0 = 24 ✓. *Diff:* H.

Q18 → (b) A. The order is E > C > A > B > D. *Diff:* M.

Q19 → (b) 45 cm × 10 cm. Circumference = 2 × 22/7 × 7 = 44 cm; adding 1 cm of overlap gives 45 cm. Height 10 cm. *Diff:* M.

Q20 → (b). A continuous guiding cue (floor lines) plus signs at the decision points matches how people navigate. (c) overloads memory; (a) and (d) don't solve wayfinding. *Diff:* M.

Q21 → (b). *Diff:* E.

Q22 → (c) 12. The large gear's 4 turns pass 4 × 45 = 180 teeth; 180 ÷ 15 = 12 turns of the small gear. *Diff:* M.

DETAILED SOLUTIONS — PART II

Q23. Expressive typography + BAARISH logo (15 marks)

(a) Model treatments (2 marks each)

- **MELT:** bold letters whose lower halves drip and pool into a puddle; the "T" bends.

- **FLY:** light letters rising diagonally upward; the "Y" becomes a bird's wing shape, with small motion lines.
- **CRACK:** solid block letters split by a jagged crack running through all five; small fragments falling.
- **ECHO:** the word repeated 3–4 times, each repetition lighter and smaller, receding (a fading repeat).

(b) Logo "BAARISH" (7 marks)

- *Concept:* the two **A**s become two **raindrops** (inverted teardrops with the crossbar as a ripple). Alternatively, the dot of the **i** becomes a falling drop that splashes into ripple lines under the word.
- *Colour version:* monsoon teal and deep indigo, with a fresh green accent.
- *Single-colour version:* all black on white, to prove it works without colour.
- *Typeface:* a rounded sans for friendliness.

Rubric: (a) 8 = 4 words × 2 (meaning conveyed + legibility). (b) 7 = concept 3 + execution 2 + mono version working 2.

Mistakes: illegible distortions; simply colouring the words instead of transforming their form; a logo that collapses in black and white.

Q24. Pigeon's-eye railway station (20 marks)

Approach

- **Perspective:** 3-point, looking down. The third VP is below the page, the platforms converge, and we see the tops of heads, umbrellas, luggage, the train roof and the overhead wires in the foreground (the pigeon's girder can frame the top edge).
- **Crowd:** 25+ simplified figures shown as head-and-shoulder ovals from above, with long luggage shadows (for example, late-afternoon light).
- **Hidden story options:**
  - a child's balloon escaping, with someone reaching for it
  - a porter helping an elderly woman with a heavy trunk across the footbridge
  - two friends running towards each other for a reunion
  - a pickpocket being chased by a policeman, shown as a line of displaced people
- **Findability:** place the story at a thirds intersection; give the main characters a single accent colour (e.g. a red dupatta) while the rest of the crowd is muted.

Rubric (20)

Criterion	Marks
Correct bird's-eye perspective	6
Crowd handling & variety	4
Hidden story: clarity & cleverness	4
Composition & use of accent	3
Station details (tracks, platform, signs, train roof)	3

Mistakes: drawing an eye-level view; making the story so central that it isn't "hidden"; repeating identical figures.

Q25. Kulhad packaging (15 marks)

Model concept: "Matti-Pack"

- **Structure:** a corrugated-cardboard box (recycled kraft) with an internal **die-cut honeycomb insert**: 6 conical cut-outs arranged 3 × 2 that cradle each kulhad upside down. The kulhads' rims rest on a **molded-pulp (egg-tray type) bed** or on crumpled newspaper/rice straw.
- **Dieline:** the box net shows solid lines = cut, dashed lines = fold, glue flaps marked; a separate insert net with 6 circular cut-outs and slit tabs that lock into the walls.
- **Assembled view:** 3/4 view with the lid open, showing the kulhads nested.
- **Protection:** each kulhad is isolated (no clay-on-clay contact); there is a 1.5 cm air gap from the walls; a "FRAGILE – this side up" pictogram; no plastic.
- **Graphic identity:** a terracotta-orange band with a simple line icon of a kulhad and steam, a handmade-craft story line ("Made by the potters of Khurja"), and a recycle symbol.
- **Reuse:** the insert doubles as a 6-cup serving tray.

Rubric (15)

Criterion	Marks
Correct, readable dieline	4
Protection logic	4
Sustainability (no plastic, reuse)	3
Graphic identity	2
Clarity of assembled drawing	2

Mistakes: bubble wrap (contradicts eco-friendly); a dieline that cannot fold into the shown box; ignoring breakage risk.

Q26. Festival lost-and-found system (20 marks)

(a) User groups & problems (4)

1. **Young children:** cannot state their address or phone number.
2. **Elderly people:** get confused, may have poor hearing or memory, may not carry a phone.
3. **Rural first-time visitors:** language barriers; unfamiliar with announcements and digital systems.
4. *(optional)* **Volunteers/police:** need quick identification without a common language.

(b) System design (10)

- **ID: "Milan Band"** (milan = reunion). At the entry gates, families receive matching coloured wristbands. Each band carries a family number printed in large digits, plus a QR code linked to the guardian's phone number. A colour indicates the sector of the camp where the family is staying. The bands are tear-resistant Tyvek and waterproof.
- **Booths:** tall, visible "Milan Kendra" towers every 500 m, with a beacon light and a large handshake icon. Each has a seating area, water and a public-address system; volunteers scan the band's QR code, which sends an SMS to the guardian; a loudspeaker announcement is made in 3 languages.
- **Signage:** a consistent icon (two hands joining) in bright yellow-and-black, directional arrows at intersections ("Milan Kendra 200 m →"), and floor stencils.
- **Low-tech backup:** if the phone network fails, families have pre-agreed meeting points (numbered pillars).

(c) 3-frame storyboard (6)

1. A child, crying, is separated in the crowd; a volunteer notices the yellow band.
2. At the Milan Kendra, the volunteer scans the band; the mother's phone buzzes with "Your child is at Milan Kendra 12".
3. A reunion hug at the booth; the band numbers match; a calm environment.

Rubric (20): problems 4; ID design 3; booth 3; signage 2; low-tech/backup thinking 2; storyboard clarity 4; overall feasibility 2.

Mistakes: an app-only solution (ignores the elderly, rural users and network failure); unclear storyboard continuity; no signage.

SCORING INTERPRETATION — MODEL PAPER 3

Practice benchmarks only, not NID cutoffs.

Score	Band
75–100	**Excellent**
60–74	**Strong**
45–59	**Good**
30–44	**Needs improvement**
< 30	**Major improvement required**

Diagnostic tip: Q15–17 are the hardest spatial items in the papers so far. If you missed 2 or more, redo Vol 4B §C–E.

NEXT VOLUME: Volume 9 — Model Paper 4 (`09-VOL9-MODEL-PAPER-4.md`)

VOLUME 9 — MODEL PAPER 4 (Difficulty: Difficult) · 2026-STYLE ALL-SUBJECTIVE FORMAT

NID DAT (B.Des) Prelims: Practice Model Paper 4
*This paper mirrors the **reported structure of the actual DAT 2026 Prelims (Dec 2025)**: no objective section; 6 long questions with sub-parts (13 tasks here) in 3 hours. That structure is [REPORTED] from candidate memory and is not officially confirmed. This practice paper was written for this package; it is not an official NID paper.*

INSTRUCTIONS

1. Time: **3 hours (180 min)**. Total: **100 marks**. Attempt **all** questions.
2. Use one A4 sheet (or answer-booklet page) per question. Pencil, colour pencils, pastels and pens are allowed.
3. Read every question fully before starting. Note the counts, viewpoints and users.
4. Suggested timing (≈ 1.65 min per mark, plus a 10-minute buffer):

Q	Marks	Time
Q1	18	30 min
Q2	18	30 min
Q3	14	23 min
Q4	16	26 min
Q5	18	30 min
Q6	16	26 min
Reading + buffer	—	15 min

QUESTIONS

Q1. Observation, Perspective & Critique (18 marks)
(a) Draw the inside of a **crowded city bus on a rainy morning**, as seen from the **conductor's seat near the rear door**. Show at least **10 passengers** (12 marks).
(b) In your drawing, mark and number **three design problems** you can see (3 marks).
(c) In **two lines each**, suggest a fix for any **two** of them (3 marks).

Q2. Sequential Storytelling (18 marks)
In **five frames**, show what happens when a **street vendor's cart breaks a wheel in the middle of a busy road**: how people around respond, and how the situation is resolved. The **last frame** must show a **design improvement** that would prevent or reduce the problem in future.

Q3. Empathy Kit (14 marks)
(a) Draw, as a neat **flat-lay (knolled) arrangement**, everything a **visually impaired Class 9 student** needs for a school day. Label each item (8 marks).
(b) Choose **one** item and redesign it to work better for this student. Show the redesign with labels (6 marks).

Q4. Food, Culture & Writing (16 marks)
(a) Illustrate a **regional festival dish being prepared** (for example: modak, pitha, puran poli, sarson ka saag, appam). Show the dish, the cooking vessel, the key ingredients and the hands of the person cooking (10 marks).
(b) In about **150 words**, describe this dish to someone who has never seen or tasted it (6 marks).

Q5. Visual Communication System (18 marks)
Your town is introducing a **ban on single-use plastic bags at its weekly market (haat)**.
(a) Design **four signboards** for the market that work as a consistent system (8 marks).
(b) Design a **poster** announcing the ban (6 marks).
(c) Design a graphic for a **reusable cloth bag** that the municipality will distribute (4 marks).

Q6. Imagination (16 marks)
(a) **Object completion.** You are given this starting shape: *a circle with a short vertical line rising from its top*. Complete it into **three different objects**, one per box (6 marks).
(b) **"What if the footpaths of your city could change shape depending on who walks on them?"** Draw one scene that shows this idea working for at least **three different users** (10 marks).

ANSWER KEY (Compact: what must appear for full credit)

Q	Must-have elements
1(a)	Rear-door eye level; 1-point perspective down the aisle; 10+ passengers (standing, holding grab handles, seated); umbrellas dripping, wet floor, fogged windows, raincoats; the conductor's ticket machine
1(b)	Three numbered problems visible in the drawing (e.g. wet slippery floor, grab handles out of reach of short passengers, no umbrella storage, a high entry step)
1(c)	Two specific, feasible fixes (drainage-grooved anti-slip floor; hanging straps at two heights; umbrella sleeves or holders)
2	5 frames with continuity; escalating then resolving traffic; cooperation shown; frame 5 = design improvement (spare-wheel clamp, a stronger axle, a quick-jack kit, a wheel guard)
3(a)	10–14 labelled items (white cane, Braille slate and stylus, Braille books, audio player / screen-reader phone, tactile ruler, talking calculator, tiffin with tactile markers, water bottle with a textured grip, ID card with Braille, wide-ruled notebook for low vision, magnifier if low-vision) in a neat grid
3(b)	One redesigned item with labels: a clear user benefit, tactile or audio features
4(a)	The dish shown being prepared (e.g. modak being pleated), a vessel (steamer / ukadiche patra), ingredients labelled (rice flour, jaggery, coconut, cardamom), hands in action, colour, appetite appeal
4(b)	~150 words covering appearance, texture, taste, aroma, the festival context and how it is eaten; vivid sensory language
5(a)	4 consistent signs (shape, colour code, icon style); bilingual; e.g. "No plastic bags", "Cloth bag counter →", "Return & reuse point", "Fine ₹___ for plastic"; placement shown
5(b)	Strong headline + one big visual idea + date/venue + call to action
5(c)	A bag graphic that people want to reuse: local identity, a durable 1–2 colour print, a message
6(a)	Three distinct, recognisable objects from different categories (e.g. an apple with a stem, a wall clock on its hook, a cartoon bomb with a fuse, a magnifying glass, a lollipop)
6(b)	A coherent speculative scene; 3+ users with different needs (wheelchair user → ramp forms; child → playful bumps or guidance; elderly → handrails rise; blind → tactile strips); mechanism hinted; perspective

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DETAILED SOLUTIONS & MODEL APPROACHES

Q1 (18 marks)

(a) Drawing approach (12 marks)

- **Viewpoint:** the conductor sits near the rear door, at seated eye level (about 1.1 m), looking forward up the aisle. Use one-point perspective with the VP near the front windscreen. The ceiling and its grab rails converge toward the VP, and so do the floor lines and seat rows.
- **Passengers (10+):**
  - a standing office-goer holding a strap with a dripping umbrella
  - a schoolgirl with a wet raincoat
  - an elderly man seated in the reserved seat
  - a mother holding a toddler
  - a man with a vegetable sack
  - two college students sharing earphones
  - a woman wringing her dupatta
  - a man reading a phone
  - a passenger on the footboard
  - the driver's back in the distance
- **Rain details:**
  - fogged windows with trickles
  - a puddle on the floor
  - folded umbrellas dripping
  - wet footprints
  - plastic covers over bags
  - the grey light outside
- **Composition:** strong converging lines lead to the windscreen; foreground figures overlap the frame edges, which creates a sense of crowding.

(b) Three problems (3 marks). Example set:

1. Wet, slippery floor near the door, a fall risk.
2. Overhead grab rails too high for short passengers and children.
3. No place for wet umbrellas, so passengers drip onto others.
   Others that count: a high first step; ticket counting in a crush; no priority-seat signage visible.

(c) Two fixes (3 marks):

- "Fit a grooved, anti-slip rubber floor near the doors that drains water into side channels."
- "Add hanging straps at two heights (1.6 m and 1.8 m) so children and shorter adults can hold on."
- *Alternatives:* umbrella sleeves (a dispenser of recyclable sleeves at the door) or a vertical umbrella rack; a kneeling or low-floor bus.

Rubric: (a) perspective 4 + crowd/proportion 4 + rain observation 2 + composition 2; (b) 1 per valid, visible problem; (c) 1.5 per specific, feasible fix.

Common mistakes: drawing from the front; fewer than 10 people; problems that are not visible in the drawing; vague fixes ("make it better").

Q2 (18 marks)

Model 5-frame plan

Frame	Shot	Content
1	Wide	A vendor pushes a fruit cart across a busy junction; vehicles all around
2	Medium	Crack! A wheel snaps; bananas roll onto the road; the cart tilts; horns blare
3	Wide	Traffic jams; an auto driver gets out to help; a student picks up the fruit; a traffic policeman signals the vehicles
4	Medium	Three people lift the cart to the footpath; a mechanic from a nearby cycle shop brings a spare wheel; the vendor's relieved face
5	Close-up + inset	**Design improvement:** the redesigned cart has a **built-in fold-down kick-stand jack** and a **spare-wheel clip** under the base, plus **thicker solid-rubber wheels on a steel-hub axle**. A label reads: "Jack + spare: 2 min fix, no road block."

Rubric (18): narrative logic 4; continuity (same vendor, cart and location) 3; shot variety and composition 3; emotion and community response 3; design improvement quality 4; finish 1.
Mistakes: a final frame without a design idea; inconsistent characters; overcrowded, unreadable frames.

Q3 (14 marks)

(a) Knolling layout (8 marks). Arrange the items on a clean grid, all aligned at 90°, with labels below:

1. White cane (foldable)
2. Braille slate and stylus
3. Braille textbooks
4. Smartphone with screen reader + earphones
5. Tactile geometry kit (raised-line ruler, protractor)
6. Talking calculator
7. Tiffin with tactile-dot lids (to identify the compartments)
8. Water bottle with a textured grip band
9. Braille ID card on a lanyard
10. Bag with distinct textured pockets
11. Abacus (Cranmer type)
12. Bold-line notebook + magnifier (if low vision)

Rubric: item relevance 4, layout neatness 2, labels 2.

(b) Redesign (6 marks), model: "Feel-Find School Bag"

- Each pocket has a different texture patch: ridges = books, dots = tiffin, smooth = electronics.
- A **sound tag** on the zipper pull helps locate the bag; a magnetic flap replaces fiddly zips on the front pocket.
- Reflective strips for low-vision students and for safety.

Rubric: user insight 2, feature clarity 2, drawing and labels 2.

Mistakes: treating every visually impaired student as totally blind (low vision exists, so the magnifier and bold lines count); a cluttered layout.

Q4 (16 marks)

(a) Illustration (10 marks), example: ukadiche modak for Ganesh Chaturthi

- **Scene:** a close-up of a kitchen counter. One pair of hands pleats a modak around the filling, while finished modaks sit in a steel steamer plate lined with banana leaf.
- **Labelled ingredients in small bowls:** rice-flour dough, grated coconut + jaggery filling, cardamom, ghee. The steamer (ukadiche patra) gives off steam.

- **Composition:** the hands and modak at a thirds point, with the ingredients arranged in an arc around them.
- **Colour:** warm creams, golden jaggery and banana-leaf green.

Rubric: accuracy of the dish and process 3; hands in action 2; ingredients labelled 2; composition 2; appetite appeal and colour 1.

(b) Model ~150-word description:

> Ukadiche modak is a small steamed dumpling made for Ganesh Chaturthi in Maharashtra. It looks like a white garlic bulb or a closed lotus bud, with about eleven neat pleats meeting at a pointed tip. The outer shell is made from rice flour cooked into a soft dough, so it is thin, smooth and slightly sticky. Inside is a warm, golden filling of freshly grated coconut cooked with jaggery and a pinch of cardamom, which smells sweet and earthy. When you bite into it, the soft shell gives way to a juicy, gently crunchy centre. Families serve it hot, with a spoon of ghee poured into the opened top. Making modaks is a family ritual: grandmothers teach children to pleat them, because a perfect modak is a sign of patience and care. It is believed to be Lord Ganesha's favourite sweet.

(About 150 words.)

Rubric: sensory detail 2, clarity for a newcomer 2, cultural context 1, word limit 1.

Q5 (18 marks)

(a) Signage system (8 marks)

- **System rules:** 60 × 40 cm boards on bamboo poles (local and cheap); a green border = positive action, a red circle-slash = prohibition; a uniform icon style (thick line); bilingual (local language + English).
- **Four signs:**
    1. "No plastic bags": a bag with a red slash.
    2. "Cloth bag counter →": a cloth-bag icon with an arrow.
    3. "Bring your bag. Save ₹5": a bag + coin icon.
    4. "Plastic return point": a bin with returning arrows (for collecting old bags).
- **Placement sketch:** at the haat entrance and at the decision points between stall rows.

(b) Poster (6 marks), concept: a vegetable seller's basket shaped like the Earth, with produce on top and a cloth bag strap as the handle. Headline: "**Haat goes plastic-free from 1 January.**" Sub-line: "Carry a cloth bag. Every bag counts." A municipality logo and a QR code for details. Colours: green, earth brown, white.

(c) Cloth bag graphic (4 marks): a one-colour screen-print of a stylised market scene (stall umbrellas and baskets) in a block-print style inspired by local textiles, with the tagline "Mera Thaila, Meri Shaan" (My bag, my pride). It is durable, appealing, and has a local identity people are proud to carry.

Rubric: (a) consistency 3 + clarity 3 + placement 2; (b) concept 3 + hierarchy 2 + details 1; (c) appeal 2 + reusability logic 2.

Q6 (16 marks)

(a) Object completion (6 marks). Three objects from different categories score better than three similar ones. Examples:

1. **Apple** (a circle with a stem and a leaf added)

2. **Wall clock** (the line becomes the hanging hook; numbers and hands inside)
3. **Cartoon bomb with a lit fuse**
   Also acceptable: a balloon (with the line as the string, flipped), a magnifying glass (with the line as the handle), a lollipop, an onion, a planet with an antenna (a satellite), a cherry, a musical note, a sitar gourd.

Rubric: 2 per object = recognisability 1 + originality or category spread 1.

(b) Speculative footpath scene (10 marks)

- **Concept:** smart modular pavement tiles that sense the user (via a footstep pattern, a cane tap or a wheel load) and reshape.
- **Users and responses:**
  - **wheelchair user:** the kerb tiles slide down into a ramp; rough tiles flatten
  - **visually impaired person:** raised tactile guidance lines appear and lead to the crossing
  - **elderly woman:** a handrail rises from the edge; a bench folds out at intervals
  - **child:** a playful hopscotch pattern lights up, but only in the safe zone
- **Drawing:** a street in 2-point perspective; callouts with arrows explain each transformation; a small inset shows the tile mechanism (pistons or modules).

Rubric: idea coherence 3, users' needs addressed 3, perspective and scene quality 2, explanation and callouts 2.

Mistakes: an imaginative scene that doesn't show a single user benefit; vague magic without even a hinted mechanism.

SCORING INTERPRETATION — MODEL PAPER 4

Practice benchmarks only, not NID cutoffs.

Score	Band	Note
72–100	**Excellent**	You can handle the 2026-style format
58–71	**Strong**	Polish time management
45–57	**Good**	Work on finishing all 13 tasks
30–44	**Needs improvement**	Speed drills: 12-minute sketches
< 30	**Major improvement required**	Revisit Vol 3 fundamentals

Key metric for this format: *completion rate*. Track how many of the 13 tasks you fully completed. The target is 13/13 with at least 70% polish each.

NEXT VOLUME: Volume 10 — Model Paper 5 (Full Mock Simulation) (`10-VOL10-MODEL-PAPER-5-FULL-MOCK.md`)

VOLUME 10 — MODEL PAPER 5 · FULL MOCK EXAM SIMULATION (Toughest Realistic Paper)

NID DAT (B.Des) Prelims: Full-Length Practice Mock (Model Paper 5)
Handbook-scheme format (22 objective + 4 subjective). This practice paper was written for this package; it is not an official NID paper.
*Because the real DAT 2026 paper was reportedly all subjective, take **Model Paper 4** as your second full simulation as well. Between them, you cover both possible formats.*

MOCK-EXAM PROTOCOL (follow it exactly)

When: a Sunday, **10:00 am – 1:00 pm**, the same slot as the real Prelims (last cycle's timing was 10 am–1 pm [REPORTED]).
Where: a quiet table; phone in another room; a wall clock or a simple watch.
Kit: exactly what you'll carry on exam day (Vol 1 §8.1).
Rules:

1. No breaks, no references, no music.
2. Start the timer at 10:00. Pens down at 13:00 sharp.
3. Use the Answer Sheet below for Part I. Use 4 separate A4 sheets for Part II, labelled Q23–Q26.
4. Afterwards, wait 30 minutes, then evaluate using the Answer Key and the Self-Evaluation System. (Better still, ask an art teacher or a design student to score Part II using the rubrics.)

INSTRUCTIONS (as printed on the paper)

1. Duration: **3 hours**. Maximum marks: **100**.
2. **Part I (Q1–22):** objective. Q1–14 carry **1 mark** each; Q15–22 carry **2 marks** each. **No negative marking.** Mark one option per question.
3. **Part II (Q23–26):** subjective. Marks: Q23 = 15, Q24 = 20, Q25 = 15, Q26 = 20. Answer in the space provided. Follow every count and constraint in the question.
4. You may use pencils, colour pencils, crayons, oil pastels and sketch pens. Do not use any electronic device.

SUGGESTED SECTION TIMING

Clock time	Task	Minutes
10:00–10:05	Read the entire paper; jot idea keywords beside Q23–26	5
10:05–10:30	Part I (skip anything over 90 seconds; return later)	25
10:30–10:58	Q23	28
10:58–11:36	Q24	38
11:36–12:04	Q25	28
12:04–12:42	Q26	38
12:42–13:00	Buffer: unanswered MCQs, labels, colour on your best answer	18

ANSWER SHEET — PART I (copy onto paper or print)

```
Q   a b c d       Q   a b c d       Q   a b c d
1   ○ ○ ○ ○       9   ○ ○ ○ ○       17  ○ ○ ○ ○
2   ○ ○ ○ ○       10  ○ ○ ○ ○       18  ○ ○ ○ ○
3   ○ ○ ○ ○       11  ○ ○ ○ ○       19  ○ ○ ○ ○
4   ○ ○ ○ ○       12  ○ ○ ○ ○       20  ○ ○ ○ ○
5   ○ ○ ○ ○       13  ○ ○ ○ ○       21  ○ ○ ○ ○
6   ○ ○ ○ ○       14  ○ ○ ○ ○       22  ○ ○ ○ ○
7   ○ ○ ○ ○       15  ○ ○ ○ ○
8   ○ ○ ○ ○       16  ○ ○ ○ ○       Part I score: ____ / 30
```

ANSWER SHEET — PART II (one A4 per question)

```
+-------------------------------------------------------------+
| Q__  | Time started: ____  Time ended: ____                 |
|-------------------------------------------------------------|
|                                                             |
|                    [ drawing / writing area ]               |
|                                                             |
|-------------------------------------------------------------|
| Self-score: ___ / ___   Evaluator score: ___ / ___          |
+-------------------------------------------------------------+
```

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. The famous miniature "Bani Thani" belongs to which school of painting?
(a) Kangra (b) Kishangarh (c) Mewar (d) Basohli

Q2. A clock shows 10:25. What time does its vertical-mirror image show?
(a) 1:35 (b) 2:35 (c) 1:25 (d) 2:25

Q3. How many letters of the word **DESIGN** are symmetric about a horizontal axis (i.e. look the same in a water image)?
(a) 2 (b) 3 (c) 4 (d) 5

Q4. The Brihadeeswara Temple at Thanjavur was built under:
(a) Rajendra Chola I (b) Rajaraja Chola I (c) Narasimhavarman II (d) Krishnadeva Raya

Q5. The typeface designed in 1916 for the London Underground is:
(a) Gill Sans (b) Johnston (c) Helvetica (d) Futura

Q6. Mixing equal amounts of yellow and violet paint gives approximately:
(a) bright green (b) a neutral greyish-brown (c) orange (d) pink

Q7. Odd one out (think: architecture awards): Zaha Hadid, Kazuyo Sejima, Anne Lacaton, Gae Aulenti
(a) Zaha Hadid (b) Kazuyo Sejima (c) Anne Lacaton (d) Gae Aulenti

Q8. How many diagonals does a decagon (10 sides) have?
(a) 20 (b) 35 (c) 40 (d) 45

Q9. 2, 6, 12, 20, 30, 42, ?
(a) 52 (b) 54 (c) 56 (d) 60

Q10. Which Indian site was inscribed on the UNESCO World Heritage List in 2024?
(a) Santiniketan (b) Moidams of Charaideo (c) Ramappa Temple (d) Dholavira

Q11. Modulor : Le Corbusier :: Vitruvian Man : ?
(a) Michelangelo (b) Leonardo da Vinci (c) Vitruvius only (d) Raphael

Q12. At a certain time, a 2 m pole casts a 3 m shadow. How long is the shadow of a 6 m pole at the same time and place?
(a) 4 m (b) 6 m (c) 9 m (d) 12 m

Q13. Which is the clearest example of poka-yoke (mistake-proofing)?
(a) A decorative handle (b) A SIM-card tray with a cut corner that fits only one way (c) A brighter screen (d) A bigger logo

Q14. Traditional Ajrakh printing is identified by which colour pair?
(a) indigo and madder red (b) pink and lime (c) gold and silver (d) purple and orange

Q15–Q22 (2 marks each)

Q15. How many squares of all sizes are there in a 4 × 5 grid of unit squares (4 rows, 5 columns)?
(a) 30 (b) 36 (c) 40 (d) 50

Q16. This net folds into a cube. Which face is opposite face 1?

```
[1][2]
   [3][4]
      [5][6]
```

(a) 3 (b) 4 (c) 5 (d) 6

Q17. An ant walks on the **outside surface** of a solid cube of edge 1 m, from one corner to the diagonally opposite corner (the far corner of the cube). What is the length of the shortest possible path?
(a) √2 m ≈ 1.41 m (b) √3 m ≈ 1.73 m (c) √5 m ≈ 2.24 m (d) 3 m

Q18. Ira, Jai, Kavi and Lata each designed one different object: a lamp, a chair, a kettle and a clock. Jai designed the chair. Ira designed neither the lamp nor the clock. Kavi designed the clock. Who designed the lamp?
(a) Ira (b) Jai (c) Kavi (d) Lata

Q19. A 1 : 20 scale model of a cylindrical water tank holds 0.5 litres. How much does the real tank hold?
(a) 10 L (b) 200 L (c) 400 L (d) 4,000 L

Q20. At metro stations, boarding passengers crowd the centre of the doors and block those trying to exit. Which design intervention is MOST effective?
(a) Louder announcements only
(b) Floor markings showing waiting zones on both sides of each door and an "exit lane" arrow in the middle
(c) Fewer trains
(d) Removing seats from the platform

Q21. Statements: All ceramics are fired. Some fired objects are glazed. No glazed object is porous. Conclusions: I. Some ceramics are not porous. II. Some fired objects are not porous.
(a) Only I follows (b) Only II follows (c) Both follow (d) Neither follows

Q22. In an eye-level street drawing (the viewer is a standing adult), an adult in the distance has their head exactly on the horizon line. A child half the adult's height stands at the same distance. Where should the child's head be drawn?

(a) On the horizon line
(b) Above the horizon line
(c) Halfway between the horizon line and the point where the child's feet touch the ground
(d) At the vanishing point

PART II — SUBJECTIVE (70 marks)

Q23. Critique & Redesign (15 marks)
The monobloc plastic chair is found everywhere in India.
(a) Draw it **from memory** in a 3/4 view (5 marks).
(b) List **three strengths** and **three weaknesses** of its design (4 marks).
(c) Redesign it for use in a **primary-school classroom**. Show the redesign with labels (6 marks).

Q24. One Place, Three Times (20 marks)
Draw the **same street corner at 6 am, 1 pm and 9 pm** in three frames. Show how its **activities, light, users and mood** change. The viewpoint must be **identical** in all three frames.

Q25. Identity Design (15 marks)
"**Kala Setu**" is a new platform connecting **rural artisans** with **city buyers**.
(a) Design its **logo** (7 marks).
(b) Apply the logo to a **shopping bag** and a **mobile-app icon** (4 marks).
(c) Explain your concept in **about 40 words** (4 marks).

Q26. Designing for Heatwaves (20 marks)
Heatwaves make afternoons dangerous for **delivery riders, traffic police and street vendors**.
(a) Choose **one** user group. Show their day as a **journey map** with at least **four pain points** (6 marks).
(b) Design a **solution system** (a product and a service) for this group (10 marks).
(c) Draw **one "after" scene** showing your solution in use (4 marks).

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	d	13	b	19	d
2	a	8	b	14	a	20	b
3	b	9	c	15	c	21	b
4	b	10	b	16	b	22	c
5	b	11	b	17	c		
6	b	12	c	18	d		

DETAILED SOLUTIONS — PART I

Q1 → (b) Kishangarh. Painted by Nihal Chand, 18th century. *Diff:* M.

Q2 → (a) 1:35. 11:60 − 10:25 = 1:35. *Check:* the minute hand at "5" (25 min) mirrors to "7" (35 min); the hour hand at about 10.4 mirrors to about 1.6, i.e. between 1 and 2 ✓. *Diff:* M.

Q3 → (b) 3. D, E and I are symmetric about a horizontal axis. S is not (it has rotational symmetry only); G and N are not. *Diff:* M.

Q4 → (b) Rajaraja Chola I (c. 1010 CE). Rajendra Chola I built Gangaikonda Cholapuram. *Diff:* M.

Q5 → (b) Johnston, by Edward Johnston. Gill Sans (1928) was designed by his former student Eric Gill. *Diff:* M.

Q6 → (b) neutral greyish-brown. Complementary pigments neutralise each other. *Diff:* M.

Q7 → (d) Gae Aulenti. The other three are Pritzker laureates (Hadid 2004, Sejima 2010 with Nishizawa, Lacaton 2021 with Vassal). Aulenti was a distinguished Italian architect and designer (the Musée d'Orsay conversion) but did not win the Pritzker. *Diff:* H.

Q8 → (b) 35. $n(n-3)/2 = 10 \times 7 / 2 = 35$. *Diff:* M.

Q9 → (c) 56. The differences are 4, 6, 8, 10, 12, so the next is +14: 42 + 14 = 56. (The terms follow n(n + 1).) *Diff:* E.

Q10 → (b) Moidams of Charaideo (Assam, 2024). Santiniketan was inscribed in 2023; Ramappa and Dholavira in 2021. *Diff:* M.

Q11 → (b) Leonardo da Vinci. Each is a proportion study by its creator. Vitruvius wrote the text that Leonardo illustrated, but the drawing is Leonardo's. *Diff:* M.

Q12 → (c) 9 m. At the same time and place, shadow length is proportional to height: a 3 : 2 ratio, so 6 × 1.5 = 9. *Diff:* E.

Q13 → (b). The geometry physically prevents a wrong insertion. *Diff:* E.

Q14 → (a) indigo and madder red. *Diff:* M.

Q15 → (c) 40. 1×1 squares: 4 × 5 = 20. 2×2: 3 × 4 = 12. 3×3: 2 × 3 = 6. 4×4: 1 × 2 = 2. Total 40. *Diff:* M.

Q16 → (b) 4. Fold with face 3 as the base:

- Face 2 (above 3 in the net) becomes the north wall.
- Face 1 (left of 2) folds down to become the west wall.
- Face 4 (right of 3) becomes the east wall.
- Face 5 (below 4) folds around the south-east corner edge to become the south wall.
- Face 6 (right of 5) folds over to become the top.

So the opposite pairs are 3–6, 2–5 and **1–4**. *Diff:* H. (The 2-2-2 "staircase" is one of the 11 valid cube nets.)

Q17 → (c) √5 ≈ 2.24 m. Unfold two adjacent faces into a flat 2 × 1 rectangle; the shortest path is its diagonal, $\sqrt{(2^2 + 1^2)} = \sqrt{5}$. Option (b), √3, is the straight line *through* the cube, which an ant on the surface cannot take. Walking along three edges gives 3 m. *Diff:* H.

Q18 → (d) Lata. Jai = chair and Kavi = clock. Ira cannot have the lamp or the clock, so Ira = kettle. That leaves Lata = lamp. *Diff:* M.

Q19 → (d) 4,000 L. Volume scales by $20^3 = 8{,}000$, and 0.5 × 8,000 = 4,000 L. *Diff:* M.

Q20 → (b). It is a visual nudge at the exact point of action, separating the two flows. *Diff:* M.

Q21 → (b) Only II. Glazed objects are fired (they are among "some fired objects") and are not porous, so some fired objects are not porous (II ✓). We cannot tell whether any glazed fired objects are ceramics, so I is not certain. *Diff:* H.

Q22 → (c). At a given distance, height scales uniformly between the ground point and the eye level. An adult's head reaches the horizon (the viewer's eye height), so a half-height child's head reaches halfway between the ground contact point and the horizon. *Diff:* H. (This is a very useful rule for drawing crowds of mixed ages.)

DETAILED SOLUTIONS — PART II (model approaches + rubrics)

Q23. Monobloc chair critique & redesign (15)

(a) Memory drawing (5):

- a one-piece moulded form
- a backrest with horizontal slots or a fan pattern
- splayed legs, wider at the bottom, which allows stacking
- curved armrests flowing into the seat
- a slight seat dip
- moulding ribs under the seat (an observation bonus)

Rubric: form accuracy 2, proportion 1, details 1, line quality 1.

(b) Strengths and weaknesses (4).

- *Strengths:* cheap; lightweight; stackable; weatherproof and easy to clean; made in one moulding.
- *Weaknesses:* legs can crack or splay under load (safety); it gets hot in the sun; it is not ergonomic for long sitting; it is not easily recyclable once it has UV damage; it looks visually monotonous; it slides on smooth floors.

Rubric: 4 valid points minimum (about 0.67 each, rounded).

(c) Redesign for a primary classroom (6), model: "Grow-Seat"

- Sized for children aged 6–10: seat height 32–35 cm.
- **Reversible base:** flipped one way it gives a 32 cm seat; flipped the other it gives 36 cm, so it grows with the child.
- A **rounded-edge** recycled-plastic shell with **non-slip rubber feet**; a **hook at the back** for the school bag; a **cut-out handle** in the backrest so children can lift and carry it easily.
- **Rocking prevention:** wide-stance feet. **Stackable** to clear the floor for activities.
- **Colours:** muted primaries that can be colour-coded by group.

Rubric: child ergonomics 2, classroom-specific features 2, drawing and labels 2.

Q24. One street corner at three times (20)

Plan. Fixed viewpoint: a 2-point perspective of a corner with a tea stall, a medical shop, a bus stop, a tree, a streetlight and a temple gate.

	6 am	1 pm	9 pm
Light	Low warm sun from the east; long shadows pointing west; mist	Harsh overhead sun; short, dark shadows under objects	Artificial light: streetlight cones, shop glow, headlights; deep blue sky

	6 am	1 pm	9 pm
Activities	Milkman, newspaper boy, morning walkers, the tea stall opening, temple bells	Office crowd at the stall, a delivery rider sheltering under the tree, a vendor's umbrella, few walkers (heat)	Families eating street food, the medical shop busy, a security guard, a dog sleeping, shutters half-down
Mood / colours	Fresh pale yellows and pinks	Bleached, high contrast, hot	Warm pools of light against a cool dark

Execution tips.

- Lightly trace the same construction lines in each frame (or draw the outlines with a scale for consistency).
- Change only the light, the people and the props.
- Shadow direction must follow the sun: pointing west in the morning, short at noon, and spreading out from the lamps at night.

Rubric (20).

Criterion	Marks
Identical viewpoint and consistency	4
Light logic (sun angle, shadows, artificial light)	5
Activity and user change: observation richness	5
Mood via colour / value	3
Drawing quality	3

Mistakes: the viewpoint drifts between frames; the same people appear in all three frames; ignoring shadows.

Q25. "Kala Setu" identity (15)

(a) Logo (7), model concept: *setu* means "bridge". Draw an **arch bridge** formed by two hands: one decorated with a mehendi-like craft pattern (the artisan), the other plain with a shopping-bag strap (the buyer). Their fingertips meet at the keystone. Beneath the arch, three short lines suggest a river, which double as a loom's warp threads. The wordmark "KALA SETU" is set in a warm humanist sans, with a small Devanagari "कला सेतु" sub-line. Colours: terracotta (craft, earth) + indigo (trust, textile).

(b) Applications (4):

- *Shopping bag:* a jute or kraft bag with the logo printed large in indigo; a strip of craft motifs along the bottom edge.
- *App icon:* the bridge-hands symbol alone, simplified, on a terracotta rounded square; it still reads at 48 px.

(c) 40-word explanation (4):

> "Kala Setu's logo is a bridge made of two hands: an artisan's patterned hand and a buyer's hand meeting at the keystone. Beneath flow loom-like river lines. Terracotta and indigo signal earth, craft and trust, uniting village skill with city demand."

(40 words.)

Rubric: concept relevance 3, simplicity and scalability 2, craft/culture reference 2; applications 4 (2 each); explanation 4 (clarity 2, word limit 1, link to visuals 1).

Q26. Heatwave design (20)

(a) Journey map (6), user: delivery riders

```
Time:   8am      11am         1pm            3pm             6pm
Step:  Login → Pickups → Peak-heat drops → Waiting at → Evening rush
                                          restaurants
Mood:   🙂       😐             😫             😣              😐
Pain:        (1) no shade   (2) dehydration  (3) no place to  (4) phone
             at pickup      & no water       rest/toilet      overheats,
             points         refill           while waiting    battery dies
```

Rubric: steps 2, at least 4 pain points 2, emotional curve 1, clarity 1.

(b) Solution system (10), model: "Thanda Stop" + "CoolKit"

- **Product, CoolKit:**
  - a helmet with a ventilated inner liner and a detachable neck-shade flap
  - an evaporative cooling vest (water-soaked inserts)
  - an insulated 1.5 L bottle mounted on the delivery box
  - a phone mount with a sun visor and air gap
- **Service, Thanda Stops:**
  - shaded kiosks at major food clusters, co-funded by delivery platforms and the municipality
  - free water refills, toilets, charging points and seating
  - an app feature that shows the nearest stop and allows "heat breaks" without ranking penalties between 1 and 4 pm
- **Policy hook:** rider ratings are not penalised for heat breaks during red-alert days.

Rubric: product feasibility 3, service design 3, links to the pain points 2, drawing and labels 2.

(c) "After" scene (4): a shaded kiosk with a green roof; riders refilling bottles; one charging a phone; a heat-index board showing "44 °C – Take a break"; riders wearing the CoolKit.
Rubric: shows use 2, clarity 1, context 1.

SELF-EVALUATION SYSTEM

Step 1: Score Part I using the key (out of 30).

Step 2: Score Part II using the rubrics (out of 70)

- Score each criterion honestly.
- If unsure between two marks, choose the lower one.
- If possible, get a second evaluator and average the two scores.

Step 3: Complete this evaluation sheet

Item	Score	Max	%	Notes
Part I		30		Which types were wrong? (spatial / GK / logic)
Q23		15		
Q24		20		

Item	Score	Max	%	Notes
Q25		15		
Q26		20		
Total		**100**		
Time overrun? (Y/N)				Which question took longest?
Instructions missed?				Count / colour / word-limit violations

Step 4: Log every error in the Error Log (Vol 13 §6) within 24 hours.

SCORING INTERPRETATION — FULL MOCK (Model Paper 5)

*These bands are **practice benchmarks** for this deliberately tough paper. They are **not** NID cutoffs: NID does not publish verifiable cutoffs, and Prelims marks only shortlist for Mains.*

Total	Band	What it suggests	Action for the next 7 days
70–100	**Excellent**	Exam-ready on both objective and subjective skills	Maintain; focus on originality and Mains prep (Studio Test)
56–69	**Strong**	Competitive; minor gaps	Target the lowest-scoring question type twice more
42–55	**Good**	On track but inconsistent	Daily 25-min timed subjective drills + 15 MCQs
28–41	**Needs improvement**	Major time or skill gaps	Switch to the 60-day plan intensity (Vol 13); redo Papers 1–3
< 28	**Major improvement required**	Fundamentals need rebuilding	Restart Vol 2A & 3 drills; skip mocks for 2 weeks, then retry

Red flags, whatever your total:

1. Any subjective question left blank. Fix your time management first.
2. Part I below 15/30. Do visual reasoning drills (Vol 4B).
3. Instruction violations (counts, colours, word limits). Practise "circle the constraints" on every prompt.

NEXT VOLUME: Volume 11 — Current Affairs (`11-VOL11-CURRENT-AFFAIRS.md`)

VOLUME 11 — DESIGN-ORIENTED CURRENT AFFAIRS (≈ mid-2025 to Oct 2026) + 100 MCQs

Read this first.

- **Reliability labels.** [OFFICIAL] = from an official body's own publication. [REPORTED] = reported by reputable media or institutional websites, found through search (I could not open every primary page from my research environment). Facts in **Part C** are older, confirmed history kept here for context.
- **How much does this matter?** Current affairs have rarely appeared in recent Prelims papers, and none were reported in the all-subjective 2026 paper. They matter more for (1) the **In-Person Sensitivity Test** at Mains, (2) giving your drawings contemporary relevance, and (3) a possible objective Part I. Spend about 30 minutes a week here, not more.
- **Keep it updated.** Between now and the exam, add new items to the blank table at the end, from design sites (Dezeen, ArchDaily, Domus, Design Week India) and the newspaper's culture pages.

PART A — EVENT FILE (each entry gives: what happened, when, who, why it matters, design relevance, and a possible question)

A1. NID's own news

#	Event	Date	People / organisation	Why it matters	Design relevance	Possible question
1	**NID Admissions 2027-28 open; DAT Prelims on 20 Dec 2026** [OFFICIAL]	Applications 10 Sep – 30 Nov 2026	NID	Your exam	—	Interview: "How has the DAT changed recently?"
2	**New 5.5-year "Professional Education M.Des (Integrated Pathway)"**: 11 semesters, 15 disciplines under 5 streams, at Ahmedabad, Gandhinagar and Bengaluru [OFFICIAL title; details REPORTED]	2027-28 cycle	NID	A new UG entry route through the same DAT	Interdisciplinary design education	"Which new programme did NID introduce for 2027-28?"
3	**Union Budget 2026-27 proposes a new NID in eastern India** through a "challenge route" [REPORTED]	1 Feb 2026	Govt of India	Expanding design education	The Orange Economy / creative-economy push	"Where is the newly proposed NID to be located?"
4	**Budget 2026-27: AVGC Content Creator Labs** through IICT Mumbai in 15,000 schools and 500 colleges [REPORTED]	Feb 2026	IICT, Mumbai	Animation, VFX, gaming and comics	Communication and animation design careers	"AVGC stands for?" (Animation, Visual Effects, Gaming, Comics)
5	**DAT 2026 Prelims had no objective section** [REPORTED]	21 Dec 2025	NID	A pattern shift	Drawing and ideation emphasis	(Exam awareness)

A2. Architecture

#	Event	Date	People	Why it matters	Design relevance	Possible question
6	**Pritzker Prize 2026: Smiljan Radić Clarke** (Chile), the 55th laureate [REPORTED]	Announced Mar 2026	Radić; known for the 2014 Serpentine Pavilion and the Biobío Regional Theatre (Concepción)	Architecture's top honour	Poetic, experimental, landscape-and-memory architecture	"Who won the Pritzker 2026?"
7	**Pritzker 2025: Liu Jiakun** (China), the 2nd Chinese laureate after Wang Shu (2012) [REPORTED]	Mar 2025	Jiakun Architects (Chengdu)	Recognition of humane, local architecture	**West Village, Chengdu** (a five-storey block around a courtyard); "rebirth bricks" made from 2008 Wenchuan earthquake rubble	"Which architect used earthquake rubble to make bricks?"
8	**Frank Gehry dies at 96** [REPORTED]	5 Dec 2025	Gehry: Guggenheim Bilbao, Walt Disney Concert Hall, Fondation Louis Vuitton	End of an era of deconstructivist icons	The "Bilbao effect"	"Which architect of the Guggenheim Bilbao died in 2025?"
9	**Venice Architecture Biennale 2025 (19th)**: "Intelligens. Natural. Artificial. Collective.", curated by **Carlo Ratti** [REPORTED, La Biennale]	10 May – 23 Nov 2025	Carlo Ratti	Architecture responding to climate and AI	**Golden Lion for national participation: Bahrain, "Heatwave"** (curated by Andrea Faraguna; passive cooling). Best participant: **Canal Café by Diller Scofidio + Renfro**	"Which country won the Golden Lion at the 2025 Architecture Biennale, and for what theme?"
10	**Serpentine Pavilion 2026 (the 25th) by LANZA atelier** (Isabel Abascal & Alessandro Arienzo, Mexico City), titled **"A Serpentine"**: an undulating brick wall [REPORTED]	6 Jun – 25 Oct 2026	Serpentine Galleries, London	An annual showcase for emerging architects	Brick, handmade, natural environment	"Who designed the 2026 Serpentine Pavilion?"
11	**Navi Mumbai International Airport inaugurated**; lotus-inspired terminal by **Zaha Hadid Architects** [REPORTED]	Inaugurated 8 Oct 2025; commercial flights from 25 Dec 2025	PM; ZHA; Adani Airports	A major Indian infrastructure icon	Biomorphic form (lotus petals, a central "pond" atrium), jaali-inspired lattice	"Which flower inspired the NMIA terminal?"

#	Event	Date	People	Why it matters	Design relevance	Possible question
12	**Expo 2025 Osaka: the "Grand Ring" by Sou Fujimoto** was recognised by Guinness as the **world's largest wooden architectural structure** (~2 km circumference, 61,000+ m² roof, traditional *nuki* joints) [REPORTED]	Expo 13 Apr – 13 Oct 2025	Sou Fujimoto	Mass timber and traditional joinery	Sustainable timber construction	"The Grand Ring at Expo 2025 is made mainly of what material?"

A3. Heritage & culture

#	Event	Date	People	Why it matters	Design relevance	Possible question
13	**Maratha Military Landscapes** inscribed as India's **44th** World Heritage Site: 12 forts (11 in Maharashtra plus **Gingee, Tamil Nadu**) [REPORTED]	Jul 2025, 47th WHC session (Paris)	ASI / Maharashtra	Military architecture and landscape	Fortification design on coastal and hill terrain	"Which Tamil Nadu fort is part of the Maratha Military Landscapes?"
14	**48th WHC session, Busan (Republic of Korea)**: 25 new sites; total 1,273 [REPORTED]. **Ancient Buddhist Site of Sarnath** reported as India's **45th** WHS [REPORTED; verify]	Jul 2026	UNESCO	Heritage	Buddhist architecture (Dhamek and Chaukhandi stupas)	"Where was the 48th World Heritage Committee session held?"
15	**Lucknow named a UNESCO Creative City of Gastronomy**; 58 new cities joined; the network now has 408 cities [REPORTED]	31 Oct 2025 (World Cities Day)	UNESCO	Awadhi culinary heritage	Food culture as design heritage (food illustration prompts!)	"In which category did Lucknow join the UCCN?"
16	**Kolhapuri chappal–Prada row**: Prada's Men's SS26 show (Milan, 23 Jun 2025) featured sandals resembling Kolhapuris; after backlash, Prada acknowledged the Indian inspiration [REPORTED]	Jun 2025	Prada; Kolhapur artisans; GI holders	Cultural appropriation vs credit; the value of GI	Craft IP, ethical design collaboration	"Which Indian GI craft was at the centre of a 2025 global fashion controversy?"
17	**GI registrations 2025-26**: about 125 new GI tags; **Madhya Pradesh topped with 26**, West Bengal second with 24 [REPORTED]	FY 2025-26	GI Registry, Chennai	Craft recognition	Craft identity and branding	"Which state received the most GI tags in 2025-26?"
18	**Jharkhand's 4 new GI tags**: Bhagaiya silk (Godda), Kuchai silk (tasar, Seraikela-Kharsawan), Munda jewellery, Jharkhand bamboo craft [REPORTED]	Jun 2026	GI Registry	Tribal craft recognition	Textile and jewellery design	"Kuchai silk belongs to which state?"

#	Event	Date	People	Why it matters	Design relevance	Possible question
19	**Padma Awards 2026**: 131 awardees, many folk, tribal and visual artists among the Padma Shri recipients; Padma Vibhushan (Art) to **Dharmendra (posthumous)** and violinist **N. Rajam**; Padma Bhushan to Alka Yagnik and Mammootty [REPORTED]	Announced 25 Jan 2026	Govt of India	Recognition of artists	Living craft and art traditions	"Name a 2026 Padma Vibhushan awardee in art."
20	**Maha Kumbh 2025, Prayagraj** [REPORTED]	13 Jan – 26 Feb 2025	UP Govt	A temporary city for crores of pilgrims	Crowd management, wayfinding, temporary architecture (lost-and-found design!)	"Design a wayfinding system for a Kumbh-scale event."

A4. Art & exhibitions

#	Event	Date	People	Why it matters	Design relevance	Possible question
21	**Venice Art Biennale 2026 (61st): "In Minor Keys"**, conceived by **Koyo Kouoh** (who died in May 2025); La Biennale carried out her exhibition [REPORTED]	9 May – 22 Nov 2026	La Biennale	The world's leading art exhibition	Curation, themes of quietness and care	"Who conceived the 2026 Venice Art Biennale?"
22	**India Pavilion returns to Venice** (first time since 2019): "**Geographies of Distance: remembering home**", curated by **Amin Jaffer**; artists Alwar Balasubramaniam, Sumakshi Singh, Ranjani Shettar, Asim Waqif, Skarma Sonam Tashi; presented by the Ministry of Culture with NMACC and Serendipity Arts Foundation [REPORTED]	2026	Ministry of Culture	India's global art presence	Material-based installation art	"What is the title of India's 2026 Venice pavilion?"
23	**2026 Biennale awards controversy**: reports say the international jury resigned and the Golden Lions were to be decided by public vote at the close [REPORTED; verify]	2026	La Biennale	Art and politics	—	(Interview discussion only)
24	**Kochi-Muziris Biennale, 6th edition: "For the Time Being"**, curated by artist **Nikhil Chopra** with HH Art Spaces (Goa) [REPORTED]	12 Dec 2025 – 31 Mar 2026 (110 days)	Kochi Biennale Foundation	India's largest contemporary art event	Ecosystem-based curation	"Who curated the 2025-26 Kochi-Muziris Biennale?"

A5. Design events & awards

#	Event	Date	People	Why it matters	Design relevance	Possible question
25	**World Design Capital 2026: Frankfurt RheinMain**, the first in Germany; theme "**Design for Democracy. Atmospheres for a better life**"; ~2,000 events [REPORTED]	Throughout 2026	World Design Organization	Design for social and civic good	Participatory design	"Which region is World Design Capital 2026?"
26	**India Design ID 2026 (14th edition)**, NSIC Grounds, New Delhi; theme "**The Age of Design Syncretism**"; 150+ brands [REPORTED]	19–22 Feb 2026	India Design ID	India's leading design week	Tradition meets technology	"What was the theme of India Design ID 2026?"
27	**XXIX ADI Compasso d'Oro**: 20 awards and 38 honourable mentions, presented at the ADI Design Museum, Milan; emphasis on social impact and research [REPORTED]	22 May 2026	ADI (Italy)	The "Oscar of design"	Design beyond objects	"Compasso d'Oro is awarded in which country?"
28	**Milano Cortina 2026 Winter Olympics torch "Essential"** by **Carlo Ratti Associati** (with Eni and Versalis): runs on **bio-LPG**, has an open top that reveals the flame, and an iridescent finish [REPORTED]	Unveiled Apr 2025; Games Feb 2026	Carlo Ratti	Sustainable object design	Material honesty, minimalism	"Who designed the Milano Cortina 2026 torch?"
29	**Apple's "Liquid Glass" design language** (translucent, glass-like UI across its platforms) [REPORTED]	Announced Jun 2025 (WWDC)	Apple	A major UI shift after flat design	Translucency, depth, accessibility debates	"Name Apple's 2025 UI design language."
30	**Giorgio Armani dies** [REPORTED]	Sep 2025	Armani	Fashion legend	Unstructured tailoring	"Which Italian designer, famous for unstructured suits, died in 2025?"

A6. Technology, AI, science & environment

#	Event	Date	People	Why it matters	Design relevance	Possible question
31	**India AI Impact Summit**, Bharat Mandapam, New Delhi; adopted the **New Delhi Declaration on AI Impact**; anchored on the "sutras" **People, Planet, Progress**; showcased Indian foundation models [REPORTED, NeGD/PIB]	Feb 2026	MeitY / IndiaAI	Global South AI governance	AI and design ethics; multilingual AI	"What three 'sutras' anchored the AI Impact Summit 2026?"
32	**Shubhanshu Shukla**: the first Indian to visit the ISS (Axiom-4) [REPORTED]	Jun–Jul 2025	ISRO, Axiom, NASA, SpaceX	Gaganyaan preparation	Space habitat design	"Who was the first Indian astronaut to visit the ISS?"

#	Event	Date	People	Why it matters	Design relevance	Possible question
33	**COP30, Belém, Brazil** [REPORTED]	Nov 2025	UNFCCC	Climate negotiations in the Amazon	Climate-responsive design	"Where was COP30 held?"
34	**Grand Egyptian Museum** fully opened near the Giza pyramids [REPORTED]	Nov 2025	Egypt; architects Heneghan Peng	One of the world's largest archaeology museums	Museum architecture	"Near which monument is the Grand Egyptian Museum?"

PART B — HOW TO USE CURRENT AFFAIRS IN YOUR ANSWERS

1. **Drawing prompts get richer:** a heatwave scene can reference the Bahrain "Heatwave" pavilion's passive cooling; a festival wayfinding answer can cite the Maha Kumbh.
2. **Interview:** prepare 3 items you can discuss for 60 seconds each, e.g. the Navi Mumbai Airport (biomimicry and identity), the Kolhapuri–Prada row (craft IP and ethics), AI and design (authorship, the human-in-the-loop).
3. **Opinion structure:** what happened → why it matters to design → your view → a question it raises.

PART C — CONFIRMED BACKGROUND FACTS OFTEN LINKED TO NEWS (older, stable)

- Garba of Gujarat: UNESCO ICH, 2023. Durga Puja in Kolkata: ICH, 2021.
- Moidams of Charaideo: WHS, 2024. Hoysala temples and Santiniketan: 2023.
- Chandrayaan-3 landed on 23 Aug 2023; that date became National Space Day.
- Air India's new identity ("The Vista"): 2023. Twitter was rebranded as X in 2023.
- The new Parliament building (Bimal Patel): 2023.
- Zaha Hadid: the first woman Pritzker laureate (2004). B.V. Doshi: the first Indian (2018).

PART D — 100 CURRENT-AFFAIRS MCQs

Format: Q → **Ans** → explanation [label] · *Concept · Diff · Why NID*.

CA-1. The NID DAT 2027 Prelims is scheduled for: (a) 14 Dec 2026 (b) 20 Dec 2026 (c) 10 Jan 2027 (d) 21 Dec 2025
Ans: (b). 21 Dec 2025 was DAT 2026. [OFFICIAL] *Exam awareness · E · Planning.*

CA-2. The last date to apply for NID 2027-28 admissions is: (a) 30 Nov 2026 (b) 31 Oct 2026 (c) 15 Dec 2026 (d) 10 Sep 2026
Ans: (a). [OFFICIAL] *E.*

CA-3. NID's new undergraduate-entry programme for 2027-28 lasts: (a) 4 years (b) 5 years (c) 5.5 years (d) 6 years
Ans: (c). It is the Professional Education M.Des (Integrated Pathway), with 11 semesters. [OFFICIAL] *M · Interview.*

CA-4. The Integrated Pathway programme is reported to offer how many disciplines? (a) 9 (b) 12 (c) 15 (d) 19
Ans: (c). 15 disciplines under 5 streams. [REPORTED] *M.*

CA-5. The Union Budget 2026-27 proposed a new NID in which region? (a) North-east (b) Eastern India (c) South India (d) Ladakh
Ans: (b). [REPORTED] *E · Design-education policy.*

CA-6. "AVGC", as in the Budget's content-creator labs, stands for: (a) Art, Video, Graphics, Craft (b) Animation, Visual Effects, Gaming, Comics (c) Audio, Visual, Graphic, Cinema (d) Animation, Video, Games, Culture
Ans: (b). [REPORTED] *E · Communication-design careers.*

CA-7. The Budget's AVGC labs are to be set up with support from: (a) IICT Mumbai (b) NID Ahmedabad (c) NIFT Delhi (d) IIT Bombay
Ans: (a). Indian Institute of Creative Technologies. [REPORTED] *M.*

CA-8. The reported DAT 2026 Prelims (Dec 2025) contained: (a) only MCQs (b) 22 MCQs + 4 drawing questions (c) only subjective questions (d) an online test
Ans: (c). [REPORTED] *E · Exam trend.*

CA-9. Who won the 2026 Pritzker Architecture Prize? (a) Liu Jiakun (b) Smiljan Radić Clarke (c) Kengo Kuma (d) Bjarke Ingels
Ans: (b). [REPORTED] *E.*

CA-10. Smiljan Radić is from: (a) Mexico (b) Chile (c) Argentina (d) Spain
Ans: (b). [REPORTED] *E.*

CA-11. Which earlier work is Radić known for? (a) The 2014 Serpentine Pavilion (b) The Shard (c) The Louvre Abu Dhabi (d) The CCTV HQ
Ans: (a). [REPORTED] *M.*

CA-12. The 2025 Pritzker laureate Liu Jiakun is based in: (a) Beijing (b) Shanghai (c) Chengdu (d) Hong Kong
Ans: (c). [REPORTED] *M.*

CA-13. Liu Jiakun made "rebirth bricks" from: (a) plastic waste (b) rubble of the 2008 Wenchuan earthquake (c) rice husk (d) coal ash only
Ans: (b). [REPORTED] *M · Sustainable material reuse.*

CA-14. Which Chinese architect won the Pritzker before Liu Jiakun? (a) I.M. Pei (b) Wang Shu (c) Ma Yansong (d) Zhang Ke
Ans: (b). In 2012. I.M. Pei (1983) was Chinese-born but American. [REPORTED] *H · A careful-fact habit.*

CA-15. Frank Gehry died in December 2025 at the age of: (a) 88 (b) 92 (c) 96 (d) 101
Ans: (c). [REPORTED] *E.*

CA-16. Which is NOT a Frank Gehry building? (a) Guggenheim Bilbao (b) Walt Disney Concert Hall (c) Fondation Louis Vuitton (d) Heydar Aliyev Center
Ans: (d). That is by Zaha Hadid. *E.*

CA-17. The curator of the 2025 Venice Architecture Biennale was: (a) Hashim Sarkis (b) Lesley Lokko (c) Carlo Ratti (d) Rem Koolhaas
Ans: (c). Sarkis curated 2021 and Lokko 2023. [REPORTED] *M.*

CA-18. The 2025 Architecture Biennale's title was: (a) "How will we live together?" (b) "The Laboratory of the Future" (c) "Intelligens. Natural. Artificial. Collective." (d) "Freespace"
Ans: (c). (a) = 2021, (b) = 2023, (d) = 2018. [REPORTED] *M.*

CA-19. The Golden Lion for Best National Participation (Architecture 2025) went to: (a) Japan (b) Bahrain (c) India (d) Brazil
Ans: (b). "Heatwave". [REPORTED] *M.*

CA-20. Bahrain's "Heatwave" pavilion explored: (a) solar panels only (b) passive cooling for outdoor work in extreme heat (c) desert tourism (d) glass skyscrapers
Ans: (b). Wind towers, shaded courtyards, solar chimneys. [REPORTED] *M · Climate design.*

CA-21. Diller Scofidio + Renfro won Best Participant (Architecture 2025) for: (a) Canal Café (b) Floating Library (c) Water Tower (d) Glass Bridge
Ans: (a). [REPORTED] *H.*

CA-22. The 2026 Serpentine Pavilion was designed by: (a) LANZA atelier (b) Frida Escobedo (c) Minsuk Cho (d) Sumayya Vally
Ans: (a). Escobedo did 2018, Cho 2025, Vally 2021. [REPORTED] *M.*

CA-23. LANZA atelier is based in: (a) Mexico City (b) Madrid (c) Lima (d) Bogotá
Ans: (a). [REPORTED] *M.*

CA-24. The 2026 Serpentine Pavilion, titled "A Serpentine", features mainly a(n): (a) undulating brick wall (b) glass dome (c) steel mesh tower (d) inflatable roof
Ans: (a). [REPORTED] *M · Materials.*

CA-25. The 2026 Serpentine Pavilion is the series' ___ edition. (a) 20th (b) 25th (c) 30th (d) 15th
Ans: (b). The series began in 2000. [REPORTED] *M.*

CA-26. The Navi Mumbai International Airport terminal design is inspired by the: (a) peacock (b) lotus (c) banyan tree (d) conch
Ans: (b). [REPORTED] *E · Biomorphic design.*

CA-27. The NMIA terminal was designed by: (a) Foster + Partners (b) Zaha Hadid Architects (c) SOM (d) Hafeez Contractor
Ans: (b). [REPORTED] *E.*

CA-28. NMIA was inaugurated on: (a) 8 Oct 2025 (b) 26 Jan 2025 (c) 15 Aug 2025 (d) 25 Dec 2025
Ans: (a). Commercial operations began on 25 Dec 2025. [REPORTED] *M · Trap: (d).*

CA-29. The "Grand Ring" at Expo 2025 Osaka was designed by: (a) Kengo Kuma (b) Sou Fujimoto (c) Tadao Ando (d) Shigeru Ban
Ans: (b). [REPORTED] *M.*

CA-30. The Grand Ring set a Guinness record as the world's largest: (a) steel arch (b) wooden architectural structure (c) glass roof (d) 3D-printed building
Ans: (b). [REPORTED] *E · Mass timber.*

CA-31. The Grand Ring uses a traditional Japanese joinery technique called: (a) nuki (b) origami (c) kintsugi (d) tatami
Ans: (a). Kintsugi is gold-repair of ceramics. [REPORTED] *H.*

CA-32. Expo 2025 was held in: (a) Tokyo (b) Osaka (c) Kyoto (d) Nagoya
Ans: (b). *E.*

CA-33. India's 44th World Heritage Site (2025) is: (a) Sarnath (b) Maratha Military Landscapes (c) Moidams (d) Hoysala temples
Ans: (b). [REPORTED] *E.*

CA-34. Which fort in Tamil Nadu is part of the Maratha Military Landscapes? (a) Vellore (b) Gingee (c) Dindigul (d) Thanjavur
Ans: (b). [REPORTED] *M.*

CA-35. The Maratha Military Landscapes serial site includes how many forts? (a) 8 (b) 10 (c) 12 (d) 14
Ans: (c). [REPORTED] *M.*

CA-36. The 48th session of the World Heritage Committee (Jul 2026) was held in: (a) Paris (b) New Delhi (c) Busan (d) Riyadh
Ans: (c). New Delhi hosted the 46th (2024). [REPORTED] *M.*

CA-37. At the 48th session, how many new sites were reported inscribed? (a) 12 (b) 19 (c) 25 (d) 32
Ans: (c). [REPORTED] *H.*

CA-38. Which Indian site was reported as inscribed in 2026, becoming India's 45th WHS? (a) Sarnath (b) Majuli (c) Lepakshi (d) Bishnupur
Ans: (a). [REPORTED: verify on the UNESCO site] *M.*

CA-39. Sarnath is associated with: (a) the Buddha's first sermon (b) Mahavira's birth (c) Guru Nanak's birth (d) the Mughal capital
Ans: (a). *E.*

CA-40. Which monument at Sarnath gave India its national emblem? (a) The Lion Capital of Ashoka (b) The Dhamek Stupa (c) A Gupta temple (d) The Sanchi gateway
Ans: (a). *E.*

CA-41. Lucknow joined the UNESCO Creative Cities Network in 2025 in the category: (a) Crafts and Folk Art (b) Gastronomy (c) Music (d) Literature
Ans: (b). [REPORTED] *E.*

CA-42. Lucknow's UCCN designation was announced on: (a) World Heritage Day (b) World Cities Day, 31 Oct (c) Republic Day (d) World Food Day
Ans: (b). [REPORTED] *M.*

CA-43. Which other Indian city is a UNESCO Creative City of Gastronomy? (a) Hyderabad (b) Kolkata (c) Amritsar (d) Indore
Ans: (a). Since 2019. *M.*

CA-44. How many new cities joined the UCCN in 2025? (a) 25 (b) 58 (c) 100 (d) 12
Ans: (b). [REPORTED] *H.*

CA-45. The 2025 Prada controversy involved which Indian footwear? (a) Mojari (b) Kolhapuri chappal (c) Paduka (d) Jutti
Ans: (b). [REPORTED] *E · Craft ethics.*

CA-46. The Prada show where the sandals appeared was its: (a) Women's FW25, Paris (b) Men's SS26, Milan (c) Resort 2026, Tokyo (d) Couture, New York
Ans: (b). 23 June 2025. [REPORTED] *M.*

CA-47. The Kolhapuri chappal's GI registration covers districts of: (a) Maharashtra and Karnataka (b) Gujarat (c) Goa (d) Rajasthan
Ans: (a). *M.*

CA-48. The central design issue raised by the Prada–Kolhapuri case was: (a) cultural appropriation without credit to craft communities (b) colour choice (c) price (d) sizing
Ans: (a). *E · Ethics.*

CA-49. In FY 2025-26, which state reportedly received the most new GI tags? (a) Uttar Pradesh (b) Madhya Pradesh (c) Tamil Nadu (d) West Bengal
Ans: (b). 26 tags. [REPORTED] *M.*

CA-50. About how many new GI tags were granted in India in FY 2025-26? (a) 25 (b) 75 (c) 125 (d) 250
Ans: (c). [REPORTED] *H.*

CA-51. Kuchai silk, which received a GI tag in 2026, is a variety of: (a) Muga (b) Tasar (c) Eri (d) Mulberry
Ans: (b). [REPORTED] *M.*

CA-52. Which of these was among Jharkhand's 2026 GI tags? (a) Munda jewellery (b) Bidriware (c) Rogan art (d) Pattachitra
Ans: (a). [REPORTED] *M.*

CA-53. Bhagaiya silk is from which Jharkhand district? (a) Ranchi (b) Godda (c) Dumka (d) Bokaro
Ans: (b). [REPORTED] *H.*

CA-54. How many Padma awards were announced in 2026? (a) 106 (b) 119 (c) 131 (d) 139
Ans: (c). [REPORTED] *H.*

CA-55. Which veteran violinist received the Padma Vibhushan (2026)? (a) N. Rajam (b) L. Subramaniam (c) Kala Ramnath (d) M.S. Gopalakrishnan
Ans: (a). [REPORTED] *M.*

CA-56. The Maha Kumbh 2025 was held at: (a) Haridwar (b) Prayagraj (c) Nashik (d) Ujjain
Ans: (b). [REPORTED] *E.*

CA-57. The biggest design challenge of a Kumbh-scale event is: (a) choosing logo colours (b) crowd flow, wayfinding, sanitation and lost-and-found for crores of people (c) ticket printing (d) parking only
Ans: (b). *E · Systems design.*

CA-58. The 2026 Venice Art Biennale is titled: (a) "Foreigners Everywhere" (b) "In Minor Keys" (c) "The Milk of Dreams" (d) "May You Live in Interesting Times"
Ans: (b). The others were 2024, 2022 and 2019. [REPORTED] *M.*

CA-59. "In Minor Keys" was conceived by curator: (a) Adriano Pedrosa (b) Koyo Kouoh (c) Cecilia Alemani (d) Okwui Enwezor
Ans: (b). [REPORTED] *M.*

CA-60. Koyo Kouoh was a curator from: (a) Cameroon / Switzerland, known for leading Zeitz MOCAA, Cape Town (b) Brazil (c) Japan (d) India
Ans: (a). [REPORTED] *H.*

CA-61. India's 2026 Venice pavilion is titled: (a) "Geographies of Distance: remembering home" (b) "Our Time for a Future Caring" (c) "Everyone Agrees" (d) "Home is Where"
Ans: (a). [REPORTED] *M.*

CA-62. The curator of India's 2026 Venice pavilion is: (a) Amin Jaffer (b) Bose Krishnamachari (c) Nikhil Chopra (d) Roobina Karode
Ans: (a). [REPORTED] *M.*

CA-63. India last had a national pavilion at the Venice Art Biennale in: (a) 2011 (b) 2015 (c) 2019 (d) 2022
Ans: (c). [REPORTED] *M.*

CA-64. Which artist was among those in India's 2026 Venice pavilion? (a) Ranjani Shettar (b) Subodh Gupta (c) Bharti Kher (d) Anish Kapoor
Ans: (a). Also Alwar Balasubramaniam, Sumakshi Singh, Asim Waqif and Skarma Sonam Tashi. [REPORTED] *M.*

CA-65. The theme of the 6th Kochi-Muziris Biennale (2025-26) is: (a) "For the Time Being" (b) "In Our Veins Flow Ink and Fire" (c) "Whorled Explorations" (d) "Possibilities for a Non-Alienated Life"
Ans: (a). (b) = 2022, (c) = 2014, (d) = 2018. [REPORTED] *M.*

CA-66. The 2025-26 Kochi Biennale was curated by: (a) Nikhil Chopra (b) Shubigi Rao (c) Anita Dube (d) Jitish Kallat
Ans: (a). Shubigi Rao curated 2022, Anita Dube 2018 and Jitish Kallat 2014. [REPORTED] *M.*

CA-67. The Kochi-Muziris Biennale 2025-26 ran for about: (a) 30 days (b) 60 days (c) 110 days (d) 365 days
Ans: (c). 12 Dec 2025 – 31 Mar 2026. [REPORTED] *H.*

CA-68. World Design Capital 2026 is: (a) Valencia (b) San Diego–Tijuana (c) Frankfurt RheinMain (d) Lille
Ans: (c). [REPORTED] *E.*

CA-69. The WDC 2026 theme is: (a) "Design for Democracy. Atmospheres for a better life." (b) "Design for Health" (c) "Smart Cities" (d) "Design Everywhere"
Ans: (a). [REPORTED] *M.*

CA-70. World Design Capital 2026 is the first time the title has gone to: (a) Germany (b) Asia (c) Africa (d) a capital city
Ans: (a). [REPORTED] *M.*

CA-71. India Design ID 2026 was held at: (a) Pragati Maidan (b) NSIC Grounds, Okhla, New Delhi (c) Jio World Centre (d) Bengaluru Palace
Ans: (b). [REPORTED] *M.*

CA-72. The India Design ID 2026 theme was: (a) "The Age of Design Syncretism" (b) "Made in India" (c) "Future Craft" (d) "Design Democracy"
Ans: (a). [REPORTED] *M.*

CA-73. "Syncretism" in that theme means: (a) the blending of different traditions and influences (b) synchronised lights (c) synthetic materials (d) symmetry
Ans: (a). *E · Vocabulary.*

CA-74. The XXIX Compasso d'Oro (2026) was presented at the: (a) ADI Design Museum, Milan (b) Triennale, Rome (c) V&A, London (d) MoMA
Ans: (a). [REPORTED] *M.*

CA-75. The Compasso d'Oro was originally conceived by: (a) Gio Ponti (b) Ettore Sottsass (c) Achille Castiglioni (d) Bruno Munari
Ans: (a). In 1954, with La Rinascente. *M.*

CA-76. The Milano Cortina 2026 Olympic torch is named: (a) Fiamma (b) Essential (c) Luce (d) Aurora
Ans: (b). [REPORTED] *M.*

CA-77. The torch was designed by: (a) Philippe Starck (b) Carlo Ratti Associati (c) Marc Newson (d) Patricia Urquiola
Ans: (b). [REPORTED] *M.*

CA-78. The "Essential" torch burns: (a) hydrogen (b) bio-LPG (c) kerosene (d) natural gas only
Ans: (b). [REPORTED] *M · Sustainable fuel.*

CA-79. A distinctive design feature of the torch is: (a) an open top revealing how the flame is generated (b) a wooden handle (c) LED-only light (d) a solar panel
Ans: (a). [REPORTED] *M · Design honesty.*

CA-80. Apple's 2025 design language is called: (a) Liquid Glass (b) Aqua 2 (c) Material You (d) Fluent
Ans: (a). [REPORTED] *E.*

CA-81. A common critique of highly translucent UI is: (a) reduced legibility and contrast, an accessibility concern (b) too much colour (c) a slow internet (d) small icons
Ans: (a). *M · Usability thinking.*

CA-82. The Italian fashion designer who died in September 2025 was: (a) Valentino Garavani (b) Giorgio Armani (c) Miuccia Prada (d) Donatella Versace
Ans: (b). [REPORTED] *E.*

CA-83. The India AI Impact Summit (Feb 2026) was held at: (a) Bharat Mandapam, New Delhi (b) Yashobhoomi (c) Bengaluru Palace (d) Mumbai BKC
Ans: (a). [REPORTED] *M.*

CA-84. The summit's outcome document was the: (a) Bletchley Declaration (b) New Delhi Declaration on AI Impact (c) Paris AI Charter (d) Seoul Commitments
Ans: (b). Bletchley was the UK summit of 2023. [REPORTED] *M.*

CA-85. The three "sutras" of the summit were: (a) People, Planet, Progress (b) Safety, Speed, Scale (c) Data, Compute, Talent (d) Trust, Truth, Tech
Ans: (a). [REPORTED] *M.*

CA-86. Which is the BEST example of "human-in-the-loop" AI in design? (a) AI generates 50 logo sketches; the designer selects, refines and tests them with users (b) AI publishes final designs automatically (c) No AI use (d) AI replaces user research
Ans: (a). *E · AI ethics.*

CA-87. Shubhanshu Shukla flew to the ISS on: (a) Gaganyaan-1 (b) Axiom Mission 4 (c) Soyuz MS-25 (d) Artemis II
Ans: (b). [REPORTED] *E.*

CA-88. Before Shubhanshu Shukla, the only Indian in space was: (a) Kalpana Chawla (b) Rakesh Sharma (c) Sunita Williams (d) Ravish Malhotra
Ans: (b). In 1984. Kalpana Chawla and Sunita Williams were US astronauts of Indian origin. *M · Trap: (a).*

CA-89. COP30 took place in: (a) Belém, Brazil (b) Baku (c) Bonn (d) Rio de Janeiro
Ans: (a). [REPORTED] *E.*

CA-90. Belém's choice as COP30 host emphasised the importance of the: (a) Amazon rainforest (b) Arctic (c) Sahara (d) Himalayas
Ans: (a). *E.*

CA-91. The Grand Egyptian Museum is located near: (a) the Pyramids of Giza (b) Luxor (c) Alexandria's lighthouse (d) Abu Simbel
Ans: (a). [REPORTED] *E.*

CA-92. Which design idea links the Bahrain pavilion, khus curtains and Indian stepwells? (a) Passive cooling (b) Neon lighting (c) Air-conditioning (d) Glass façades
Ans: (a). *M · Cross-topic synthesis.*

CA-93. Which design idea links the NMIA terminal (lotus), Burj Khalifa (a desert flower) and the Lotus Temple? (a) Biomorphic, nature-inspired form (b) Brutalism (c) Art Deco (d) Pop Art
Ans: (a). *M · Synthesis.*

CA-94. The Grand Ring, Shigeru Ban's paper shelters and Liu Jiakun's rubble bricks all show: (a) material innovation for sustainability (b) the use of gold (c) digital-only design (d) temporary signage
Ans: (a). *M · Synthesis.*

CA-95. Which 2025-26 event is MOST relevant to a "craft and intellectual property" interview question? (a) The Kolhapuri–Prada row (b) COP30 (c) Expo 2025 (d) The Axiom-4 mission
Ans: (a). *E · Interview relevance.*

CA-96. A question about "design for heat" could cite which 2025 architecture award winner? (a) Bahrain's "Heatwave" pavilion (b) The Serpentine 2026 (c) Navi Mumbai Airport (d) The Grand Ring
Ans: (a). *E.*

CA-97. The World Design Organization (which awards the World Design Capital title) was formerly called: (a) ICSID (b) UIA (c) RIBA (d) AIGA
Ans: (a). *M.*

CA-98. Which country's architects designed the 2026 Serpentine Pavilion? (a) Mexico (b) Chile (c) South Korea (d) Japan
Ans: (a). LANZA atelier. Chile is the trap: Radić, the 2026 Pritzker laureate, designed the 2014 pavilion. [REPORTED] *M.*

CA-99. Which event marked India's first in-orbit stay aboard the ISS? (a) Axiom-4 (2025) (b) Chandrayaan-3 (2023) (c) Aditya-L1 (2023) (d) Gaganyaan G1
Ans: (a). *E.*

CA-100. The best way to keep current affairs useful for NID is to: (a) memorise every date (b) link each event to a design idea you can draw or discuss (c) ignore it (d) read only sports news
Ans: (b). *E · Strategy.*

PART E — YOUR UPDATE LOG (fill in weekly until the exam)

Date added	Event	Source	Design link	Possible question

NEXT VOLUME: Volume 12 — Revision Book: Last-Minute Notes, Flashcards, Top-100 Mistakes (`12-VOL12-REVISION-BOOK.md`)

VOLUME 12 — NID DAT LAST-MINUTE REVISION BOOK

Contents: 1. One-page exam strategy · 2. Visual & design principles · 3. Colour · 4. Typography & visual communication · 5. Perspective & drawing rules · 6. Reasoning formulas · 7. Indian art, crafts & textiles · 8. Architecture · 9. Designers & movements · 10. Logos & products · 11. Terms · 12. **Flashcards (160)** · 13. **Top 100 NID DAT mistakes**

1. ONE-PAGE EXAM STRATEGY (read this the night before)

- **Exam:** Sunday 20 Dec 2026 · 3 hrs · 100 marks · pen and paper · no negative marking · Prelims = **shortlisting only**.
- **Two possible formats:** (A) 22 objective + 4 subjective (the Handbook scheme); (B) all-subjective, about 6 questions with sub-parts (reported for 2026). Adapt within the first 5 minutes.
- **First 5–10 minutes:** read everything; **circle the counts, viewpoints, users and colour limits**; write a time budget next to each question; jot 2 idea keywords per question.
- **Time rule:** minutes ≈ marks × 1.65. Use the **70% rule**: move on when a drawing is 70% done.
- **Never leave a question blank.** A 3-minute labelled sketch scores marks; a blank scores zero.
- **The order of value:** idea → clarity → composition → proportion → rendering.
- **MCQs:** answer all of them (no negative marking). Skip anything over 90 seconds, then come back.
- **Last 15 minutes:** fill unanswered MCQs, add labels and titles, add colour to the strongest answer, check the counts.

2. VISUAL & DESIGN PRINCIPLES (core list)

- **Elements:** point, line, shape (2D), form (3D), colour, texture (actual or visual), space (positive or negative), value.
- **Principles:** balance (symmetrical / asymmetrical / radial), contrast, emphasis, hierarchy, proportion (part to part), scale (relative to the human or another object), rhythm (regular / alternating / progressive / flowing), repetition, movement (the eye path), unity, variety, harmony.
- **Gestalt:** proximity, similarity, closure, continuity, figure–ground, common fate, symmetry, prägnanz.
- **Composition:**
  - the rule of thirds
  - one focal point
  - odd numbers of subjects
  - lead room in front of a moving subject
  - leading lines
  - framing
  - foreground, middle ground and background
  - avoid tangents
  - don't put the horizon in the middle
- **Design thinking (Stanford):** Empathise → Define → Ideate → Prototype → Test. **Double Diamond:** Discover → Define → Develop → Deliver.
- **Don Norman:** affordance, signifier, feedback, mapping, constraint. **Dieter Rams:** "Less, but better", 10 principles.
- **Universal design:** works for the widest range of users (the curb-cut effect). **Ergonomics:** design for the 5th–95th percentile.

- **Sustainability:** refuse, reduce, reuse, repair, recycle, rot; circular economy; cradle to cradle; design for disassembly; mono-material packaging; avoid greenwashing.

3. COLOUR (essentials)

- **RYB primaries:** red, yellow, blue → secondaries orange, green, violet → tertiaries such as red-orange.
- **RGB (light, additive):** R+G = yellow, G+B = cyan, R+B = magenta, all three = white.
- **CMYK (print, subtractive):** C+M = blue, M+Y = red, C+Y = green.
- **Complements (RYB):** red–green, blue–orange, yellow–violet.
- **Tint** = + white; **shade** = + black; **tone** = + grey.
- **Hue** = the colour family; **saturation** = intensity; **value** = lightness.
- **Schemes:** monochromatic, analogous, complementary, split-complementary, triadic, tetradic.
- **60-30-10 rule.** Warm colours advance and cool colours recede. **Atmospheric perspective:** distant objects look paler and bluer.
- **Accessibility:** about 8% of men are red–green colour-blind. Use value contrast plus shape, never colour alone.
- **Safety colours:** red = prohibition, yellow = warning, blue = mandatory, green = safe or exit.
- **India:** saffron = sacred and courage; white = mourning (in many communities); red = bridal and auspicious.

4. TYPOGRAPHY & VISUAL COMMUNICATION

- **Serif** (Times, Garamond, Bodoni, Baskerville) · **Sans** (Helvetica, Futura, Gill Sans, Arial, Univers, Frutiger) · **Slab** (Rockwell) · Script · Display · Monospace.
- **Kerning** = one letter pair · **Tracking** = a whole word or line · **Leading** = line spacing · **Counter** = the enclosed space in a letter · **x-height** · baseline · ascender and descender.
- **Legibility** = telling characters apart (a property of the typeface); **readability** = reading blocks of text easily (a property of the layout).
- **Rules:** at most 2 typefaces; 45–75 characters per line; use sans serif with high contrast for signage.
- **Pictograms:** a simple silhouette, a consistent stroke and grid, one idea per sign, no text needed, a red circle-and-slash for prohibition.
- **Poster:** headline (3–6 words) → one strong image → supporting line → call to action → logo, date and venue.
- **Logo types:** wordmark, lettermark, pictorial, abstract, mascot, emblem, combination. A good logo is simple, memorable, scalable, works in mono, and is appropriate.
- **Packaging functions:** protect, preserve, contain, communicate, convenience, sustainability.

5. PERSPECTIVE & DRAWING RULES

- **HL** = eye level. **1-point** (frontal: roads, rooms) · **2-point** (corners; verticals stay vertical) · **3-point** (looking up or down).
- **Standing viewer:** the heads of all standing adults sit on the HL, at any distance. A child half an adult's height has its head **halfway between its feet and the HL**.
- **Bird's-eye** = a high horizon with a third VP below; **worm's-eye** = a low horizon with a third VP above.
- **Human proportions:** an adult is 7.5–8 heads tall; the midpoint is the hips; the fingertips reach mid-thigh; the eyes sit at half the head's height.
- **Size references:**

Object	Approximate size
Door	2.1 m
Adult	1.6–1.75 m
Chair seat	0.45 m
Table	0.75 m
Kitchen counter	0.85 m
Bus	3 m tall
Auto-rickshaw	1.7 m tall

- **Shadows:** fall away from the light; are long in the morning and evening and shortest at noon; morning shadows point west.
- **Materials:** glass = sharp highlights and see-through; steel = high contrast with reflections; wood = grain; fabric = folds from tension points.
- **Storyboards:** plan the beats; keep characters and setting consistent; vary wide, medium and close shots; follow the 180° rule; number the frames; make the last frame show the resolution or a design improvement.
- **Product sketch:** a 3/4 view + 1–2 detail views + labels + a hand for scale + a small in-use scene + a 2-line justification.

6. REASONING FORMULAS & RULES

- **Mirror clock:** 11:60 − time (use 23:60 − time for times from 11:00 to 12:59).
- **Vertically symmetric letters:** A H I M O T U V W X Y. **Horizontally symmetric letters:** B C D E H I K O X. **Both:** H I O X.
- **Letters that look the same after a 180° rotation:** H I N O S X Z.
- **Folding:** n folds → 2^n layers and holes. Unfold in **reverse** order. A fold along y = x swaps the coordinates; a fold along x + y = s gives (s − y, s − x).
- **Painted cube (n):** 8 corners (3 faces); 12(n−2) edges (2 faces); $6(n-2)^2$ face-centres (1 face); $(n-2)^3$ inside (0 faces).
- **Dice:** opposite faces sum to 7. **Nets:** in a row, faces one apart are opposite; in a 1-4-1 net, the square above and the square below are opposite. A cube has **11** nets.
- **Euler:** V − E + F = 2.
- **Counting:**
  - squares in an n×n grid = Σk^2; in a 4×4 grid = 30; in a 5×5 grid = 55
  - rectangles in an m×n grid = C(m+1, 2) × C(n+1, 2)
  - triangles from a vertex with k lines and b bases = C(k, 2) × b
  - a square with both diagonals = 8 triangles; adding both midlines gives 16
- **Gears:** meshing gears alternate direction; teeth × turns is constant; an odd closed loop jams. An open belt keeps the direction; a crossed belt reverses it.
- **Clock angle:** $|30H - 5.5M|$. Handshakes: n(n−1)/2. Diagonals of a polygon: n(n−3)/2.
- **Interior angle sum:** (n−2) × 180°. Only triangles, squares and hexagons tessellate alone.
- **Scaling:** lengths × k → areas × k^2 → volumes × k^3.
- **A-paper:** A0 = 1 m^2; the sides are in the ratio 1 : √2; 1 A1 = 8 A4. **Golden ratio** ≈ 1.618.
- **Percentages:** successive changes a and b give a + b + ab/100; +20% then −20% = −4%. Profit % is on cost.
- **Speed:** km/h × 5/18 = m/s. Average speed over equal distances = 2ab/(a + b).

7. INDIAN ART, CRAFTS & TEXTILES (rapid list)

- **Paintings:**
  - Madhubani (Bihar: double lines, filled space)
  - Warli (Maharashtra: white on mud; circle, triangle, square)
  - Gond (MP: dots and dashes; Jangarh Singh Shyam)
  - Pithora (Gujarat/MP)
  - Bhil (dots; Bhuri Bai)
  - Saura (Odisha)
  - Pattachitra (Odisha: Raghurajpur)
  - Kalighat (Kolkata: bold lines, satire)
  - Kalamkari (AP: Srikalahasti pen / Machilipatnam block)
  - Phad (Rajasthan: Pabuji scrolls, Bhopas)
  - Tanjore (TN: gold foil)
  - Cheriyal (Telangana)
  - Manjusha (Bihar: snakes)
  - Sohrai-Khovar (Jharkhand)
  - Thangka (Buddhist)
- **Miniatures:** Mughal (Akbar's *Hamzanama*, Mansur's birds), Rajput (Kishangarh "Bani Thani" by Nihal Chand), Pahari (Kangra, Basohli).
- **Modern art:**
  - Raja Ravi Varma (oleographs)
  - Abanindranath (*Bharat Mata*)
  - Nandalal Bose (the Constitution; the Haripura posters)
  - Amrita Sher-Gil
  - Jamini Roy
  - the Progressive Artists' Group (Bombay, 1947: Souza, Raza with his "Bindu", Husain)
  - Ram Kinkar Baij (*Santhal Family*)
- **Saris and weaves:**
  - Banarasi (UP)
  - Kanjeevaram (TN, korvai borders)
  - Paithani (Maharashtra, peacocks)
  - **Patola** (Patan, double ikat)
  - Pochampally (Telangana, ikat)
  - Chanderi and Maheshwari (MP)
  - Baluchari (WB, narrative pallu)
  - Jamdani (supplementary weft)
  - Muga (Assam, golden)
  - Eri ("ahimsa" silk)
  - Kota Doria (checks)
  - Ilkal (Karnataka)
  - Kasavu (Kerala)
- **Embroidery:** Chikankari (Lucknow, white on white), Phulkari (Punjab), Kantha (Bengal, running stitch, upcycling), Kasuti (Karnataka), Zardozi (metallic), Toda (Nilgiris), Pipli appliqué (Odisha), Lambani (Banjara).
- **Dyeing and printing:** Bandhani (dots), Leheriya (waves), Ajrakh (indigo + madder), Bagru and Dabu (mud resist), Sanganeri, Bagh, Batik (wax). **Ikat dyes the yarn before weaving.**
- **Crafts:**
  - Blue Pottery (Jaipur, quartz)
  - Bidriware (Bidar, zinc alloy + silver)
  - Dhokra (lost wax)

    - Channapatna (lacquered wood toys)
    - Kondapalli
    - Aranmula Kannadi (a metal mirror)
    - Pietra dura (Agra)
    - Bankura horse
    - Longpi (Manipur, no wheel)
    - Thewa
    - Meenakari
    - Rogan (Nirona, Kutch)
    - Sankheda furniture
    - Sikki grass
    - Kathputli
- **GI:** Act of 1999, in force from 2003; the first GI was **Darjeeling Tea (2004)**; the GI Registry is in **Chennai**.

8. ARCHITECTURE (rapid list)

- **Temple styles: Nagara** (shikhara: Khajuraho, Konark, Lingaraja), **Dravida** (vimana and gopuram: Thanjavur, Madurai, Mahabalipuram), **Vesara** (Hoysala: Belur, Halebidu).
- **Key works:**
    - **Kailasa, Ellora** (monolithic, Rashtrakuta Krishna I)
    - **Brihadeeswara** (Rajaraja Chola I, about 1010)
    - **Konark** (Narasimhadeva I; a chariot with 24 wheels)
    - **Rani ki Vav** (shown on the ₹100 note)
    - **Humayun's Tomb** (the first garden tomb)
    - **Taj Mahal** (Ustad Ahmad Lahauri, 1632–53)
    - **Gol Gumbaz** (a whispering gallery)
    - **Charminar** (1591)
    - **Hawa Mahal** (1799)
    - **Jantar Mantar**
- **Colonial:** CSMT (F.W. Stevens), Gateway of India (Wittet), Victoria Memorial (Emerson), Rashtrapati Bhavan and India Gate (Lutyens), North and South Blocks (Baker).
- **Modern:**
    - Chandigarh (Le Corbusier)
    - IIM Ahmedabad (Louis Kahn)
    - **B.V. Doshi**: Sangath, IIM Bangalore, Aranya; Pritzker 2018
    - **Charles Correa**: Gandhi Smarak Sangrahalaya, Jawahar Kala Kendra (navagraha plan), Kanchanjunga Apartments
    - **Laurie Baker**: low-cost brick, the rat-trap bond
    - the Lotus Temple (Fariborz Sahba)
    - the New Parliament (Bimal Patel)
    - the Statue of Unity (Ram Sutar, 182 m)
    - the NMIA (ZHA, the lotus, 2025)
- **World:**
    - Fallingwater and the Guggenheim NY (Wright)
    - Villa Savoye, the Modulor and the 5 points (Le Corbusier)
    - the Barcelona Pavilion (Mies: "less is more")
    - Bauhaus (Gropius, 1919, Weimar)
    - the Sydney Opera House (Utzon)
    - the Louvre Pyramid (Pei)
    - Guggenheim Bilbao (Gehry, who died Dec 2025)

    - Heydar Aliyev Center (Hadid: the first woman Pritzker laureate, 2004)
    - the Gherkin (Foster)
    - the Pompidou (Piano + Rogers)
    - the Shard (Piano)
    - the Church of the Light (Ando)
    - paper shelters (Shigeru Ban)
    - the geodesic dome (Fuller)
    - the Burj Khalifa (SOM, 828 m)
    - the Bird's Nest (Herzog & de Meuron)
    - the Sagrada Família (Gaudí)
- **Recent Pritzker winners:** 2022 Kéré · 2023 Chipperfield · 2024 Yamamoto · 2025 Liu Jiakun · 2026 Smiljan Radić [reported].
- **Indian World Heritage Sites:**
    - 44th: the Maratha Military Landscapes (2025)
    - reported 45th: Sarnath (2026)
    - the Moidams (2024)
    - Hoysala and Santiniketan (2023)
    - Dholavira and Ramappa (2021)
    - Ahmedabad (the first WH City, 2017), Jaipur (2019)
    - Khangchendzonga (the only mixed site)

9. DESIGNERS & MOVEMENTS (rapid list)

- **Graphic and type:**
    - Paul Rand (IBM)
    - Milton Glaser (I ❤ NY)
    - Saul Bass (film titles)
    - Vignelli (the NYC subway map, 1972)
    - Harry Beck (the Tube map)
    - Edward Johnston (the Underground typeface, 1916)
    - Otl Aicher (the Munich 1972 pictograms)
    - Müller-Brockmann (grids)
    - Miedinger (Helvetica, 1957)
    - Renner (Futura, 1927)
    - Gill (Gill Sans)
    - Morison (Times New Roman, 1932)
    - Frutiger (Univers, Frutiger)
    - Carter (Verdana, Georgia)
    - Tschichold
    - Davidson (the Nike swoosh)
    - Janoff (the Apple logo)
    - Leader (the FedEx logo)
    - Kare (the Mac icons)
    - D. Udaya Kumar (the ₹ symbol)
    - Satyajit Ray (the Ray Roman typeface)
- **Product and furniture:**
    - Dieter Rams (Braun)
    - Jony Ive (Apple)
    - Raymond Loewy (MAYA)
    - Charles & Ray Eames (the Lounge Chair; *Powers of Ten*; *The India Report*, 1958)
    - Breuer (the Wassily chair)

    - Rietveld (the Red and Blue Chair)
    - Thonet (No. 14)
    - Jacobsen (the Egg chair)
    - Saarinen (the Tulip chair)
    - Panton
    - Starck (Juicy Salif)
    - Sottsass (Valentine; Memphis)
    - Ekuan (the Kikkoman bottle)
    - Yanagi (the Butterfly stool)
    - Noguchi (Akari)
    - Porsche (the Beetle)
    - Giacosa (the Fiat 500)
    - Issigonis (the Mini)
    - D'Ascanio (the Vespa)
- **Fashion:** Chanel (the LBD), Dior (the New Look, 1947), YSL (Le Smoking), Miyake (Pleats Please), Kawakubo (CDG), Westwood (punk), Armani (who died in 2025), Ritu Kumar, Sabyasachi, Rahul Mishra, Bhanu Athaiya (the first Indian Oscar, 1983).
- **Movements:**
    - Arts & Crafts (Morris)
    - Art Nouveau (Mucha, Gaudí)
    - De Stijl (Mondrian)
    - Bauhaus (1919–33)
    - Constructivism
    - Art Deco
    - Mid-century modern
    - Swiss style
    - Pop Art (Warhol)
    - Op Art (Riley)
    - Minimalism
    - Memphis (1981)
    - Postmodernism
    - Deconstructivism
    - flat design, then skeuomorphism, then Liquid Glass (2025)
- **Art movements:**
    - Impressionism (Monet)
    - Post-Impressionism (Van Gogh, Seurat's pointillism)
    - Expressionism (Munch)
    - Fauvism (Matisse)
    - Cubism (Picasso, Braque)
    - Futurism
    - Dada (Duchamp's *Fountain*)
    - Surrealism (Dalí, Magritte)
    - Abstract Expressionism (Pollock, Rothko)

10. LOGOS & PRODUCTS (rapid list)

- **FedEx:** an arrow between the E and the x. **Amazon:** an arrow from a to z (and a smile). **Toyota:** all the letters can be found in the emblem.
- **Baskin-Robbins:** a 31. **Toblerone:** a bear. **WWF:** closure. **Unilever:** 25 icons.
- **Audi:** 4 merged companies. **Mercedes:** land, sea, air. **BMW:** the Bavarian flag.
- **Olympics:** 5 rings. **Starbucks:** a siren. **Shell:** Loewy.

- **India:**
  - the Amul girl (1966)
  - the Air India Maharaja (1946) and "The Vista" (2023)
  - the national emblem (the Sarnath Lion Capital; *Satyameva Jayate* from the Mundaka Upanishad)
  - the ₹ symbol (2010)
  - Swachh Bharat (Gandhi's glasses)
  - Make in India (a lion made of cogs)
  - Incredible India (the ! as an i)
  - Gattu (R.K. Laxman)
- **Products:**
  - the Coca-Cola bottle (1915)
  - the Walkman (1979)
  - the iPhone (2007)
  - the Vespa (1946)
  - the Ambassador (Hindustan Motors)
  - the Tata Nano (2008)
  - the Jaipur Foot (1969)
  - the steel tiffin
  - the kulhad
  - LEGO (Denmark)
  - IKEA (Sweden)
  - MUJI (Japan)
- **Currency motifs:** ₹10 Konark · ₹20 Ellora · ₹50 Hampi · ₹100 Rani ki Vav · ₹200 Sanchi · ₹500 Red Fort.

11. KEY TERMS (one line each)

Term	Meaning
Affordance	An object's form suggests how to use it
Signifier	A cue that shows how to use something
Mapping	Controls correspond to their effects
Feedback	The system responds to an action
Poka-yoke	Mistake-proofing
Persona	A research-based user profile
Journey map	The user's steps with their pain points
Prototype	A test model
Iteration	Repeated refinement
Anthropometry	Body measurement
Biomimicry	Design inspired by nature
Upcycling	Turning waste into higher value
Greenwashing	False eco-claims
LCA	Life-cycle assessment
Skeuomorphism	UI that imitates real materials
Vector / raster	Paths vs pixels
Knolling	Arranging objects at right angles in a neat grid
Dieline	The flat net of a package

Term	Meaning
Isometric	A parallel projection with axes 120° apart
Chiaroscuro	Strong light–dark contrast
Vernacular	Local and traditional
Jaali	A perforated screen
Jharokha	An overhanging balcony window
Charbagh	A four-part garden
Shikhara / vimana / gopuram	Temple towers and gateway (Nagara tower / Dravida tower / southern gateway)

12. FLASHCARDS (160)

How to use them:

1. Cover the "Back" column and answer aloud.
2. Tick the cards you get right.
3. Re-test the unticked cards after 1, 3 and 7 days (spaced repetition).
4. You can paste them into Anki or onto index cards.

12.1 Design terminology & principles (1–25)

#	Front	Back
1	Shape vs form?	Shape is 2D; form is 3D (has volume)
2	Proportion vs scale?	Proportion compares parts within an object; scale compares size to a reference (often the human)
3	Three balance types?	Symmetrical, asymmetrical, radial
4	Progressive rhythm?	Repetition with a gradual change (e.g. growing sizes)
5	Hierarchy?	The order of importance shown through size, weight, colour and position
6	Negative space?	The empty space around and between subjects (the FedEx arrow)
7	Gestalt "closure"?	The mind completes incomplete shapes (the WWF panda)
8	Gestalt "proximity"?	Things that are near each other look grouped
9	Rule of thirds?	Place subjects on the third lines or their intersections
10	Lead room?	Space in front of a moving or looking subject
11	Stanford design thinking stages?	Empathise, Define, Ideate, Prototype, Test
12	Double Diamond?	Discover, Define, Develop, Deliver (Design Council, 2005)
13	Affordance example?	A cup handle suggests gripping; a flat door plate suggests pushing
14	Mapping example?	Stove knobs laid out like the burners
15	Poka-yoke example?	A SIM tray with a cut corner
16	Curb-cut effect?	Features designed for disabled people benefit everyone
17	Ergonomic percentile range?	The 5th to the 95th percentile
18	"Form follows function" coined by?	Louis Sullivan (1896)
19	"Less is more"?	Mies van der Rohe
20	Rams' motto?	"Less, but better" (*Weniger, aber besser*)
21	Loewy's MAYA?	Most Advanced Yet Acceptable

#	Front	Back
22	Norman's famous book?	*The Design of Everyday Things* (1988)
23	SCAMPER?	Substitute, Combine, Adapt, Modify, Put to other use, Eliminate, Reverse
24	Six Thinking Hats by?	Edward de Bono
25	Biomimicry example?	Velcro (burrs); the Eastgate Centre (termite mounds)

12.2 Colour (26–40)

#	Front	Back
26	RYB complement of blue?	Orange
27	RYB complement of yellow?	Violet
28	Tint, shade, tone?	Adding white, black and grey respectively
29	RGB: red + green light?	Yellow
30	CMY: cyan + magenta ink?	Blue
31	Analogous scheme?	Neighbours on the wheel
32	Triadic scheme?	Three hues 120° apart
33	Split-complementary of red?	Red + yellow-green + blue-green
34	60-30-10?	Dominant, secondary and accent colour shares
35	Atmospheric perspective?	Distance makes things paler, bluer and less detailed
36	Saturation?	The intensity or purity of a colour
37	ISO blue sign?	A mandatory action
38	ISO yellow sign?	A warning
39	Why not red text on green?	Low value contrast; fails for colour-blind viewers
40	Colour of mourning in many Indian communities?	White

12.3 Typography & communication (41–55)

#	Front	Back
41	Kerning?	The space between one specific letter pair
42	Tracking?	Uniform spacing across a word or line
43	Leading?	The space between lines
44	Legibility vs readability?	Telling characters apart vs reading blocks of text easily
45	Counter?	The enclosed space in a letter (o, a, e)
46	Helvetica designer and year?	Max Miedinger (with Eduard Hoffmann), 1957
47	Futura?	Paul Renner, 1927
48	Times New Roman?	Stanley Morison, 1932, for *The Times*
49	London Underground typeface?	Johnston (1916)
50	Frutiger was made for?	Charles de Gaulle Airport signage
51	Munich 1972 pictograms?	Otl Aicher
52	NYC subway map (1972)?	Massimo Vignelli
53	Tube map (1931/33)?	Harry Beck
54	Prohibition sign format?	A red circle with a diagonal slash
55	Max typefaces per layout?	2

12.4 Designers, artists & architects (56–90)

#	Front	Back
56	₹ symbol designer?	D. Udaya Kumar (2010)
57	NID founded?	1961, Ahmedabad (after the Eames' *India Report*, 1958)
58	NID founders (family)?	Gira and Gautam Sarabhai
59	NID became an INI by?	The NID Act, 2014
60	The Eames' admired Indian object?	The lota
61	Paul Rand logos?	IBM, ABC, UPS
62	I ❤ NY?	Milton Glaser (1977)
63	Nike swoosh?	Carolyn Davidson (1971)
64	Apple logo?	Rob Janoff (1977)
65	Mac icons?	Susan Kare
66	Braun designer?	Dieter Rams
67	iPod and iPhone designer?	Jony Ive
68	Wassily Chair?	Marcel Breuer (1925)
69	Red and Blue Chair?	Gerrit Rietveld
70	Barcelona Chair?	Mies van der Rohe
71	Egg Chair?	Arne Jacobsen
72	Tulip Chair?	Eero Saarinen
73	Juicy Salif?	Philippe Starck (Alessi, 1990)
74	Memphis Group founder?	Ettore Sottsass (1981)
75	Kikkoman bottle?	Kenji Ekuan (1961)
76	Fallingwater?	Frank Lloyd Wright
77	Villa Savoye?	Le Corbusier
78	Bauhaus founder?	Walter Gropius (1919, Weimar)
79	Sydney Opera House?	Jørn Utzon
80	Louvre Pyramid?	I.M. Pei (1989)
81	Guggenheim Bilbao?	Frank Gehry (1997)
82	First woman Pritzker laureate?	Zaha Hadid (2004)
83	First Indian Pritzker laureate?	B.V. Doshi (2018)
84	First African Pritzker laureate?	Diébédo Francis Kéré (2022)
85	Pritzker 2025?	Liu Jiakun (China)
86	Pritzker 2026?	Smiljan Radić Clarke (Chile) [reported]
87	IIM Ahmedabad architect?	Louis Kahn
88	Jawahar Kala Kendra architect?	Charles Correa
89	Low-cost brick architect in Kerala?	Laurie Baker
90	New Parliament architect?	Bimal Patel (HCP)

12.5 Movements & art (91–110)

#	Front	Back
91	Impressionism named after?	Monet's *Impression, Sunrise*

#	Front	Back
92	Pointillism?	Seurat
93	*The Starry Night*?	Van Gogh
94	*The Scream*?	Munch (Expressionism)
95	Cubism founders?	Picasso and Braque
96	*Guernica* subject?	The bombing of a Spanish town (1937)
97	Melting clocks?	Dalí, *The Persistence of Memory*
98	"Ceci n'est pas une pipe"?	Magritte
99	*Fountain* (a urinal)?	Duchamp (Dada, 1917)
100	De Stijl palette?	Primary colours + black and white, a grid (Mondrian)
101	Pop Art leader?	Andy Warhol
102	Op Art?	Bridget Riley, Victor Vasarely
103	Drip painting?	Jackson Pollock
104	Fauvism?	Matisse; wild colour
105	Arts and Crafts leader?	William Morris
106	Art Nouveau posters?	Alphonse Mucha
107	Chiaroscuro?	Strong light–dark contrast (Caravaggio)
108	Linear perspective pioneer?	Brunelleschi
109	Vitruvian Man?	Leonardo da Vinci
110	Progressive Artists' Group?	Bombay, 1947: Souza, Raza, Husain

12.6 Indian crafts, textiles & art (111–140)

#	Front	Back
111	Warli state and features?	Maharashtra; white rice paste on mud; circle, triangle, square
112	Madhubani features?	Bihar; double outlines; no empty space
113	Gond features?	MP; dots and dashes fill forms; Jangarh Singh Shyam
114	Phad?	Rajasthan scroll of Pabuji, performed by Bhopas
115	Kalighat?	Kolkata; bold brush lines; satire
116	Srikalahasti vs Machilipatnam Kalamkari?	Freehand pen vs block print
117	"Bani Thani"?	Kishangarh, by Nihal Chand ("Indian Mona Lisa")
118	Patola?	Patan; double ikat
119	Ikat key fact?	The yarn is dyed before weaving
120	Kanjeevaram border technique?	Korvai
121	Paithani motifs?	Peacocks and parrots
122	Baluchari?	Narrative scenes on the pallu (WB)
123	Muga silk?	Assam; naturally golden
124	Eri silk?	"Ahimsa" silk
125	Chikankari?	Lucknow; white-on-white embroidery
126	Phulkari?	Punjab; "flower work"
127	Kantha?	Bengal; running stitch on layered old saris
128	Ajrakh colours?	Indigo and madder red
129	Leheriya vs Bandhani?	Waves vs dots

#	Front	Back
130	Dabu?	Mud-resist printing
131	Blue Pottery base?	Quartz (not clay), Jaipur
132	Bidriware?	Bidar; blackened zinc alloy with silver inlay
133	Dhokra?	Lost-wax metal casting
134	Channapatna?	Karnataka; lacquered wooden toys
135	Aranmula Kannadi?	Kerala; a metal-alloy mirror
136	Longpi pottery?	Manipur; black; made without a wheel
137	Rogan art?	Castor-oil paint; Nirona (Kutch)
138	First GI in India?	Darjeeling Tea (2004)
139	GI Registry location?	Chennai
140	Garba UNESCO ICH year?	2023

12.7 Indian architecture & heritage (141–155)

#	Front	Back
141	Nagara vs Dravida tower?	Shikhara vs vimana (+ gopuram gateways)
142	Kailasa Temple?	Ellora Cave 16; carved top-down from one rock; Rashtrakuta Krishna I
143	Brihadeeswara?	Thanjavur; Rajaraja Chola I, about 1010
144	Konark?	A sun-temple chariot with 24 wheels; Narasimhadeva I
145	First garden tomb?	Humayun's Tomb
146	Taj Mahal chief architect?	Ustad Ahmad Lahauri
147	Gol Gumbaz?	Bijapur; a whispering gallery
148	Hawa Mahal?	Jaipur, 1799; Lal Chand Ustad
149	Rani ki Vav?	Patan stepwell; on the ₹100 note
150	CSMT architect?	F.W. Stevens (1888)
151	India's first WH City?	Ahmedabad (2017)
152	India's 44th WHS?	The Maratha Military Landscapes (2025)
153	Moidams?	Ahom mound burials, Assam (WHS 2024)
154	Only mixed WHS in India?	Khangchendzonga NP
155	Chandigarh planner?	Le Corbusier (with Pierre Jeanneret)

12.8 Logos, symbols & general awareness (156–160)

#	Front	Back
156	National emblem source?	The Lion Capital of Ashoka, Sarnath
157	Flag ratio and chakra spokes?	2:3; 24 spokes
158	Amazon logo arrow?	From a to z (and a smile)
159	Air India 2023 logo name?	"The Vista"
160	World Design Capital 2026?	Frankfurt RheinMain

13. TOP 100 NID DAT MISTAKES STUDENTS MAKE (and the fix for each)

A. Time management (1–12)

1. Spending 40+ minutes on the first drawing. **Fix:** write a time budget per question before starting.
2. Polishing one drawing and leaving the last question blank. **Fix:** use the 70% rule.
3. Not reading the whole paper first. **Fix:** spend 5 minutes reading everything.
4. Getting stuck on a hard MCQ. **Fix:** move on after 90 seconds.
5. No buffer time. **Fix:** keep 15–20 minutes for finishing touches.
6. Starting to colour before all the questions are drafted. **Fix:** colour last.
7. Not wearing a watch (where allowed). **Fix:** practise with the same watch.
8. Not practising full 3-hour papers. **Fix:** at least 5 timed mocks.
9. Erasing too much. **Fix:** start with light construction lines, then commit.
10. Rewriting text answers. **Fix:** draft mentally, then write once.
11. Underestimating multi-part questions. **Fix:** count the sub-tasks and split the time.
12. Panicking after one bad question. **Fix:** it's one of 14+ tasks; reset and move on.

B. Reading instructions (13–22)

13. Drawing 3 items when 5 are asked for. **Fix:** circle the numbers.
14. Ignoring the viewpoint (e.g. "from a crow's view"). **Fix:** underline the viewpoint words.
15. Using extra colours despite a limit. **Fix:** put the allowed colours in a separate pile.
16. Exceeding or falling short of the word limit. **Fix:** count the lines (about 10–12 words per line).
17. Missing "label" or "explain" instructions. **Fix:** tick each instruction when done.
18. Answering a different question (e.g. drawing a product when a poster was asked for). **Fix:** restate the task in the margin.
19. Ignoring the specified user (elderly, child). **Fix:** draw the user in the solution.
20. Drawing outside the box or margins. **Fix:** plan the layout first.
21. Not numbering the frames. **Fix:** always number them.
22. Forgetting the title or caption when asked. **Fix:** add a checklist line at the end.

C. Drawing & proportion (23–40)

23. People taller than doors. **Fix:** use size references.
24. Floating people. **Fix:** anchor their feet on the ground plane; add cast shadows.
25. Several horizon lines in one scene. **Fix:** one HL, drawn first.
26. Tilted verticals in 1- and 2-point perspective. **Fix:** use a scale for the verticals.
27. VPs too close together. **Fix:** put the VPs far apart, even off the page.
28. Stiff, mannequin figures. **Fix:** start with the line of action and gesture.
29. Identical faces and poses for every person. **Fix:** vary age, posture and clothing.
30. Tiny drawings in huge boxes. **Fix:** fill about 70–80% of the box.
31. Over-detailing one corner. **Fix:** spread the detail across the focal hierarchy.
32. Shading with no consistent light source. **Fix:** draw a small sun or arrow first.
33. Messy, hairy lines. **Fix:** ghost the stroke, then draw it confidently.
34. Not showing materials. **Fix:** use texture conventions for each material.
35. Hands hidden behind backs to avoid drawing them. **Fix:** practise mitten hands.
36. Ignoring depth layers. **Fix:** foreground, middle and background.
37. Wrong object details (e.g. a bicycle chain on the left). **Fix:** do weekly memory-vs-reality checks.
38. Drawing from imagination only, never from observation. **Fix:** daily sketching from life.
39. Perspective on circles (ellipses) going wrong. **Fix:** practise ellipses in boxes.
40. Rendering photorealistically at the cost of the idea. **Fix:** idea first.

D. Composition & communication (41–55)

41. Centring everything. **Fix:** use the rule of thirds.
42. No focal point. **Fix:** create one with contrast, size or isolation.
43. Clutter. **Fix:** remove elements that don't serve the story.
44. Tangents. **Fix:** overlap the objects or separate them clearly.
45. Poor hierarchy in posters. **Fix:** headline > image > text > logo.
46. Too much text on a poster or sign. **Fix:** 3–6-word headlines; use icons.
47. Illegible handwriting. **Fix:** print in capitals for labels.
48. Labels with no arrows. **Fix:** use thin leader lines.
49. Unclear storyboards. **Fix:** vary the shots; keep the characters consistent.
50. Changing the character's look between frames. **Fix:** give them one distinctive accessory.
51. A story with no resolution. **Fix:** plan the beats: setup, problem, action, resolution.
52. Ignoring negative space. **Fix:** let the subject breathe.
53. Random colours. **Fix:** choose a scheme (analogous, complementary).
54. Low contrast in signs. **Fix:** dark on light or light on dark.
55. A logo that fails in black and white. **Fix:** test a mono version.

E. Creativity & originality (56–68)

56. The first cliché (a crying Earth, a melting globe). **Fix:** generate 5 ideas and pick the 3rd or 4th.
57. Copying a famous logo or poster from memory. **Fix:** start from the brief's keywords.
58. Being "creative" but irrelevant. **Fix:** check the idea against the brief.
59. Only one category of ideas (low flexibility). **Fix:** vary the categories (tool, toy, art, shelter).
60. Overusing technology (apps, drones, robots). **Fix:** think physical, low-cost and contextual first.
61. Unrealistic magic solutions. **Fix:** hint at the mechanism or material.
62. Fear of simple ideas. **Fix:** a simple idea done brilliantly beats a complex mess.
63. Not using the Indian context. **Fix:** add local details (chai, autos, rangoli).
64. Ignoring humour or emotion where appropriate. **Fix:** a touch of wit makes answers memorable.
65. Not personalising "product you love" answers. **Fix:** a real story plus specific details.
66. Repeating the same idea style in every answer. **Fix:** vary your approaches.
67. Ignoring viewpoint as a creative lever. **Fix:** try unusual angles.
68. Not sketching thumbnails. **Fix:** 3 thumbnails in 3 minutes.

F. Problem-solving & user focus (69–80)

69. Jumping to a solution without defining the problem. **Fix:** write a one-line problem statement.
70. Designing for yourself, not the user. **Fix:** draw the user.
71. Ignoring constraints (cost, electricity, water). **Fix:** list the constraints first.
72. Solving one stakeholder's problem only (e.g. in the neem-branch dispute). **Fix:** consider every party.
73. Destructive solutions (ban it, cut it down). **Fix:** look for win-win, sustainable fixes.
74. Ignoring safety, especially for children's products. **Fix:** add one safety note.
75. Ignoring maintenance and cleaning. **Fix:** think about the whole life cycle.
76. Ignoring accessibility. **Fix:** consider low vision, mobility and literacy.
77. No explanation of how the solution works. **Fix:** labels plus a 2-line note.
78. No in-use scene. **Fix:** a small inset showing the user using it.
79. Over-complex systems for a small problem. **Fix:** proportionate solutions.
80. Not considering the systemic fix (e.g. repairing the pothole, not just helping the person who fell). **Fix:** end with the system improvement.

G. Visual reasoning & objective questions (81–92)

81. Forgetting to reverse the letter order in mirror words. **Fix:** reverse, then flip each letter.
82. Confusing a mirror image with a rotation. **Fix:** a rotation never produces a mirror image.
83. Unfolding in the wrong order. **Fix:** unfold in reverse.
84. Reflecting across the wrong axis in folding. **Fix:** draw the fold line.
85. Calling faces opposite when they are adjacent in nets. **Fix:** the one-apart rule.
86. Forgetting the downward triangles when counting. **Fix:** count systematically by size and orientation.
87. Mixing up the RGB and RYB wheels. **Fix:** check whether the question is about paint or light.
88. Overthinking simple GK. **Fix:** go with the first well-founded answer.
89. Leaving MCQs blank (there is no negative marking). **Fix:** guess intelligently.
90. Unit errors (cm vs m). **Fix:** convert first.
91. Accepting a syllogism conclusion that is only possible. **Fix:** it must be true in every diagram.
92. Careless option marking. **Fix:** re-check the question number against the answer sheet.

H. General awareness & preparation habits (93–100)

93. Memorising GK without visuals. **Fix:** look at images of every craft and building you learn.
94. Ignoring Indian crafts and textiles. **Fix:** 3 crafts a day from Vol 2C.
95. Cramming current affairs. **Fix:** link each event to a design idea.
96. Not keeping an error log. **Fix:** Vol 13 §6, updated daily.
97. Practising only favourite topics. **Fix:** follow the plan's rotation.
98. Not getting feedback on drawings. **Fix:** a weekly review by a teacher or design student.
99. Skipping sleep before the exam. **Fix:** 7–8 hours; no all-nighters.
100. Forgetting the admit card, ID or kit. **Fix:** pack the night before using the checklist.

NEXT VOLUME: Volume 13 — Study Plans, Daily Routines, Revision System & Error Log (`13-VOL13-STUDY-PLANS.md`)

VOLUME 13 — STUDY PLANS, DAILY ROUTINES, REVISION SYSTEM & ERROR LOG

> **Your real timeline:** today is **7 Oct 2026**, and Prelims is on **20 Dec 2026**, which leaves **74 days**. Use the **dated 10-week calendar (§1)** as your main plan. The 30/60/90-day plans (§2–4) are the requested generic templates: use the 60-day plan for the core, or the 30-day plan if you are starting late. Keep the 90-day plan for the Mains phase (it runs to April 2027) or for the next cycle.

Allocation principle (from Vol 0 §9):

Area	Share of time
Drawing	40%
Ideation / problem-solving	20%
Mocks	15%
Visual reasoning	10%
Design / culture GK	10%
Current affairs	5%

1. THE DATED 10-WEEK CALENDAR (7 Oct → 20 Dec 2026)

Week	Dates	Theme	Daily drawing focus	Ideation / problem drills	Objective / GK	Mock
1	7–13 Oct	Foundations	Lines, ellipses, boxes; basic forms; 20 gestures/day	SCAMPER on 1 object/day	Vol 2A Part 1; 15 VR MCQs/day	—
2	14–20 Oct	Perspective + people	1-point and 2-point scenes (1/day); proportion studies	1 Vol 3 §13 prompt/day (D or J)	Vol 2A Part 2; Vol 4B §A–B	—
3	21–27 Oct	Scenes + memory	Scene with 6+ people (1/day); 3 memory objects	Posters (E) on alternate days	Vol 2C §2–4 (crafts and textiles); Vol 4F §B–C	**Paper 1** (Sun 25 Oct)
4	28 Oct–3 Nov	Storytelling	1 storyboard every 2 days; faces/expressions grid	Signage and pictograms (F, G)	Vol 4B §C–D; Vol 4A §A–D	—
5	4–10 Nov	Product + redesign	Product sketching (3/4 views, hands, labels)	PUCIS redesign drill daily	Vol 2C §5–8; Vol 4G §C–D	**Paper 2** (Sun 8 Nov)
6	11–17 Nov	Viewpoints + imagination	Bird's/worm's-eye; "what if"	Kits / knolling (B4), packaging	Vol 4B §E–H; Vol 4G §E–F	—
7	18–24 Nov	Integration I	Mixed prompts at exam timing (25 min each)	Vol 5 drill set, items 1–7	Vol 4C (selected); Vol 11 Parts A + D	**Paper 3** (Sun 22 Nov)
8	25 Nov–1 Dec	All-subjective format	Two 12-minute sketches back to back, ×4	Vol 5 drills, items 8–14	Vol 4D/4E light; flashcards	**Paper 4** (Sun 29 Nov)
9	2–8 Dec	Full simulations	Weakest drawing type, ×2/day	Vol 5 drills, items 15–20	Error log review; flashcards	**Paper 5** (Sun 6 Dec, 10–1)

Week	Dates	Theme	Daily drawing focus	Ideation / problem drills	Objective / GK	Mock
10	9–20 Dec	Revision & taper	Light daily sketching (45 min)	Re-attempt the 6 lowest-scored prompts	Vol 12 revision book	**Paper 4 retake** (Sun 13 Dec); rest 19 Dec

Administrative reminders:

- **Submit the application before 30 Nov** (aim for 31 Oct).
- Check for the admit card around 10 Dec.
- Read the admit-card instructions on the materials allowed.

2. 30-DAY PLAN (late-start or compressed version) · 5–6 hours/day

Days	Daily hours	Topics	Drawing practice	Question practice	Mocks	Revision	Current affairs	Error analysis
1–5	6	Vol 1, Vol 2A (fundamentals, perspective)	2 hrs: lines, boxes, 1-pt/2-pt scenes, gestures	30 VR MCQs/day	—	15 min flashcards	10 min	15 min
6–10	6	Vol 3 §§4–10 (scenes, story, product)	2.5 hrs: 1 scene + 1 storyboard/day	20 MCQs (4A/4B)	Day 10: Paper 1	15 min	10 min	20 min
11–15	6	Vol 2C (crafts, architecture, designers)	2 hrs: product redesign + posters	20 MCQs (4F/4G)	Day 15: Paper 2	20 min	10 min	20 min
16–20	6	Vol 5 expected patterns	2.5 hrs: Tier-A drills at exam timing	15 MCQs	Day 20: Paper 3	20 min	10 min	20 min
21–25	6	Integration	3 hrs: mixed prompts	15 MCQs	Day 23: Paper 4; Day 25: Paper 5	25 min	10 min	30 min
26–29	4	Weak areas + revision	1.5 hrs: weakest types	10 MCQs	Day 27: Paper 4 retake (another set from Vol 5)	1 hr (Vol 12)	10 min	20 min
30	1–2	Final 24-hour strategy (§9)	30 min of light sketching	—	—	Vol 12 §1	—	Read the error log

3. 60-DAY PLAN (recommended core) · 4–5 hours/day

Week	Topics	Drawing (daily)	Questions	Mock	Revision	Current affairs	Error log
1	Elements, principles, colour; mirror/rotation	90 min: lines, shapes, forms, value scales	20 VR + 10 DA	—	10 min flashcards	10 min (Mon/Thu)	Daily
2	Perspective; folding; cubes	90 min: 1-pt and 2-pt scenes; people in perspective	20 VR	—	Weekly recap Sunday	10 min	Daily
3	Typography, composition, visual communication	90 min: posters, lettering; memory objects	15 DA + 10 GK	Paper 1 (end of the week)	Sunday recap	10 min	Daily + mock analysis

Week	Topics	Drawing (daily)	Questions	Mock	Revision	Current affairs	Error log
4	HCD and design thinking; product sketching	90 min: redesigns (PUCIS)	15 DA	—	Sunday	10 min	Daily
5	Indian art, crafts, textiles	90 min: storyboards, faces, hands	20 GK (4F)	Paper 2	Sunday	10 min	Daily
6	Architecture, designers, movements	90 min: viewpoints, imagination	20 GK (4G)	—	Sunday	10 min	Daily
7	Expected patterns (Vol 5)	120 min: Tier-A at exam timing	15 mixed	Paper 3	Sunday	15 min (Vol 11)	Daily
8	All-subjective format training	120 min: 12-min sprints	10 mixed	Paper 4	Sunday	15 min	Daily
9	Full simulations; weak areas	90 min: weakest type	10 mixed	Paper 5 (full protocol)	2× a week	15 min	Daily
10 (days 64–70; trim if short)	Revision (§7)	45 min light	flashcards	Retake of the weakest paper	Daily	10 min	Review

4. 90-DAY PLAN (for the Mains phase or the next cycle) · 3–4 hours/day

Phase	Days	Focus	Drawing	Questions	Mocks	Revision / CA / errors
Foundation	1–30	Vol 1, 2A, 2B; basics of drawing	75 min/day: lines → forms → perspective → people	20 MCQs/day across 4A–4E	1 diagnostic (Paper 1) on day 30	10 min flashcards; CA weekly; daily error log
Build	31–60	Vol 2C, Vol 3 complete; Vol 5	90 min/day: scenes, stories, products, posters	15 MCQs/day (4F, 4G, 11)	Papers 2 & 3 (days 45, 60)	Weekly recap; CA weekly; error log
Perform	61–80	Exam timing, mixed prompts, both formats	120 min/day timed	10 MCQs/day	Papers 4 & 5 (days 70, 77) + one Vol 5 drill paper	2× weekly revision
Peak & taper	81–90	Revision system (§7–9)	45–60 min/day	Flashcards	One retake	The final 7-day strategy

Mains add-on (after the Prelims result, about March): 3× a week clay, paper and wire model-making (Vol 3 §12); 2× a week mock interviews with 8 prepared answers; current-affairs talking points.

5. DAILY ROUTINES

5.1 Daily 2-hour plan (school days)

Time	Activity
0:00–0:10	Warm-up: 20 gestures (30 sec each) + lines/ellipses
0:10–0:55	**Main drawing task** (one scene, storyboard or product redesign, rotating daily)
0:55–1:15	**Ideation drill**: one Vol 3 §13 prompt as 3 thumbnails

Time	Activity
1:15–1:35	**15 MCQs** (rotate 4A/4B/4F/4G)
1:35–1:50	Flashcards (20 cards)
1:50–2:00	Error log update + plan tomorrow
Weekly:	Sunday = a 3-hour mock instead (or a half mock on busy weeks)

5.2 Daily 4-hour plan

Time	Activity
0:00–0:15	Warm-up gestures + perspective boxes
0:15–1:15	**Drawing block 1**: a scene or storyboard
1:15–1:30	Break
1:30–2:15	**Drawing block 2**: product, poster or signage
2:15–2:45	**Theory**: one Vol 2 topic (notes + practice)
2:45–3:15	**MCQs**: 25 questions + review
3:15–3:35	Memory drawing (1 object) + reality check
3:35–3:50	Flashcards + current affairs (2–3 items)
3:50–4:00	Error log

5.3 Daily 6-hour plan (holidays and weekends)

Time	Activity
0:00–0:20	Warm-up
0:20–1:30	**Timed subjective set**: 2 prompts at exam timing
1:30–1:45	Break
1:45–2:45	**Observation drawing outdoors** (a market, station, park), 3 quick scenes
2:45–3:30	Lunch
3:30–4:15	**Theory**: Vol 2 topic
4:15–5:00	**MCQs**: 40 questions + review
5:00–5:30	**Storyboard or product drill**
5:30–5:50	Flashcards + current affairs
5:50–6:00	Error log + tomorrow's plan

Health rules: stand and stretch every 45 minutes; get 7–8 hours of sleep; spend 20 minutes outdoors daily, sketching counts. Avoid burnout: take one lighter day a week (2 hours).

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6. ERROR LOG SYSTEM

6.1 Template

Date	Question (ref)	Topic	My answer	Correct answer	Why I got it wrong	Lesson	Reattempt date
12 Oct	VR-42	Paper folding	(c)	(b)	Reflected across the wrong axis	Draw the fold line; unfold in reverse	15 Oct

Date	Question (ref)	Topic	My answer	Correct answer	Why I got it wrong	Lesson	Reattempt date
25 Oct	Paper 1 Q24	Storyboard	11/20	16+/20	Tree grew but the child didn't age; viewpoint changed	Fix the camera; age the character	28 Oct

6.2 How to maintain it

1. **Log every error within 24 hours.** That includes MCQs you got wrong, MCQs you guessed right (mark them "G"), and every subjective question scoring below 70% of its marks.
2. **Classify the cause** using one of these codes:
   - **K** = knowledge gap
   - **C** = concept misunderstanding
   - **R** = reading or instruction error
   - **T** = time pressure
   - **S** = skill (drawing) weakness
   - **X** = careless
3. **Write the lesson as a rule** ("Always unfold in reverse order"), not as a feeling.
4. **Schedule reattempts** after 3 days and again after 10 days (spaced repetition). On a reattempt, solve the question cold before looking at your notes.
5. **Every Sunday:** count the codes. If R + X > 30% of your errors, slow down and use the "circle the constraints" habit. If S dominates, add 30 minutes of daily drawing for the weakest skill. If K dominates, add flashcards.
6. **Before each mock:** read the last 2 weeks of lessons (5 minutes).
7. **The final week:** read only the "Lesson" column.

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7. 15-DAY REVISION PLAN (use 5–20 Dec, or any 15-day window)

Day	Revise	Practice	Don't waste time on
1	Vol 12 §2–3 (principles, colour)	1 poster (20 min)	Re-reading full theory chapters
2	Vol 12 §5 (perspective rules)	1 scene (25 min)	New techniques
3	Vol 12 §4 (type, visual communication)	Signage set (20 min)	—
4	Vol 12 §7 (crafts and textiles)	Flashcards 111–140	Obscure crafts beyond the list
5	Vol 12 §8 (architecture)	Flashcards 141–155	Memorising every date
6	Vol 12 §9 (designers and movements)	Flashcards 56–110	—
7	**Mock** (Paper 4 or 5, whichever you didn't do last)	Evaluate it	—
8	Error log (all lessons)	Reattempt 10 logged MCQs	—
9	Vol 12 §6 (reasoning formulas)	25 VR MCQs	Hard maths
10	Vol 5 Tier A	2 Tier-A prompts timed	—
11	Vol 11 Part A	CA MCQs 1–50	Unverified gossip
12	Vol 5 Tier B	2 prompts timed	—
13	**Mock** (a new subjective set from Vol 5 §6, all-subjective format)	Evaluate it	—
14	Weakest 2 topics from the error log	Light practice	Heavy new practice

Day	Revise	Practice	Don't waste time on
15	§9 final 24-hour strategy	30 min of light sketching	Anything new

8. 7-DAY REVISION PLAN (if you only have one week)

Day	Morning (2 h)	Afternoon (2 h)	Evening (1 h)
1	Vol 12 §1–3	Scene + storyboard timed	Flashcards 1–55
2	Vol 12 §4–5	Product redesign + poster timed	Flashcards 56–110
3	Vol 12 §6 + 25 VR MCQs	Signage + memory drawing	Flashcards 111–160
4	**Full mock** (all-subjective format)	Evaluate it + error log	Rest
5	Vol 12 §7–9	Two Tier-A prompts	CA Part A
6	Error log lessons	One light timed set	Pack the kit
7	Final 24-hour strategy (§9)	—	Sleep early

9. FINAL 7-DAY STRATEGY (13–19 Dec) & FINAL 24-HOUR STRATEGY (19–20 Dec)

Final 7 days: what to revise

- **Revise:** Vol 12 (the whole revision book), your error-log lessons, the Vol 5 Tier-A patterns, and the formulas in Vol 12 §6.
- **Practise:** one timed subjective set every 2 days; 20 minutes of light sketching on the other days.
- **Do one mock** on Sunday 13 Dec at 10:00 (the exam slot) to set your body clock.
- **Don't waste time on:**
  - new books or channels
  - learning new rendering media (e.g. watercolour, if you haven't practised it)
  - obscure GK
  - hard maths
  - comparing yourself with others on social media
  - all-night sessions

Final 24 hours (19 Dec)

Time	Do
Morning	30 minutes of light sketching (gestures + one quick scene) to stay warm
Midday	Read Vol 12 §1 (exam strategy) and the last page of your error log
Afternoon	**Pack the kit:** admit card (printed, 2 copies), photo ID, pencils (HB, 2B, 4B, 6B + spares), eraser, kneaded eraser, sharpener, scale, colour pencils, oil pastels, sketch pens, black fine-liners, a watch (if allowed), water, a light snack. Check the admit card's materials list.
Evening	Confirm the centre route and travel time; set 2 alarms; eat light; no screens after 9 pm
Night	Sleep 7–8 hours

Exam morning (20 Dec)

- A light breakfast; reach the centre **45 minutes early**.
- Before the paper is handed out, do 5 slow breaths.
- **In the first 5 minutes:** read everything, circle the constraints, write your time budget.
- Remember: **finish everything, idea first, 70% rule, no blanks.**

NEXT: Sources & Self-Audit (`99-SOURCES-AND-SELF-AUDIT.md`)

SOURCES USED, RESEARCH LOG & QUALITY SELF-AUDIT

A. Research method & limitations (read this)

- **Research date:** 7 October 2026.
- **Method:** web search across official NID document listings and education and news portals. Where possible, each claim that can change from year to year was cross-checked in two or more sources.
- **Limitation:** my research environment's network policy **blocked direct access to `admissions.nid.edu` and to most third-party sites**. Official facts were therefore taken from search-engine extracts of NID's own PDF documents (the URLs are listed below) and corroborated by education portals. **Please verify all dates, fees and rules on https://admissions.nid.edu before acting on them.**
- **Labels used in the package:** [OFFICIAL] · [REPORTED] · [TREND-BASED / INFERRED] · [PREDICTION] · [UNVERIFIED].

B. SOURCES

B1. Official NID sources (Tier 1)

Source	Used for
NID Admissions portal: https://admissions.nid.edu/ ("NID Admission: 2027-28")	Cycle identification, application window
Admissions 2027-28 Handbook: 5.5-Year Professional Education M.Des (Integrated Pathway) and B.Des, https://admissions.nid.edu/NIDA2027/download/BDES_IntegratedPathway_AdmissionsHandbook2027_28.pdf	2027-28 programmes, dates (Prelims 20 Dec 2026; last date 30 Nov 2026), DAT stages
B.Des Admissions Handbook 2026-27, https://admissions.nid.edu/NIDA2026/download/BDes_AdmissionsHandbook2026-27.pdf	Prelims marking scheme (Part I 22 objective; Part II 4 subjective; 100 marks; 3 hours; Prelims not carried forward)

Source	Used for
B.Des DAT Prelims Sample Test Paper, https://admissions.nid.edu/NIDA2026/download/SamplePaper_DATPrelims_BDes.pdf	Confirms that an official sample paper exists and gives its format (contents could not be opened directly)

B2. Previous-paper analyses & exam pattern (Tier 2–3)

- Careers360, *NID DAT 2026 Prelims Question Paper with Solution*: https://design.careers360.com/articles/nid-dat-2026-prelims-question-paper-with-solution
- Careers360, *NID DAT 2026 Prelims Question Paper Part B*: https://design.careers360.com/articles/nid-dat-2026-prelims-question-paper-part-b
- Careers360, *NID 2026 Paper Solution, Memory-Based Questions*: https://design.careers360.com/articles/nid-2026-paper-solution
- Shiksha, *NID DAT 2026 B.Des Exam Analysis*: https://www.shiksha.com/design/articles/nid-dat-2026-exam-analysis-for-b-des-blogId-216840
- Shiksha, *NID DAT 2026 exam-day analysis*: https://www.shiksha.com/news/design-nid-dat-2026-exam-day-paper-analysis-live-updates-blogId-216832
- GetMyUni, *NID DAT Exam Analysis 2026*: https://www.getmyuni.com/articles/nid-dat-exam-analysis-2026
- Careers360, *NID DAT 2025: lengthy and moderately difficult; memory-based questions*: https://design.careers360.com/articles/nid-dat-2025-lengthy-and-moderately-difficult-weightage-memory-based-questions
- PW Live, *NID Exam Analysis 2025*: https://www.pw.live/design-architecture/exams/nid-exam-analysis-2025
- Careers360 News, *NID DAT 2024 live updates and analysis*: https://news.careers360.com/nid-dat-2024-exam-live-updates-question-paper-analysis-answer-key-student-reactions-result-date-latest-updates
- Shiksha, *NID exam analysis 2024 for B.Des*: https://www.shiksha.com/design/articles/nid-exam-analysis-2024-for-b-des-course-blogId-147653
- Careers360, *NID 2023 question paper*: https://design.careers360.com/articles/nid-2023-question-paper
- Toprankers, *NID Exam Pattern 2027* and *NID 2027 Exam*: https://www.toprankers.com/nid-exam-pattern ; https://www.toprankers.com/nid-exam
- Collegedunia, *NID DAT 2027*: https://collegedunia.com/exams/nid

B3. Dates, eligibility, seats, Mains

- PW Live, *NID DAT Notification 2027*: https://www.pw.live/design-architecture/exams/nid-dat-notification-2027
- PW Live, *NID Admission 2027-28 / 5.5-year Integrated M.Des*: https://www.pw.live/design-architecture/exams/nid-admission-2027 ; https://www.pw.live/design-architecture/exams/nid-announces-5-5-year-integrated-m-des-admissions-2027-28
- Shiksha, *NID DAT Eligibility Criteria 2027*: https://www.shiksha.com/design/articles/nid-dat-eligibility-criteria-blogId-23559

- Shiksha, *NID DAT Registration 2027*: https://www.shiksha.com/design/communication-design/nid-entrance-exam-application-form
- Toprankers, *NID Seats*: https://www.toprankers.com/news/cuet-exam/nid-seats
- Careers360, *NID B.Des Studio Sensitivity Test 2026* and *DAT Mains 2026*: https://design.careers360.com/articles/nid-bdes-studio-sensitivity-test-2026-date-out ; https://design.careers360.com/articles/nid-dat-mains-2026
- Careers360, *NID DAT Result 2026*: https://design.careers360.com/articles/nid-dat-result

B4. Design, art & architecture references (current and background)

- La Biennale di Venezia, *Awards of the Biennale Architettura 2025*: https://www.labiennale.org/en/news/awards-biennale-architettura-2025
- La Biennale di Venezia, *Biennale Arte 2026: In Minor Keys*: https://www.labiennale.org/en/news/biennale-arte-2026-minor-keys-0
- Dezeen, *Canal Café / Bahrain Golden Lion*: https://www.dezeen.com/2025/05/10/diller-scofidio-renfro-bahrain-golden-lion-awards-venice-architecture-biennale/
- Metalocus, *Smiljan Radić, 2026 Pritzker*: https://www.metalocus.es/en/news/smiljan-radic-clarke-new-2026-pritzker-architecture-prize
- UIA, *Smiljan Radić wins the 2026 Pritzker*: https://www.uia-architectes.org/?p=40207
- Dezeen, *Liu Jiakun key projects*: https://www.dezeen.com/2025/03/04/liu-jiakun-key-projects/amp/
- Building Design, *Liu Jiakun awarded 2025 Pritzker*: https://bdonline.co.uk/news/liu-jiakun-awarded-2025-pritzker-prize/5134794.article
- PBS NewsHour, *Frank Gehry dies at 96*: https://www.pbs.org/newshour/arts/frank-gehry-celebrated-architect-of-wildly-imaginative-buildings-dies-at-96
- Serpentine Galleries, *Serpentine Pavilion 2026, LANZA atelier*: https://www.serpentinegalleries.org/watch/in-the-studio-with-serpentine-pavilion-2026-architects-lanza-atelier/
- Dezeen, *Serpentine Pavilion 2026 architect revealed*: https://www.dezeen.com/2026/01/24/serpentine-pavilion-2026-architect-revealed/amp/
- New Atlas, *Grand Ring, Expo 2025 Osaka*: https://newatlas.com/architecture/grand-ring-osaka-expo-2025/
- Dezeen, *Carlo Ratti torches for Milano Cortina 2026*: https://www.dezeen.com/?p=2193590 ; MIT, *"Essential" torch*: https://mad.mit.edu/news/essential-torch-heralds-the-start-of-the-2026-winter-olympics
- Domus, *Compasso d'Oro 2026 winners*: https://www.domusweb.it/en/news/gallery/2026/05/23/compasso-doro-2026-winners.html
- German Design Council, *World Design Capital Frankfurt RheinMain 2026*: https://www.german-design-council.de/en/world-design-capital-2026
- India Design ID: https://indiadesignid.com/id-delhi-programme/ ; Elle India, *India Design ID 2026*: https://elle.in/life-culture/top-things-to-see-at-india-design-id-2026-11110959
- e-flux, *Kochi-Muziris Biennale 2025–26*: https://www.e-flux.com/announcements/6783901/for-the-time-being
- ArtAsiaPacific, *India names artists and curator for 2026 Venice Biennale*: https://www.artasiapacific.com/news/india-names-artists-and-curator-for-2026-venice-biennale/
- Architexturez / Business Today, *Navi Mumbai International Airport*: https://architexturez.net/node/246554 ; https://www.businesstoday.in/latest/trends/photo/inside-navi-mumbai-international-airport-indias-largest-greenfield-aviation-project-497534-2025-10-09
- *Background reference facts* (designers, movements, historic architecture, crafts): standard design-history knowledge, cross-checked against the author's knowledge base. These are stable, long-established facts; any you doubt can be checked in encyclopaedic sources (e.g. Britannica, museum websites).

B5. Current affairs & policy

- NeGD (Govt of India), *AI Impact Summit 2026 concludes with New Delhi Declaration*: https://negd.gov.in/press_release/ai-impact-summit-2026-concludes-with-adoption-of-new-delhi-declaration/
- Entrepreneur India, *Union Budget 2026-27: Education, Design, Science*: https://india.entrepreneur.com/news-and-trends/union-budget-2026-27-proposes-key-initiatives-in-education/502338
- Vision IAS, *Orange Economy in Union Budget 2026-27*: https://visionias.in/current-affairs/news-today/2026-02-02/economy/finance-minister-announces-plans-to-boost-for-orange-economy-in-union-budget-2026-27
- Time Out, *25 new UNESCO World Heritage Sites 2026*: https://www.timeout.com/news/25-new-unesco-world-heritage-sites-2026-072726
- Drishti IAS, *Maratha Military Landscapes added to the UNESCO list*: https://www.drishtiias.com/daily-updates/daily-news-analysis/maratha-military-landscapes-added-to-unesco-world-heritage-list/print_manually
- Deccan Chronicle / ETV Bharat, *Lucknow named UNESCO Creative City of Gastronomy*: https://www.deccanchronicle.com/lifestyle/food/lucknow-named-unesco-creative-city-for-its-rich-culinary-heritage-1914061
- Adda247, *GI registrations 2025-26*: https://currentaffairs.adda247.com/eastern-india-leads-gi-registrations-madhya-pradesh-tops-india-in-2025-26/amp/
- Anantam IAS, *Jharkhand's four new GI tags*: https://anantamias.com/current-affairs/jharkhand-four-gi-tags-tribal-crafts-2026/?pdf=1
- Vajiram & Ravi / IASGyan, *Padma Awards 2026*: https://vajiramandravi.com/current-affairs/padma-awards-2026-winners-list/
- The Week / Business Today, *Prada acknowledges Kolhapuri inspiration*: https://www.theweek.in/wire-updates/national/2025/06/28/bom5-mh-kolhapuri-prada.html

C. QUALITY-CONTROL SELF-AUDIT

Check	Status	Note
Verified the correct NID admission cycle?	✅	2027-28 cycle ("NID DAT 2027"); Prelims on 20 Dec 2026
Used the latest official NID information?	✅ with caveat	Handbook extracts for 2026-27 and 2027-28; the portal itself was blocked in my environment, so verify there
Analysed official sample papers?	⚠️ Partial	The official sample paper's existence and format are confirmed; its full contents could not be opened
Analysed previous papers where available?	✅ (memory-based)	2023–2026 reported papers analysed; no official past papers are published by NID
Identified recurring question types?	✅	Vol 1 §6–7, Vol 5
Covered every major topic?	✅	Syllabus A–H in Vol 1 §5; notes in Vol 2A–2C, 3
Created detailed notes?	✅	12-point format in Vol 2A; tables in 2B and 2C
Created a large question bank?	✅	150 + 150 + 100 + 100 + 100 + 200 + 200 = **1,100 MCQs** (Vol 4A–4G) + **100 current-affairs MCQs** (Vol 11) + **120 subjective prompts** (Vol 3) + 20 drill prompts (Vol 5)
Created expected-question categories?	✅	Tiers A, B, C in Vol 5, with reasons
Created exactly 5 complete model papers?	✅	Vol 6–10 (Papers 1–3 and 5 in Handbook format; Paper 4 in the 2026 all-subjective format)
Realistic NID-style questions in every paper?	✅	Mirror the reported themes: empathy, Indian context, signage, viewpoint, redesign

Check	Status	Note
Every question has an answer?	✓	Verified by count; every subjective question has a model approach and rubric
Every question has an explanation?	✓	Every MCQ carries an explanation line
Included subjective/drawing questions?	✓	Vol 3 §13; Part II of every paper
Design awareness, Indian art/culture/crafts, architecture, current affairs, visual reasoning, design thinking?	✓	Vol 2A, 2C, 4A, 4B, 4F, 4G, 11
Avoided unsupported claims; distinguished facts from predictions?	✓	Labels throughout; cutoffs marked unverified and not used as targets
Avoided repetitive questions?	✓ (best effort)	The banks are split by section and angle; some deliberate overlap for reinforcement (e.g. the ₹ symbol appears in notes, GK and flashcards)
Checked all answer keys?	✓	Hand-verified, with arithmetic re-computed (e.g. painted-cube, folding-coordinate and counting items)
Identified and fixed ambiguous questions?	✓	Rewritten during the audit: VR-4, 22, 30, 47, 48, 52, 58, 67, 79, 95, 108, 150; LR-52, LR-81; DA-21; GA-94; Paper 4 key 6(a)
No question without an answer, no answer without an explanation?	✓	—

Items to re-verify closer to the exam (time-sensitive)

1. The Mains weight split (60/40 last cycle) for 2027-28.
2. Whether Prelims 2027 keeps the 22 + 4 scheme or the 2026 all-subjective format (watch for any NID notice or updated sample paper).
3. Allowed materials (check the admit card).
4. Sarnath's World Heritage status (reported July 2026): confirm on whc.unesco.org.
5. The 2026 Venice Biennale award outcome (Golden Lions by public vote, reported).